WHAT IS THE IoT DESIGN DECISION ASSISTANT?

Comprehensive Knowledge Base: Access a base of knowledge of IoT domains, architectural solutions, quality requirements, and technologies, all curated from peer-reviewed research and industry best practices.
Intelligent Decision Support: Leverage our advanced technology to match your project requirements with optimal architectural solutions and technologies.
Quality-Driven Approach: Ensure your designs meet the highest standards by aligning them with established quality attributes and requirements specific to IoT systems.
Stay Current: Benefit from regularly updated content from the literature, reflecting the latest advancements in IoT technology and design methodologies.

Key Features:

•Interactive Design Explorer: Visually navigate through IoT domains, solutions, and technologies.
•Requirements Analyzer: Define and prioritize your project's quality requirements with ease.
•Solution Recommender: Receive tailored architectural recommendations based on your specific needs.
•Technology Evaluator: Compare and assess various IoT technologies to find the perfect fit for your project.
•Knowledge Contribution: Submit your own experiences and solutions to enrich the community's collective wisdom.

IoT Architecture Solution Knowledge Base

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WHAT IS QUALITY REQUIREMENT OR NOM-FUNCTIONAL REQURIMENTS?

Based on ISO 25010:2023, quality requirements are defined by eight main quality characteristics for software products and systems.

Here's the complete list with definitions for all quality characteristics and sub-characteristics from ISO 25010:2023:

1. Functional Suitability:

capability of a product to provide functions that meet stated and implied needs of intended users when it is used under specified conditions. Functional suitability is concerned with whether the functions meet not only stated and implied needs, but also the functional specification.
- Functional completeness: Degree to which functions cover all specified tasks/objectives
- Functional correctness: Degree to which functions provide correct results with needed precision
- Functional appropriateness: Degree to which functions facilitate task accomplishment

2. Performance Efficiency:

capability of a product to perform its functions within specified time and throughput parameters and be efficient in the use of resources under specified conditions. Resources can be CPU, memory, storage, and network devices. It also can include other software products, the software and hardware configuration of the system, energy, and materials (e.g. print paper, storage media).
- Time behavior: Response and processing times and throughput rates
- Resource utilization: Resources used (materials, CPU, memory, I/O devices)
- Capacity: Maximum limits (data storage, users, bandwidth) that meet requirements

3. Compatibility:

capability of a product to exchange information with other products, and/or to perform its required functions while sharing the same common environment and resources.
- Co-existence: Efficient performance while sharing environment with other products
- Interoperability: Exchange information and use information that has been exchanged

4. Interaction capability:

capability of a product to be interacted with by specified users to exchange information between a user and a system via the user interface to complete the intended task.
- Appropriateness recognizability: Users recognize if product meets their needs
- Learnability: Product can be used for specified learning goals
- Operability: Ease of operation and control
- User error protection: System protects against user errors
- User interface aesthetics: Pleasant and satisfying interaction
- Accessibility: Usable by people with widest range of characteristics
- Technical accessibility: Meets technical accessibility standards/guidelines

5. Reliability:

capability of a product to perform specified functions under specified conditions for a specified period of time without interruptions and failures.
- Maturity: Meets reliability needs under normal operation
- Availability: Operational and accessible when required
- Fault tolerance: Operates despite hardware/software faults
- Recoverability: Recover data and restore state after interruption

6. Security:

capability of a product to protect information and data so that persons or other products have the degree of data access appropriate to their types and levels of authorization, and to defend against attack patterns by malicious actors.
- Confidentiality: Data accessible only to authorized access
- Integrity: Prevents unauthorized data modification
- Non-repudiation: Actions can be proven to have occurred
- Accountability: Actions can be traced to specific entity
- Authenticity: Identity of subject/resource can be proven
- Auditability: Records can be analyzed to investigate events

- Prevention: Prevents unauthorized access, attacks, data breaches
- Privacy: Personal/private information protected per regulations

7. Maintainability:

capability of a product to be modified by the intended maintainers with effectiveness and efficiency.
- Modularity: Component changes have minimal impact on others
- Reusability: Assets can be used in multiple systems
- Analysability: Impact assessment effectiveness for modification
- Modifiability: Can be modified without quality degradation
- Testability: Test criteria can be established and tests performed

8. Flexibility:

capability of a product to be adapted to changes in its requirements, contexts of use, or system environment.
- Adaptability: Can adapt to different environments
- Installability: Can be installed/uninstalled effectively
- Replaceability: Can replace other specified software
- Scalability: capability of a product to handle growing or shrinking workloads or to adapt its capacity to handle variability

9. Safety:

capability of a product under defined conditions to avoid a state in which human life, health, property, or the environment is endangered.
- Operational Constraint: capability of a product to constrain its operation to within safe parameters or states when encountering operational hazard
- Risk Identification: capability of a product to identify a course of events or operations that can expose life, property or environment to unacceptable risk
- Fail Safe: capability of a product to automatically place itself in a safe operating mode, or to revert to a safe condition in the event of a failure
- Hazard Warning: capability of a product to provide warnings of unacceptable risks to operations or internal controls so that they can react in sufficient time to sustain safe operations
- Safe Integration: capability of a product to maintain safety (3.9) during and after integration with one or more components

Each characteristic should be specified with measurable criteria and acceptable thresholds when used as quality requirements.

For more details, visit - *https://www.iso.org/obp/ui/en/#iso:std:iso-iec:25010:ed-2:v1:en*

SOLUTIONS FOUND BY THIS RESEARCH

The solution is composed by the architecture named as A4IoT (Anyplace 4.0 IoT), proposed in the paper The Anyplace 4.0 IoT Localization Architecture, 2020, with D.O.I. 10.1109/MDM48529.2020.00045 (link https://ieeexplore.ieee.org/document/9162205). This solution addresses the following Quality Requirements: Flexibility,Interaction Capability,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and A4IoT (Anyplace 4.0 IoT).
Solution Architecture A4IoT (Anyplace 4.0 IoT): described as: It is an open-source architecture designed for IoT-based indoor localization using technologies like Wi-Fi, BLE, Cellular, UWB, and Computer Vision (CV). The system supports edge computing, timeseries databases, and crowdsourcing to manage localization tasks across various smart spaces, such as factories, healthcare facilities, and emergency response environments. Its modular and scalable design ensures integration across platforms like Android, Linux, Windows, and Robot OS, making it adaptable to diverse IoT ecosystems..
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Reliability is reached/addressed using/by Multi-node Cluster Deployment (Fault Tolerance - The production environment uses a 3-node cluster with two replicas, for both the database engines and the DFS (Distributed Filesystem). This setup allows for a full operation with just a single node active at any time.").
Security is reached/addressed using/by SSL certificates - Encrypted Communications (Confidentiality - All communication is encrypted even in internal networks, as A4IoT image automatically creates and uses SSL certificates.).
Interaction Capability is reached/addressed using/by Web-based Dashboard (Operability - Web Apps built with HTML5, CSS3, and AngularJS.").
Flexibility is reached/addressed using/by Docker containers (Adaptability - It utilizes Docker to enable the deployment of single-node or multi-node configurations, regardless of any dependencies or the underlying OS.).
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Solution Architecture Agricultural IoT Reference Architecture (AITRA): described as: .
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The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Blockchain-based Zero Trust.
Solution Architecture Blockchain-based Zero Trust: described as: The architecture proposed in "Blockchain-Based Zero Trust on the Edge" integrates Zero Trust principles with Blockchain to ensure secure access and trust management for Edge Computing in IoT environments. It focuses on continuous verification of users and devices, ensuring that all access is authenticated before being granted. Blockchain is used for decentralized trust management, maintaining an immutable record of activities, which enhances security and transparency. This setup is optimized for edge computing, reducing latency by processing data locally while maintaining the flexibility to scale across diverse IoT applications, such as smart cities and industrial IoT..
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Asynchronous processing for write operations improves system responsiveness under high loads.).
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The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.
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The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Blockchain-enabled.
Solution Architecture Blockchain-enabled: described as: .
Performance/Efficiency is reached/addressed using/by Routing protocol designed for IoT devices in the SDN controller [Yazdinejad et al. 2020] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, https://doi.org/10.1109/TSC.2020.2966970.

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Reference: An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, https://doi.org/10.1109/TSC.2020.2966970.
The solution is composed by the architecture named as Cloud-based Secure Biometric, proposed in the paper BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, with D.O.I. doi.org/10.1016/j.jksuci.2017.07.001 (link https://doi.org/10.1016/j.jksuci.2017.07.001). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Cloud-based Secure Biometric.
Solution Architecture Cloud-based Secure Biometric: described as: .
Flexibility is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Reliability is reached/addressed using/by MapReduce framework (No description.).
Security is reached/addressed using/by Cloud-based biometric authentication [Shakil et al. 2020] (No description.).
Reference: BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, https://doi.org/10.1016/j.jksuci.2017.07.001.

The solution is composed by the architecture named as A4IoT (Anyplace 4.0 IoT), proposed in the paper The Anyplace 4.0 IoT Localization Architecture, 2020, with D.O.I. 10.1109/MDM48529.2020.00045 (link https://ieeexplore.ieee.org/document/9162205). This solution addresses the following Quality Requirements: Flexibility,Interaction Capability,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and A4IoT (Anyplace 4.0 IoT).
Solution Architecture A4IoT (Anyplace 4.0 IoT): described as: It is an open-source architecture designed for IoT-based indoor localization using technologies like Wi-Fi, BLE, Cellular, UWB, and Computer Vision (CV). The system supports edge computing, timeseries databases, and crowdsourcing to manage localization tasks across various smart spaces, such as factories, healthcare facilities, and emergency response environments. Its modular and scalable design ensures integration across platforms like Android, Linux, Windows, and Robot OS, making it adaptable to diverse IoT ecosystems..
Performance/Efficiency is reached/addressed using/by Time-Series Database (InfluxDB) (Time Behavior - With temporal queries we can now consume high-volume input streams to provide real-time IoT tracking.").
Reliability is reached/addressed using/by Multi-node Cluster Deployment (Fault Tolerance - The production environment uses a 3-node cluster with two replicas, for both the database engines and the DFS (Distributed Filesystem). This setup allows for a full operation with just a single node active at any time.").
Security is reached/addressed using/by SSL certificates - Encrypted Communications (Confidentiality - All communication is encrypted even in internal networks, as A4IoT image automatically creates and uses SSL certificates.).
Interaction Capability is reached/addressed using/by Web-based Dashboard (Operability - Web Apps built with HTML5, CSS3, and AngularJS.").
Flexibility is reached/addressed using/by Docker containers (Adaptability - It utilizes Docker to enable the deployment of single-node or multi-node configurations, regardless of any dependencies or the underlying OS.).
Reference: The Anyplace 4.0 IoT Localization Architecture, 2020, https://ieeexplore.ieee.org/document/9162205.
The solution is composed by the architecture named as Agricultural IoT Reference Architecture (AITRA), proposed in the paper Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3031634 (link https://doi.org/10.1109/ACCESS.2020.3031634). This solution addresses the following Quality Requirements: Compatibility,Reliability,Security,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Agricultural IoT Reference Architecture (AITRA).
Solution Architecture Agricultural IoT Reference Architecture (AITRA): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Reliability is reached/addressed using/by Edge gateway (No description.).
Security is reached/addressed using/by Authentication, authorization, and data access control (No description.).
Reference: Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, https://doi.org/10.1109/ACCESS.2020.3031634.
The solution is composed by the architecture named as Blockchain-based Zero Trust, proposed in the paper Blockchain-Based Zero Trust on the Edge, 2023, with D.O.I. 10.1109/CSCI62032.2023.00167 (link https://ieeexplore.ieee.org/document/10590526). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Performance/Efficiency,Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Blockchain-based Zero Trust.
Solution Architecture Blockchain-based Zero Trust: described as: The architecture proposed in "Blockchain-Based Zero Trust on the Edge" integrates Zero Trust principles with Blockchain to ensure secure access and trust management for Edge Computing in IoT environments. It focuses on continuous verification of users and devices, ensuring that all access is authenticated before being granted. Blockchain is used for decentralized trust management, maintaining an immutable record of activities, which enhances security and transparency. This setup is optimized for edge computing, reducing latency by processing data locally while maintaining the flexibility to scale across diverse IoT applications, such as smart cities and industrial IoT..
Security is reached/addressed using/by Zero Trust Architecture (ZTA) (Confidentiality - Zero Trust Architecture (ZTA)

ZTA ensures continuous verification and monitoring of all network traffic, restricting unauthorized access.).
Security is reached/addressed using/by Blockchain Technology (Integrity - Blockchain serves as an immutable ledger for storing requests, ensuring data integrity and trustworthiness.).
Performance/Efficiency is reached/addressed using/by Redis Message Broker (Time Behavior - Redis Message Broker for Asynchronous Processing [redis.io]
Asynchronous processing for write operations improves system responsiveness under high loads.).
Maintainability is reached/addressed using/by Component isolation through microservices (Modularity -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.
).
Flexibility is reached/addressed using/by Component isolation through microservices (Scalability -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.).
Reference: Blockchain-Based Zero Trust on the Edge, 2023, https://ieeexplore.ieee.org/document/10590526.
The solution is composed by the architecture named as Blockchain-enabled, proposed in the paper An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, with D.O.I. doi.org/10.1109/TSC.2020.2966970 (link https://doi.org/10.1109/TSC.2020.2966970). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Blockchain-enabled.
Solution Architecture Blockchain-enabled: described as: .
Performance/Efficiency is reached/addressed using/by Routing protocol designed for IoT devices in the SDN controller [Yazdinejad et al. 2020] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, https://doi.org/10.1109/TSC.2020.2966970.
The solution is composed by the architecture named as Cloud-based Secure Biometric, proposed in the paper BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, with D.O.I. doi.org/10.1016/j.jksuci.2017.07.001 (link https://doi.org/10.1016/j.jksuci.2017.07.001). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Cloud-based Secure Biometric.
Solution Architecture Cloud-based Secure Biometric: described as: .
Flexibility is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Reliability is reached/addressed using/by MapReduce framework (No description.).
Security is reached/addressed using/by Cloud-based biometric authentication [Shakil et al. 2020] (No description.).
Reference: BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, https://doi.org/10.1016/j.jksuci.2017.07.001.
The solution is composed by the architecture named as Deep Learning based, proposed in the paper HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, with D.O.I. doi.org/10.1016/j.future.2019.10.043 (link https://doi.org/10.1016/j.future.2019.10.043). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Deep Learning based.
Solution Architecture Deep Learning based: described as: .
Performance/Efficiency is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Security is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Reference: HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, https://doi.org/10.1016/j.future.2019.10.043.

The solution is composed by the architecture named as A4IoT (Anyplace 4.0 IoT), proposed in the paper The Anyplace 4.0 IoT Localization Architecture, 2020, with D.O.I. 10.1109/MDM48529.2020.00045 (link https://ieeexplore.ieee.org/document/9162205). This solution addresses the following Quality Requirements: Flexibility,Interaction Capability,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and A4IoT (Anyplace 4.0 IoT).
Solution Architecture A4IoT (Anyplace 4.0 IoT): described as: It is an open-source architecture designed for IoT-based indoor localization using technologies like Wi-Fi, BLE, Cellular, UWB, and Computer Vision (CV). The system supports edge computing, timeseries databases, and crowdsourcing to manage localization tasks across various smart spaces, such as factories, healthcare facilities, and emergency response environments. Its modular and scalable design ensures integration across platforms like Android, Linux, Windows, and Robot OS, making it adaptable to diverse IoT ecosystems..
Performance/Efficiency is reached/addressed using/by Time-Series Database (InfluxDB) (Time Behavior - With temporal queries we can now consume high-volume input streams to provide real-time IoT tracking.").
Reliability is reached/addressed using/by Multi-node Cluster Deployment (Fault Tolerance - The production environment uses a 3-node cluster with two replicas, for both the database engines and the DFS (Distributed Filesystem). This setup allows for a full operation with just a single node active

at any time.").
Security is reached/addressed using/by SSL certificates - Encrypted Communications (Confidentiality - All communication is encrypted even in internal networks, as A4IoT image automatically creates and uses SSL certificates.).
Interaction Capability is reached/addressed using/by Web-based Dashboard (Operability - Web Apps built with HTML5, CSS3, and AngularJS.").
Flexibility is reached/addressed using/by Docker containers (Adaptability - It utilizes Docker to enable the deployment of single-node or multi-node configurations, regardless of any dependencies or the underlying OS.).
Reference: The Anyplace 4.0 IoT Localization Architecture, 2020, https://ieeexplore.ieee.org/document/9162205.
The solution is composed by the architecture named as Agricultural IoT Reference Architecture (AITRA), proposed in the paper Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3031634 (link https://doi.org/10.1109/ACCESS.2020.3031634). This solution addresses the following Quality Requirements: Compatibility,Reliability,Security,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Agricultural IoT Reference Architecture (AITRA).
Solution Architecture Agricultural IoT Reference Architecture (AITRA): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Reliability is reached/addressed using/by Edge gateway (No description.).
Security is reached/addressed using/by Authentication, authorization, and data access control (No description.).
Reference: Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, https://doi.org/10.1109/ACCESS.2020.3031634.
The solution is composed by the architecture named as Blockchain-based Zero Trust, proposed in the paper Blockchain-Based Zero Trust on the Edge, 2023, with D.O.I. 10.1109/CSCI62032.2023.00167 (link https://ieeexplore.ieee.org/document/10590526). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Performance/Efficiency,Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Blockchain-based Zero Trust.
Solution Architecture Blockchain-based Zero Trust: described as: The architecture proposed in "Blockchain-Based Zero Trust on the Edge" integrates Zero Trust principles with Blockchain to ensure secure access and trust management for Edge Computing in IoT environments. It focuses on continuous verification of users and devices, ensuring that all access is authenticated before being granted. Blockchain is used for decentralized trust management, maintaining an immutable record of activities, which enhances security and transparency. This setup is optimized for edge computing, reducing latency by processing data locally while maintaining the flexibility to scale across diverse IoT applications, such as smart cities and industrial IoT..
Security is reached/addressed using/by Zero Trust Architecture (ZTA) (Confidentiality - Zero Trust Architecture (ZTA)
ZTA ensures continuous verification and monitoring of all network traffic, restricting unauthorized access.).
Security is reached/addressed using/by Blockchain Technology (Integrity - Blockchain serves as an immutable ledger for storing requests, ensuring data integrity and trustworthiness.).
Performance/Efficiency is reached/addressed using/by Redis Message Broker (Time Behavior - Redis Message Broker for Asynchronous Processing [redis.io]
Asynchronous processing for write operations improves system responsiveness under high loads.).
Maintainability is reached/addressed using/by Component isolation through microservices (Modularity -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.
).
Flexibility is reached/addressed using/by Component isolation through microservices (Scalability -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.).
Reference: Blockchain-Based Zero Trust on the Edge, 2023, https://ieeexplore.ieee.org/document/10590526.
The solution is composed by the architecture named as Blockchain-enabled, proposed in the paper An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, with D.O.I. doi.org/10.1109/TSC.2020.2966970 (link https://doi.org/10.1109/TSC.2020.2966970). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Blockchain-enabled.
Solution Architecture Blockchain-enabled: described as: .
Performance/Efficiency is reached/addressed using/by Routing protocol designed for IoT devices in the SDN controller [Yazdinejad et al. 2020] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, https://doi.org/10.1109/TSC.2020.2966970.
The solution is composed by the architecture named as Cloud-based Secure Biometric, proposed in the paper BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, with D.O.I. doi.org/10.1016/j.jksuci.2017.07.001 (link https://doi.org/10.1016/j.jksuci.2017.07.001). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Cloud-based Secure Biometric.
Solution Architecture Cloud-based Secure Biometric: described as: .
Flexibility is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Reliability is reached/addressed using/by MapReduce framework (No description.).

Security is reached/addressed using/by Cloud-based biometric authentication [Shakil et al. 2020] (No description.).
Reference: BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, https://doi.org/10.1016/j.jksuci.2017.07.001.
The solution is composed by the architecture named as Deep Learning based, proposed in the paper HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, with D.O.I. doi.org/10.1016/j.future.2019.10.043 (link https://doi.org/10.1016/j.future.2019.10.043). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Deep Learning based.
Solution Architecture Deep Learning based: described as: .
Performance/Efficiency is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Security is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Reference: HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, https://doi.org/10.1016/j.future.2019.10.043.
The solution is composed by the architecture named as Distributed Blockchain-SDN based, proposed in the paper DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3039113 (link https://doi.org/10.1109/ACCESS.2020.3039113). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Distributed Blockchain-SDN based.
Solution Architecture Distributed Blockchain-SDN based: described as: .
Performance/Efficiency is reached/addressed using/by Decentralized blockchain nodes, dynamic SDN controllers, and edge computing (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, https://doi.org/10.1109/ACCESS.2020.3039113.

The solution is composed by the architecture named as A4IoT (Anyplace 4.0 IoT), proposed in the paper The Anyplace 4.0 IoT Localization Architecture, 2020, with D.O.I. 10.1109/MDM48529.2020.00045 (link https://ieeexplore.ieee.org/document/9162205). This solution addresses the following Quality Requirements: Flexibility,Interaction Capability,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and A4IoT (Anyplace 4.0 IoT).
Solution Architecture A4IoT (Anyplace 4.0 IoT): described as: It is an open-source architecture designed for IoT-based indoor localization using technologies like Wi-Fi, BLE, Cellular, UWB, and Computer Vision (CV). The system supports edge computing, timeseries databases, and crowdsourcing to manage localization tasks across various smart spaces, such as factories, healthcare facilities, and emergency response environments. Its modular and scalable design ensures integration across platforms like Android, Linux, Windows, and Robot OS, making it adaptable to diverse IoT ecosystems..
Performance/Efficiency is reached/addressed using/by Time-Series Database (InfluxDB) (Time Behavior - With temporal queries we can now consume high-volume input streams to provide real-time IoT tracking.").
Reliability is reached/addressed using/by Multi-node Cluster Deployment (Fault Tolerance - The production environment uses a 3-node cluster with two replicas, for both the database engines and the DFS (Distributed Filesystem). This setup allows for a full operation with just a single node active at any time.").
Security is reached/addressed using/by SSL certificates - Encrypted Communications (Confidentiality - All communication is encrypted even in internal networks, as A4IoT image automatically creates and uses SSL certificates.).
Interaction Capability is reached/addressed using/by Web-based Dashboard (Operability - Web Apps built with HTML5, CSS3, and AngularJS.").
Flexibility is reached/addressed using/by Docker containers (Adaptability - It utilizes Docker to enable the deployment of single-node or multi-node configurations, regardless of any dependencies or the underlying OS.).
Reference: The Anyplace 4.0 IoT Localization Architecture, 2020, https://ieeexplore.ieee.org/document/9162205.
The solution is composed by the architecture named as Agricultural IoT Reference Architecture (AITRA), proposed in the paper Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3031634 (link https://doi.org/10.1109/ACCESS.2020.3031634). This solution addresses the following Quality Requirements: Compatibility,Reliability,Security,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Agricultural IoT Reference Architecture (AITRA).
Solution Architecture Agricultural IoT Reference Architecture (AITRA): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Reliability is reached/addressed using/by Edge gateway (No description.).
Security is reached/addressed using/by Authentication, authorization, and data access control (No description.).
Reference: Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, https://doi.org/10.1109/ACCESS.2020.3031634.
The solution is composed by the architecture named as Blockchain-based Zero Trust, proposed in the paper Blockchain-Based Zero Trust on the Edge, 2023, with D.O.I. 10.1109/CSCI62032.2023.00167 (link https://ieeexplore.ieee.org/document/10590526). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Performance/Efficiency,Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Blockchain-

based Zero Trust.
Solution Architecture Blockchain-based Zero Trust: described as: The architecture proposed in "Blockchain-Based Zero Trust on the Edge" integrates Zero Trust principles with Blockchain to ensure secure access and trust management for Edge Computing in IoT environments. It focuses on continuous verification of users and devices, ensuring that all access is authenticated before being granted. Blockchain is used for decentralized trust management, maintaining an immutable record of activities, which enhances security and transparency. This setup is optimized for edge computing, reducing latency by processing data locally while maintaining the flexibility to scale across diverse IoT applications, such as smart cities and industrial IoT..
Security is reached/addressed using/by Zero Trust Architecture (ZTA) (Confidentiality - Zero Trust Architecture (ZTA)
ZTA ensures continuous verification and monitoring of all network traffic, restricting unauthorized access.).
Security is reached/addressed using/by Blockchain Technology (Integrity - Blockchain serves as an immutable ledger for storing requests, ensuring data integrity and trustworthiness.).
Performance/Efficiency is reached/addressed using/by Redis Message Broker (Time Behavior - Redis Message Broker for Asynchronous Processing [redis.io]
Asynchronous processing for write operations improves system responsiveness under high loads.).
Maintainability is reached/addressed using/by Component isolation through microservices (Modularity -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.
).
Flexibility is reached/addressed using/by Component isolation through microservices (Scalability -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.).
Reference: Blockchain-Based Zero Trust on the Edge, 2023, https://ieeexplore.ieee.org/document/10590526.
The solution is composed by the architecture named as Blockchain-enabled, proposed in the paper An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, with D.O.I. doi.org/10.1109/TSC.2020.2966970 (link https://doi.org/10.1109/TSC.2020.2966970). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Blockchain-enabled.
Solution Architecture Blockchain-enabled: described as: .
Performance/Efficiency is reached/addressed using/by Routing protocol designed for IoT devices in the SDN controller [Yazdinejad et al. 2020] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, https://doi.org/10.1109/TSC.2020.2966970.
The solution is composed by the architecture named as Cloud-based Secure Biometric, proposed in the paper BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, with D.O.I. doi.org/10.1016/j.jksuci.2017.07.001 (link https://doi.org/10.1016/j.jksuci.2017.07.001). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Cloud-based Secure Biometric.
Solution Architecture Cloud-based Secure Biometric: described as: .
Flexibility is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Reliability is reached/addressed using/by MapReduce framework (No description.).
Security is reached/addressed using/by Cloud-based biometric authentication [Shakil et al. 2020] (No description.).
Reference: BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, https://doi.org/10.1016/j.jksuci.2017.07.001.
The solution is composed by the architecture named as Deep Learning based, proposed in the paper HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, with D.O.I. doi.org/10.1016/j.future.2019.10.043 (link https://doi.org/10.1016/j.future.2019.10.043). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Deep Learning based.
Solution Architecture Deep Learning based: described as: .
Performance/Efficiency is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Security is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Reference: HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, https://doi.org/10.1016/j.future.2019.10.043.
The solution is composed by the architecture named as Distributed Blockchain-SDN based, proposed in the paper DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3039113 (link https://doi.org/10.1109/ACCESS.2020.3039113). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Distributed Blockchain-SDN based.
Solution Architecture Distributed Blockchain-SDN based: described as: .
Performance/Efficiency is reached/addressed using/by Decentralized blockchain nodes, dynamic SDN controllers, and edge computing (No description.).

Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, https://doi.org/10.1109/ACCESS.2020.3039113.
The solution is composed by the architecture named as Distributed Edge-Cloud, proposed in the paper Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, with D.O.I. doi.org/10.1016/j.future.2020.05.042 (link https://doi.org/10.1016/j.future.2020.05.042). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Distributed Edge-Cloud.
Solution Architecture Distributed Edge-Cloud: described as: .
Security is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Reference: Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, https://doi.org/10.1016/j.future.2020.05.042.

The solution is composed by the architecture named as A4IoT (Anyplace 4.0 IoT), proposed in the paper The Anyplace 4.0 IoT Localization Architecture, 2020, with D.O.I. 10.1109/MDM48529.2020.00045 (link https://ieeexplore.ieee.org/document/9162205). This solution addresses the following Quality Requirements: Flexibility,Interaction Capability,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and A4IoT (Anyplace 4.0 IoT).
Solution Architecture A4IoT (Anyplace 4.0 IoT): described as: It is an open-source architecture designed for IoT-based indoor localization using technologies like Wi-Fi, BLE, Cellular, UWB, and Computer Vision (CV). The system supports edge computing, timeseries databases, and crowdsourcing to manage localization tasks across various smart spaces, such as factories, healthcare facilities, and emergency response environments. Its modular and scalable design ensures integration across platforms like Android, Linux, Windows, and Robot OS, making it adaptable to diverse IoT ecosystems..
Performance/Efficiency is reached/addressed using/by Time-Series Database (InfluxDB) (Time Behavior - With temporal queries we can now consume high-volume input streams to provide real-time IoT tracking.").
Reliability is reached/addressed using/by Multi-node Cluster Deployment (Fault Tolerance - The production environment uses a 3-node cluster with two replicas, for both the database engines and the DFS (Distributed Filesystem). This setup allows for a full operation with just a single node active at any time.").
Security is reached/addressed using/by SSL certificates - Encrypted Communications (Confidentiality - All communication is encrypted even in internal networks, as A4IoT image automatically creates and uses SSL certificates.).
Interaction Capability is reached/addressed using/by Web-based Dashboard (Operability - Web Apps built with HTML5, CSS3, and AngularJS.").
Flexibility is reached/addressed using/by Docker containers (Adaptability - It utilizes Docker to enable the deployment of single-node or multi-node configurations, regardless of any dependencies or the underlying OS.).
Reference: The Anyplace 4.0 IoT Localization Architecture, 2020, https://ieeexplore.ieee.org/document/9162205.
The solution is composed by the architecture named as Agricultural IoT Reference Architecture (AITRA), proposed in the paper Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3031634 (link https://doi.org/10.1109/ACCESS.2020.3031634). This solution addresses the following Quality Requirements: Compatibility,Reliability,Security,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Agricultural IoT Reference Architecture (AITRA).
Solution Architecture Agricultural IoT Reference Architecture (AITRA): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Reliability is reached/addressed using/by Edge gateway (No description.).
Security is reached/addressed using/by Authentication, authorization, and data access control (No description.).
Reference: Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, https://doi.org/10.1109/ACCESS.2020.3031634.
The solution is composed by the architecture named as Blockchain-based Zero Trust, proposed in the paper Blockchain-Based Zero Trust on the Edge, 2023, with D.O.I. 10.1109/CSCI62032.2023.00167 (link https://ieeexplore.ieee.org/document/10590526). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Performance/Efficiency,Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Blockchain-based Zero Trust.
Solution Architecture Blockchain-based Zero Trust: described as: The architecture proposed in "Blockchain-Based Zero Trust on the Edge" integrates Zero Trust principles with Blockchain to ensure secure access and trust management for Edge Computing in IoT environments. It focuses on continuous verification of users and devices, ensuring that all access is authenticated before being granted. Blockchain is used for decentralized trust management, maintaining an immutable record of activities, which enhances security and transparency. This setup is optimized for edge computing, reducing latency by processing data locally while maintaining the flexibility to scale across diverse IoT applications, such as smart cities and industrial IoT..
Security is reached/addressed using/by Zero Trust Architecture (ZTA) (Confidentiality - Zero Trust Architecture (ZTA)
ZTA ensures continuous verification and monitoring of all network traffic, restricting unauthorized access.).
Security is reached/addressed using/by Blockchain Technology (Integrity - Blockchain serves as an immutable ledger for storing requests, ensuring data integrity and trustworthiness.).

Performance/Efficiency is reached/addressed using/by Redis Message Broker (Time Behavior - Redis Message Broker for Asynchronous Processing [redis.io]
Asynchronous processing for write operations improves system responsiveness under high loads.).
Maintainability is reached/addressed using/by Component isolation through microservices (Modularity -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.
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Flexibility is reached/addressed using/by Component isolation through microservices (Scalability -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.).
Reference: Blockchain-Based Zero Trust on the Edge, 2023, https://ieeexplore.ieee.org/document/10590526.
The solution is composed by the architecture named as Blockchain-enabled, proposed in the paper An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, with D.O.I. doi.org/10.1109/TSC.2020.2966970 (link https://doi.org/10.1109/TSC.2020.2966970). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Blockchain-enabled.
Solution Architecture Blockchain-enabled: described as: .
Performance/Efficiency is reached/addressed using/by Routing protocol designed for IoT devices in the SDN controller [Yazdinejad et al. 2020] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, https://doi.org/10.1109/TSC.2020.2966970.
The solution is composed by the architecture named as Cloud-based Secure Biometric, proposed in the paper BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, with D.O.I. doi.org/10.1016/j.jksuci.2017.07.001 (link https://doi.org/10.1016/j.jksuci.2017.07.001). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Cloud-based Secure Biometric.
Solution Architecture Cloud-based Secure Biometric: described as: .
Flexibility is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Reliability is reached/addressed using/by MapReduce framework (No description.).
Security is reached/addressed using/by Cloud-based biometric authentication [Shakil et al. 2020] (No description.).
Reference: BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, https://doi.org/10.1016/j.jksuci.2017.07.001.
The solution is composed by the architecture named as Deep Learning based, proposed in the paper HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, with D.O.I. doi.org/10.1016/j.future.2019.10.043 (link https://doi.org/10.1016/j.future.2019.10.043). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Deep Learning based.
Solution Architecture Deep Learning based: described as: .
Performance/Efficiency is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Security is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Reference: HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, https://doi.org/10.1016/j.future.2019.10.043.
The solution is composed by the architecture named as Distributed Blockchain-SDN based, proposed in the paper DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3039113 (link https://doi.org/10.1109/ACCESS.2020.3039113). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Distributed Blockchain-SDN based.
Solution Architecture Distributed Blockchain-SDN based: described as: .
Performance/Efficiency is reached/addressed using/by Decentralized blockchain nodes, dynamic SDN controllers, and edge computing (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, https://doi.org/10.1109/ACCESS.2020.3039113.
The solution is composed by the architecture named as Distributed Edge-Cloud, proposed in the paper Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, with D.O.I. doi.org/10.1016/j.future.2020.05.042 (link https://doi.org/10.1016/j.future.2020.05.042). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Distributed Edge-Cloud.
Solution Architecture Distributed Edge-Cloud: described as: .
Security is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).

Performance/Efficiency is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Reference: Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, https://doi.org/10.1016/j.future.2020.05.042.
The solution is composed by the architecture named as Edge Computing-based Fault Tolerant Framework, proposed in the paper Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, with D.O.I. doi.org/10.1109/IWCMC48107.2020.9148269 (link https://doi.org/10.1109/IWCMC48107.2020.9148269). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Edge Computing-based Fault Tolerant Framework.
Solution Architecture Edge Computing-based Fault Tolerant Framework: described as: .
Performance/Efficiency is reached/addressed using/by Kubernetes framework (No description.).
Flexibility is reached/addressed using/by Docker containers (No description.).
Reference: Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, https://doi.org/10.1109/IWCMC48107.2020.9148269.

The solution is composed by the architecture named as A4IoT (Anyplace 4.0 IoT), proposed in the paper The Anyplace 4.0 IoT Localization Architecture, 2020, with D.O.I. 10.1109/MDM48529.2020.00045 (link https://ieeexplore.ieee.org/document/9162205). This solution addresses the following Quality Requirements: Flexibility,Interaction Capability,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and A4IoT (Anyplace 4.0 IoT).
Solution Architecture A4IoT (Anyplace 4.0 IoT): described as: It is an open-source architecture designed for IoT-based indoor localization using technologies like Wi-Fi, BLE, Cellular, UWB, and Computer Vision (CV). The system supports edge computing, timeseries databases, and crowdsourcing to manage localization tasks across various smart spaces, such as factories, healthcare facilities, and emergency response environments. Its modular and scalable design ensures integration across platforms like Android, Linux, Windows, and Robot OS, making it adaptable to diverse IoT ecosystems..
Performance/Efficiency is reached/addressed using/by Time-Series Database (InfluxDB) (Time Behavior - With temporal queries we can now consume high-volume input streams to provide real-time IoT tracking.").
Reliability is reached/addressed using/by Multi-node Cluster Deployment (Fault Tolerance - The production environment uses a 3-node cluster with two replicas, for both the database engines and the DFS (Distributed Filesystem). This setup allows for a full operation with just a single node active at any time.").
Security is reached/addressed using/by SSL certificates - Encrypted Communications (Confidentiality - All communication is encrypted even in internal networks, as A4IoT image automatically creates and uses SSL certificates.).
Interaction Capability is reached/addressed using/by Web-based Dashboard (Operability - Web Apps built with HTML5, CSS3, and AngularJS.").
Flexibility is reached/addressed using/by Docker containers (Adaptability - It utilizes Docker to enable the deployment of single-node or multi-node configurations, regardless of any dependencies or the underlying OS.).
Reference: The Anyplace 4.0 IoT Localization Architecture, 2020, https://ieeexplore.ieee.org/document/9162205.
The solution is composed by the architecture named as Agricultural IoT Reference Architecture (AITRA), proposed in the paper Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3031634 (link https://doi.org/10.1109/ACCESS.2020.3031634). This solution addresses the following Quality Requirements: Compatibility,Reliability,Security,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Agricultural IoT Reference Architecture (AITRA).
Solution Architecture Agricultural IoT Reference Architecture (AITRA): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Reliability is reached/addressed using/by Edge gateway (No description.).
Security is reached/addressed using/by Authentication, authorization, and data access control (No description.).
Reference: Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, https://doi.org/10.1109/ACCESS.2020.3031634.
The solution is composed by the architecture named as Blockchain-based Zero Trust, proposed in the paper Blockchain-Based Zero Trust on the Edge, 2023, with D.O.I. 10.1109/CSCI62032.2023.00167 (link https://ieeexplore.ieee.org/document/10590526). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Performance/Efficiency,Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Blockchain-based Zero Trust.
Solution Architecture Blockchain-based Zero Trust: described as: The architecture proposed in "Blockchain-Based Zero Trust on the Edge" integrates Zero Trust principles with Blockchain to ensure secure access and trust management for Edge Computing in IoT environments. It focuses on continuous verification of users and devices, ensuring that all access is authenticated before being granted. Blockchain is used for decentralized trust management, maintaining an immutable record of activities, which enhances security and transparency. This setup is optimized for edge computing, reducing latency by processing data locally while maintaining the flexibility to scale across diverse IoT applications, such as smart cities and industrial IoT..
Security is reached/addressed using/by Zero Trust Architecture (ZTA) (Confidentiality - Zero Trust Architecture (ZTA)
ZTA ensures continuous verification and monitoring of all network traffic, restricting unauthorized access.).
Security is reached/addressed using/by Blockchain Technology (Integrity - Blockchain serves as an immutable ledger for storing requests, ensuring data integrity and trustworthiness.).

Performance/Efficiency is reached/addressed using/by Redis Message Broker (Time Behavior - Redis Message Broker for Asynchronous Processing [redis.io]
Asynchronous processing for write operations improves system responsiveness under high loads.).
Maintainability is reached/addressed using/by Component isolation through microservices (Modularity -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.
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Flexibility is reached/addressed using/by Component isolation through microservices (Scalability -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.).
Reference: Blockchain-Based Zero Trust on the Edge, 2023, https://ieeexplore.ieee.org/document/10590526.
The solution is composed by the architecture named as Blockchain-enabled, proposed in the paper An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, with D.O.I. doi.org/10.1109/TSC.2020.2966970 (link https://doi.org/10.1109/TSC.2020.2966970). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Blockchain-enabled.
Solution Architecture Blockchain-enabled: described as: .
Performance/Efficiency is reached/addressed using/by Routing protocol designed for IoT devices in the SDN controller [Yazdinejad et al. 2020] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, https://doi.org/10.1109/TSC.2020.2966970.
The solution is composed by the architecture named as Cloud-based Secure Biometric, proposed in the paper BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, with D.O.I. doi.org/10.1016/j.jksuci.2017.07.001 (link https://doi.org/10.1016/j.jksuci.2017.07.001). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Cloud-based Secure Biometric.
Solution Architecture Cloud-based Secure Biometric: described as: .
Flexibility is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Reliability is reached/addressed using/by MapReduce framework (No description.).
Security is reached/addressed using/by Cloud-based biometric authentication [Shakil et al. 2020] (No description.).
Reference: BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, https://doi.org/10.1016/j.jksuci.2017.07.001.
The solution is composed by the architecture named as Deep Learning based, proposed in the paper HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, with D.O.I. doi.org/10.1016/j.future.2019.10.043 (link https://doi.org/10.1016/j.future.2019.10.043). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Deep Learning based.
Solution Architecture Deep Learning based: described as: .
Performance/Efficiency is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Security is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Reference: HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, https://doi.org/10.1016/j.future.2019.10.043.
The solution is composed by the architecture named as Distributed Blockchain-SDN based, proposed in the paper DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3039113 (link https://doi.org/10.1109/ACCESS.2020.3039113). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Distributed Blockchain-SDN based.
Solution Architecture Distributed Blockchain-SDN based: described as: .
Performance/Efficiency is reached/addressed using/by Decentralized blockchain nodes, dynamic SDN controllers, and edge computing (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, https://doi.org/10.1109/ACCESS.2020.3039113.
The solution is composed by the architecture named as Distributed Edge-Cloud, proposed in the paper Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, with D.O.I. doi.org/10.1016/j.future.2020.05.042 (link https://doi.org/10.1016/j.future.2020.05.042). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Distributed Edge-Cloud.
Solution Architecture Distributed Edge-Cloud: described as: .
Security is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).

Performance/Efficiency is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Reference: Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, https://doi.org/10.1016/j.future.2020.05.042.
The solution is composed by the architecture named as Edge Computing-based Fault Tolerant Framework, proposed in the paper Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, with D.O.I. doi.org/10.1109/IWCMC48107.2020.9148269 (link https://doi.org/10.1109/IWCMC48107.2020.9148269). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Edge Computing-based Fault Tolerant Framework.
Solution Architecture Edge Computing-based Fault Tolerant Framework: described as: .
Performance/Efficiency is reached/addressed using/by Kubernetes framework (No description.).
Flexibility is reached/addressed using/by Docker containers (No description.).
Reference: Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, https://doi.org/10.1109/IWCMC48107.2020.9148269.
The solution is composed by the architecture named as Event-driven IoT, proposed in the paper Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, with D.O.I. doi.org/10.1007/s10586-020-03189-w (link https://doi.org/10.1007/s10586-020-03189-w). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Event-driven IoT.
Solution Architecture Event-driven IoT: described as: .
Performance/Efficiency is reached/addressed using/by Complex Event Processing (CEP) [Rahmani et al. 2020] (No description.).
Reliability is reached/addressed using/by TCP-IP protocol on Client/Server architecture for data communication (No description.).
Reference: Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, https://doi.org/10.1007/s10586-020-03189-w.

The solution is composed by the architecture named as A4IoT (Anyplace 4.0 IoT), proposed in the paper The Anyplace 4.0 IoT Localization Architecture, 2020, with D.O.I. 10.1109/MDM48529.2020.00045 (link https://ieeexplore.ieee.org/document/9162205). This solution addresses the following Quality Requirements: Flexibility,Interaction Capability,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and A4IoT (Anyplace 4.0 IoT).
Solution Architecture A4IoT (Anyplace 4.0 IoT): described as: It is an open-source architecture designed for IoT-based indoor localization using technologies like Wi-Fi, BLE, Cellular, UWB, and Computer Vision (CV). The system supports edge computing, timeseries databases, and crowdsourcing to manage localization tasks across various smart spaces, such as factories, healthcare facilities, and emergency response environments. Its modular and scalable design ensures integration across platforms like Android, Linux, Windows, and Robot OS, making it adaptable to diverse IoT ecosystems..
Performance/Efficiency is reached/addressed using/by Time-Series Database (InfluxDB) (Time Behavior - With temporal queries we can now consume high-volume input streams to provide real-time IoT tracking.").
Reliability is reached/addressed using/by Multi-node Cluster Deployment (Fault Tolerance - The production environment uses a 3-node cluster with two replicas, for both the database engines and the DFS (Distributed Filesystem). This setup allows for a full operation with just a single node active at any time.").
Security is reached/addressed using/by SSL certificates - Encrypted Communications (Confidentiality - All communication is encrypted even in internal networks, as A4IoT image automatically creates and uses SSL certificates.).
Interaction Capability is reached/addressed using/by Web-based Dashboard (Operability - Web Apps built with HTML5, CSS3, and AngularJS.").
Flexibility is reached/addressed using/by Docker containers (Adaptability - It utilizes Docker to enable the deployment of single-node or multi-node configurations, regardless of any dependencies or the underlying OS.).
Reference: The Anyplace 4.0 IoT Localization Architecture, 2020, https://ieeexplore.ieee.org/document/9162205.
The solution is composed by the architecture named as Agricultural IoT Reference Architecture (AITRA), proposed in the paper Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3031634 (link https://doi.org/10.1109/ACCESS.2020.3031634). This solution addresses the following Quality Requirements: Compatibility,Reliability,Security,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Agricultural IoT Reference Architecture (AITRA).
Solution Architecture Agricultural IoT Reference Architecture (AITRA): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Reliability is reached/addressed using/by Edge gateway (No description.).
Security is reached/addressed using/by Authentication, authorization, and data access control (No description.).
Reference: Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, https://doi.org/10.1109/ACCESS.2020.3031634.
The solution is composed by the architecture named as Blockchain-based Zero Trust, proposed in the paper Blockchain-Based Zero Trust on the Edge, 2023, with D.O.I. 10.1109/CSCI62032.2023.00167 (link https://ieeexplore.ieee.org/document/10590526). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Performance/Efficiency,Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Blockchain-based Zero Trust.

Solution Architecture Blockchain-based Zero Trust: described as: The architecture proposed in "Blockchain-Based Zero Trust on the Edge" integrates Zero Trust principles with Blockchain to ensure secure access and trust management for Edge Computing in IoT environments. It focuses on continuous verification of users and devices, ensuring that all access is authenticated before being granted. Blockchain is used for decentralized trust management, maintaining an immutable record of activities, which enhances security and transparency. This setup is optimized for edge computing, reducing latency by processing data locally while maintaining the flexibility to scale across diverse IoT applications, such as smart cities and industrial IoT..
Security is reached/addressed using/by Zero Trust Architecture (ZTA) (Confidentiality - Zero Trust Architecture (ZTA)
ZTA ensures continuous verification and monitoring of all network traffic, restricting unauthorized access.).
Security is reached/addressed using/by Blockchain Technology (Integrity - Blockchain serves as an immutable ledger for storing requests, ensuring data integrity and trustworthiness.).
Performance/Efficiency is reached/addressed using/by Redis Message Broker (Time Behavior - Redis Message Broker for Asynchronous Processing [redis.io]
Asynchronous processing for write operations improves system responsiveness under high loads.).
Maintainability is reached/addressed using/by Component isolation through microservices (Modularity -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.
).
Flexibility is reached/addressed using/by Component isolation through microservices (Scalability -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.).
Reference: Blockchain-Based Zero Trust on the Edge, 2023, https://ieeexplore.ieee.org/document/10590526.
The solution is composed by the architecture named as Blockchain-enabled, proposed in the paper An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, with D.O.I. doi.org/10.1109/TSC.2020.2966970 (link https://doi.org/10.1109/TSC.2020.2966970). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Blockchain-enabled.
Solution Architecture Blockchain-enabled: described as: .
Performance/Efficiency is reached/addressed using/by Routing protocol designed for IoT devices in the SDN controller [Yazdinejad et al. 2020] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, https://doi.org/10.1109/TSC.2020.2966970.
The solution is composed by the architecture named as Cloud-based Secure Biometric, proposed in the paper BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, with D.O.I. doi.org/10.1016/j.jksuci.2017.07.001 (link https://doi.org/10.1016/j.jksuci.2017.07.001). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Cloud-based Secure Biometric.
Solution Architecture Cloud-based Secure Biometric: described as: .
Flexibility is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Reliability is reached/addressed using/by MapReduce framework (No description.).
Security is reached/addressed using/by Cloud-based biometric authentication [Shakil et al. 2020] (No description.).
Reference: BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, https://doi.org/10.1016/j.jksuci.2017.07.001.
The solution is composed by the architecture named as Deep Learning based, proposed in the paper HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, with D.O.I. doi.org/10.1016/j.future.2019.10.043 (link https://doi.org/10.1016/j.future.2019.10.043). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Deep Learning based.
Solution Architecture Deep Learning based: described as: .
Performance/Efficiency is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Security is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Reference: HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, https://doi.org/10.1016/j.future.2019.10.043.
The solution is composed by the architecture named as Distributed Blockchain-SDN based, proposed in the paper DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3039113 (link https://doi.org/10.1109/ACCESS.2020.3039113). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Distributed Blockchain-SDN based.
Solution Architecture Distributed Blockchain-SDN based: described as: .
Performance/Efficiency is reached/addressed using/by Decentralized blockchain nodes, dynamic SDN controllers, and edge computing (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).

Reference: DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, https://doi.org/10.1109/ACCESS.2020.3039113.
The solution is composed by the architecture named as Distributed Edge-Cloud, proposed in the paper Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, with D.O.I. doi.org/10.1016/j.future.2020.05.042 (link https://doi.org/10.1016/j.future.2020.05.042). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Distributed Edge-Cloud.
Solution Architecture Distributed Edge-Cloud: described as: .
Security is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Reference: Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, https://doi.org/10.1016/j.future.2020.05.042.
The solution is composed by the architecture named as Edge Computing-based Fault Tolerant Framework, proposed in the paper Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, with D.O.I. doi.org/10.1109/IWCMC48107.2020.9148269 (link https://doi.org/10.1109/IWCMC48107.2020.9148269). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Edge Computing-based Fault Tolerant Framework.
Solution Architecture Edge Computing-based Fault Tolerant Framework: described as: .
Performance/Efficiency is reached/addressed using/by Kubernetes framework (No description.).
Flexibility is reached/addressed using/by Docker containers (No description.).
Reference: Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, https://doi.org/10.1109/IWCMC48107.2020.9148269.
The solution is composed by the architecture named as Event-driven IoT, proposed in the paper Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, with D.O.I. doi.org/10.1007/s10586-020-03189-w (link https://doi.org/10.1007/s10586-020-03189-w). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Event-driven IoT.
Solution Architecture Event-driven IoT: described as: .
Performance/Efficiency is reached/addressed using/by Complex Event Processing (CEP) [Rahmani et al. 2020] (No description.).
Reliability is reached/addressed using/by TCP-IP protocol on Client/Server architecture for data communication (No description.).
Reference: Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, https://doi.org/10.1007/s10586-020-03189-w.
The solution is composed by the architecture named as Fog Computing based, proposed in the paper FOG-based architecture and load balancing methodology for health monitoring systems, 2021, with D.O.I. doi.org/10.1109/ACCESS.2021.3094033 (link https://doi.org/10.1109/ACCESS.2021.3094033). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Fog Computing based.
Solution Architecture Fog Computing based: described as: .
Performance/Efficiency is reached/addressed using/by Load Balancing Scheme (LBS) - Distributed workload to all fog nodes [Asghar et al. 2021] (No description.).
Reference: FOG-based architecture and load balancing methodology for health monitoring systems, 2021, https://doi.org/10.1109/ACCESS.2021.3094033.

The solution is composed by the architecture named as A4IoT (Anyplace 4.0 IoT), proposed in the paper The Anyplace 4.0 IoT Localization Architecture, 2020, with D.O.I. 10.1109/MDM48529.2020.00045 (link https://ieeexplore.ieee.org/document/9162205). This solution addresses the following Quality Requirements: Flexibility,Interaction Capability,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and A4IoT (Anyplace 4.0 IoT).
Solution Architecture A4IoT (Anyplace 4.0 IoT): described as: It is an open-source architecture designed for IoT-based indoor localization using technologies like Wi-Fi, BLE, Cellular, UWB, and Computer Vision (CV). The system supports edge computing, timeseries databases, and crowdsourcing to manage localization tasks across various smart spaces, such as factories, healthcare facilities, and emergency response environments. Its modular and scalable design ensures integration across platforms like Android, Linux, Windows, and Robot OS, making it adaptable to diverse IoT ecosystems..
Performance/Efficiency is reached/addressed using/by Time-Series Database (InfluxDB) (Time Behavior - With temporal queries we can now consume high-volume input streams to provide real-time IoT tracking.").
Reliability is reached/addressed using/by Multi-node Cluster Deployment (Fault Tolerance - The production environment uses a 3-node cluster with two replicas, for both the database engines and the DFS (Distributed Filesystem). This setup allows for a full operation with just a single node active at any time.").
Security is reached/addressed using/by SSL certificates - Encrypted Communications (Confidentiality - All communication is encrypted even in internal networks, as A4IoT image automatically creates and uses SSL certificates.).

Interaction Capability is reached/addressed using/by Web-based Dashboard (Operability - Web Apps built with HTML5, CSS3, and AngularJS.").
Flexibility is reached/addressed using/by Docker containers (Adaptability - It utilizes Docker to enable the deployment of single-node or multi-node configurations, regardless of any dependencies or the underlying OS.).
Reference: The Anyplace 4.0 IoT Localization Architecture, 2020, https://ieeexplore.ieee.org/document/9162205.
The solution is composed by the architecture named as Agricultural IoT Reference Architecture (AITRA), proposed in the paper Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3031634 (link https://doi.org/10.1109/ACCESS.2020.3031634). This solution addresses the following Quality Requirements: Compatibility,Reliability,Security,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Agricultural IoT Reference Architecture (AITRA).
Solution Architecture Agricultural IoT Reference Architecture (AITRA): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Reliability is reached/addressed using/by Edge gateway (No description.).
Security is reached/addressed using/by Authentication, authorization, and data access control (No description.).
Reference: Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, https://doi.org/10.1109/ACCESS.2020.3031634.
The solution is composed by the architecture named as Blockchain-based Zero Trust, proposed in the paper Blockchain-Based Zero Trust on the Edge, 2023, with D.O.I. 10.1109/CSCI62032.2023.00167 (link https://ieeexplore.ieee.org/document/10590526). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Performance/Efficiency,Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Blockchain-based Zero Trust.
Solution Architecture Blockchain-based Zero Trust: described as: The architecture proposed in "Blockchain-Based Zero Trust on the Edge" integrates Zero Trust principles with Blockchain to ensure secure access and trust management for Edge Computing in IoT environments. It focuses on continuous verification of users and devices, ensuring that all access is authenticated before being granted. Blockchain is used for decentralized trust management, maintaining an immutable record of activities, which enhances security and transparency. This setup is optimized for edge computing, reducing latency by processing data locally while maintaining the flexibility to scale across diverse IoT applications, such as smart cities and industrial IoT..
Security is reached/addressed using/by Zero Trust Architecture (ZTA) (Confidentiality - Zero Trust Architecture (ZTA)
ZTA ensures continuous verification and monitoring of all network traffic, restricting unauthorized access.).
Security is reached/addressed using/by Blockchain Technology (Integrity - Blockchain serves as an immutable ledger for storing requests, ensuring data integrity and trustworthiness.).
Performance/Efficiency is reached/addressed using/by Redis Message Broker (Time Behavior - Redis Message Broker for Asynchronous Processing [redis.io]
Asynchronous processing for write operations improves system responsiveness under high loads.).
Maintainability is reached/addressed using/by Component isolation through microservices (Modularity -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.
).
Flexibility is reached/addressed using/by Component isolation through microservices (Scalability -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.).
Reference: Blockchain-Based Zero Trust on the Edge, 2023, https://ieeexplore.ieee.org/document/10590526.
The solution is composed by the architecture named as Blockchain-enabled, proposed in the paper An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, with D.O.I. doi.org/10.1109/TSC.2020.2966970 (link https://doi.org/10.1109/TSC.2020.2966970). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Blockchain-enabled.
Solution Architecture Blockchain-enabled: described as: .
Performance/Efficiency is reached/addressed using/by Routing protocol designed for IoT devices in the SDN controller [Yazdinejad et al. 2020] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, https://doi.org/10.1109/TSC.2020.2966970.
The solution is composed by the architecture named as Cloud-based Secure Biometric, proposed in the paper BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, with D.O.I. doi.org/10.1016/j.jksuci.2017.07.001 (link https://doi.org/10.1016/j.jksuci.2017.07.001). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Cloud-based Secure Biometric.
Solution Architecture Cloud-based Secure Biometric: described as: .
Flexibility is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Reliability is reached/addressed using/by MapReduce framework (No description.).
Security is reached/addressed using/by Cloud-based biometric authentication [Shakil et al. 2020] (No description.).
Reference: BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, https://doi.org/10.1016/j.jksuci.2017.07.001.

The solution is composed by the architecture named as Deep Learning based, proposed in the paper HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, with D.O.I. doi.org/10.1016/j.future.2019.10.043 (link https://doi.org/10.1016/j.future.2019.10.043). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Deep Learning based.
Solution Architecture Deep Learning based: described as: .
Performance/Efficiency is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Security is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Reference: HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, https://doi.org/10.1016/j.future.2019.10.043.
The solution is composed by the architecture named as Distributed Blockchain-SDN based, proposed in the paper DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3039113 (link https://doi.org/10.1109/ACCESS.2020.3039113). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Distributed Blockchain-SDN based.
Solution Architecture Distributed Blockchain-SDN based: described as: .
Performance/Efficiency is reached/addressed using/by Decentralized blockchain nodes, dynamic SDN controllers, and edge computing (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, https://doi.org/10.1109/ACCESS.2020.3039113.
The solution is composed by the architecture named as Distributed Edge-Cloud, proposed in the paper Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, with D.O.I. doi.org/10.1016/j.future.2020.05.042 (link https://doi.org/10.1016/j.future.2020.05.042). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Distributed Edge-Cloud.
Solution Architecture Distributed Edge-Cloud: described as: .
Security is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Reference: Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, https://doi.org/10.1016/j.future.2020.05.042.
The solution is composed by the architecture named as Edge Computing-based Fault Tolerant Framework, proposed in the paper Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, with D.O.I. doi.org/10.1109/IWCMC48107.2020.9148269 (link https://doi.org/10.1109/IWCMC48107.2020.9148269). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Edge Computing-based Fault Tolerant Framework.
Solution Architecture Edge Computing-based Fault Tolerant Framework: described as: .
Performance/Efficiency is reached/addressed using/by Kubernetes framework (No description.).
Flexibility is reached/addressed using/by Docker containers (No description.).
Reference: Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, https://doi.org/10.1109/IWCMC48107.2020.9148269.
The solution is composed by the architecture named as Event-driven IoT, proposed in the paper Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, with D.O.I. doi.org/10.1007/s10586-020-03189-w (link https://doi.org/10.1007/s10586-020-03189-w). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Event-driven IoT.
Solution Architecture Event-driven IoT: described as: .
Performance/Efficiency is reached/addressed using/by Complex Event Processing (CEP) [Rahmani et al. 2020] (No description.).
Reliability is reached/addressed using/by TCP-IP protocol on Client/Server architecture for data communication (No description.).
Reference: Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, https://doi.org/10.1007/s10586-020-03189-w.
The solution is composed by the architecture named as Fog Computing based, proposed in the paper FOG-based architecture and load balancing methodology for health monitoring systems, 2021, with D.O.I. doi.org/10.1109/ACCESS.2021.3094033 (link https://doi.org/10.1109/ACCESS.2021.3094033). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Fog Computing based.
Solution Architecture Fog Computing based: described as: .
Performance/Efficiency is reached/addressed using/by Load Balancing Scheme (LBS) - Distributed workload to all fog nodes [Asghar et al. 2021] (No description.).
Reference: FOG-based architecture and load balancing methodology for health monitoring systems, 2021,

https://doi.org/10.1109/ACCESS.2021.3094033.
The solution is composed by the architecture named as Geographic-based, proposed in the paper A platform for integrating heterogeneous data and developing smart city applications, 2022, with D.O.I. doi.org/10.1016/j.future.2021.10.030 (link https://doi.org/10.1016/j.future.2021.10.030). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Geographic-based.
Solution Architecture Geographic-based: described as: .
Security is reached/addressed using/by FIWARE Platform [www.fiware.org] (Authentication, authorization, and data access control) (No description.).
Flexibility is reached/addressed using/by Virtualization (No description.).
Compatibility is reached/addressed using/by RESTFul APIs (No description.).
Reference: A platform for integrating heterogeneous data and developing smart city applications, 2022, https://doi.org/10.1016/j.future.2021.10.030.

The solution is composed by the architecture named as A4IoT (Anyplace 4.0 IoT), proposed in the paper The Anyplace 4.0 IoT Localization Architecture, 2020, with D.O.I. 10.1109/MDM48529.2020.00045 (link https://ieeexplore.ieee.org/document/9162205). This solution addresses the following Quality Requirements: Flexibility,Interaction Capability,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and A4IoT (Anyplace 4.0 IoT).
Solution Architecture A4IoT (Anyplace 4.0 IoT): described as: It is an open-source architecture designed for IoT-based indoor localization using technologies like Wi-Fi, BLE, Cellular, UWB, and Computer Vision (CV). The system supports edge computing, timeseries databases, and crowdsourcing to manage localization tasks across various smart spaces, such as factories, healthcare facilities, and emergency response environments. Its modular and scalable design ensures integration across platforms like Android, Linux, Windows, and Robot OS, making it adaptable to diverse IoT ecosystems..
Performance/Efficiency is reached/addressed using/by Time-Series Database (InfluxDB) (Time Behavior - With temporal queries we can now consume high-volume input streams to provide real-time IoT tracking.").
Reliability is reached/addressed using/by Multi-node Cluster Deployment (Fault Tolerance - The production environment uses a 3-node cluster with two replicas, for both the database engines and the DFS (Distributed Filesystem). This setup allows for a full operation with just a single node active at any time.").
Security is reached/addressed using/by SSL certificates - Encrypted Communications (Confidentiality - All communication is encrypted even in internal networks, as A4IoT image automatically creates and uses SSL certificates.).
Interaction Capability is reached/addressed using/by Web-based Dashboard (Operability - Web Apps built with HTML5, CSS3, and AngularJS.").
Flexibility is reached/addressed using/by Docker containers (Adaptability - It utilizes Docker to enable the deployment of single-node or multi-node configurations, regardless of any dependencies or the underlying OS.).
Reference: The Anyplace 4.0 IoT Localization Architecture, 2020, https://ieeexplore.ieee.org/document/9162205.
The solution is composed by the architecture named as Agricultural IoT Reference Architecture (AITRA), proposed in the paper Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3031634 (link https://doi.org/10.1109/ACCESS.2020.3031634). This solution addresses the following Quality Requirements: Compatibility,Reliability,Security,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Agricultural IoT Reference Architecture (AITRA).
Solution Architecture Agricultural IoT Reference Architecture (AITRA): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Reliability is reached/addressed using/by Edge gateway (No description.).
Security is reached/addressed using/by Authentication, authorization, and data access control (No description.).
Reference: Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, https://doi.org/10.1109/ACCESS.2020.3031634.
The solution is composed by the architecture named as Blockchain-based Zero Trust, proposed in the paper Blockchain-Based Zero Trust on the Edge, 2023, with D.O.I. 10.1109/CSCI62032.2023.00167 (link https://ieeexplore.ieee.org/document/10590526). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Performance/Efficiency,Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Blockchain-based Zero Trust.
Solution Architecture Blockchain-based Zero Trust: described as: The architecture proposed in "Blockchain-Based Zero Trust on the Edge" integrates Zero Trust principles with Blockchain to ensure secure access and trust management for Edge Computing in IoT environments. It focuses on continuous verification of users and devices, ensuring that all access is authenticated before being granted. Blockchain is used for decentralized trust management, maintaining an immutable record of activities, which enhances security and transparency. This setup is optimized for edge computing, reducing latency by processing data locally while maintaining the flexibility to scale across diverse IoT applications, such as smart cities and industrial IoT..
Security is reached/addressed using/by Zero Trust Architecture (ZTA) (Confidentiality - Zero Trust Architecture (ZTA)
ZTA ensures continuous verification and monitoring of all network traffic, restricting unauthorized access.).
Security is reached/addressed using/by Blockchain Technology (Integrity - Blockchain serves as an immutable ledger for storing requests, ensuring data integrity and trustworthiness.).
Performance/Efficiency is reached/addressed using/by Redis Message Broker (Time Behavior - Redis Message Broker for Asynchronous Processing [redis.io]

Asynchronous processing for write operations improves system responsiveness under high loads.).
Maintainability is reached/addressed using/by Component isolation through microservices (Modularity -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.
).
Flexibility is reached/addressed using/by Component isolation through microservices (Scalability -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.).
Reference: Blockchain-Based Zero Trust on the Edge, 2023, https://ieeexplore.ieee.org/document/10590526.
The solution is composed by the architecture named as Blockchain-enabled, proposed in the paper An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, with D.O.I. doi.org/10.1109/TSC.2020.2966970 (link https://doi.org/10.1109/TSC.2020.2966970). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Blockchain-enabled.
Solution Architecture Blockchain-enabled: described as: .
Performance/Efficiency is reached/addressed using/by Routing protocol designed for IoT devices in the SDN controller [Yazdinejad et al. 2020] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, https://doi.org/10.1109/TSC.2020.2966970.
The solution is composed by the architecture named as Cloud-based Secure Biometric, proposed in the paper BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, with D.O.I. doi.org/10.1016/j.jksuci.2017.07.001 (link https://doi.org/10.1016/j.jksuci.2017.07.001). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Cloud-based Secure Biometric.
Solution Architecture Cloud-based Secure Biometric: described as: .
Flexibility is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Reliability is reached/addressed using/by MapReduce framework (No description.).
Security is reached/addressed using/by Cloud-based biometric authentication [Shakil et al. 2020] (No description.).
Reference: BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, https://doi.org/10.1016/j.jksuci.2017.07.001.
The solution is composed by the architecture named as Deep Learning based, proposed in the paper HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, with D.O.I. doi.org/10.1016/j.future.2019.10.043 (link https://doi.org/10.1016/j.future.2019.10.043). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Deep Learning based.
Solution Architecture Deep Learning based: described as: .
Performance/Efficiency is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Security is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Reference: HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, https://doi.org/10.1016/j.future.2019.10.043.
The solution is composed by the architecture named as Distributed Blockchain-SDN based, proposed in the paper DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3039113 (link https://doi.org/10.1109/ACCESS.2020.3039113). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Distributed Blockchain-SDN based.
Solution Architecture Distributed Blockchain-SDN based: described as: .
Performance/Efficiency is reached/addressed using/by Decentralized blockchain nodes, dynamic SDN controllers, and edge computing (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, https://doi.org/10.1109/ACCESS.2020.3039113.
The solution is composed by the architecture named as Distributed Edge-Cloud, proposed in the paper Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, with D.O.I. doi.org/10.1016/j.future.2020.05.042 (link https://doi.org/10.1016/j.future.2020.05.042). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Distributed Edge-Cloud.
Solution Architecture Distributed Edge-Cloud: described as: .
Security is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).

Reference: Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, https://doi.org/10.1016/j.future.2020.05.042.
The solution is composed by the architecture named as Edge Computing-based Fault Tolerant Framework, proposed in the paper Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, with D.O.I. doi.org/10.1109/IWCMC48107.2020.9148269 (link https://doi.org/10.1109/IWCMC48107.2020.9148269). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Edge Computing-based Fault Tolerant Framework.
Solution Architecture Edge Computing-based Fault Tolerant Framework: described as: .
Performance/Efficiency is reached/addressed using/by Kubernetes framework (No description.).
Flexibility is reached/addressed using/by Docker containers (No description.).
Reference: Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, https://doi.org/10.1109/IWCMC48107.2020.9148269.
The solution is composed by the architecture named as Event-driven IoT, proposed in the paper Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, with D.O.I. doi.org/10.1007/s10586-020-03189-w (link https://doi.org/10.1007/s10586-020-03189-w). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Event-driven IoT.
Solution Architecture Event-driven IoT: described as: .
Performance/Efficiency is reached/addressed using/by Complex Event Processing (CEP) [Rahmani et al. 2020] (No description.).
Reliability is reached/addressed using/by TCP-IP protocol on Client/Server architecture for data communication (No description.).
Reference: Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, https://doi.org/10.1007/s10586-020-03189-w.
The solution is composed by the architecture named as Fog Computing based, proposed in the paper FOG-based architecture and load balancing methodology for health monitoring systems, 2021, with D.O.I. doi.org/10.1109/ACCESS.2021.3094033 (link https://doi.org/10.1109/ACCESS.2021.3094033). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Fog Computing based.
Solution Architecture Fog Computing based: described as: .
Performance/Efficiency is reached/addressed using/by Load Balancing Scheme (LBS) - Distributed workload to all fog nodes [Asghar et al. 2021] (No description.).
Reference: FOG-based architecture and load balancing methodology for health monitoring systems, 2021, https://doi.org/10.1109/ACCESS.2021.3094033.
The solution is composed by the architecture named as Geographic-based, proposed in the paper A platform for integrating heterogeneous data and developing smart city applications, 2022, with D.O.I. doi.org/10.1016/j.future.2021.10.030 (link https://doi.org/10.1016/j.future.2021.10.030). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Geographic-based.
Solution Architecture Geographic-based: described as: .
Security is reached/addressed using/by FIWARE Platform [www.fiware.org] (Authentication, authorization, and data access control) (No description.).
Flexibility is reached/addressed using/by Virtualization (No description.).
Compatibility is reached/addressed using/by RESTFul APIs (No description.).
Reference: A platform for integrating heterogeneous data and developing smart city applications, 2022, https://doi.org/10.1016/j.future.2021.10.030.
The solution is composed by the architecture named as Hybrid (Centralized and Distributed), proposed in the paper Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, with D.O.I. doi.org/10.1016/j.procs.2021.03.030 (link https://doi.org/10.1016/j.procs.2021.03.030). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid (Centralized and Distributed).
Solution Architecture Hybrid (Centralized and Distributed): described as: .
Performance/Efficiency is reached/addressed using/by Dynamic allocation of the resources using Software Defined Networking (SDN) [Chudhary and Sharma 2021] (No description.).
Reference: Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, https://doi.org/10.1016/j.procs.2021.03.030.

The solution is composed by the architecture named as A4IoT (Anyplace 4.0 IoT), proposed in the paper The Anyplace 4.0 IoT Localization Architecture, 2020, with D.O.I. 10.1109/MDM48529.2020.00045 (link https://ieeexplore.ieee.org/document/9162205). This solution addresses the following Quality Requirements: Flexibility,Interaction Capability,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and A4IoT (Anyplace 4.0 IoT).
Solution Architecture A4IoT (Anyplace 4.0 IoT): described as: It is an open-source architecture designed for IoT-based indoor localization using technologies like Wi-Fi, BLE, Cellular, UWB, and Computer Vision (CV). The system supports edge computing, timeseries databases, and crowdsourcing to manage localization tasks across various smart spaces, such as factories, healthcare facilities, and emergency response environments. Its modular and scalable design ensures integration across platforms like Android, Linux, Windows, and Robot OS, making it

adaptable to diverse IoT ecosystems..
Performance/Efficiency is reached/addressed using/by Time-Series Database (InfluxDB) (Time Behavior - With temporal queries we can now consume high-volume input streams to provide real-time IoT tracking.").
Reliability is reached/addressed using/by Multi-node Cluster Deployment (Fault Tolerance - The production environment uses a 3-node cluster with two replicas, for both the database engines and the DFS (Distributed Filesystem). This setup allows for a full operation with just a single node active at any time.").
Security is reached/addressed using/by SSL certificates - Encrypted Communications (Confidentiality - All communication is encrypted even in internal networks, as A4IoT image automatically creates and uses SSL certificates.).
Interaction Capability is reached/addressed using/by Web-based Dashboard (Operability - Web Apps built with HTML5, CSS3, and AngularJS.").
Flexibility is reached/addressed using/by Docker containers (Adaptability - It utilizes Docker to enable the deployment of single-node or multi-node configurations, regardless of any dependencies or the underlying OS.).
Reference: The Anyplace 4.0 IoT Localization Architecture, 2020, https://ieeexplore.ieee.org/document/9162205.
The solution is composed by the architecture named as Agricultural IoT Reference Architecture (AITRA), proposed in the paper Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3031634 (link https://doi.org/10.1109/ACCESS.2020.3031634). This solution addresses the following Quality Requirements: Compatibility,Reliability,Security,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Agricultural IoT Reference Architecture (AITRA).
Solution Architecture Agricultural IoT Reference Architecture (AITRA): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Reliability is reached/addressed using/by Edge gateway (No description.).
Security is reached/addressed using/by Authentication, authorization, and data access control (No description.).
Reference: Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, https://doi.org/10.1109/ACCESS.2020.3031634.
The solution is composed by the architecture named as Blockchain-based Zero Trust, proposed in the paper Blockchain-Based Zero Trust on the Edge, 2023, with D.O.I. 10.1109/CSCI62032.2023.00167 (link https://ieeexplore.ieee.org/document/10590526). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Performance/Efficiency,Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Blockchain-based Zero Trust.
Solution Architecture Blockchain-based Zero Trust: described as: The architecture proposed in "Blockchain-Based Zero Trust on the Edge" integrates Zero Trust principles with Blockchain to ensure secure access and trust management for Edge Computing in IoT environments. It focuses on continuous verification of users and devices, ensuring that all access is authenticated before being granted. Blockchain is used for decentralized trust management, maintaining an immutable record of activities, which enhances security and transparency. This setup is optimized for edge computing, reducing latency by processing data locally while maintaining the flexibility to scale across diverse IoT applications, such as smart cities and industrial IoT..
Security is reached/addressed using/by Zero Trust Architecture (ZTA) (Confidentiality - Zero Trust Architecture (ZTA)
ZTA ensures continuous verification and monitoring of all network traffic, restricting unauthorized access.).
Security is reached/addressed using/by Blockchain Technology (Integrity - Blockchain serves as an immutable ledger for storing requests, ensuring data integrity and trustworthiness.).
Performance/Efficiency is reached/addressed using/by Redis Message Broker (Time Behavior - Redis Message Broker for Asynchronous Processing [redis.io]
Asynchronous processing for write operations improves system responsiveness under high loads.).
Maintainability is reached/addressed using/by Component isolation through microservices (Modularity -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.
).
Flexibility is reached/addressed using/by Component isolation through microservices (Scalability -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.).
Reference: Blockchain-Based Zero Trust on the Edge, 2023, https://ieeexplore.ieee.org/document/10590526.
The solution is composed by the architecture named as Blockchain-enabled, proposed in the paper An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, with D.O.I. doi.org/10.1109/TSC.2020.2966970 (link https://doi.org/10.1109/TSC.2020.2966970). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Blockchain-enabled.
Solution Architecture Blockchain-enabled: described as: .
Performance/Efficiency is reached/addressed using/by Routing protocol designed for IoT devices in the SDN controller [Yazdinejad et al. 2020] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, https://doi.org/10.1109/TSC.2020.2966970.
The solution is composed by the architecture named as Cloud-based Secure Biometric, proposed in the paper BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, with D.O.I. doi.org/10.1016/j.jksuci.2017.07.001 (link https://doi.org/10.1016/j.jksuci.2017.07.001). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Cloud-

based Secure Biometric.
Solution Architecture Cloud-based Secure Biometric: described as: .
Flexibility is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Reliability is reached/addressed using/by MapReduce framework (No description.).
Security is reached/addressed using/by Cloud-based biometric authentication [Shakil et al. 2020] (No description.).
Reference: BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, https://doi.org/10.1016/j.jksuci.2017.07.001.
The solution is composed by the architecture named as Deep Learning based, proposed in the paper HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, with D.O.I. doi.org/10.1016/j.future.2019.10.043 (link https://doi.org/10.1016/j.future.2019.10.043). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Deep Learning based.
Solution Architecture Deep Learning based: described as: .
Performance/Efficiency is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Security is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Reference: HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, https://doi.org/10.1016/j.future.2019.10.043.
The solution is composed by the architecture named as Distributed Blockchain-SDN based, proposed in the paper DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3039113 (link https://doi.org/10.1109/ACCESS.2020.3039113). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Distributed Blockchain-SDN based.
Solution Architecture Distributed Blockchain-SDN based: described as: .
Performance/Efficiency is reached/addressed using/by Decentralized blockchain nodes, dynamic SDN controllers, and edge computing (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, https://doi.org/10.1109/ACCESS.2020.3039113.
The solution is composed by the architecture named as Distributed Edge-Cloud, proposed in the paper Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, with D.O.I. doi.org/10.1016/j.future.2020.05.042 (link https://doi.org/10.1016/j.future.2020.05.042). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Distributed Edge-Cloud.
Solution Architecture Distributed Edge-Cloud: described as: .
Security is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Reference: Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, https://doi.org/10.1016/j.future.2020.05.042.
The solution is composed by the architecture named as Edge Computing-based Fault Tolerant Framework, proposed in the paper Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, with D.O.I. doi.org/10.1109/IWCMC48107.2020.9148269 (link https://doi.org/10.1109/IWCMC48107.2020.9148269). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Edge Computing-based Fault Tolerant Framework.
Solution Architecture Edge Computing-based Fault Tolerant Framework: described as: .
Performance/Efficiency is reached/addressed using/by Kubernetes framework (No description.).
Flexibility is reached/addressed using/by Docker containers (No description.).
Reference: Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, https://doi.org/10.1109/IWCMC48107.2020.9148269.
The solution is composed by the architecture named as Event-driven IoT, proposed in the paper Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, with D.O.I. doi.org/10.1007/s10586-020-03189-w (link https://doi.org/10.1007/s10586-020-03189-w). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Event-driven IoT.
Solution Architecture Event-driven IoT: described as: .
Performance/Efficiency is reached/addressed using/by Complex Event Processing (CEP) [Rahmani et al. 2020] (No description.).
Reliability is reached/addressed using/by TCP-IP protocol on Client/Server architecture for data communication (No description.).
Reference: Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, https://doi.org/10.1007/s10586-020-03189-w.
The solution is composed by the architecture named as Fog Computing based, proposed in the paper FOG-based architecture and load balancing

methodology for health monitoring systems, 2021, with D.O.I. doi.org/10.1109/ACCESS.2021.3094033 (link https://doi.org/10.1109/ACCESS.2021.3094033). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Fog Computing based.
Solution Architecture Fog Computing based: described as: .
Performance/Efficiency is reached/addressed using/by Load Balancing Scheme (LBS) - Distributed workload to all fog nodes [Asghar et al. 2021] (No description.).
Reference: FOG-based architecture and load balancing methodology for health monitoring systems, 2021, https://doi.org/10.1109/ACCESS.2021.3094033.
The solution is composed by the architecture named as Geographic-based, proposed in the paper A platform for integrating heterogeneous data and developing smart city applications, 2022, with D.O.I. doi.org/10.1016/j.future.2021.10.030 (link https://doi.org/10.1016/j.future.2021.10.030). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Geographic-based.
Solution Architecture Geographic-based: described as: .
Security is reached/addressed using/by FIWARE Platform [www.fiware.org] (Authentication, authorization, and data access control) (No description.).
Flexibility is reached/addressed using/by Virtualization (No description.).
Compatibility is reached/addressed using/by RESTFul APIs (No description.).
Reference: A platform for integrating heterogeneous data and developing smart city applications, 2022, https://doi.org/10.1016/j.future.2021.10.030.
The solution is composed by the architecture named as Hybrid (Centralized and Distributed), proposed in the paper Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, with D.O.I. doi.org/10.1016/j.procs.2021.03.030 (link https://doi.org/10.1016/j.procs.2021.03.030). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid (Centralized and Distributed).
Solution Architecture Hybrid (Centralized and Distributed): described as: .
Performance/Efficiency is reached/addressed using/by Dynamic allocation of the resources using Software Defined Networking (SDN) [Chudhary and Sharma 2021] (No description.).
Reference: Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, https://doi.org/10.1016/j.procs.2021.03.030.
The solution is composed by the architecture named as Hybrid Cloud-Fog Computing, proposed in the paper Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, with D.O.I. doi.org/10.1007/s10586-021-03375-4 (link https://link.springer.com/article/10.1007/s10586-021-03375-4). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid Cloud-Fog Computing.
Solution Architecture Hybrid Cloud-Fog Computing: described as: .
Reliability is reached/addressed using/by Fog layer in two levels: 1-Intra mobility (User network): Nodes (Data transfer, device management, temporary local storage, notifications, and data consistency). 2-Extra mobility (Different domains): Smart Gateways (Local processing and database) (No description.).
Performance/Efficiency is reached/addressed using/by Fog computing (No description.).
Reference: Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, https://link.springer.com/article/10.1007/s10586-021-03375-4.

The solution is composed by the architecture named as A4IoT (Anyplace 4.0 IoT), proposed in the paper The Anyplace 4.0 IoT Localization Architecture, 2020, with D.O.I. 10.1109/MDM48529.2020.00045 (link https://ieeexplore.ieee.org/document/9162205). This solution addresses the following Quality Requirements: Flexibility,Interaction Capability,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and A4IoT (Anyplace 4.0 IoT).
Solution Architecture A4IoT (Anyplace 4.0 IoT): described as: It is an open-source architecture designed for IoT-based indoor localization using technologies like Wi-Fi, BLE, Cellular, UWB, and Computer Vision (CV). The system supports edge computing, timeseries databases, and crowdsourcing to manage localization tasks across various smart spaces, such as factories, healthcare facilities, and emergency response environments. Its modular and scalable design ensures integration across platforms like Android, Linux, Windows, and Robot OS, making it adaptable to diverse IoT ecosystems..
Performance/Efficiency is reached/addressed using/by Time-Series Database (InfluxDB) (Time Behavior - With temporal queries we can now consume high-volume input streams to provide real-time IoT tracking.").
Reliability is reached/addressed using/by Multi-node Cluster Deployment (Fault Tolerance - The production environment uses a 3-node cluster with two replicas, for both the database engines and the DFS (Distributed Filesystem). This setup allows for a full operation with just a single node active at any time.").
Security is reached/addressed using/by SSL certificates - Encrypted Communications (Confidentiality - All communication is encrypted even in internal networks, as A4IoT image automatically creates and uses SSL certificates.).
Interaction Capability is reached/addressed using/by Web-based Dashboard (Operability - Web Apps built with HTML5, CSS3, and AngularJS.").
Flexibility is reached/addressed using/by Docker containers (Adaptability - It utilizes Docker to enable the deployment of single-node or multi-node

configurations, regardless of any dependencies or the underlying OS.).
Reference: The Anyplace 4.0 IoT Localization Architecture, 2020, https://ieeexplore.ieee.org/document/9162205.
The solution is composed by the architecture named as Agricultural IoT Reference Architecture (AITRA), proposed in the paper Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3031634 (link https://doi.org/10.1109/ACCESS.2020.3031634). This solution addresses the following Quality Requirements: Compatibility,Reliability,Security,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Agricultural IoT Reference Architecture (AITRA).
Solution Architecture Agricultural IoT Reference Architecture (AITRA): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Reliability is reached/addressed using/by Edge gateway (No description.).
Security is reached/addressed using/by Authentication, authorization, and data access control (No description.).
Reference: Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, https://doi.org/10.1109/ACCESS.2020.3031634.
The solution is composed by the architecture named as Blockchain-based Zero Trust, proposed in the paper Blockchain-Based Zero Trust on the Edge, 2023, with D.O.I. 10.1109/CSCI62032.2023.00167 (link https://ieeexplore.ieee.org/document/10590526). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Performance/Efficiency,Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Blockchain-based Zero Trust.
Solution Architecture Blockchain-based Zero Trust: described as: The architecture proposed in "Blockchain-Based Zero Trust on the Edge" integrates Zero Trust principles with Blockchain to ensure secure access and trust management for Edge Computing in IoT environments. It focuses on continuous verification of users and devices, ensuring that all access is authenticated before being granted. Blockchain is used for decentralized trust management, maintaining an immutable record of activities, which enhances security and transparency. This setup is optimized for edge computing, reducing latency by processing data locally while maintaining the flexibility to scale across diverse IoT applications, such as smart cities and industrial IoT..
Security is reached/addressed using/by Zero Trust Architecture (ZTA) (Confidentiality - Zero Trust Architecture (ZTA)
ZTA ensures continuous verification and monitoring of all network traffic, restricting unauthorized access.).
Security is reached/addressed using/by Blockchain Technology (Integrity - Blockchain serves as an immutable ledger for storing requests, ensuring data integrity and trustworthiness.).
Performance/Efficiency is reached/addressed using/by Redis Message Broker (Time Behavior - Redis Message Broker for Asynchronous Processing [redis.io]
Asynchronous processing for write operations improves system responsiveness under high loads.).
Maintainability is reached/addressed using/by Component isolation through microservices (Modularity -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.
).
Flexibility is reached/addressed using/by Component isolation through microservices (Scalability -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.).
Reference: Blockchain-Based Zero Trust on the Edge, 2023, https://ieeexplore.ieee.org/document/10590526.
The solution is composed by the architecture named as Blockchain-enabled, proposed in the paper An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, with D.O.I. doi.org/10.1109/TSC.2020.2966970 (link https://doi.org/10.1109/TSC.2020.2966970). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Blockchain-enabled.
Solution Architecture Blockchain-enabled: described as: .
Performance/Efficiency is reached/addressed using/by Routing protocol designed for IoT devices in the SDN controller [Yazdinejad et al. 2020] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, https://doi.org/10.1109/TSC.2020.2966970.
The solution is composed by the architecture named as Cloud-based Secure Biometric, proposed in the paper BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, with D.O.I. doi.org/10.1016/j.jksuci.2017.07.001 (link https://doi.org/10.1016/j.jksuci.2017.07.001). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Cloud-based Secure Biometric.
Solution Architecture Cloud-based Secure Biometric: described as: .
Flexibility is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Reliability is reached/addressed using/by MapReduce framework (No description.).
Security is reached/addressed using/by Cloud-based biometric authentication [Shakil et al. 2020] (No description.).
Reference: BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, https://doi.org/10.1016/j.jksuci.2017.07.001.
The solution is composed by the architecture named as Deep Learning based, proposed in the paper HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, with D.O.I.

doi.org/10.1016/j.future.2019.10.043 (link https://doi.org/10.1016/j.future.2019.10.043). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Deep Learning based.
Solution Architecture Deep Learning based: described as: .
Performance/Efficiency is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Security is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Reference: HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, https://doi.org/10.1016/j.future.2019.10.043.
The solution is composed by the architecture named as Distributed Blockchain-SDN based, proposed in the paper DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3039113 (link https://doi.org/10.1109/ACCESS.2020.3039113). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Distributed Blockchain-SDN based.
Solution Architecture Distributed Blockchain-SDN based: described as: .
Performance/Efficiency is reached/addressed using/by Decentralized blockchain nodes, dynamic SDN controllers, and edge computing (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, https://doi.org/10.1109/ACCESS.2020.3039113.
The solution is composed by the architecture named as Distributed Edge-Cloud, proposed in the paper Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, with D.O.I. doi.org/10.1016/j.future.2020.05.042 (link https://doi.org/10.1016/j.future.2020.05.042). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Distributed Edge-Cloud.
Solution Architecture Distributed Edge-Cloud: described as: .
Security is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Reference: Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, https://doi.org/10.1016/j.future.2020.05.042.
The solution is composed by the architecture named as Edge Computing-based Fault Tolerant Framework, proposed in the paper Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, with D.O.I. doi.org/10.1109/IWCMC48107.2020.9148269 (link https://doi.org/10.1109/IWCMC48107.2020.9148269). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Edge Computing-based Fault Tolerant Framework.
Solution Architecture Edge Computing-based Fault Tolerant Framework: described as: .
Performance/Efficiency is reached/addressed using/by Kubernetes framework (No description.).
Flexibility is reached/addressed using/by Docker containers (No description.).
Reference: Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, https://doi.org/10.1109/IWCMC48107.2020.9148269.
The solution is composed by the architecture named as Event-driven IoT, proposed in the paper Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, with D.O.I. doi.org/10.1007/s10586-020-03189-w (link https://doi.org/10.1007/s10586-020-03189-w). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Event-driven IoT.
Solution Architecture Event-driven IoT: described as: .
Performance/Efficiency is reached/addressed using/by Complex Event Processing (CEP) [Rahmani et al. 2020] (No description.).
Reliability is reached/addressed using/by TCP-IP protocol on Client/Server architecture for data communication (No description.).
Reference: Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, https://doi.org/10.1007/s10586-020-03189-w.
The solution is composed by the architecture named as Fog Computing based, proposed in the paper FOG-based architecture and load balancing methodology for health monitoring systems, 2021, with D.O.I. doi.org/10.1109/ACCESS.2021.3094033 (link https://doi.org/10.1109/ACCESS.2021.3094033). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Fog Computing based.
Solution Architecture Fog Computing based: described as: .
Performance/Efficiency is reached/addressed using/by Load Balancing Scheme (LBS) - Distributed workload to all fog nodes [Asghar et al. 2021] (No description.).
Reference: FOG-based architecture and load balancing methodology for health monitoring systems, 2021, https://doi.org/10.1109/ACCESS.2021.3094033.
The solution is composed by the architecture named as Geographic-based, proposed in the paper A platform for integrating heterogeneous data

and developing smart city applications, 2022, with D.O.I. doi.org/10.1016/j.future.2021.10.030 (link https://doi.org/10.1016/j.future.2021.10.030). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Security,.

The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Geographic-based.

Solution Architecture Geographic-based: described as: .

Security is reached/addressed using/by FIWARE Platform [www.fiware.org] (Authentication, authorization, and data access control) (No description.).

Flexibility is reached/addressed using/by Virtualization (No description.).

Compatibility is reached/addressed using/by RESTFul APIs (No description.).

Reference: A platform for integrating heterogeneous data and developing smart city applications, 2022, https://doi.org/10.1016/j.future.2021.10.030.

The solution is composed by the architecture named as Hybrid (Centralized and Distributed), proposed in the paper Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, with D.O.I. doi.org/10.1016/j.procs.2021.03.030 (link https://doi.org/10.1016/j.procs.2021.03.030). This solution addresses the following Quality Requirements: Performance/Efficiency,.

The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid (Centralized and Distributed).

Solution Architecture Hybrid (Centralized and Distributed): described as: .

Performance/Efficiency is reached/addressed using/by Dynamic allocation of the resources using Software Defined Networking (SDN) [Chudhary and Sharma 2021] (No description.).

Reference: Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, https://doi.org/10.1016/j.procs.2021.03.030.

The solution is composed by the architecture named as Hybrid Cloud-Fog Computing, proposed in the paper Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, with D.O.I. doi.org/10.1007/s10586-021-03375-4 (link https://link.springer.com/article/10.1007/s10586-021-03375-4). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.

The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid Cloud-Fog Computing.

Solution Architecture Hybrid Cloud-Fog Computing: described as: .

Reliability is reached/addressed using/by Fog layer in two levels: 1-Intra mobility (User network): Nodes (Data transfer, device management, temporary local storage, notifications, and data consistency). 2-Extra mobility (Different domains): Smart Gateways (Local processing and database) (No description.).

Performance/Efficiency is reached/addressed using/by Fog computing (No description.).

Reference: Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, https://link.springer.com/article/10.1007/s10586-021-03375-4.

The solution is composed by the architecture named as Industrial Internet of Things (IIoT), proposed in the paper A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Performance/Efficiency,Security,.

The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and Industrial Internet of Things (IIoT).

Solution Architecture Industrial Internet of Things (IIoT): described as: .

Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).

Flexibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).

Performance/Efficiency is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).

Security is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).

Reference: A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, https://doi.org/10.3390/s20195603.

The solution is composed by the architecture named as A4IoT (Anyplace 4.0 IoT), proposed in the paper The Anyplace 4.0 IoT Localization Architecture, 2020, with D.O.I. 10.1109/MDM48529.2020.00045 (link https://ieeexplore.ieee.org/document/9162205). This solution addresses the following Quality Requirements: Flexibility,Interaction Capability,Performance/Efficiency,Reliability,Security,.

The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and A4IoT (Anyplace 4.0 IoT).

Solution Architecture A4IoT (Anyplace 4.0 IoT): described as: It is an open-source architecture designed for IoT-based indoor localization using technologies like Wi-Fi, BLE, Cellular, UWB, and Computer Vision (CV). The system supports edge computing, timeseries databases, and crowdsourcing to manage localization tasks across various smart spaces, such as factories, healthcare facilities, and emergency response environments. Its modular and scalable design ensures integration across platforms like Android, Linux, Windows, and Robot OS, making it adaptable to diverse IoT ecosystems..

Performance/Efficiency is reached/addressed using/by Time-Series Database (InfluxDB) (Time Behavior - With temporal queries we can now consume high-volume input streams to provide real-time IoT tracking.").

Reliability is reached/addressed using/by Multi-node Cluster Deployment (Fault Tolerance - The production environment uses a 3-node cluster with two replicas, for both the database engines and the DFS (Distributed Filesystem). This setup allows for a full operation with just a single node active at any time.").

Security is reached/addressed using/by SSL certificates - Encrypted Communications (Confidentiality - All communication is encrypted even in internal networks, as A4IoT image automatically creates and uses SSL certificates.).

Interaction Capability is reached/addressed using/by Web-based Dashboard (Operability - Web Apps built with HTML5, CSS3, and AngularJS.").

Flexibility is reached/addressed using/by Docker containers (Adaptability - It utilizes Docker to enable the deployment of single-node or multi-node configurations, regardless of any dependencies or the underlying OS.).
Reference: The Anyplace 4.0 IoT Localization Architecture, 2020, https://ieeexplore.ieee.org/document/9162205.
The solution is composed by the architecture named as Agricultural IoT Reference Architecture (AITRA), proposed in the paper Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3031634 (link https://doi.org/10.1109/ACCESS.2020.3031634). This solution addresses the following Quality Requirements: Compatibility,Reliability,Security,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Agricultural IoT Reference Architecture (AITRA).
Solution Architecture Agricultural IoT Reference Architecture (AITRA): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Reliability is reached/addressed using/by Edge gateway (No description.).
Security is reached/addressed using/by Authentication, authorization, and data access control (No description.).
Reference: Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, https://doi.org/10.1109/ACCESS.2020.3031634.
The solution is composed by the architecture named as Blockchain-based Zero Trust, proposed in the paper Blockchain-Based Zero Trust on the Edge, 2023, with D.O.I. 10.1109/CSCI62032.2023.00167 (link https://ieeexplore.ieee.org/document/10590526). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Performance/Efficiency,Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Blockchain-based Zero Trust.
Solution Architecture Blockchain-based Zero Trust: described as: The architecture proposed in "Blockchain-Based Zero Trust on the Edge" integrates Zero Trust principles with Blockchain to ensure secure access and trust management for Edge Computing in IoT environments. It focuses on continuous verification of users and devices, ensuring that all access is authenticated before being granted. Blockchain is used for decentralized trust management, maintaining an immutable record of activities, which enhances security and transparency. This setup is optimized for edge computing, reducing latency by processing data locally while maintaining the flexibility to scale across diverse IoT applications, such as smart cities and industrial IoT..
Security is reached/addressed using/by Zero Trust Architecture (ZTA) (Confidentiality - Zero Trust Architecture (ZTA)
ZTA ensures continuous verification and monitoring of all network traffic, restricting unauthorized access.).
Security is reached/addressed using/by Blockchain Technology (Integrity - Blockchain serves as an immutable ledger for storing requests, ensuring data integrity and trustworthiness.).
Performance/Efficiency is reached/addressed using/by Redis Message Broker (Time Behavior - Redis Message Broker for Asynchronous Processing [redis.io]
Asynchronous processing for write operations improves system responsiveness under high loads.).
Maintainability is reached/addressed using/by Component isolation through microservices (Modularity -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.
).
Flexibility is reached/addressed using/by Component isolation through microservices (Scalability -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.).
Reference: Blockchain-Based Zero Trust on the Edge, 2023, https://ieeexplore.ieee.org/document/10590526.
The solution is composed by the architecture named as Blockchain-enabled, proposed in the paper An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, with D.O.I. doi.org/10.1109/TSC.2020.2966970 (link https://doi.org/10.1109/TSC.2020.2966970). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Blockchain-enabled.
Solution Architecture Blockchain-enabled: described as: .
Performance/Efficiency is reached/addressed using/by Routing protocol designed for IoT devices in the SDN controller [Yazdinejad et al. 2020] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, https://doi.org/10.1109/TSC.2020.2966970.
The solution is composed by the architecture named as Cloud-based Secure Biometric, proposed in the paper BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, with D.O.I. doi.org/10.1016/j.jksuci.2017.07.001 (link https://doi.org/10.1016/j.jksuci.2017.07.001). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Cloud-based Secure Biometric.
Solution Architecture Cloud-based Secure Biometric: described as: .
Flexibility is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Reliability is reached/addressed using/by MapReduce framework (No description.).
Security is reached/addressed using/by Cloud-based biometric authentication [Shakil et al. 2020] (No description.).
Reference: BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, https://doi.org/10.1016/j.jksuci.2017.07.001.
The solution is composed by the architecture named as Deep Learning based, proposed in the paper HealthFog: An ensemble deep learning-based

Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, with D.O.I. doi.org/10.1016/j.future.2019.10.043 (link https://doi.org/10.1016/j.future.2019.10.043). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Deep Learning based.
Solution Architecture Deep Learning based: described as: .
Performance/Efficiency is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Security is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Reference: HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, https://doi.org/10.1016/j.future.2019.10.043.
The solution is composed by the architecture named as Distributed Blockchain-SDN based, proposed in the paper DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3039113 (link https://doi.org/10.1109/ACCESS.2020.3039113). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Distributed Blockchain-SDN based.
Solution Architecture Distributed Blockchain-SDN based: described as: .
Performance/Efficiency is reached/addressed using/by Decentralized blockchain nodes, dynamic SDN controllers, and edge computing (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, https://doi.org/10.1109/ACCESS.2020.3039113.
The solution is composed by the architecture named as Distributed Edge-Cloud, proposed in the paper Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, with D.O.I. doi.org/10.1016/j.future.2020.05.042 (link https://doi.org/10.1016/j.future.2020.05.042). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Distributed Edge-Cloud.
Solution Architecture Distributed Edge-Cloud: described as: .
Security is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Reference: Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, https://doi.org/10.1016/j.future.2020.05.042.
The solution is composed by the architecture named as Edge Computing-based Fault Tolerant Framework, proposed in the paper Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, with D.O.I. doi.org/10.1109/IWCMC48107.2020.9148269 (link https://doi.org/10.1109/IWCMC48107.2020.9148269). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Edge Computing-based Fault Tolerant Framework.
Solution Architecture Edge Computing-based Fault Tolerant Framework: described as: .
Performance/Efficiency is reached/addressed using/by Kubernetes framework (No description.).
Flexibility is reached/addressed using/by Docker containers (No description.).
Reference: Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, https://doi.org/10.1109/IWCMC48107.2020.9148269.
The solution is composed by the architecture named as Event-driven IoT, proposed in the paper Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, with D.O.I. doi.org/10.1007/s10586-020-03189-w (link https://doi.org/10.1007/s10586-020-03189-w). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Event-driven IoT.
Solution Architecture Event-driven IoT: described as: .
Performance/Efficiency is reached/addressed using/by Complex Event Processing (CEP) [Rahmani et al. 2020] (No description.).
Reliability is reached/addressed using/by TCP-IP protocol on Client/Server architecture for data communication (No description.).
Reference: Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, https://doi.org/10.1007/s10586-020-03189-w.
The solution is composed by the architecture named as Fog Computing based, proposed in the paper FOG-based architecture and load balancing methodology for health monitoring systems, 2021, with D.O.I. doi.org/10.1109/ACCESS.2021.3094033 (link https://doi.org/10.1109/ACCESS.2021.3094033). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Fog Computing based.
Solution Architecture Fog Computing based: described as: .
Performance/Efficiency is reached/addressed using/by Load Balancing Scheme (LBS) - Distributed workload to all fog nodes [Asghar et al. 2021] (No description.).
Reference: FOG-based architecture and load balancing methodology for health monitoring systems, 2021, https://doi.org/10.1109/ACCESS.2021.3094033.

The solution is composed by the architecture named as Geographic-based, proposed in the paper A platform for integrating heterogeneous data and developing smart city applications, 2022, with D.O.I. doi.org/10.1016/j.future.2021.10.030 (link https://doi.org/10.1016/j.future.2021.10.030). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Security,.

The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Geographic-based.

Solution Architecture Geographic-based: described as: .

Security is reached/addressed using/by FIWARE Platform [www.fiware.org] (Authentication, authorization, and data access control) (No description.).

Flexibility is reached/addressed using/by Virtualization (No description.).

Compatibility is reached/addressed using/by RESTFul APIs (No description.).

Reference: A platform for integrating heterogeneous data and developing smart city applications, 2022, https://doi.org/10.1016/j.future.2021.10.030.

The solution is composed by the architecture named as Hybrid (Centralized and Distributed), proposed in the paper Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, with D.O.I. doi.org/10.1016/j.procs.2021.03.030 (link https://doi.org/10.1016/j.procs.2021.03.030). This solution addresses the following Quality Requirements: Performance/Efficiency,.

The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid (Centralized and Distributed).

Solution Architecture Hybrid (Centralized and Distributed): described as: .

Performance/Efficiency is reached/addressed using/by Dynamic allocation of the resources using Software Defined Networking (SDN) [Chudhary and Sharma 2021] (No description.).

Reference: Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, https://doi.org/10.1016/j.procs.2021.03.030.

The solution is composed by the architecture named as Hybrid Cloud-Fog Computing, proposed in the paper Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, with D.O.I. doi.org/10.1007/s10586-021-03375-4 (link https://link.springer.com/article/10.1007/s10586-021-03375-4). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.

The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid Cloud-Fog Computing.

Solution Architecture Hybrid Cloud-Fog Computing: described as: .

Reliability is reached/addressed using/by Fog layer in two levels: 1-Intra mobility (User network): Nodes (Data transfer, device management, temporary local storage, notifications, and data consistency). 2-Extra mobility (Different domains): Smart Gateways (Local processing and database) (No description.).

Performance/Efficiency is reached/addressed using/by Fog computing (No description.).

Reference: Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, https://link.springer.com/article/10.1007/s10586-021-03375-4.

The solution is composed by the architecture named as Industrial Internet of Things (IIoT), proposed in the paper A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Performance/Efficiency,Security,.

The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and Industrial Internet of Things (IIoT).

Solution Architecture Industrial Internet of Things (IIoT): described as: .

Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).

Flexibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).

Performance/Efficiency is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).

Security is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).

Reference: A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, https://doi.org/10.3390/s20195603.

The solution is composed by the architecture named as Layered, proposed in the paper An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, with D.O.I. doi.org/10.1016/j.future.2020.07.047 (link https://doi.org/10.1016/j.future.2020.07.047). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Security,.

The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.

Solution Architecture Layered: described as: .

Flexibility is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).

Performance/Efficiency is reached/addressed using/by Big data analytics engine [Ali et al. 2021] (No description.).

Security is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).

Reference: An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, https://doi.org/10.1016/j.future.2020.07.047.

The solution is composed by the architecture named as A4IoT (Anyplace 4.0 IoT), proposed in the paper The Anyplace 4.0 IoT Localization Architecture, 2020, with D.O.I. 10.1109/MDM48529.2020.00045 (link https://ieeexplore.ieee.org/document/9162205). This solution addresses the following Quality Requirements: Flexibility,Interaction Capability,Performance/Efficiency,Reliability,Security,.

The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and A4IoT (Anyplace 4.0 IoT).

Solution Architecture A4IoT (Anyplace 4.0 IoT): described as: It is an open-source architecture designed for IoT-based indoor localization using technologies like Wi-Fi, BLE, Cellular, UWB, and Computer Vision (CV). The system supports edge computing, timeseries databases, and crowdsourcing to manage localization tasks across various smart spaces, such as factories, healthcare facilities, and emergency response environments. Its modular and scalable design ensures integration across platforms like Android, Linux, Windows, and Robot OS, making it adaptable to diverse IoT ecosystems..
Performance/Efficiency is reached/addressed using/by Time-Series Database (InfluxDB) (Time Behavior - With temporal queries we can now consume high-volume input streams to provide real-time IoT tracking.").
Reliability is reached/addressed using/by Multi-node Cluster Deployment (Fault Tolerance - The production environment uses a 3-node cluster with two replicas, for both the database engines and the DFS (Distributed Filesystem). This setup allows for a full operation with just a single node active at any time.").
Security is reached/addressed using/by SSL certificates - Encrypted Communications (Confidentiality - All communication is encrypted even in internal networks, as A4IoT image automatically creates and uses SSL certificates.).
Interaction Capability is reached/addressed using/by Web-based Dashboard (Operability - Web Apps built with HTML5, CSS3, and AngularJS.").
Flexibility is reached/addressed using/by Docker containers (Adaptability - It utilizes Docker to enable the deployment of single-node or multi-node configurations, regardless of any dependencies or the underlying OS.).
Reference: The Anyplace 4.0 IoT Localization Architecture, 2020, https://ieeexplore.ieee.org/document/9162205.
The solution is composed by the architecture named as Agricultural IoT Reference Architecture (AITRA), proposed in the paper Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3031634 (link https://doi.org/10.1109/ACCESS.2020.3031634). This solution addresses the following Quality Requirements: Compatibility,Reliability,Security,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Agricultural IoT Reference Architecture (AITRA).
Solution Architecture Agricultural IoT Reference Architecture (AITRA): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Reliability is reached/addressed using/by Edge gateway (No description.).
Security is reached/addressed using/by Authentication, authorization, and data access control (No description.).
Reference: Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, https://doi.org/10.1109/ACCESS.2020.3031634.
The solution is composed by the architecture named as Blockchain-based Zero Trust, proposed in the paper Blockchain-Based Zero Trust on the Edge, 2023, with D.O.I. 10.1109/CSCI62032.2023.00167 (link https://ieeexplore.ieee.org/document/10590526). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Performance/Efficiency,Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Blockchain-based Zero Trust.
Solution Architecture Blockchain-based Zero Trust: described as: The architecture proposed in "Blockchain-Based Zero Trust on the Edge" integrates Zero Trust principles with Blockchain to ensure secure access and trust management for Edge Computing in IoT environments. It focuses on continuous verification of users and devices, ensuring that all access is authenticated before being granted. Blockchain is used for decentralized trust management, maintaining an immutable record of activities, which enhances security and transparency. This setup is optimized for edge computing, reducing latency by processing data locally while maintaining the flexibility to scale across diverse IoT applications, such as smart cities and industrial IoT..
Security is reached/addressed using/by Zero Trust Architecture (ZTA) (Confidentiality - Zero Trust Architecture (ZTA)
ZTA ensures continuous verification and monitoring of all network traffic, restricting unauthorized access.).
Security is reached/addressed using/by Blockchain Technology (Integrity - Blockchain serves as an immutable ledger for storing requests, ensuring data integrity and trustworthiness.).
Performance/Efficiency is reached/addressed using/by Redis Message Broker (Time Behavior - Redis Message Broker for Asynchronous Processing [redis.io]
Asynchronous processing for write operations improves system responsiveness under high loads.).
Maintainability is reached/addressed using/by Component isolation through microservices (Modularity -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.
).
Flexibility is reached/addressed using/by Component isolation through microservices (Scalability -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.).
Reference: Blockchain-Based Zero Trust on the Edge, 2023, https://ieeexplore.ieee.org/document/10590526.
The solution is composed by the architecture named as Blockchain-enabled, proposed in the paper An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, with D.O.I. doi.org/10.1109/TSC.2020.2966970 (link https://doi.org/10.1109/TSC.2020.2966970). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Blockchain-enabled.
Solution Architecture Blockchain-enabled: described as: .
Performance/Efficiency is reached/addressed using/by Routing protocol designed for IoT devices in the SDN controller [Yazdinejad et al. 2020] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, https://doi.org/10.1109/TSC.2020.2966970.
The solution is composed by the architecture named as Cloud-based Secure Biometric, proposed in the paper BAMHealthCloud: A biometric

authentication and data management system for healthcare data in the cloud, 2020, with D.O.I. doi.org/10.1016/j.jksuci.2017.07.001 (link https://doi.org/10.1016/j.jksuci.2017.07.001). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Reliability,Security,.

The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Cloud-based Secure Biometric.

Solution Architecture Cloud-based Secure Biometric: described as: .

Flexibility is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).

Performance/Efficiency is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).

Reliability is reached/addressed using/by MapReduce framework (No description.).

Security is reached/addressed using/by Cloud-based biometric authentication [Shakil et al. 2020] (No description.).

Reference: BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, https://doi.org/10.1016/j.jksuci.2017.07.001.

The solution is composed by the architecture named as Deep Learning based, proposed in the paper HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, with D.O.I. doi.org/10.1016/j.future.2019.10.043 (link https://doi.org/10.1016/j.future.2019.10.043). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.

The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Deep Learning based.

Solution Architecture Deep Learning based: described as: .

Performance/Efficiency is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).

Security is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).

Reference: HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, https://doi.org/10.1016/j.future.2019.10.043.

The solution is composed by the architecture named as Distributed Blockchain-SDN based, proposed in the paper DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3039113 (link https://doi.org/10.1109/ACCESS.2020.3039113). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.

The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Distributed Blockchain-SDN based.

Solution Architecture Distributed Blockchain-SDN based: described as: .

Performance/Efficiency is reached/addressed using/by Decentralized blockchain nodes, dynamic SDN controllers, and edge computing (No description.).

Security is reached/addressed using/by Blockchain Technology (No description.).

Reference: DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, https://doi.org/10.1109/ACCESS.2020.3039113.

The solution is composed by the architecture named as Distributed Edge-Cloud, proposed in the paper Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, with D.O.I. doi.org/10.1016/j.future.2020.05.042 (link https://doi.org/10.1016/j.future.2020.05.042). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.

The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Distributed Edge-Cloud.

Solution Architecture Distributed Edge-Cloud: described as: .

Security is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).

Performance/Efficiency is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).

Reference: Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, https://doi.org/10.1016/j.future.2020.05.042.

The solution is composed by the architecture named as Edge Computing-based Fault Tolerant Framework, proposed in the paper Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, with D.O.I. doi.org/10.1109/IWCMC48107.2020.9148269 (link https://doi.org/10.1109/IWCMC48107.2020.9148269). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,.

The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Edge Computing-based Fault Tolerant Framework.

Solution Architecture Edge Computing-based Fault Tolerant Framework: described as: .

Performance/Efficiency is reached/addressed using/by Kubernetes framework (No description.).

Flexibility is reached/addressed using/by Docker containers (No description.).

Reference: Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, https://doi.org/10.1109/IWCMC48107.2020.9148269.

The solution is composed by the architecture named as Event-driven IoT, proposed in the paper Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, with D.O.I. doi.org/10.1007/s10586-020-03189-w (link https://doi.org/10.1007/s10586-020-03189-w). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.

The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Event-driven IoT.

Solution Architecture Event-driven IoT: described as: .

Performance/Efficiency is reached/addressed using/by Complex Event Processing (CEP) [Rahmani et al. 2020] (No description.).

Reliability is reached/addressed using/by TCP-IP protocol on Client/Server architecture for data communication (No description.).
Reference: Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, https://doi.org/10.1007/s10586-020-03189-w.
The solution is composed by the architecture named as Fog Computing based, proposed in the paper FOG-based architecture and load balancing methodology for health monitoring systems, 2021, with D.O.I. doi.org/10.1109/ACCESS.2021.3094033 (link https://doi.org/10.1109/ACCESS.2021.3094033). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Fog Computing based.
Solution Architecture Fog Computing based: described as: .
Performance/Efficiency is reached/addressed using/by Load Balancing Scheme (LBS) - Distributed workload to all fog nodes [Asghar et al. 2021] (No description.).
Reference: FOG-based architecture and load balancing methodology for health monitoring systems, 2021, https://doi.org/10.1109/ACCESS.2021.3094033.
The solution is composed by the architecture named as Geographic-based, proposed in the paper A platform for integrating heterogeneous data and developing smart city applications, 2022, with D.O.I. doi.org/10.1016/j.future.2021.10.030 (link https://doi.org/10.1016/j.future.2021.10.030). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Geographic-based.
Solution Architecture Geographic-based: described as: .
Security is reached/addressed using/by FIWARE Platform [www.fiware.org] (Authentication, authorization, and data access control) (No description.).
Flexibility is reached/addressed using/by Virtualization (No description.).
Compatibility is reached/addressed using/by RESTFul APIs (No description.).
Reference: A platform for integrating heterogeneous data and developing smart city applications, 2022, https://doi.org/10.1016/j.future.2021.10.030.
The solution is composed by the architecture named as Hybrid (Centralized and Distributed), proposed in the paper Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, with D.O.I. doi.org/10.1016/j.procs.2021.03.030 (link https://doi.org/10.1016/j.procs.2021.03.030). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid (Centralized and Distributed).
Solution Architecture Hybrid (Centralized and Distributed): described as: .
Performance/Efficiency is reached/addressed using/by Dynamic allocation of the resources using Software Defined Networking (SDN) [Chudhary and Sharma 2021] (No description.).
Reference: Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, https://doi.org/10.1016/j.procs.2021.03.030.
The solution is composed by the architecture named as Hybrid Cloud-Fog Computing, proposed in the paper Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, with D.O.I. doi.org/10.1007/s10586-021-03375-4 (link https://link.springer.com/article/10.1007/s10586-021-03375-4). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid Cloud-Fog Computing.
Solution Architecture Hybrid Cloud-Fog Computing: described as: .
Reliability is reached/addressed using/by Fog layer in two levels: 1-Intra mobility (User network): Nodes (Data transfer, device management, temporary local storage, notifications, and data consistency). 2-Extra mobility (Different domains): Smart Gateways (Local processing and database) (No description.).
Performance/Efficiency is reached/addressed using/by Fog computing (No description.).
Reference: Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, https://link.springer.com/article/10.1007/s10586-021-03375-4.
The solution is composed by the architecture named as Industrial Internet of Things (IIoT), proposed in the paper A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Performance/Efficiency,Security,.
The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and Industrial Internet of Things (IIoT).
Solution Architecture Industrial Internet of Things (IIoT): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Flexibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Performance/Efficiency is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).
Security is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).
Reference: A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, https://doi.org/10.3390/s20195603.
The solution is composed by the architecture named as Layered, proposed in the paper An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, with D.O.I. doi.org/10.1016/j.future.2020.07.047 (link https://doi.org/10.1016/j.future.2020.07.047). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.

Solution Architecture Layered: described as: .
Flexibility is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).
Performance/Efficiency is reached/addressed using/by Big data analytics engine [Ali et al. 2021] (No description.).
Security is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).
Reference: An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, https://doi.org/10.1016/j.future.2020.07.047.
The solution is composed by the architecture named as Layered, proposed in the paper A novel architecture for the Internet of Things based E-Health systems, 2020, with D.O.I. doi.org/10.1166/jmihi.2020.3184 (link http://dx.doi.org/10.1166/jmihi.2020.3184). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.
Solution Architecture Layered: described as: .
Performance/Efficiency is reached/addressed using/by IoT objects connected at the edge of the network (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: A novel architecture for the Internet of Things based E-Health systems, 2020, http://dx.doi.org/10.1166/jmihi.2020.3184.

The solution is composed by the architecture named as A4IoT (Anyplace 4.0 IoT), proposed in the paper The Anyplace 4.0 IoT Localization Architecture, 2020, with D.O.I. 10.1109/MDM48529.2020.00045 (link https://ieeexplore.ieee.org/document/9162205). This solution addresses the following Quality Requirements: Flexibility,Interaction Capability,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and A4IoT (Anyplace 4.0 IoT).
Solution Architecture A4IoT (Anyplace 4.0 IoT): described as: It is an open-source architecture designed for IoT-based indoor localization using technologies like Wi-Fi, BLE, Cellular, UWB, and Computer Vision (CV). The system supports edge computing, timeseries databases, and crowdsourcing to manage localization tasks across various smart spaces, such as factories, healthcare facilities, and emergency response environments. Its modular and scalable design ensures integration across platforms like Android, Linux, Windows, and Robot OS, making it adaptable to diverse IoT ecosystems..
Performance/Efficiency is reached/addressed using/by Time-Series Database (InfluxDB) (Time Behavior - With temporal queries we can now consume high-volume input streams to provide real-time IoT tracking.").
Reliability is reached/addressed using/by Multi-node Cluster Deployment (Fault Tolerance - The production environment uses a 3-node cluster with two replicas, for both the database engines and the DFS (Distributed Filesystem). This setup allows for a full operation with just a single node active at any time.").
Security is reached/addressed using/by SSL certificates - Encrypted Communications (Confidentiality - All communication is encrypted even in internal networks, as A4IoT image automatically creates and uses SSL certificates.).
Interaction Capability is reached/addressed using/by Web-based Dashboard (Operability - Web Apps built with HTML5, CSS3, and AngularJS.").
Flexibility is reached/addressed using/by Docker containers (Adaptability - It utilizes Docker to enable the deployment of single-node or multi-node configurations, regardless of any dependencies or the underlying OS.).
Reference: The Anyplace 4.0 IoT Localization Architecture, 2020, https://ieeexplore.ieee.org/document/9162205.
The solution is composed by the architecture named as Agricultural IoT Reference Architecture (AITRA), proposed in the paper Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3031634 (link https://doi.org/10.1109/ACCESS.2020.3031634). This solution addresses the following Quality Requirements: Compatibility,Reliability,Security,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Agricultural IoT Reference Architecture (AITRA).
Solution Architecture Agricultural IoT Reference Architecture (AITRA): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Reliability is reached/addressed using/by Edge gateway (No description.).
Security is reached/addressed using/by Authentication, authorization, and data access control (No description.).
Reference: Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, https://doi.org/10.1109/ACCESS.2020.3031634.
The solution is composed by the architecture named as Blockchain-based Zero Trust, proposed in the paper Blockchain-Based Zero Trust on the Edge, 2023, with D.O.I. 10.1109/CSCI62032.2023.00167 (link https://ieeexplore.ieee.org/document/10590526). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Performance/Efficiency,Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Blockchain-based Zero Trust.
Solution Architecture Blockchain-based Zero Trust: described as: The architecture proposed in "Blockchain-Based Zero Trust on the Edge" integrates Zero Trust principles with Blockchain to ensure secure access and trust management for Edge Computing in IoT environments. It focuses on continuous verification of users and devices, ensuring that all access is authenticated before being granted. Blockchain is used for decentralized trust management, maintaining an immutable record of activities, which enhances security and transparency. This setup is optimized for edge computing, reducing latency by processing data locally while maintaining the flexibility to scale across diverse IoT applications, such as smart cities and industrial IoT..
Security is reached/addressed using/by Zero Trust Architecture (ZTA) (Confidentiality - Zero Trust Architecture (ZTA)
ZTA ensures continuous verification and monitoring of all network traffic, restricting unauthorized access.).
Security is reached/addressed using/by Blockchain Technology (Integrity - Blockchain serves as an immutable ledger for storing requests, ensuring

data integrity and trustworthiness.).
Performance/Efficiency is reached/addressed using/by Redis Message Broker (Time Behavior - Redis Message Broker for Asynchronous Processing [redis.io]
Asynchronous processing for write operations improves system responsiveness under high loads.).
Maintainability is reached/addressed using/by Component isolation through microservices (Modularity -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.
).
Flexibility is reached/addressed using/by Component isolation through microservices (Scalability -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.).
Reference: Blockchain-Based Zero Trust on the Edge, 2023, https://ieeexplore.ieee.org/document/10590526.
The solution is composed by the architecture named as Blockchain-enabled, proposed in the paper An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, with D.O.I. doi.org/10.1109/TSC.2020.2966970 (link https://doi.org/10.1109/TSC.2020.2966970). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Blockchain-enabled.
Solution Architecture Blockchain-enabled: described as: .
Performance/Efficiency is reached/addressed using/by Routing protocol designed for IoT devices in the SDN controller [Yazdinejad et al. 2020] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, https://doi.org/10.1109/TSC.2020.2966970.
The solution is composed by the architecture named as Cloud-based Secure Biometric, proposed in the paper BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, with D.O.I. doi.org/10.1016/j.jksuci.2017.07.001 (link https://doi.org/10.1016/j.jksuci.2017.07.001). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Cloud-based Secure Biometric.
Solution Architecture Cloud-based Secure Biometric: described as: .
Flexibility is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Reliability is reached/addressed using/by MapReduce framework (No description.).
Security is reached/addressed using/by Cloud-based biometric authentication [Shakil et al. 2020] (No description.).
Reference: BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, https://doi.org/10.1016/j.jksuci.2017.07.001.
The solution is composed by the architecture named as Deep Learning based, proposed in the paper HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, with D.O.I. doi.org/10.1016/j.future.2019.10.043 (link https://doi.org/10.1016/j.future.2019.10.043). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Deep Learning based.
Solution Architecture Deep Learning based: described as: .
Performance/Efficiency is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Security is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Reference: HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, https://doi.org/10.1016/j.future.2019.10.043.
The solution is composed by the architecture named as Distributed Blockchain-SDN based, proposed in the paper DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3039113 (link https://doi.org/10.1109/ACCESS.2020.3039113). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Distributed Blockchain-SDN based.
Solution Architecture Distributed Blockchain-SDN based: described as: .
Performance/Efficiency is reached/addressed using/by Decentralized blockchain nodes, dynamic SDN controllers, and edge computing (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, https://doi.org/10.1109/ACCESS.2020.3039113.
The solution is composed by the architecture named as Distributed Edge-Cloud, proposed in the paper Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, with D.O.I. doi.org/10.1016/j.future.2020.05.042 (link https://doi.org/10.1016/j.future.2020.05.042). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Distributed Edge-Cloud.
Solution Architecture Distributed Edge-Cloud: described as: .
Security is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020]

(No description.).
Performance/Efficiency is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Reference: Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, https://doi.org/10.1016/j.future.2020.05.042.
The solution is composed by the architecture named as Edge Computing-based Fault Tolerant Framework, proposed in the paper Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, with D.O.I. doi.org/10.1109/IWCMC48107.2020.9148269 (link https://doi.org/10.1109/IWCMC48107.2020.9148269). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Edge Computing-based Fault Tolerant Framework.
Solution Architecture Edge Computing-based Fault Tolerant Framework: described as: .
Performance/Efficiency is reached/addressed using/by Kubernetes framework (No description.).
Flexibility is reached/addressed using/by Docker containers (No description.).
Reference: Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, https://doi.org/10.1109/IWCMC48107.2020.9148269.
The solution is composed by the architecture named as Event-driven IoT, proposed in the paper Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, with D.O.I. doi.org/10.1007/s10586-020-03189-w (link https://doi.org/10.1007/s10586-020-03189-w). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Event-driven IoT.
Solution Architecture Event-driven IoT: described as: .
Performance/Efficiency is reached/addressed using/by Complex Event Processing (CEP) [Rahmani et al. 2020] (No description.).
Reliability is reached/addressed using/by TCP-IP protocol on Client/Server architecture for data communication (No description.).
Reference: Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, https://doi.org/10.1007/s10586-020-03189-w.
The solution is composed by the architecture named as Fog Computing based, proposed in the paper FOG-based architecture and load balancing methodology for health monitoring systems, 2021, with D.O.I. doi.org/10.1109/ACCESS.2021.3094033 (link https://doi.org/10.1109/ACCESS.2021.3094033). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Fog Computing based.
Solution Architecture Fog Computing based: described as: .
Performance/Efficiency is reached/addressed using/by Load Balancing Scheme (LBS) - Distributed workload to all fog nodes [Asghar et al. 2021] (No description.).
Reference: FOG-based architecture and load balancing methodology for health monitoring systems, 2021, https://doi.org/10.1109/ACCESS.2021.3094033.
The solution is composed by the architecture named as Geographic-based, proposed in the paper A platform for integrating heterogeneous data and developing smart city applications, 2022, with D.O.I. doi.org/10.1016/j.future.2021.10.030 (link https://doi.org/10.1016/j.future.2021.10.030). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Geographic-based.
Solution Architecture Geographic-based: described as: .
Security is reached/addressed using/by FIWARE Platform [www.fiware.org] (Authentication, authorization, and data access control) (No description.).
Flexibility is reached/addressed using/by Virtualization (No description.).
Compatibility is reached/addressed using/by RESTFul APIs (No description.).
Reference: A platform for integrating heterogeneous data and developing smart city applications, 2022, https://doi.org/10.1016/j.future.2021.10.030.
The solution is composed by the architecture named as Hybrid (Centralized and Distributed), proposed in the paper Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, with D.O.I. doi.org/10.1016/j.procs.2021.03.030 (link https://doi.org/10.1016/j.procs.2021.03.030). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid (Centralized and Distributed).
Solution Architecture Hybrid (Centralized and Distributed): described as: .
Performance/Efficiency is reached/addressed using/by Dynamic allocation of the resources using Software Defined Networking (SDN) [Chudhary and Sharma 2021] (No description.).
Reference: Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, https://doi.org/10.1016/j.procs.2021.03.030.
The solution is composed by the architecture named as Hybrid Cloud-Fog Computing, proposed in the paper Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, with D.O.I. doi.org/10.1007/s10586-021-03375-4 (link https://link.springer.com/article/10.1007/s10586-021-03375-4). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid Cloud-Fog Computing.
Solution Architecture Hybrid Cloud-Fog Computing: described as: .
Reliability is reached/addressed using/by Fog layer in two levels: 1-Intra mobility (User network): Nodes (Data transfer, device management,

temporary local storage, notifications, and data consistency). 2-Extra mobility (Different domains): Smart Gateways (Local processing and database) (No description.).
Performance/Efficiency is reached/addressed using/by Fog computing (No description.).
Reference: Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, https://link.springer.com/article/10.1007/s10586-021-03375-4.
The solution is composed by the architecture named as Industrial Internet of Things (IIoT), proposed in the paper A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Performance/Efficiency,Security,.
The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and Industrial Internet of Things (IIoT).
Solution Architecture Industrial Internet of Things (IIoT): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Flexibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Performance/Efficiency is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).
Security is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).
Reference: A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, https://doi.org/10.3390/s20195603.
The solution is composed by the architecture named as Layered, proposed in the paper An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, with D.O.I. doi.org/10.1016/j.future.2020.07.047 (link https://doi.org/10.1016/j.future.2020.07.047). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.
Solution Architecture Layered: described as: .
Flexibility is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).
Performance/Efficiency is reached/addressed using/by Big data analytics engine [Ali et al. 2021] (No description.).
Security is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).
Reference: An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, https://doi.org/10.1016/j.future.2020.07.047.
The solution is composed by the architecture named as Layered, proposed in the paper A novel architecture for the Internet of Things based E-Health systems, 2020, with D.O.I. doi.org/10.1166/jmihi.2020.3184 (link http://dx.doi.org/10.1166/jmihi.2020.3184). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.
Solution Architecture Layered: described as: .
Performance/Efficiency is reached/addressed using/by IoT objects connected at the edge of the network (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: A novel architecture for the Internet of Things based E-Health systems, 2020, http://dx.doi.org/10.1166/jmihi.2020.3184.
The solution is composed by the architecture named as Layered, proposed in the paper Software architecture quality attributes of a layered sensor-based IoT system, 2021, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Reliability,Security,.
The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and Layered.
Solution Architecture Layered: described as: .
Flexibility is reached/addressed using/by Messaging protocol brokers and at the client side, web-based applications (No description.).
Maintainability is reached/addressed using/by Loosely coupled components (No description.).
Reliability is reached/addressed using/by Open source software solutions and modular architecture design with clear relationships between components (No description.).
Security is reached/addressed using/by IoT devices authentication and data encryption (No description.).
Reference: Software architecture quality attributes of a layered sensor-based IoT system, 2021, https://doi.org/10.3390/s20195603.

The solution is composed by the architecture named as A4IoT (Anyplace 4.0 IoT), proposed in the paper The Anyplace 4.0 IoT Localization Architecture, 2020, with D.O.I. 10.1109/MDM48529.2020.00045 (link https://ieeexplore.ieee.org/document/9162205). This solution addresses the following Quality Requirements: Flexibility,Interaction Capability,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and A4IoT (Anyplace 4.0 IoT).
Solution Architecture A4IoT (Anyplace 4.0 IoT): described as: It is an open-source architecture designed for IoT-based indoor localization using technologies like Wi-Fi, BLE, Cellular, UWB, and Computer Vision (CV). The system supports edge computing, timeseries databases, and crowdsourcing to manage localization tasks across various smart spaces, such as factories, healthcare facilities, and emergency response environments. Its modular and scalable design ensures integration across platforms like Android, Linux, Windows, and Robot OS, making it adaptable to diverse IoT ecosystems..
Performance/Efficiency is reached/addressed using/by Time-Series Database (InfluxDB) (Time Behavior - With temporal queries we can now consume high-volume input streams to provide real-time IoT tracking.").

Reliability is reached/addressed using/by Multi-node Cluster Deployment (Fault Tolerance - The production environment uses a 3-node cluster with two replicas, for both the database engines and the DFS (Distributed Filesystem). This setup allows for a full operation with just a single node active at any time.").
Security is reached/addressed using/by SSL certificates - Encrypted Communications (Confidentiality - All communication is encrypted even in internal networks, as A4IoT image automatically creates and uses SSL certificates.).
Interaction Capability is reached/addressed using/by Web-based Dashboard (Operability - Web Apps built with HTML5, CSS3, and AngularJS.").
Flexibility is reached/addressed using/by Docker containers (Adaptability - It utilizes Docker to enable the deployment of single-node or multi-node configurations, regardless of any dependencies or the underlying OS.).
Reference: The Anyplace 4.0 IoT Localization Architecture, 2020, https://ieeexplore.ieee.org/document/9162205.
The solution is composed by the architecture named as Agricultural IoT Reference Architecture (AITRA), proposed in the paper Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3031634 (link https://doi.org/10.1109/ACCESS.2020.3031634). This solution addresses the following Quality Requirements: Compatibility,Reliability,Security,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Agricultural IoT Reference Architecture (AITRA).
Solution Architecture Agricultural IoT Reference Architecture (AITRA): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Reliability is reached/addressed using/by Edge gateway (No description.).
Security is reached/addressed using/by Authentication, authorization, and data access control (No description.).
Reference: Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, https://doi.org/10.1109/ACCESS.2020.3031634.
The solution is composed by the architecture named as Blockchain-based Zero Trust, proposed in the paper Blockchain-Based Zero Trust on the Edge, 2023, with D.O.I. 10.1109/CSCI62032.2023.00167 (link https://ieeexplore.ieee.org/document/10590526). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Performance/Efficiency,Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Blockchain-based Zero Trust.
Solution Architecture Blockchain-based Zero Trust: described as: The architecture proposed in "Blockchain-Based Zero Trust on the Edge" integrates Zero Trust principles with Blockchain to ensure secure access and trust management for Edge Computing in IoT environments. It focuses on continuous verification of users and devices, ensuring that all access is authenticated before being granted. Blockchain is used for decentralized trust management, maintaining an immutable record of activities, which enhances security and transparency. This setup is optimized for edge computing, reducing latency by processing data locally while maintaining the flexibility to scale across diverse IoT applications, such as smart cities and industrial IoT..
Security is reached/addressed using/by Zero Trust Architecture (ZTA) (Confidentiality - Zero Trust Architecture (ZTA)
ZTA ensures continuous verification and monitoring of all network traffic, restricting unauthorized access.).
Security is reached/addressed using/by Blockchain Technology (Integrity - Blockchain serves as an immutable ledger for storing requests, ensuring data integrity and trustworthiness.).
Performance/Efficiency is reached/addressed using/by Redis Message Broker (Time Behavior - Redis Message Broker for Asynchronous Processing [redis.io]
Asynchronous processing for write operations improves system responsiveness under high loads.).
Maintainability is reached/addressed using/by Component isolation through microservices (Modularity -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.
).
Flexibility is reached/addressed using/by Component isolation through microservices (Scalability -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.).
Reference: Blockchain-Based Zero Trust on the Edge, 2023, https://ieeexplore.ieee.org/document/10590526.
The solution is composed by the architecture named as Blockchain-enabled, proposed in the paper An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, with D.O.I. doi.org/10.1109/TSC.2020.2966970 (link https://doi.org/10.1109/TSC.2020.2966970). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Blockchain-enabled.
Solution Architecture Blockchain-enabled: described as: .
Performance/Efficiency is reached/addressed using/by Routing protocol designed for IoT devices in the SDN controller [Yazdinejad et al. 2020] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, https://doi.org/10.1109/TSC.2020.2966970.
The solution is composed by the architecture named as Cloud-based Secure Biometric, proposed in the paper BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, with D.O.I. doi.org/10.1016/j.jksuci.2017.07.001 (link https://doi.org/10.1016/j.jksuci.2017.07.001). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Cloud-based Secure Biometric.
Solution Architecture Cloud-based Secure Biometric: described as: .
Flexibility is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).

Performance/Efficiency is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Reliability is reached/addressed using/by MapReduce framework (No description.).
Security is reached/addressed using/by Cloud-based biometric authentication [Shakil et al. 2020] (No description.).
Reference: BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, https://doi.org/10.1016/j.jksuci.2017.07.001.
The solution is composed by the architecture named as Deep Learning based, proposed in the paper HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, with D.O.I. doi.org/10.1016/j.future.2019.10.043 (link https://doi.org/10.1016/j.future.2019.10.043). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Deep Learning based.
Solution Architecture Deep Learning based: described as: .
Performance/Efficiency is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Security is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Reference: HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, https://doi.org/10.1016/j.future.2019.10.043.
The solution is composed by the architecture named as Distributed Blockchain-SDN based, proposed in the paper DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3039113 (link https://doi.org/10.1109/ACCESS.2020.3039113). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Distributed Blockchain-SDN based.
Solution Architecture Distributed Blockchain-SDN based: described as: .
Performance/Efficiency is reached/addressed using/by Decentralized blockchain nodes, dynamic SDN controllers, and edge computing (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, https://doi.org/10.1109/ACCESS.2020.3039113.
The solution is composed by the architecture named as Distributed Edge-Cloud, proposed in the paper Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, with D.O.I. doi.org/10.1016/j.future.2020.05.042 (link https://doi.org/10.1016/j.future.2020.05.042). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Distributed Edge-Cloud.
Solution Architecture Distributed Edge-Cloud: described as: .
Security is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Reference: Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, https://doi.org/10.1016/j.future.2020.05.042.
The solution is composed by the architecture named as Edge Computing-based Fault Tolerant Framework, proposed in the paper Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, with D.O.I. doi.org/10.1109/IWCMC48107.2020.9148269 (link https://doi.org/10.1109/IWCMC48107.2020.9148269). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Edge Computing-based Fault Tolerant Framework.
Solution Architecture Edge Computing-based Fault Tolerant Framework: described as: .
Performance/Efficiency is reached/addressed using/by Kubernetes framework (No description.).
Flexibility is reached/addressed using/by Docker containers (No description.).
Reference: Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, https://doi.org/10.1109/IWCMC48107.2020.9148269.
The solution is composed by the architecture named as Event-driven IoT, proposed in the paper Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, with D.O.I. doi.org/10.1007/s10586-020-03189-w (link https://doi.org/10.1007/s10586-020-03189-w). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Event-driven IoT.
Solution Architecture Event-driven IoT: described as: .
Performance/Efficiency is reached/addressed using/by Complex Event Processing (CEP) [Rahmani et al. 2020] (No description.).
Reliability is reached/addressed using/by TCP-IP protocol on Client/Server architecture for data communication (No description.).
Reference: Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, https://doi.org/10.1007/s10586-020-03189-w.
The solution is composed by the architecture named as Fog Computing based, proposed in the paper FOG-based architecture and load balancing methodology for health monitoring systems, 2021, with D.O.I. doi.org/10.1109/ACCESS.2021.3094033 (link https://doi.org/10.1109/ACCESS.2021.3094033). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Fog

Computing based.
Solution Architecture Fog Computing based: described as: .
Performance/Efficiency is reached/addressed using/by Load Balancing Scheme (LBS) - Distributed workload to all fog nodes [Asghar et al. 2021] (No description.).
Reference: FOG-based architecture and load balancing methodology for health monitoring systems, 2021, https://doi.org/10.1109/ACCESS.2021.3094033.
The solution is composed by the architecture named as Geographic-based, proposed in the paper A platform for integrating heterogeneous data and developing smart city applications, 2022, with D.O.I. doi.org/10.1016/j.future.2021.10.030 (link https://doi.org/10.1016/j.future.2021.10.030). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Geographic-based.
Solution Architecture Geographic-based: described as: .
Security is reached/addressed using/by FIWARE Platform [www.fiware.org] (Authentication, authorization, and data access control) (No description.).
Flexibility is reached/addressed using/by Virtualization (No description.).
Compatibility is reached/addressed using/by RESTFul APIs (No description.).
Reference: A platform for integrating heterogeneous data and developing smart city applications, 2022, https://doi.org/10.1016/j.future.2021.10.030.
The solution is composed by the architecture named as Hybrid (Centralized and Distributed), proposed in the paper Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, with D.O.I. doi.org/10.1016/j.procs.2021.03.030 (link https://doi.org/10.1016/j.procs.2021.03.030). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid (Centralized and Distributed).
Solution Architecture Hybrid (Centralized and Distributed): described as: .
Performance/Efficiency is reached/addressed using/by Dynamic allocation of the resources using Software Defined Networking (SDN) [Chudhary and Sharma 2021] (No description.).
Reference: Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, https://doi.org/10.1016/j.procs.2021.03.030.
The solution is composed by the architecture named as Hybrid Cloud-Fog Computing, proposed in the paper Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, with D.O.I. doi.org/10.1007/s10586-021-03375-4 (link https://link.springer.com/article/10.1007/s10586-021-03375-4). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid Cloud-Fog Computing.
Solution Architecture Hybrid Cloud-Fog Computing: described as: .
Reliability is reached/addressed using/by Fog layer in two levels: 1-Intra mobility (User network): Nodes (Data transfer, device management, temporary local storage, notifications, and data consistency). 2-Extra mobility (Different domains): Smart Gateways (Local processing and database) (No description.).
Performance/Efficiency is reached/addressed using/by Fog computing (No description.).
Reference: Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, https://link.springer.com/article/10.1007/s10586-021-03375-4.
The solution is composed by the architecture named as Industrial Internet of Things (IIoT), proposed in the paper A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Performance/Efficiency,Security,.
The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and Industrial Internet of Things (IIoT).
Solution Architecture Industrial Internet of Things (IIoT): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Flexibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Performance/Efficiency is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).
Security is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).
Reference: A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, https://doi.org/10.3390/s20195603.
The solution is composed by the architecture named as Layered, proposed in the paper An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, with D.O.I. doi.org/10.1016/j.future.2020.07.047 (link https://doi.org/10.1016/j.future.2020.07.047). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.
Solution Architecture Layered: described as: .
Flexibility is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).
Performance/Efficiency is reached/addressed using/by Big data analytics engine [Ali et al. 2021] (No description.).
Security is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).
Reference: An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, https://doi.org/10.1016/j.future.2020.07.047.
The solution is composed by the architecture named as Layered, proposed in the paper A novel architecture for the Internet of Things based E-

Health systems, 2020, with D.O.I. doi.org/10.1166/jmihi.2020.3184 (link http://dx.doi.org/10.1166/jmihi.2020.3184). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.
Solution Architecture Layered: described as: .
Performance/Efficiency is reached/addressed using/by IoT objects connected at the edge of the network (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: A novel architecture for the Internet of Things based E-Health systems, 2020, http://dx.doi.org/10.1166/jmihi.2020.3184.
The solution is composed by the architecture named as Layered, proposed in the paper Software architecture quality attributes of a layered sensor-based IoT system, 2021, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Reliability,Security,.
The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and Layered.
Solution Architecture Layered: described as: .
Flexibility is reached/addressed using/by Messaging protocol brokers and at the client side, web-based applications (No description.).
Maintainability is reached/addressed using/by Loosely coupled components (No description.).
Reliability is reached/addressed using/by Open source software solutions and modular architecture design with clear relationships between components (No description.).
Security is reached/addressed using/by IoT devices authentication and data encryption (No description.).
Reference: Software architecture quality attributes of a layered sensor-based IoT system, 2021, https://doi.org/10.3390/s20195603.
The solution is composed by the architecture named as Layered, proposed in the paper Cloud-FOG assisted Energy Efficient Architectural Paradigm for disaster evacuation, 2022, with D.O.I. doi.org/10.1016/j.is.2021.101732 (link https://doi.org/10.1016/j.is.2021.101732). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Layered.
Solution Architecture Layered: described as: .
Performance/Efficiency is reached/addressed using/by Algorithm for Fog layer data selection (Data uniqueness and only important data) for transmission [Kaur et al. 2022] (No description.).
Reference: Cloud-FOG assisted Energy Efficient Architectural Paradigm for disaster evacuation, 2022, https://doi.org/10.1016/j.is.2021.101732.

The solution is composed by the architecture named as A4IoT (Anyplace 4.0 IoT), proposed in the paper The Anyplace 4.0 IoT Localization Architecture, 2020, with D.O.I. 10.1109/MDM48529.2020.00045 (link https://ieeexplore.ieee.org/document/9162205). This solution addresses the following Quality Requirements: Flexibility,Interaction Capability,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and A4IoT (Anyplace 4.0 IoT).
Solution Architecture A4IoT (Anyplace 4.0 IoT): described as: It is an open-source architecture designed for IoT-based indoor localization using technologies like Wi-Fi, BLE, Cellular, UWB, and Computer Vision (CV). The system supports edge computing, timeseries databases, and crowdsourcing to manage localization tasks across various smart spaces, such as factories, healthcare facilities, and emergency response environments. Its modular and scalable design ensures integration across platforms like Android, Linux, Windows, and Robot OS, making it adaptable to diverse IoT ecosystems..
Performance/Efficiency is reached/addressed using/by Time-Series Database (InfluxDB) (Time Behavior - With temporal queries we can now consume high-volume input streams to provide real-time IoT tracking.").
Reliability is reached/addressed using/by Multi-node Cluster Deployment (Fault Tolerance - The production environment uses a 3-node cluster with two replicas, for both the database engines and the DFS (Distributed Filesystem). This setup allows for a full operation with just a single node active at any time.").
Security is reached/addressed using/by SSL certificates - Encrypted Communications (Confidentiality - All communication is encrypted even in internal networks, as A4IoT image automatically creates and uses SSL certificates.).
Interaction Capability is reached/addressed using/by Web-based Dashboard (Operability - Web Apps built with HTML5, CSS3, and AngularJS.").
Flexibility is reached/addressed using/by Docker containers (Adaptability - It utilizes Docker to enable the deployment of single-node or multi-node configurations, regardless of any dependencies or the underlying OS.).
Reference: The Anyplace 4.0 IoT Localization Architecture, 2020, https://ieeexplore.ieee.org/document/9162205.
The solution is composed by the architecture named as Agricultural IoT Reference Architecture (AITRA), proposed in the paper Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3031634 (link https://doi.org/10.1109/ACCESS.2020.3031634). This solution addresses the following Quality Requirements: Compatibility,Reliability,Security,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Agricultural IoT Reference Architecture (AITRA).
Solution Architecture Agricultural IoT Reference Architecture (AITRA): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Reliability is reached/addressed using/by Edge gateway (No description.).
Security is reached/addressed using/by Authentication, authorization, and data access control (No description.).
Reference: Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, https://doi.org/10.1109/ACCESS.2020.3031634.

The solution is composed by the architecture named as Blockchain-based Zero Trust, proposed in the paper Blockchain-Based Zero Trust on the Edge, 2023, with D.O.I. 10.1109/CSCI62032.2023.00167 (link https://ieeexplore.ieee.org/document/10590526). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Performance/Efficiency,Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Blockchain-based Zero Trust.
Solution Architecture Blockchain-based Zero Trust: described as: The architecture proposed in "Blockchain-Based Zero Trust on the Edge" integrates Zero Trust principles with Blockchain to ensure secure access and trust management for Edge Computing in IoT environments. It focuses on continuous verification of users and devices, ensuring that all access is authenticated before being granted. Blockchain is used for decentralized trust management, maintaining an immutable record of activities, which enhances security and transparency. This setup is optimized for edge computing, reducing latency by processing data locally while maintaining the flexibility to scale across diverse IoT applications, such as smart cities and industrial IoT..
Security is reached/addressed using/by Zero Trust Architecture (ZTA) (Confidentiality - Zero Trust Architecture (ZTA)
ZTA ensures continuous verification and monitoring of all network traffic, restricting unauthorized access.).
Security is reached/addressed using/by Blockchain Technology (Integrity - Blockchain serves as an immutable ledger for storing requests, ensuring data integrity and trustworthiness.).
Performance/Efficiency is reached/addressed using/by Redis Message Broker (Time Behavior - Redis Message Broker for Asynchronous Processing [redis.io]
Asynchronous processing for write operations improves system responsiveness under high loads.).
Maintainability is reached/addressed using/by Component isolation through microservices (Modularity -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.
).
Flexibility is reached/addressed using/by Component isolation through microservices (Scalability -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.).
Reference: Blockchain-Based Zero Trust on the Edge, 2023, https://ieeexplore.ieee.org/document/10590526.
The solution is composed by the architecture named as Blockchain-enabled, proposed in the paper An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, with D.O.I. doi.org/10.1109/TSC.2020.2966970 (link https://doi.org/10.1109/TSC.2020.2966970). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Blockchain-enabled.
Solution Architecture Blockchain-enabled: described as: .
Performance/Efficiency is reached/addressed using/by Routing protocol designed for IoT devices in the SDN controller [Yazdinejad et al. 2020] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, https://doi.org/10.1109/TSC.2020.2966970.
The solution is composed by the architecture named as Cloud-based Secure Biometric, proposed in the paper BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, with D.O.I. doi.org/10.1016/j.jksuci.2017.07.001 (link https://doi.org/10.1016/j.jksuci.2017.07.001). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Cloud-based Secure Biometric.
Solution Architecture Cloud-based Secure Biometric: described as: .
Flexibility is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Reliability is reached/addressed using/by MapReduce framework (No description.).
Security is reached/addressed using/by Cloud-based biometric authentication [Shakil et al. 2020] (No description.).
Reference: BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, https://doi.org/10.1016/j.jksuci.2017.07.001.
The solution is composed by the architecture named as Deep Learning based, proposed in the paper HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, with D.O.I. doi.org/10.1016/j.future.2019.10.043 (link https://doi.org/10.1016/j.future.2019.10.043). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Deep Learning based.
Solution Architecture Deep Learning based: described as: .
Performance/Efficiency is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Security is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Reference: HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, https://doi.org/10.1016/j.future.2019.10.043.
The solution is composed by the architecture named as Distributed Blockchain-SDN based, proposed in the paper DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3039113 (link https://doi.org/10.1109/ACCESS.2020.3039113). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and

Distributed Blockchain-SDN based.
Solution Architecture Distributed Blockchain-SDN based: described as: .
Performance/Efficiency is reached/addressed using/by Decentralized blockchain nodes, dynamic SDN controllers, and edge computing (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, https://doi.org/10.1109/ACCESS.2020.3039113.
The solution is composed by the architecture named as Distributed Edge-Cloud, proposed in the paper Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, with D.O.I. doi.org/10.1016/j.future.2020.05.042 (link https://doi.org/10.1016/j.future.2020.05.042). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Distributed Edge-Cloud.
Solution Architecture Distributed Edge-Cloud: described as: .
Security is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Reference: Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, https://doi.org/10.1016/j.future.2020.05.042.
The solution is composed by the architecture named as Edge Computing-based Fault Tolerant Framework, proposed in the paper Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, with D.O.I. doi.org/10.1109/IWCMC48107.2020.9148269 (link https://doi.org/10.1109/IWCMC48107.2020.9148269). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Edge Computing-based Fault Tolerant Framework.
Solution Architecture Edge Computing-based Fault Tolerant Framework: described as: .
Performance/Efficiency is reached/addressed using/by Kubernetes framework (No description.).
Flexibility is reached/addressed using/by Docker containers (No description.).
Reference: Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, https://doi.org/10.1109/IWCMC48107.2020.9148269.
The solution is composed by the architecture named as Event-driven IoT, proposed in the paper Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, with D.O.I. doi.org/10.1007/s10586-020-03189-w (link https://doi.org/10.1007/s10586-020-03189-w). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Event-driven IoT.
Solution Architecture Event-driven IoT: described as: .
Performance/Efficiency is reached/addressed using/by Complex Event Processing (CEP) [Rahmani et al. 2020] (No description.).
Reliability is reached/addressed using/by TCP-IP protocol on Client/Server architecture for data communication (No description.).
Reference: Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, https://doi.org/10.1007/s10586-020-03189-w.
The solution is composed by the architecture named as Fog Computing based, proposed in the paper FOG-based architecture and load balancing methodology for health monitoring systems, 2021, with D.O.I. doi.org/10.1109/ACCESS.2021.3094033 (link https://doi.org/10.1109/ACCESS.2021.3094033). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Fog Computing based.
Solution Architecture Fog Computing based: described as: .
Performance/Efficiency is reached/addressed using/by Load Balancing Scheme (LBS) - Distributed workload to all fog nodes [Asghar et al. 2021] (No description.).
Reference: FOG-based architecture and load balancing methodology for health monitoring systems, 2021, https://doi.org/10.1109/ACCESS.2021.3094033.
The solution is composed by the architecture named as Geographic-based, proposed in the paper A platform for integrating heterogeneous data and developing smart city applications, 2022, with D.O.I. doi.org/10.1016/j.future.2021.10.030 (link https://doi.org/10.1016/j.future.2021.10.030). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Geographic-based.
Solution Architecture Geographic-based: described as: .
Security is reached/addressed using/by FIWARE Platform [www.fiware.org] (Authentication, authorization, and data access control) (No description.).
Flexibility is reached/addressed using/by Virtualization (No description.).
Compatibility is reached/addressed using/by RESTFul APIs (No description.).
Reference: A platform for integrating heterogeneous data and developing smart city applications, 2022, https://doi.org/10.1016/j.future.2021.10.030.
The solution is composed by the architecture named as Hybrid (Centralized and Distributed), proposed in the paper Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, with D.O.I. doi.org/10.1016/j.procs.2021.03.030 (link https://doi.org/10.1016/j.procs.2021.03.030). This solution addresses the following Quality Requirements: Performance/Efficiency,.

The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid (Centralized and Distributed).
Solution Architecture Hybrid (Centralized and Distributed): described as: .
Performance/Efficiency is reached/addressed using/by Dynamic allocation of the resources using Software Defined Networking (SDN) [Chudhary and Sharma 2021] (No description.).
Reference: Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, https://doi.org/10.1016/j.procs.2021.03.030.
The solution is composed by the architecture named as Hybrid Cloud-Fog Computing, proposed in the paper Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, with D.O.I. doi.org/10.1007/s10586-021-03375-4 (link https://link.springer.com/article/10.1007/s10586-021-03375-4). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid Cloud-Fog Computing.
Solution Architecture Hybrid Cloud-Fog Computing: described as: .
Reliability is reached/addressed using/by Fog layer in two levels: 1-Intra mobility (User network): Nodes (Data transfer, device management, temporary local storage, notifications, and data consistency). 2-Extra mobility (Different domains): Smart Gateways (Local processing and database) (No description.).
Performance/Efficiency is reached/addressed using/by Fog computing (No description.).
Reference: Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, https://link.springer.com/article/10.1007/s10586-021-03375-4.
The solution is composed by the architecture named as Industrial Internet of Things (IIoT), proposed in the paper A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Performance/Efficiency,Security,.
The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and Industrial Internet of Things (IIoT).
Solution Architecture Industrial Internet of Things (IIoT): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Flexibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Performance/Efficiency is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).
Security is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).
Reference: A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, https://doi.org/10.3390/s20195603.
The solution is composed by the architecture named as Layered, proposed in the paper An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, with D.O.I. doi.org/10.1016/j.future.2020.07.047 (link https://doi.org/10.1016/j.future.2020.07.047). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.
Solution Architecture Layered: described as: .
Flexibility is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).
Performance/Efficiency is reached/addressed using/by Big data analytics engine [Ali et al. 2021] (No description.).
Security is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).
Reference: An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, https://doi.org/10.1016/j.future.2020.07.047.
The solution is composed by the architecture named as Layered, proposed in the paper A novel architecture for the Internet of Things based E-Health systems, 2020, with D.O.I. doi.org/10.1166/jmihi.2020.3184 (link http://dx.doi.org/10.1166/jmihi.2020.3184). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.
Solution Architecture Layered: described as: .
Performance/Efficiency is reached/addressed using/by IoT objects connected at the edge of the network (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: A novel architecture for the Internet of Things based E-Health systems, 2020, http://dx.doi.org/10.1166/jmihi.2020.3184.
The solution is composed by the architecture named as Layered, proposed in the paper Software architecture quality attributes of a layered sensor-based IoT system, 2021, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Reliability,Security,.
The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and Layered.
Solution Architecture Layered: described as: .
Flexibility is reached/addressed using/by Messaging protocol brokers and at the client side, web-based applications (No description.).
Maintainability is reached/addressed using/by Loosely coupled components (No description.).
Reliability is reached/addressed using/by Open source software solutions and modular architecture design with clear relationships between components (No description.).
Security is reached/addressed using/by IoT devices authentication and data encryption (No description.).
Reference: Software architecture quality attributes of a layered sensor-based IoT system, 2021, https://doi.org/10.3390/s20195603.

The solution is composed by the architecture named as Layered, proposed in the paper Cloud-FOG assisted Energy Efficient Architectural Paradigm for disaster evacuation, 2022, with D.O.I. doi.org/10.1016/j.is.2021.101732 (link https://doi.org/10.1016/j.is.2021.101732). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Layered.
Solution Architecture Layered: described as: .
Performance/Efficiency is reached/addressed using/by Algorithm for Fog layer data selection (Data uniqueness and only important data) for transmission [Kaur et al. 2022] (No description.).
Reference: Cloud-FOG assisted Energy Efficient Architectural Paradigm for disaster evacuation, 2022, https://doi.org/10.1016/j.is.2021.101732.
The solution is composed by the architecture named as Layered, proposed in the paper An agile AI and IoT-Augmented Smart Farming: a Cost-Effective cognitive weather station, 2021, with D.O.I. doi.org/10.3390/agriculture12010035 (link https://doi.org/10.3390/agriculture12010035). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Interaction Capability,Maintainability,Performance/Efficiency,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Layered.
Solution Architecture Layered: described as: .
Compatibility is reached/addressed using/by Docker containers (No description.).
Flexibility is reached/addressed using/by Docker containers (No description.).
Interaction Capability is reached/addressed using/by Remote access to the deployed platform and its components (Web App and SSH) (No description.).
Maintainability is reached/addressed using/by Docker containers (No description.).
Performance/Efficiency is reached/addressed using/by Docker containers (No description.).
Reference: An agile AI and IoT-Augmented Smart Farming: a Cost-Effective cognitive weather station, 2021, https://doi.org/10.3390/agriculture12010035.

The solution is composed by the architecture named as A4IoT (Anyplace 4.0 IoT), proposed in the paper The Anyplace 4.0 IoT Localization Architecture, 2020, with D.O.I. 10.1109/MDM48529.2020.00045 (link https://ieeexplore.ieee.org/document/9162205). This solution addresses the following Quality Requirements: Flexibility,Interaction Capability,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and A4IoT (Anyplace 4.0 IoT).
Solution Architecture A4IoT (Anyplace 4.0 IoT): described as: It is an open-source architecture designed for IoT-based indoor localization using technologies like Wi-Fi, BLE, Cellular, UWB, and Computer Vision (CV). The system supports edge computing, timeseries databases, and crowdsourcing to manage localization tasks across various smart spaces, such as factories, healthcare facilities, and emergency response environments. Its modular and scalable design ensures integration across platforms like Android, Linux, Windows, and Robot OS, making it adaptable to diverse IoT ecosystems..
Performance/Efficiency is reached/addressed using/by Time-Series Database (InfluxDB) (Time Behavior - With temporal queries we can now consume high-volume input streams to provide real-time IoT tracking.").
Reliability is reached/addressed using/by Multi-node Cluster Deployment (Fault Tolerance - The production environment uses a 3-node cluster with two replicas, for both the database engines and the DFS (Distributed Filesystem). This setup allows for a full operation with just a single node active at any time.").
Security is reached/addressed using/by SSL certificates - Encrypted Communications (Confidentiality - All communication is encrypted even in internal networks, as A4IoT image automatically creates and uses SSL certificates.).
Interaction Capability is reached/addressed using/by Web-based Dashboard (Operability - Web Apps built with HTML5, CSS3, and AngularJS.").
Flexibility is reached/addressed using/by Docker containers (Adaptability - It utilizes Docker to enable the deployment of single-node or multi-node configurations, regardless of any dependencies or the underlying OS.).
Reference: The Anyplace 4.0 IoT Localization Architecture, 2020, https://ieeexplore.ieee.org/document/9162205.
The solution is composed by the architecture named as Agricultural IoT Reference Architecture (AITRA), proposed in the paper Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3031634 (link https://doi.org/10.1109/ACCESS.2020.3031634). This solution addresses the following Quality Requirements: Compatibility,Reliability,Security,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Agricultural IoT Reference Architecture (AITRA).
Solution Architecture Agricultural IoT Reference Architecture (AITRA): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Reliability is reached/addressed using/by Edge gateway (No description.).
Security is reached/addressed using/by Authentication, authorization, and data access control (No description.).
Reference: Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, https://doi.org/10.1109/ACCESS.2020.3031634.
The solution is composed by the architecture named as Blockchain-based Zero Trust, proposed in the paper Blockchain-Based Zero Trust on the Edge, 2023, with D.O.I. 10.1109/CSCI62032.2023.00167 (link https://ieeexplore.ieee.org/document/10590526). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Performance/Efficiency,Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Blockchain-based Zero Trust.
Solution Architecture Blockchain-based Zero Trust: described as: The architecture proposed in "Blockchain-Based Zero Trust on the Edge"

integrates Zero Trust principles with Blockchain to ensure secure access and trust management for Edge Computing in IoT environments. It focuses on continuous verification of users and devices, ensuring that all access is authenticated before being granted. Blockchain is used for decentralized trust management, maintaining an immutable record of activities, which enhances security and transparency. This setup is optimized for edge computing, reducing latency by processing data locally while maintaining the flexibility to scale across diverse IoT applications, such as smart cities and industrial IoT..
Security is reached/addressed using/by Zero Trust Architecture (ZTA) (Confidentiality - Zero Trust Architecture (ZTA)
ZTA ensures continuous verification and monitoring of all network traffic, restricting unauthorized access.).
Security is reached/addressed using/by Blockchain Technology (Integrity - Blockchain serves as an immutable ledger for storing requests, ensuring data integrity and trustworthiness.).
Performance/Efficiency is reached/addressed using/by Redis Message Broker (Time Behavior - Redis Message Broker for Asynchronous Processing [redis.io]
Asynchronous processing for write operations improves system responsiveness under high loads.).
Maintainability is reached/addressed using/by Component isolation through microservices (Modularity -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.
).
Flexibility is reached/addressed using/by Component isolation through microservices (Scalability -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.).
Reference: Blockchain-Based Zero Trust on the Edge, 2023, https://ieeexplore.ieee.org/document/10590526.
The solution is composed by the architecture named as Blockchain-enabled, proposed in the paper An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, with D.O.I. doi.org/10.1109/TSC.2020.2966970 (link https://doi.org/10.1109/TSC.2020.2966970). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Blockchain-enabled.
Solution Architecture Blockchain-enabled: described as: .
Performance/Efficiency is reached/addressed using/by Routing protocol designed for IoT devices in the SDN controller [Yazdinejad et al. 2020] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, https://doi.org/10.1109/TSC.2020.2966970.
The solution is composed by the architecture named as Cloud-based Secure Biometric, proposed in the paper BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, with D.O.I. doi.org/10.1016/j.jksuci.2017.07.001 (link https://doi.org/10.1016/j.jksuci.2017.07.001). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Cloud-based Secure Biometric.
Solution Architecture Cloud-based Secure Biometric: described as: .
Flexibility is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Reliability is reached/addressed using/by MapReduce framework (No description.).
Security is reached/addressed using/by Cloud-based biometric authentication [Shakil et al. 2020] (No description.).
Reference: BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, https://doi.org/10.1016/j.jksuci.2017.07.001.
The solution is composed by the architecture named as Deep Learning based, proposed in the paper HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, with D.O.I. doi.org/10.1016/j.future.2019.10.043 (link https://doi.org/10.1016/j.future.2019.10.043). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Deep Learning based.
Solution Architecture Deep Learning based: described as: .
Performance/Efficiency is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Security is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Reference: HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, https://doi.org/10.1016/j.future.2019.10.043.
The solution is composed by the architecture named as Distributed Blockchain-SDN based, proposed in the paper DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3039113 (link https://doi.org/10.1109/ACCESS.2020.3039113). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Distributed Blockchain-SDN based.
Solution Architecture Distributed Blockchain-SDN based: described as: .
Performance/Efficiency is reached/addressed using/by Decentralized blockchain nodes, dynamic SDN controllers, and edge computing (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020,

https://doi.org/10.1109/ACCESS.2020.3039113.
The solution is composed by the architecture named as Distributed Edge-Cloud, proposed in the paper Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, with D.O.I. doi.org/10.1016/j.future.2020.05.042 (link https://doi.org/10.1016/j.future.2020.05.042). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Distributed Edge-Cloud.
Solution Architecture Distributed Edge-Cloud: described as: .
Security is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Reference: Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, https://doi.org/10.1016/j.future.2020.05.042.
The solution is composed by the architecture named as Edge Computing-based Fault Tolerant Framework, proposed in the paper Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, with D.O.I. doi.org/10.1109/IWCMC48107.2020.9148269 (link https://doi.org/10.1109/IWCMC48107.2020.9148269). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Edge Computing-based Fault Tolerant Framework.
Solution Architecture Edge Computing-based Fault Tolerant Framework: described as: .
Performance/Efficiency is reached/addressed using/by Kubernetes framework (No description.).
Flexibility is reached/addressed using/by Docker containers (No description.).
Reference: Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, https://doi.org/10.1109/IWCMC48107.2020.9148269.
The solution is composed by the architecture named as Event-driven IoT, proposed in the paper Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, with D.O.I. doi.org/10.1007/s10586-020-03189-w (link https://doi.org/10.1007/s10586-020-03189-w). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Event-driven IoT.
Solution Architecture Event-driven IoT: described as: .
Performance/Efficiency is reached/addressed using/by Complex Event Processing (CEP) [Rahmani et al. 2020] (No description.).
Reliability is reached/addressed using/by TCP-IP protocol on Client/Server architecture for data communication (No description.).
Reference: Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, https://doi.org/10.1007/s10586-020-03189-w.
The solution is composed by the architecture named as Fog Computing based, proposed in the paper FOG-based architecture and load balancing methodology for health monitoring systems, 2021, with D.O.I. doi.org/10.1109/ACCESS.2021.3094033 (link https://doi.org/10.1109/ACCESS.2021.3094033). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Fog Computing based.
Solution Architecture Fog Computing based: described as: .
Performance/Efficiency is reached/addressed using/by Load Balancing Scheme (LBS) - Distributed workload to all fog nodes [Asghar et al. 2021] (No description.).
Reference: FOG-based architecture and load balancing methodology for health monitoring systems, 2021, https://doi.org/10.1109/ACCESS.2021.3094033.
The solution is composed by the architecture named as Geographic-based, proposed in the paper A platform for integrating heterogeneous data and developing smart city applications, 2022, with D.O.I. doi.org/10.1016/j.future.2021.10.030 (link https://doi.org/10.1016/j.future.2021.10.030). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Geographic-based.
Solution Architecture Geographic-based: described as: .
Security is reached/addressed using/by FIWARE Platform [www.fiware.org] (Authentication, authorization, and data access control) (No description.).
Flexibility is reached/addressed using/by Virtualization (No description.).
Compatibility is reached/addressed using/by RESTFul APIs (No description.).
Reference: A platform for integrating heterogeneous data and developing smart city applications, 2022, https://doi.org/10.1016/j.future.2021.10.030.
The solution is composed by the architecture named as Hybrid (Centralized and Distributed), proposed in the paper Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, with D.O.I. doi.org/10.1016/j.procs.2021.03.030 (link https://doi.org/10.1016/j.procs.2021.03.030). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid (Centralized and Distributed).
Solution Architecture Hybrid (Centralized and Distributed): described as: .
Performance/Efficiency is reached/addressed using/by Dynamic allocation of the resources using Software Defined Networking (SDN) [Chudhary and Sharma 2021] (No description.).
Reference: Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, https://doi.org/10.1016/j.procs.2021.03.030.

The solution is composed by the architecture named as Hybrid Cloud-Fog Computing, proposed in the paper Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, with D.O.I. doi.org/10.1007/s10586-021-03375-4 (link https://link.springer.com/article/10.1007/s10586-021-03375-4). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid Cloud-Fog Computing.
Solution Architecture Hybrid Cloud-Fog Computing: described as: .
Reliability is reached/addressed using/by Fog layer in two levels: 1-Intra mobility (User network): Nodes (Data transfer, device management, temporary local storage, notifications, and data consistency). 2-Extra mobility (Different domains): Smart Gateways (Local processing and database) (No description.).
Performance/Efficiency is reached/addressed using/by Fog computing (No description.).
Reference: Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, https://link.springer.com/article/10.1007/s10586-021-03375-4.
The solution is composed by the architecture named as Industrial Internet of Things (IIoT), proposed in the paper A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Performance/Efficiency,Security,.
The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and Industrial Internet of Things (IIoT).
Solution Architecture Industrial Internet of Things (IIoT): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Flexibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Performance/Efficiency is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).
Security is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).
Reference: A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, https://doi.org/10.3390/s20195603.
The solution is composed by the architecture named as Layered, proposed in the paper An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, with D.O.I. doi.org/10.1016/j.future.2020.07.047 (link https://doi.org/10.1016/j.future.2020.07.047). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.
Solution Architecture Layered: described as: .
Flexibility is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).
Performance/Efficiency is reached/addressed using/by Big data analytics engine [Ali et al. 2021] (No description.).
Security is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).
Reference: An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, https://doi.org/10.1016/j.future.2020.07.047.
The solution is composed by the architecture named as Layered, proposed in the paper A novel architecture for the Internet of Things based E-Health systems, 2020, with D.O.I. doi.org/10.1166/jmihi.2020.3184 (link http://dx.doi.org/10.1166/jmihi.2020.3184). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.
Solution Architecture Layered: described as: .
Performance/Efficiency is reached/addressed using/by IoT objects connected at the edge of the network (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: A novel architecture for the Internet of Things based E-Health systems, 2020, http://dx.doi.org/10.1166/jmihi.2020.3184.
The solution is composed by the architecture named as Layered, proposed in the paper Software architecture quality attributes of a layered sensor-based IoT system, 2021, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Reliability,Security,.
The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and Layered.
Solution Architecture Layered: described as: .
Flexibility is reached/addressed using/by Messaging protocol brokers and at the client side, web-based applications (No description.).
Maintainability is reached/addressed using/by Loosely coupled components (No description.).
Reliability is reached/addressed using/by Open source software solutions and modular architecture design with clear relationships between components (No description.).
Security is reached/addressed using/by IoT devices authentication and data encryption (No description.).
Reference: Software architecture quality attributes of a layered sensor-based IoT system, 2021, https://doi.org/10.3390/s20195603.
The solution is composed by the architecture named as Layered, proposed in the paper Cloud-FOG assisted Energy Efficient Architectural Paradigm for disaster evacuation, 2022, with D.O.I. doi.org/10.1016/j.is.2021.101732 (link https://doi.org/10.1016/j.is.2021.101732). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Layered.
Solution Architecture Layered: described as: .

Performance/Efficiency is reached/addressed using/by Algorithm for Fog layer data selection (Data uniqueness and only important data) for transmission [Kaur et al. 2022] (No description.).
Reference: Cloud-FOG assisted Energy Efficient Architectural Paradigm for disaster evacuation, 2022, https://doi.org/10.1016/j.is.2021.101732.
The solution is composed by the architecture named as Layered, proposed in the paper An agile AI and IoT-Augmented Smart Farming: a Cost-Effective cognitive weather station, 2021, with D.O.I. doi.org/10.3390/agriculture12010035 (link https://doi.org/10.3390/agriculture12010035). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Interaction Capability,Maintainability,Performance/Efficiency,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Layered.
Solution Architecture Layered: described as: .
Compatibility is reached/addressed using/by Docker containers (No description.).
Flexibility is reached/addressed using/by Docker containers (No description.).
Interaction Capability is reached/addressed using/by Remote access to the deployed platform and its components (Web App and SSH) (No description.).
Maintainability is reached/addressed using/by Docker containers (No description.).
Performance/Efficiency is reached/addressed using/by Docker containers (No description.).
Reference: An agile AI and IoT-Augmented Smart Farming: a Cost-Effective cognitive weather station, 2021, https://doi.org/10.3390/agriculture12010035.
The solution is composed by the architecture named as Layered, proposed in the paper Smart Indoor Farms: Leveraging Technological Advancements to Power a Sustainable Agricultural Revolution, 2021, with D.O.I. doi.org/10.3390/agriengineering3040047 (link https://doi.org/10.3390/agriengineering3040047). This solution addresses the following Quality Requirements: .
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Layered.
Solution Architecture Layered: described as: .
Reference: Smart Indoor Farms: Leveraging Technological Advancements to Power a Sustainable Agricultural Revolution, 2021, https://doi.org/10.3390/agriengineering3040047.

The solution is composed by the architecture named as A4IoT (Anyplace 4.0 IoT), proposed in the paper The Anyplace 4.0 IoT Localization Architecture, 2020, with D.O.I. 10.1109/MDM48529.2020.00045 (link https://ieeexplore.ieee.org/document/9162205). This solution addresses the following Quality Requirements: Flexibility,Interaction Capability,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and A4IoT (Anyplace 4.0 IoT).
Solution Architecture A4IoT (Anyplace 4.0 IoT): described as: It is an open-source architecture designed for IoT-based indoor localization using technologies like Wi-Fi, BLE, Cellular, UWB, and Computer Vision (CV). The system supports edge computing, timeseries databases, and crowdsourcing to manage localization tasks across various smart spaces, such as factories, healthcare facilities, and emergency response environments. Its modular and scalable design ensures integration across platforms like Android, Linux, Windows, and Robot OS, making it adaptable to diverse IoT ecosystems..
Performance/Efficiency is reached/addressed using/by Time-Series Database (InfluxDB) (Time Behavior - With temporal queries we can now consume high-volume input streams to provide real-time IoT tracking.").
Reliability is reached/addressed using/by Multi-node Cluster Deployment (Fault Tolerance - The production environment uses a 3-node cluster with two replicas, for both the database engines and the DFS (Distributed Filesystem). This setup allows for a full operation with just a single node active at any time.").
Security is reached/addressed using/by SSL certificates - Encrypted Communications (Confidentiality - All communication is encrypted even in internal networks, as A4IoT image automatically creates and uses SSL certificates.).
Interaction Capability is reached/addressed using/by Web-based Dashboard (Operability - Web Apps built with HTML5, CSS3, and AngularJS.").
Flexibility is reached/addressed using/by Docker containers (Adaptability - It utilizes Docker to enable the deployment of single-node or multi-node configurations, regardless of any dependencies or the underlying OS.).
Reference: The Anyplace 4.0 IoT Localization Architecture, 2020, https://ieeexplore.ieee.org/document/9162205.
The solution is composed by the architecture named as Agricultural IoT Reference Architecture (AITRA), proposed in the paper Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3031634 (link https://doi.org/10.1109/ACCESS.2020.3031634). This solution addresses the following Quality Requirements: Compatibility,Reliability,Security,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Agricultural IoT Reference Architecture (AITRA).
Solution Architecture Agricultural IoT Reference Architecture (AITRA): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Reliability is reached/addressed using/by Edge gateway (No description.).
Security is reached/addressed using/by Authentication, authorization, and data access control (No description.).
Reference: Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, https://doi.org/10.1109/ACCESS.2020.3031634.
The solution is composed by the architecture named as Blockchain-based Zero Trust, proposed in the paper Blockchain-Based Zero Trust on the Edge, 2023, with D.O.I. 10.1109/CSCI62032.2023.00167 (link https://ieeexplore.ieee.org/document/10590526). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Performance/Efficiency,Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Blockchain-

based Zero Trust.
Solution Architecture Blockchain-based Zero Trust: described as: The architecture proposed in "Blockchain-Based Zero Trust on the Edge" integrates Zero Trust principles with Blockchain to ensure secure access and trust management for Edge Computing in IoT environments. It focuses on continuous verification of users and devices, ensuring that all access is authenticated before being granted. Blockchain is used for decentralized trust management, maintaining an immutable record of activities, which enhances security and transparency. This setup is optimized for edge computing, reducing latency by processing data locally while maintaining the flexibility to scale across diverse IoT applications, such as smart cities and industrial IoT..
Security is reached/addressed using/by Zero Trust Architecture (ZTA) (Confidentiality - Zero Trust Architecture (ZTA)
ZTA ensures continuous verification and monitoring of all network traffic, restricting unauthorized access.).
Security is reached/addressed using/by Blockchain Technology (Integrity - Blockchain serves as an immutable ledger for storing requests, ensuring data integrity and trustworthiness.).
Performance/Efficiency is reached/addressed using/by Redis Message Broker (Time Behavior - Redis Message Broker for Asynchronous Processing [redis.io]
Asynchronous processing for write operations improves system responsiveness under high loads.).
Maintainability is reached/addressed using/by Component isolation through microservices (Modularity -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.
).
Flexibility is reached/addressed using/by Component isolation through microservices (Scalability -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.).
Reference: Blockchain-Based Zero Trust on the Edge, 2023, https://ieeexplore.ieee.org/document/10590526.
The solution is composed by the architecture named as Blockchain-enabled, proposed in the paper An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, with D.O.I. doi.org/10.1109/TSC.2020.2966970 (link https://doi.org/10.1109/TSC.2020.2966970). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Blockchain-enabled.
Solution Architecture Blockchain-enabled: described as: .
Performance/Efficiency is reached/addressed using/by Routing protocol designed for IoT devices in the SDN controller [Yazdinejad et al. 2020] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, https://doi.org/10.1109/TSC.2020.2966970.
The solution is composed by the architecture named as Cloud-based Secure Biometric, proposed in the paper BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, with D.O.I. doi.org/10.1016/j.jksuci.2017.07.001 (link https://doi.org/10.1016/j.jksuci.2017.07.001). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Cloud-based Secure Biometric.
Solution Architecture Cloud-based Secure Biometric: described as: .
Flexibility is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Reliability is reached/addressed using/by MapReduce framework (No description.).
Security is reached/addressed using/by Cloud-based biometric authentication [Shakil et al. 2020] (No description.).
Reference: BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, https://doi.org/10.1016/j.jksuci.2017.07.001.
The solution is composed by the architecture named as Deep Learning based, proposed in the paper HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, with D.O.I. doi.org/10.1016/j.future.2019.10.043 (link https://doi.org/10.1016/j.future.2019.10.043). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Deep Learning based.
Solution Architecture Deep Learning based: described as: .
Performance/Efficiency is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Security is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Reference: HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, https://doi.org/10.1016/j.future.2019.10.043.
The solution is composed by the architecture named as Distributed Blockchain-SDN based, proposed in the paper DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3039113 (link https://doi.org/10.1109/ACCESS.2020.3039113). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Distributed Blockchain-SDN based.
Solution Architecture Distributed Blockchain-SDN based: described as: .
Performance/Efficiency is reached/addressed using/by Decentralized blockchain nodes, dynamic SDN controllers, and edge computing (No description.).

Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, https://doi.org/10.1109/ACCESS.2020.3039113.
The solution is composed by the architecture named as Distributed Edge-Cloud, proposed in the paper Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, with D.O.I. doi.org/10.1016/j.future.2020.05.042 (link https://doi.org/10.1016/j.future.2020.05.042). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Distributed Edge-Cloud.
Solution Architecture Distributed Edge-Cloud: described as: .
Security is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Reference: Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, https://doi.org/10.1016/j.future.2020.05.042.
The solution is composed by the architecture named as Edge Computing-based Fault Tolerant Framework, proposed in the paper Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, with D.O.I. doi.org/10.1109/IWCMC48107.2020.9148269 (link https://doi.org/10.1109/IWCMC48107.2020.9148269). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Edge Computing-based Fault Tolerant Framework.
Solution Architecture Edge Computing-based Fault Tolerant Framework: described as: .
Performance/Efficiency is reached/addressed using/by Kubernetes framework (No description.).
Flexibility is reached/addressed using/by Docker containers (No description.).
Reference: Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, https://doi.org/10.1109/IWCMC48107.2020.9148269.
The solution is composed by the architecture named as Event-driven IoT, proposed in the paper Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, with D.O.I. doi.org/10.1007/s10586-020-03189-w (link https://doi.org/10.1007/s10586-020-03189-w). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Event-driven IoT.
Solution Architecture Event-driven IoT: described as: .
Performance/Efficiency is reached/addressed using/by Complex Event Processing (CEP) [Rahmani et al. 2020] (No description.).
Reliability is reached/addressed using/by TCP-IP protocol on Client/Server architecture for data communication (No description.).
Reference: Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, https://doi.org/10.1007/s10586-020-03189-w.
The solution is composed by the architecture named as Fog Computing based, proposed in the paper FOG-based architecture and load balancing methodology for health monitoring systems, 2021, with D.O.I. doi.org/10.1109/ACCESS.2021.3094033 (link https://doi.org/10.1109/ACCESS.2021.3094033). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Fog Computing based.
Solution Architecture Fog Computing based: described as: .
Performance/Efficiency is reached/addressed using/by Load Balancing Scheme (LBS) - Distributed workload to all fog nodes [Asghar et al. 2021] (No description.).
Reference: FOG-based architecture and load balancing methodology for health monitoring systems, 2021, https://doi.org/10.1109/ACCESS.2021.3094033.
The solution is composed by the architecture named as Geographic-based, proposed in the paper A platform for integrating heterogeneous data and developing smart city applications, 2022, with D.O.I. doi.org/10.1016/j.future.2021.10.030 (link https://doi.org/10.1016/j.future.2021.10.030). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Geographic-based.
Solution Architecture Geographic-based: described as: .
Security is reached/addressed using/by FIWARE Platform [www.fiware.org] (Authentication, authorization, and data access control) (No description.).
Flexibility is reached/addressed using/by Virtualization (No description.).
Compatibility is reached/addressed using/by RESTFul APIs (No description.).
Reference: A platform for integrating heterogeneous data and developing smart city applications, 2022, https://doi.org/10.1016/j.future.2021.10.030.
The solution is composed by the architecture named as Hybrid (Centralized and Distributed), proposed in the paper Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, with D.O.I. doi.org/10.1016/j.procs.2021.03.030 (link https://doi.org/10.1016/j.procs.2021.03.030). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid (Centralized and Distributed).
Solution Architecture Hybrid (Centralized and Distributed): described as: .
Performance/Efficiency is reached/addressed using/by Dynamic allocation of the resources using Software Defined Networking (SDN) [Chudhary

and Sharma 2021] (No description.).
Reference: Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, https://doi.org/10.1016/j.procs.2021.03.030.
The solution is composed by the architecture named as Hybrid Cloud-Fog Computing, proposed in the paper Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, with D.O.I. doi.org/10.1007/s10586-021-03375-4 (link https://link.springer.com/article/10.1007/s10586-021-03375-4). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid Cloud-Fog Computing.
Solution Architecture Hybrid Cloud-Fog Computing: described as: .
Reliability is reached/addressed using/by Fog layer in two levels: 1-Intra mobility (User network): Nodes (Data transfer, device management, temporary local storage, notifications, and data consistency). 2-Extra mobility (Different domains): Smart Gateways (Local processing and database) (No description.).
Performance/Efficiency is reached/addressed using/by Fog computing (No description.).
Reference: Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, https://link.springer.com/article/10.1007/s10586-021-03375-4.
The solution is composed by the architecture named as Industrial Internet of Things (IIoT), proposed in the paper A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Performance/Efficiency,Security,.
The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and Industrial Internet of Things (IIoT).
Solution Architecture Industrial Internet of Things (IIoT): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Flexibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Performance/Efficiency is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).
Security is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).
Reference: A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, https://doi.org/10.3390/s20195603.
The solution is composed by the architecture named as Layered, proposed in the paper An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, with D.O.I. doi.org/10.1016/j.future.2020.07.047 (link https://doi.org/10.1016/j.future.2020.07.047). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.
Solution Architecture Layered: described as: .
Flexibility is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).
Performance/Efficiency is reached/addressed using/by Big data analytics engine [Ali et al. 2021] (No description.).
Security is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).
Reference: An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, https://doi.org/10.1016/j.future.2020.07.047.
The solution is composed by the architecture named as Layered, proposed in the paper A novel architecture for the Internet of Things based E-Health systems, 2020, with D.O.I. doi.org/10.1166/jmihi.2020.3184 (link http://dx.doi.org/10.1166/jmihi.2020.3184). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.
Solution Architecture Layered: described as: .
Performance/Efficiency is reached/addressed using/by IoT objects connected at the edge of the network (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: A novel architecture for the Internet of Things based E-Health systems, 2020, http://dx.doi.org/10.1166/jmihi.2020.3184.
The solution is composed by the architecture named as Layered, proposed in the paper Software architecture quality attributes of a layered sensor-based IoT system, 2021, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Reliability,Security,.
The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and Layered.
Solution Architecture Layered: described as: .
Flexibility is reached/addressed using/by Messaging protocol brokers and at the client side, web-based applications (No description.).
Maintainability is reached/addressed using/by Loosely coupled components (No description.).
Reliability is reached/addressed using/by Open source software solutions and modular architecture design with clear relationships between components (No description.).
Security is reached/addressed using/by IoT devices authentication and data encryption (No description.).
Reference: Software architecture quality attributes of a layered sensor-based IoT system, 2021, https://doi.org/10.3390/s20195603.
The solution is composed by the architecture named as Layered, proposed in the paper Cloud-FOG assisted Energy Efficient Architectural Paradigm for disaster evacuation, 2022, with D.O.I. doi.org/10.1016/j.is.2021.101732 (link https://doi.org/10.1016/j.is.2021.101732). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and

Layered.
Solution Architecture Layered: described as: .
Performance/Efficiency is reached/addressed using/by Algorithm for Fog layer data selection (Data uniqueness and only important data) for transmission [Kaur et al. 2022] (No description.).
Reference: Cloud-FOG assisted Energy Efficient Architectural Paradigm for disaster evacuation, 2022, https://doi.org/10.1016/j.is.2021.101732.
The solution is composed by the architecture named as Layered, proposed in the paper An agile AI and IoT-Augmented Smart Farming: a Cost-Effective cognitive weather station, 2021, with D.O.I. doi.org/10.3390/agriculture12010035 (link https://doi.org/10.3390/agriculture12010035). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Interaction Capability,Maintainability,Performance/Efficiency,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Layered.
Solution Architecture Layered: described as: .
Compatibility is reached/addressed using/by Docker containers (No description.).
Flexibility is reached/addressed using/by Docker containers (No description.).
Interaction Capability is reached/addressed using/by Remote access to the deployed platform and its components (Web App and SSH) (No description.).
Maintainability is reached/addressed using/by Docker containers (No description.).
Performance/Efficiency is reached/addressed using/by Docker containers (No description.).
Reference: An agile AI and IoT-Augmented Smart Farming: a Cost-Effective cognitive weather station, 2021, https://doi.org/10.3390/agriculture12010035.
The solution is composed by the architecture named as Layered, proposed in the paper Smart Indoor Farms: Leveraging Technological Advancements to Power a Sustainable Agricultural Revolution, 2021, with D.O.I. doi.org/10.3390/agriengineering3040047 (link https://doi.org/10.3390/agriengineering3040047). This solution addresses the following Quality Requirements: .
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Layered.
Solution Architecture Layered: described as: .
Reference: Smart Indoor Farms: Leveraging Technological Advancements to Power a Sustainable Agricultural Revolution, 2021, https://doi.org/10.3390/agriengineering3040047.
The solution is composed by the architecture named as Layered Blockchain and AI-enabled, proposed in the paper A blockchain and artificial intelligence-enabled smart IoT framework for sustainable city, 2022, with D.O.I. doi.org/10.1002/int.22852 (link https://doi.org/10.1002/int.22852). This solution addresses the following Quality Requirements: Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Layered Blockchain and AI-enabled.
Solution Architecture Layered Blockchain and AI-enabled: described as: .
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: A blockchain and artificial intelligence-enabled smart IoT framework for sustainable city, 2022, https://doi.org/10.1002/int.22852.

The solution is composed by the architecture named as A4IoT (Anyplace 4.0 IoT), proposed in the paper The Anyplace 4.0 IoT Localization Architecture, 2020, with D.O.I. 10.1109/MDM48529.2020.00045 (link https://ieeexplore.ieee.org/document/9162205). This solution addresses the following Quality Requirements: Flexibility,Interaction Capability,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and A4IoT (Anyplace 4.0 IoT).
Solution Architecture A4IoT (Anyplace 4.0 IoT): described as: It is an open-source architecture designed for IoT-based indoor localization using technologies like Wi-Fi, BLE, Cellular, UWB, and Computer Vision (CV). The system supports edge computing, timeseries databases, and crowdsourcing to manage localization tasks across various smart spaces, such as factories, healthcare facilities, and emergency response environments. Its modular and scalable design ensures integration across platforms like Android, Linux, Windows, and Robot OS, making it adaptable to diverse IoT ecosystems..
Performance/Efficiency is reached/addressed using/by Time-Series Database (InfluxDB) (Time Behavior - With temporal queries we can now consume high-volume input streams to provide real-time IoT tracking.").
Reliability is reached/addressed using/by Multi-node Cluster Deployment (Fault Tolerance - The production environment uses a 3-node cluster with two replicas, for both the database engines and the DFS (Distributed Filesystem). This setup allows for a full operation with just a single node active at any time.").
Security is reached/addressed using/by SSL certificates - Encrypted Communications (Confidentiality - All communication is encrypted even in internal networks, as A4IoT image automatically creates and uses SSL certificates.).
Interaction Capability is reached/addressed using/by Web-based Dashboard (Operability - Web Apps built with HTML5, CSS3, and AngularJS.").
Flexibility is reached/addressed using/by Docker containers (Adaptability - It utilizes Docker to enable the deployment of single-node or multi-node configurations, regardless of any dependencies or the underlying OS.).
Reference: The Anyplace 4.0 IoT Localization Architecture, 2020, https://ieeexplore.ieee.org/document/9162205.
The solution is composed by the architecture named as Agricultural IoT Reference Architecture (AITRA), proposed in the paper Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3031634 (link https://doi.org/10.1109/ACCESS.2020.3031634). This solution addresses the following Quality Requirements: Compatibility,Reliability,Security,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Agricultural IoT Reference Architecture (AITRA).

Solution Architecture Agricultural IoT Reference Architecture (AITRA): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Reliability is reached/addressed using/by Edge gateway (No description.).
Security is reached/addressed using/by Authentication, authorization, and data access control (No description.).
Reference: Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, https://doi.org/10.1109/ACCESS.2020.3031634.
The solution is composed by the architecture named as Blockchain-based Zero Trust, proposed in the paper Blockchain-Based Zero Trust on the Edge, 2023, with D.O.I. 10.1109/CSCI62032.2023.00167 (link https://ieeexplore.ieee.org/document/10590526). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Performance/Efficiency,Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Blockchain-based Zero Trust.
Solution Architecture Blockchain-based Zero Trust: described as: The architecture proposed in "Blockchain-Based Zero Trust on the Edge" integrates Zero Trust principles with Blockchain to ensure secure access and trust management for Edge Computing in IoT environments. It focuses on continuous verification of users and devices, ensuring that all access is authenticated before being granted. Blockchain is used for decentralized trust management, maintaining an immutable record of activities, which enhances security and transparency. This setup is optimized for edge computing, reducing latency by processing data locally while maintaining the flexibility to scale across diverse IoT applications, such as smart cities and industrial IoT..
Security is reached/addressed using/by Zero Trust Architecture (ZTA) (Confidentiality - Zero Trust Architecture (ZTA)
ZTA ensures continuous verification and monitoring of all network traffic, restricting unauthorized access.).
Security is reached/addressed using/by Blockchain Technology (Integrity - Blockchain serves as an immutable ledger for storing requests, ensuring data integrity and trustworthiness.).
Performance/Efficiency is reached/addressed using/by Redis Message Broker (Time Behavior - Redis Message Broker for Asynchronous Processing [redis.io]
Asynchronous processing for write operations improves system responsiveness under high loads.).
Maintainability is reached/addressed using/by Component isolation through microservices (Modularity -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.
).
Flexibility is reached/addressed using/by Component isolation through microservices (Scalability -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.).
Reference: Blockchain-Based Zero Trust on the Edge, 2023, https://ieeexplore.ieee.org/document/10590526.
The solution is composed by the architecture named as Blockchain-enabled, proposed in the paper An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, with D.O.I. doi.org/10.1109/TSC.2020.2966970 (link https://doi.org/10.1109/TSC.2020.2966970). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Blockchain-enabled.
Solution Architecture Blockchain-enabled: described as: .
Performance/Efficiency is reached/addressed using/by Routing protocol designed for IoT devices in the SDN controller [Yazdinejad et al. 2020] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, https://doi.org/10.1109/TSC.2020.2966970.
The solution is composed by the architecture named as Cloud-based Secure Biometric, proposed in the paper BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, with D.O.I. doi.org/10.1016/j.jksuci.2017.07.001 (link https://doi.org/10.1016/j.jksuci.2017.07.001). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Cloud-based Secure Biometric.
Solution Architecture Cloud-based Secure Biometric: described as: .
Flexibility is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Reliability is reached/addressed using/by MapReduce framework (No description.).
Security is reached/addressed using/by Cloud-based biometric authentication [Shakil et al. 2020] (No description.).
Reference: BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, https://doi.org/10.1016/j.jksuci.2017.07.001.
The solution is composed by the architecture named as Deep Learning based, proposed in the paper HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, with D.O.I. doi.org/10.1016/j.future.2019.10.043 (link https://doi.org/10.1016/j.future.2019.10.043). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Deep Learning based.
Solution Architecture Deep Learning based: described as: .
Performance/Efficiency is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Security is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).

Reference: HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, https://doi.org/10.1016/j.future.2019.10.043.
The solution is composed by the architecture named as Distributed Blockchain-SDN based, proposed in the paper DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3039113 (link https://doi.org/10.1109/ACCESS.2020.3039113). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Distributed Blockchain-SDN based.
Solution Architecture Distributed Blockchain-SDN based: described as: .
Performance/Efficiency is reached/addressed using/by Decentralized blockchain nodes, dynamic SDN controllers, and edge computing (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, https://doi.org/10.1109/ACCESS.2020.3039113.
The solution is composed by the architecture named as Distributed Edge-Cloud, proposed in the paper Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, with D.O.I. doi.org/10.1016/j.future.2020.05.042 (link https://doi.org/10.1016/j.future.2020.05.042). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Distributed Edge-Cloud.
Solution Architecture Distributed Edge-Cloud: described as: .
Security is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Reference: Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, https://doi.org/10.1016/j.future.2020.05.042.
The solution is composed by the architecture named as Edge Computing-based Fault Tolerant Framework, proposed in the paper Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, with D.O.I. doi.org/10.1109/IWCMC48107.2020.9148269 (link https://doi.org/10.1109/IWCMC48107.2020.9148269). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Edge Computing-based Fault Tolerant Framework.
Solution Architecture Edge Computing-based Fault Tolerant Framework: described as: .
Performance/Efficiency is reached/addressed using/by Kubernetes framework (No description.).
Flexibility is reached/addressed using/by Docker containers (No description.).
Reference: Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, https://doi.org/10.1109/IWCMC48107.2020.9148269.
The solution is composed by the architecture named as Event-driven IoT, proposed in the paper Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, with D.O.I. doi.org/10.1007/s10586-020-03189-w (link https://doi.org/10.1007/s10586-020-03189-w). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Event-driven IoT.
Solution Architecture Event-driven IoT: described as: .
Performance/Efficiency is reached/addressed using/by Complex Event Processing (CEP) [Rahmani et al. 2020] (No description.).
Reliability is reached/addressed using/by TCP-IP protocol on Client/Server architecture for data communication (No description.).
Reference: Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, https://doi.org/10.1007/s10586-020-03189-w.
The solution is composed by the architecture named as Fog Computing based, proposed in the paper FOG-based architecture and load balancing methodology for health monitoring systems, 2021, with D.O.I. doi.org/10.1109/ACCESS.2021.3094033 (link https://doi.org/10.1109/ACCESS.2021.3094033). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Fog Computing based.
Solution Architecture Fog Computing based: described as: .
Performance/Efficiency is reached/addressed using/by Load Balancing Scheme (LBS) - Distributed workload to all fog nodes [Asghar et al. 2021] (No description.).
Reference: FOG-based architecture and load balancing methodology for health monitoring systems, 2021, https://doi.org/10.1109/ACCESS.2021.3094033.
The solution is composed by the architecture named as Geographic-based, proposed in the paper A platform for integrating heterogeneous data and developing smart city applications, 2022, with D.O.I. doi.org/10.1016/j.future.2021.10.030 (link https://doi.org/10.1016/j.future.2021.10.030). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Geographic-based.
Solution Architecture Geographic-based: described as: .
Security is reached/addressed using/by FIWARE Platform [www.fiware.org] (Authentication, authorization, and data access control) (No description.).

Flexibility is reached/addressed using/by Virtualization (No description.).
Compatibility is reached/addressed using/by RESTFul APIs (No description.).
Reference: A platform for integrating heterogeneous data and developing smart city applications, 2022, https://doi.org/10.1016/j.future.2021.10.030.
The solution is composed by the architecture named as Hybrid (Centralized and Distributed), proposed in the paper Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, with D.O.I. doi.org/10.1016/j.procs.2021.03.030 (link https://doi.org/10.1016/j.procs.2021.03.030). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid (Centralized and Distributed).
Solution Architecture Hybrid (Centralized and Distributed): described as: .
Performance/Efficiency is reached/addressed using/by Dynamic allocation of the resources using Software Defined Networking (SDN) [Chudhary and Sharma 2021] (No description.).
Reference: Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, https://doi.org/10.1016/j.procs.2021.03.030.
The solution is composed by the architecture named as Hybrid Cloud-Fog Computing, proposed in the paper Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, with D.O.I. doi.org/10.1007/s10586-021-03375-4 (link https://link.springer.com/article/10.1007/s10586-021-03375-4). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid Cloud-Fog Computing.
Solution Architecture Hybrid Cloud-Fog Computing: described as: .
Reliability is reached/addressed using/by Fog layer in two levels: 1-Intra mobility (User network): Nodes (Data transfer, device management, temporary local storage, notifications, and data consistency). 2-Extra mobility (Different domains): Smart Gateways (Local processing and database) (No description.).
Performance/Efficiency is reached/addressed using/by Fog computing (No description.).
Reference: Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, https://link.springer.com/article/10.1007/s10586-021-03375-4.
The solution is composed by the architecture named as Industrial Internet of Things (IIoT), proposed in the paper A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Performance/Efficiency,Security,.
The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and Industrial Internet of Things (IIoT).
Solution Architecture Industrial Internet of Things (IIoT): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Flexibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Performance/Efficiency is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).
Security is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).
Reference: A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, https://doi.org/10.3390/s20195603.
The solution is composed by the architecture named as Layered, proposed in the paper An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, with D.O.I. doi.org/10.1016/j.future.2020.07.047 (link https://doi.org/10.1016/j.future.2020.07.047). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.
Solution Architecture Layered: described as: .
Flexibility is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).
Performance/Efficiency is reached/addressed using/by Big data analytics engine [Ali et al. 2021] (No description.).
Security is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).
Reference: An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, https://doi.org/10.1016/j.future.2020.07.047.
The solution is composed by the architecture named as Layered, proposed in the paper A novel architecture for the Internet of Things based E-Health systems, 2020, with D.O.I. doi.org/10.1166/jmihi.2020.3184 (link http://dx.doi.org/10.1166/jmihi.2020.3184). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.
Solution Architecture Layered: described as: .
Performance/Efficiency is reached/addressed using/by IoT objects connected at the edge of the network (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: A novel architecture for the Internet of Things based E-Health systems, 2020, http://dx.doi.org/10.1166/jmihi.2020.3184.
The solution is composed by the architecture named as Layered, proposed in the paper Software architecture quality attributes of a layered sensor-based IoT system, 2021, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Reliability,Security,.
The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and Layered.
Solution Architecture Layered: described as: .

Flexibility is reached/addressed using/by Messaging protocol brokers and at the client side, web-based applications (No description.).
Maintainability is reached/addressed using/by Loosely coupled components (No description.).
Reliability is reached/addressed using/by Open source software solutions and modular architecture design with clear relationships between components (No description.).
Security is reached/addressed using/by IoT devices authentication and data encryption (No description.).
Reference: Software architecture quality attributes of a layered sensor-based IoT system, 2021, https://doi.org/10.3390/s20195603.
The solution is composed by the architecture named as Layered, proposed in the paper Cloud-FOG assisted Energy Efficient Architectural Paradigm for disaster evacuation, 2022, with D.O.I. doi.org/10.1016/j.is.2021.101732 (link https://doi.org/10.1016/j.is.2021.101732). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Layered.
Solution Architecture Layered: described as: .
Performance/Efficiency is reached/addressed using/by Algorithm for Fog layer data selection (Data uniqueness and only important data) for transmission [Kaur et al. 2022] (No description.).
Reference: Cloud-FOG assisted Energy Efficient Architectural Paradigm for disaster evacuation, 2022, https://doi.org/10.1016/j.is.2021.101732.
The solution is composed by the architecture named as Layered, proposed in the paper An agile AI and IoT-Augmented Smart Farming: a Cost-Effective cognitive weather station, 2021, with D.O.I. doi.org/10.3390/agriculture12010035 (link https://doi.org/10.3390/agriculture12010035). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Interaction Capability,Maintainability,Performance/Efficiency,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Layered.
Solution Architecture Layered: described as: .
Compatibility is reached/addressed using/by Docker containers (No description.).
Flexibility is reached/addressed using/by Docker containers (No description.).
Interaction Capability is reached/addressed using/by Remote access to the deployed platform and its components (Web App and SSH) (No description.).
Maintainability is reached/addressed using/by Docker containers (No description.).
Performance/Efficiency is reached/addressed using/by Docker containers (No description.).
Reference: An agile AI and IoT-Augmented Smart Farming: a Cost-Effective cognitive weather station, 2021, https://doi.org/10.3390/agriculture12010035.
The solution is composed by the architecture named as Layered, proposed in the paper Smart Indoor Farms: Leveraging Technological Advancements to Power a Sustainable Agricultural Revolution, 2021, with D.O.I. doi.org/10.3390/agriengineering3040047 (link https://doi.org/10.3390/agriengineering3040047). This solution addresses the following Quality Requirements: .
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Layered.
Solution Architecture Layered: described as: .
Reference: Smart Indoor Farms: Leveraging Technological Advancements to Power a Sustainable Agricultural Revolution, 2021, https://doi.org/10.3390/agriengineering3040047.
The solution is composed by the architecture named as Layered Blockchain and AI-enabled, proposed in the paper A blockchain and artificial intelligence-enabled smart IoT framework for sustainable city, 2022, with D.O.I. doi.org/10.1002/int.22852 (link https://doi.org/10.1002/int.22852). This solution addresses the following Quality Requirements: Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Layered Blockchain and AI-enabled.
Solution Architecture Layered Blockchain and AI-enabled: described as: .
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: A blockchain and artificial intelligence-enabled smart IoT framework for sustainable city, 2022, https://doi.org/10.1002/int.22852.
The solution is composed by the architecture named as Layered Blockchain-Based SDN, proposed in the paper Intelligent Blockchain Based secure routing for multidomain SDN Enabled IoT networks, 2022, with D.O.I. doi.org/10.1155/2022/5693962 (link https://doi.org/10.1155/2022/5693962). This solution addresses the following Quality Requirements: Reliability,Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Layered Blockchain-Based SDN.
Solution Architecture Layered Blockchain-Based SDN: described as: .
Reliability is reached/addressed using/by Reputation concept (Blockchain-based SDN-enabled network architecture) [Zeng et al. 2022] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: Intelligent Blockchain Based secure routing for multidomain SDN Enabled IoT networks, 2022, https://doi.org/10.1155/2022/5693962.

The solution is composed by the architecture named as A4IoT (Anyplace 4.0 IoT), proposed in the paper The Anyplace 4.0 IoT Localization Architecture, 2020, with D.O.I. 10.1109/MDM48529.2020.00045 (link https://ieeexplore.ieee.org/document/9162205). This solution addresses the following Quality Requirements: Flexibility,Interaction Capability,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and A4IoT (Anyplace 4.0 IoT).
Solution Architecture A4IoT (Anyplace 4.0 IoT): described as: It is an open-source architecture designed for IoT-based indoor localization using

technologies like Wi-Fi, BLE, Cellular, UWB, and Computer Vision (CV). The system supports edge computing, timeseries databases, and crowdsourcing to manage localization tasks across various smart spaces, such as factories, healthcare facilities, and emergency response environments. Its modular and scalable design ensures integration across platforms like Android, Linux, Windows, and Robot OS, making it adaptable to diverse IoT ecosystems..
Performance/Efficiency is reached/addressed using/by Time-Series Database (InfluxDB) (Time Behavior - With temporal queries we can now consume high-volume input streams to provide real-time IoT tracking.").
Reliability is reached/addressed using/by Multi-node Cluster Deployment (Fault Tolerance - The production environment uses a 3-node cluster with two replicas, for both the database engines and the DFS (Distributed Filesystem). This setup allows for a full operation with just a single node active at any time.").
Security is reached/addressed using/by SSL certificates - Encrypted Communications (Confidentiality - All communication is encrypted even in internal networks, as A4IoT image automatically creates and uses SSL certificates.).
Interaction Capability is reached/addressed using/by Web-based Dashboard (Operability - Web Apps built with HTML5, CSS3, and AngularJS.").
Flexibility is reached/addressed using/by Docker containers (Adaptability - It utilizes Docker to enable the deployment of single-node or multi-node configurations, regardless of any dependencies or the underlying OS.).
Reference: The Anyplace 4.0 IoT Localization Architecture, 2020, https://ieeexplore.ieee.org/document/9162205.
The solution is composed by the architecture named as Agricultural IoT Reference Architecture (AITRA), proposed in the paper Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3031634 (link https://doi.org/10.1109/ACCESS.2020.3031634). This solution addresses the following Quality Requirements: Compatibility,Reliability,Security,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Agricultural IoT Reference Architecture (AITRA).
Solution Architecture Agricultural IoT Reference Architecture (AITRA): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Reliability is reached/addressed using/by Edge gateway (No description.).
Security is reached/addressed using/by Authentication, authorization, and data access control (No description.).
Reference: Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, https://doi.org/10.1109/ACCESS.2020.3031634.
The solution is composed by the architecture named as Blockchain-based Zero Trust, proposed in the paper Blockchain-Based Zero Trust on the Edge, 2023, with D.O.I. 10.1109/CSCI62032.2023.00167 (link https://ieeexplore.ieee.org/document/10590526). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Performance/Efficiency,Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Blockchain-based Zero Trust.
Solution Architecture Blockchain-based Zero Trust: described as: The architecture proposed in "Blockchain-Based Zero Trust on the Edge" integrates Zero Trust principles with Blockchain to ensure secure access and trust management for Edge Computing in IoT environments. It focuses on continuous verification of users and devices, ensuring that all access is authenticated before being granted. Blockchain is used for decentralized trust management, maintaining an immutable record of activities, which enhances security and transparency. This setup is optimized for edge computing, reducing latency by processing data locally while maintaining the flexibility to scale across diverse IoT applications, such as smart cities and industrial IoT..
Security is reached/addressed using/by Zero Trust Architecture (ZTA) (Confidentiality - Zero Trust Architecture (ZTA)
ZTA ensures continuous verification and monitoring of all network traffic, restricting unauthorized access.).
Security is reached/addressed using/by Blockchain Technology (Integrity - Blockchain serves as an immutable ledger for storing requests, ensuring data integrity and trustworthiness.).
Performance/Efficiency is reached/addressed using/by Redis Message Broker (Time Behavior - Redis Message Broker for Asynchronous Processing [redis.io]
Asynchronous processing for write operations improves system responsiveness under high loads.).
Maintainability is reached/addressed using/by Component isolation through microservices (Modularity -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.
).
Flexibility is reached/addressed using/by Component isolation through microservices (Scalability -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.).
Reference: Blockchain-Based Zero Trust on the Edge, 2023, https://ieeexplore.ieee.org/document/10590526.
The solution is composed by the architecture named as Blockchain-enabled, proposed in the paper An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, with D.O.I. doi.org/10.1109/TSC.2020.2966970 (link https://doi.org/10.1109/TSC.2020.2966970). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Blockchain-enabled.
Solution Architecture Blockchain-enabled: described as: .
Performance/Efficiency is reached/addressed using/by Routing protocol designed for IoT devices in the SDN controller [Yazdinejad et al. 2020] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, https://doi.org/10.1109/TSC.2020.2966970.
The solution is composed by the architecture named as Cloud-based Secure Biometric, proposed in the paper BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, with D.O.I. doi.org/10.1016/j.jksuci.2017.07.001 (link

https://doi.org/10.1016/j.jksuci.2017.07.001). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Reliability,Security,.

The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Cloud-based Secure Biometric.

Solution Architecture Cloud-based Secure Biometric: described as: .

Flexibility is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).

Performance/Efficiency is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).

Reliability is reached/addressed using/by MapReduce framework (No description.).

Security is reached/addressed using/by Cloud-based biometric authentication [Shakil et al. 2020] (No description.).

Reference: BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, https://doi.org/10.1016/j.jksuci.2017.07.001.

The solution is composed by the architecture named as Deep Learning based, proposed in the paper HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, with D.O.I. doi.org/10.1016/j.future.2019.10.043 (link https://doi.org/10.1016/j.future.2019.10.043). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.

The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Deep Learning based.

Solution Architecture Deep Learning based: described as: .

Performance/Efficiency is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).

Security is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).

Reference: HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, https://doi.org/10.1016/j.future.2019.10.043.

The solution is composed by the architecture named as Distributed Blockchain-SDN based, proposed in the paper DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3039113 (link https://doi.org/10.1109/ACCESS.2020.3039113). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.

The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Distributed Blockchain-SDN based.

Solution Architecture Distributed Blockchain-SDN based: described as: .

Performance/Efficiency is reached/addressed using/by Decentralized blockchain nodes, dynamic SDN controllers, and edge computing (No description.).

Security is reached/addressed using/by Blockchain Technology (No description.).

Reference: DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, https://doi.org/10.1109/ACCESS.2020.3039113.

The solution is composed by the architecture named as Distributed Edge-Cloud, proposed in the paper Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, with D.O.I. doi.org/10.1016/j.future.2020.05.042 (link https://doi.org/10.1016/j.future.2020.05.042). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.

The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Distributed Edge-Cloud.

Solution Architecture Distributed Edge-Cloud: described as: .

Security is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).

Performance/Efficiency is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).

Reference: Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, https://doi.org/10.1016/j.future.2020.05.042.

The solution is composed by the architecture named as Edge Computing-based Fault Tolerant Framework, proposed in the paper Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, with D.O.I. doi.org/10.1109/IWCMC48107.2020.9148269 (link https://doi.org/10.1109/IWCMC48107.2020.9148269). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,.

The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Edge Computing-based Fault Tolerant Framework.

Solution Architecture Edge Computing-based Fault Tolerant Framework: described as: .

Performance/Efficiency is reached/addressed using/by Kubernetes framework (No description.).

Flexibility is reached/addressed using/by Docker containers (No description.).

Reference: Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, https://doi.org/10.1109/IWCMC48107.2020.9148269.

The solution is composed by the architecture named as Event-driven IoT, proposed in the paper Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, with D.O.I. doi.org/10.1007/s10586-020-03189-w (link https://doi.org/10.1007/s10586-020-03189-w). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.

The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Event-driven IoT.

Solution Architecture Event-driven IoT: described as: .

Performance/Efficiency is reached/addressed using/by Complex Event Processing (CEP) [Rahmani et al. 2020] (No description.).

Reliability is reached/addressed using/by TCP-IP protocol on Client/Server architecture for data communication (No description.).

Reference: Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, https://doi.org/10.1007/s10586-020-03189-w.
The solution is composed by the architecture named as Fog Computing based, proposed in the paper FOG-based architecture and load balancing methodology for health monitoring systems, 2021, with D.O.I. doi.org/10.1109/ACCESS.2021.3094033 (link https://doi.org/10.1109/ACCESS.2021.3094033). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Fog Computing based.
Solution Architecture Fog Computing based: described as: .
Performance/Efficiency is reached/addressed using/by Load Balancing Scheme (LBS) - Distributed workload to all fog nodes [Asghar et al. 2021] (No description.).
Reference: FOG-based architecture and load balancing methodology for health monitoring systems, 2021, https://doi.org/10.1109/ACCESS.2021.3094033.
The solution is composed by the architecture named as Geographic-based, proposed in the paper A platform for integrating heterogeneous data and developing smart city applications, 2022, with D.O.I. doi.org/10.1016/j.future.2021.10.030 (link https://doi.org/10.1016/j.future.2021.10.030). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Geographic-based.
Solution Architecture Geographic-based: described as: .
Security is reached/addressed using/by FIWARE Platform [www.fiware.org] (Authentication, authorization, and data access control) (No description.).
Flexibility is reached/addressed using/by Virtualization (No description.).
Compatibility is reached/addressed using/by RESTFul APIs (No description.).
Reference: A platform for integrating heterogeneous data and developing smart city applications, 2022, https://doi.org/10.1016/j.future.2021.10.030.
The solution is composed by the architecture named as Hybrid (Centralized and Distributed), proposed in the paper Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, with D.O.I. doi.org/10.1016/j.procs.2021.03.030 (link https://doi.org/10.1016/j.procs.2021.03.030). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid (Centralized and Distributed).
Solution Architecture Hybrid (Centralized and Distributed): described as: .
Performance/Efficiency is reached/addressed using/by Dynamic allocation of the resources using Software Defined Networking (SDN) [Chudhary and Sharma 2021] (No description.).
Reference: Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, https://doi.org/10.1016/j.procs.2021.03.030.
The solution is composed by the architecture named as Hybrid Cloud-Fog Computing, proposed in the paper Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, with D.O.I. doi.org/10.1007/s10586-021-03375-4 (link https://link.springer.com/article/10.1007/s10586-021-03375-4). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid Cloud-Fog Computing.
Solution Architecture Hybrid Cloud-Fog Computing: described as: .
Reliability is reached/addressed using/by Fog layer in two levels: 1-Intra mobility (User network): Nodes (Data transfer, device management, temporary local storage, notifications, and data consistency). 2-Extra mobility (Different domains): Smart Gateways (Local processing and database) (No description.).
Performance/Efficiency is reached/addressed using/by Fog computing (No description.).
Reference: Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, https://link.springer.com/article/10.1007/s10586-021-03375-4.
The solution is composed by the architecture named as Industrial Internet of Things (IIoT), proposed in the paper A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Performance/Efficiency,Security,.
The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and Industrial Internet of Things (IIoT).
Solution Architecture Industrial Internet of Things (IIoT): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Flexibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Performance/Efficiency is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).
Security is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).
Reference: A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, https://doi.org/10.3390/s20195603.
The solution is composed by the architecture named as Layered, proposed in the paper An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, with D.O.I. doi.org/10.1016/j.future.2020.07.047 (link https://doi.org/10.1016/j.future.2020.07.047). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.
Solution Architecture Layered: described as: .

Flexibility is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).
Performance/Efficiency is reached/addressed using/by Big data analytics engine [Ali et al. 2021] (No description.).
Security is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).
Reference: An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, https://doi.org/10.1016/j.future.2020.07.047.
The solution is composed by the architecture named as Layered, proposed in the paper A novel architecture for the Internet of Things based E-Health systems, 2020, with D.O.I. doi.org/10.1166/jmihi.2020.3184 (link http://dx.doi.org/10.1166/jmihi.2020.3184). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.
Solution Architecture Layered: described as: .
Performance/Efficiency is reached/addressed using/by IoT objects connected at the edge of the network (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: A novel architecture for the Internet of Things based E-Health systems, 2020, http://dx.doi.org/10.1166/jmihi.2020.3184.
The solution is composed by the architecture named as Layered, proposed in the paper Software architecture quality attributes of a layered sensor-based IoT system, 2021, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Reliability,Security,.
The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and Layered.
Solution Architecture Layered: described as: .
Flexibility is reached/addressed using/by Messaging protocol brokers and at the client side, web-based applications (No description.).
Maintainability is reached/addressed using/by Loosely coupled components (No description.).
Reliability is reached/addressed using/by Open source software solutions and modular architecture design with clear relationships between components (No description.).
Security is reached/addressed using/by IoT devices authentication and data encryption (No description.).
Reference: Software architecture quality attributes of a layered sensor-based IoT system, 2021, https://doi.org/10.3390/s20195603.
The solution is composed by the architecture named as Layered, proposed in the paper Cloud-FOG assisted Energy Efficient Architectural Paradigm for disaster evacuation, 2022, with D.O.I. doi.org/10.1016/j.is.2021.101732 (link https://doi.org/10.1016/j.is.2021.101732). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Layered.
Solution Architecture Layered: described as: .
Performance/Efficiency is reached/addressed using/by Algorithm for Fog layer data selection (Data uniqueness and only important data) for transmission [Kaur et al. 2022] (No description.).
Reference: Cloud-FOG assisted Energy Efficient Architectural Paradigm for disaster evacuation, 2022, https://doi.org/10.1016/j.is.2021.101732.
The solution is composed by the architecture named as Layered, proposed in the paper An agile AI and IoT-Augmented Smart Farming: a Cost-Effective cognitive weather station, 2021, with D.O.I. doi.org/10.3390/agriculture12010035 (link https://doi.org/10.3390/agriculture12010035). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Interaction Capability,Maintainability,Performance/Efficiency,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Layered.
Solution Architecture Layered: described as: .
Compatibility is reached/addressed using/by Docker containers (No description.).
Flexibility is reached/addressed using/by Docker containers (No description.).
Interaction Capability is reached/addressed using/by Remote access to the deployed platform and its components (Web App and SSH) (No description.).
Maintainability is reached/addressed using/by Docker containers (No description.).
Performance/Efficiency is reached/addressed using/by Docker containers (No description.).
Reference: An agile AI and IoT-Augmented Smart Farming: a Cost-Effective cognitive weather station, 2021, https://doi.org/10.3390/agriculture12010035.
The solution is composed by the architecture named as Layered, proposed in the paper Smart Indoor Farms: Leveraging Technological Advancements to Power a Sustainable Agricultural Revolution, 2021, with D.O.I. doi.org/10.3390/agriengineering3040047 (link https://doi.org/10.3390/agriengineering3040047). This solution addresses the following Quality Requirements: .
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Layered.
Solution Architecture Layered: described as: .
Reference: Smart Indoor Farms: Leveraging Technological Advancements to Power a Sustainable Agricultural Revolution, 2021, https://doi.org/10.3390/agriengineering3040047.
The solution is composed by the architecture named as Layered Blockchain and AI-enabled, proposed in the paper A blockchain and artificial intelligence-enabled smart IoT framework for sustainable city, 2022, with D.O.I. doi.org/10.1002/int.22852 (link https://doi.org/10.1002/int.22852). This solution addresses the following Quality Requirements: Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Layered Blockchain and AI-enabled.
Solution Architecture Layered Blockchain and AI-enabled: described as: .

Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: A blockchain and artificial intelligence-enabled smart IoT framework for sustainable city, 2022, https://doi.org/10.1002/int.22852.
The solution is composed by the architecture named as Layered Blockchain-Based SDN, proposed in the paper Intelligent Blockchain Based secure routing for multidomain SDN Enabled IoT networks, 2022, with D.O.I. doi.org/10.1155/2022/5693962 (link https://doi.org/10.1155/2022/5693962). This solution addresses the following Quality Requirements: Reliability,Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Layered Blockchain-Based SDN.
Solution Architecture Layered Blockchain-Based SDN: described as: .
Reliability is reached/addressed using/by Reputation concept (Blockchain-based SDN-enabled network architecture) [Zeng et al. 2022] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: Intelligent Blockchain Based secure routing for multidomain SDN Enabled IoT networks, 2022, https://doi.org/10.1155/2022/5693962.
The solution is composed by the architecture named as LoRaWAN, proposed in the paper A new Low-Cost LORAWAN power switch for smart farm applications, 2021, with D.O.I. doi.org/10.1109/SMC52423.2021.9659076 (link https://doi.org/10.1109/SMC52423.2021.9659076). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and LoRaWAN.
Solution Architecture LoRaWAN: described as: .
Performance/Efficiency is reached/addressed using/by Long-Range Wide Area Network (LoRaWAN) (No description.).
Reference: A new Low-Cost LORAWAN power switch for smart farm applications, 2021, https://doi.org/10.1109/SMC52423.2021.9659076.

The solution is composed by the architecture named as A4IoT (Anyplace 4.0 IoT), proposed in the paper The Anyplace 4.0 IoT Localization Architecture, 2020, with D.O.I. 10.1109/MDM48529.2020.00045 (link https://ieeexplore.ieee.org/document/9162205). This solution addresses the following Quality Requirements: Flexibility,Interaction Capability,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and A4IoT (Anyplace 4.0 IoT).
Solution Architecture A4IoT (Anyplace 4.0 IoT): described as: It is an open-source architecture designed for IoT-based indoor localization using technologies like Wi-Fi, BLE, Cellular, UWB, and Computer Vision (CV). The system supports edge computing, timeseries databases, and crowdsourcing to manage localization tasks across various smart spaces, such as factories, healthcare facilities, and emergency response environments. Its modular and scalable design ensures integration across platforms like Android, Linux, Windows, and Robot OS, making it adaptable to diverse IoT ecosystems..
Performance/Efficiency is reached/addressed using/by Time-Series Database (InfluxDB) (Time Behavior - With temporal queries we can now consume high-volume input streams to provide real-time IoT tracking.").
Reliability is reached/addressed using/by Multi-node Cluster Deployment (Fault Tolerance - The production environment uses a 3-node cluster with two replicas, for both the database engines and the DFS (Distributed Filesystem). This setup allows for a full operation with just a single node active at any time.").
Security is reached/addressed using/by SSL certificates - Encrypted Communications (Confidentiality - All communication is encrypted even in internal networks, as A4IoT image automatically creates and uses SSL certificates.).
Interaction Capability is reached/addressed using/by Web-based Dashboard (Operability - Web Apps built with HTML5, CSS3, and AngularJS.").
Flexibility is reached/addressed using/by Docker containers (Adaptability - It utilizes Docker to enable the deployment of single-node or multi-node configurations, regardless of any dependencies or the underlying OS.).
Reference: The Anyplace 4.0 IoT Localization Architecture, 2020, https://ieeexplore.ieee.org/document/9162205.
The solution is composed by the architecture named as Agricultural IoT Reference Architecture (AITRA), proposed in the paper Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3031634 (link https://doi.org/10.1109/ACCESS.2020.3031634). This solution addresses the following Quality Requirements: Compatibility,Reliability,Security,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Agricultural IoT Reference Architecture (AITRA).
Solution Architecture Agricultural IoT Reference Architecture (AITRA): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Reliability is reached/addressed using/by Edge gateway (No description.).
Security is reached/addressed using/by Authentication, authorization, and data access control (No description.).
Reference: Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, https://doi.org/10.1109/ACCESS.2020.3031634.
The solution is composed by the architecture named as Blockchain-based Zero Trust, proposed in the paper Blockchain-Based Zero Trust on the Edge, 2023, with D.O.I. 10.1109/CSCI62032.2023.00167 (link https://ieeexplore.ieee.org/document/10590526). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Performance/Efficiency,Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Blockchain-based Zero Trust.
Solution Architecture Blockchain-based Zero Trust: described as: The architecture proposed in "Blockchain-Based Zero Trust on the Edge" integrates Zero Trust principles with Blockchain to ensure secure access and trust management for Edge Computing in IoT environments. It focuses on continuous verification of users and devices, ensuring that all access is authenticated before being granted. Blockchain is used for decentralized trust management, maintaining an immutable record of activities, which enhances security and transparency. This setup is optimized

for edge computing, reducing latency by processing data locally while maintaining the flexibility to scale across diverse IoT applications, such as smart cities and industrial IoT..
Security is reached/addressed using/by Zero Trust Architecture (ZTA) (Confidentiality - Zero Trust Architecture (ZTA)
ZTA ensures continuous verification and monitoring of all network traffic, restricting unauthorized access.).
Security is reached/addressed using/by Blockchain Technology (Integrity - Blockchain serves as an immutable ledger for storing requests, ensuring data integrity and trustworthiness.).
Performance/Efficiency is reached/addressed using/by Redis Message Broker (Time Behavior - Redis Message Broker for Asynchronous Processing [redis.io]
Asynchronous processing for write operations improves system responsiveness under high loads.).
Maintainability is reached/addressed using/by Component isolation through microservices (Modularity -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.
).
Flexibility is reached/addressed using/by Component isolation through microservices (Scalability -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.).
Reference: Blockchain-Based Zero Trust on the Edge, 2023, https://ieeexplore.ieee.org/document/10590526.
The solution is composed by the architecture named as Blockchain-enabled, proposed in the paper An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, with D.O.I. doi.org/10.1109/TSC.2020.2966970 (link https://doi.org/10.1109/TSC.2020.2966970). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Blockchain-enabled.
Solution Architecture Blockchain-enabled: described as: .
Performance/Efficiency is reached/addressed using/by Routing protocol designed for IoT devices in the SDN controller [Yazdinejad et al. 2020] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, https://doi.org/10.1109/TSC.2020.2966970.
The solution is composed by the architecture named as Cloud-based Secure Biometric, proposed in the paper BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, with D.O.I. doi.org/10.1016/j.jksuci.2017.07.001 (link https://doi.org/10.1016/j.jksuci.2017.07.001). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Cloud-based Secure Biometric.
Solution Architecture Cloud-based Secure Biometric: described as: .
Flexibility is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Reliability is reached/addressed using/by MapReduce framework (No description.).
Security is reached/addressed using/by Cloud-based biometric authentication [Shakil et al. 2020] (No description.).
Reference: BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, https://doi.org/10.1016/j.jksuci.2017.07.001.
The solution is composed by the architecture named as Deep Learning based, proposed in the paper HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, with D.O.I. doi.org/10.1016/j.future.2019.10.043 (link https://doi.org/10.1016/j.future.2019.10.043). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Deep Learning based.
Solution Architecture Deep Learning based: described as: .
Performance/Efficiency is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Security is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Reference: HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, https://doi.org/10.1016/j.future.2019.10.043.
The solution is composed by the architecture named as Distributed Blockchain-SDN based, proposed in the paper DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3039113 (link https://doi.org/10.1109/ACCESS.2020.3039113). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Distributed Blockchain-SDN based.
Solution Architecture Distributed Blockchain-SDN based: described as: .
Performance/Efficiency is reached/addressed using/by Decentralized blockchain nodes, dynamic SDN controllers, and edge computing (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, https://doi.org/10.1109/ACCESS.2020.3039113.
The solution is composed by the architecture named as Distributed Edge-Cloud, proposed in the paper Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, with D.O.I. doi.org/10.1016/j.future.2020.05.042 (link

https://doi.org/10.1016/j.future.2020.05.042). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Distributed Edge-Cloud.
Solution Architecture Distributed Edge-Cloud: described as: .
Security is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Reference: Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, https://doi.org/10.1016/j.future.2020.05.042.
The solution is composed by the architecture named as Edge Computing-based Fault Tolerant Framework, proposed in the paper Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, with D.O.I. doi.org/10.1109/IWCMC48107.2020.9148269 (link https://doi.org/10.1109/IWCMC48107.2020.9148269). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Edge Computing-based Fault Tolerant Framework.
Solution Architecture Edge Computing-based Fault Tolerant Framework: described as: .
Performance/Efficiency is reached/addressed using/by Kubernetes framework (No description.).
Flexibility is reached/addressed using/by Docker containers (No description.).
Reference: Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, https://doi.org/10.1109/IWCMC48107.2020.9148269.
The solution is composed by the architecture named as Event-driven IoT, proposed in the paper Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, with D.O.I. doi.org/10.1007/s10586-020-03189-w (link https://doi.org/10.1007/s10586-020-03189-w). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Event-driven IoT.
Solution Architecture Event-driven IoT: described as: .
Performance/Efficiency is reached/addressed using/by Complex Event Processing (CEP) [Rahmani et al. 2020] (No description.).
Reliability is reached/addressed using/by TCP-IP protocol on Client/Server architecture for data communication (No description.).
Reference: Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, https://doi.org/10.1007/s10586-020-03189-w.
The solution is composed by the architecture named as Fog Computing based, proposed in the paper FOG-based architecture and load balancing methodology for health monitoring systems, 2021, with D.O.I. doi.org/10.1109/ACCESS.2021.3094033 (link https://doi.org/10.1109/ACCESS.2021.3094033). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Fog Computing based.
Solution Architecture Fog Computing based: described as: .
Performance/Efficiency is reached/addressed using/by Load Balancing Scheme (LBS) - Distributed workload to all fog nodes [Asghar et al. 2021] (No description.).
Reference: FOG-based architecture and load balancing methodology for health monitoring systems, 2021, https://doi.org/10.1109/ACCESS.2021.3094033.
The solution is composed by the architecture named as Geographic-based, proposed in the paper A platform for integrating heterogeneous data and developing smart city applications, 2022, with D.O.I. doi.org/10.1016/j.future.2021.10.030 (link https://doi.org/10.1016/j.future.2021.10.030). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Geographic-based.
Solution Architecture Geographic-based: described as: .
Security is reached/addressed using/by FIWARE Platform [www.fiware.org] (Authentication, authorization, and data access control) (No description.).
Flexibility is reached/addressed using/by Virtualization (No description.).
Compatibility is reached/addressed using/by RESTFul APIs (No description.).
Reference: A platform for integrating heterogeneous data and developing smart city applications, 2022, https://doi.org/10.1016/j.future.2021.10.030.
The solution is composed by the architecture named as Hybrid (Centralized and Distributed), proposed in the paper Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, with D.O.I. doi.org/10.1016/j.procs.2021.03.030 (link https://doi.org/10.1016/j.procs.2021.03.030). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid (Centralized and Distributed).
Solution Architecture Hybrid (Centralized and Distributed): described as: .
Performance/Efficiency is reached/addressed using/by Dynamic allocation of the resources using Software Defined Networking (SDN) [Chudhary and Sharma 2021] (No description.).
Reference: Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, https://doi.org/10.1016/j.procs.2021.03.030.
The solution is composed by the architecture named as Hybrid Cloud-Fog Computing, proposed in the paper Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, with D.O.I. doi.org/10.1007/s10586-021-03375-4 (link https://link.springer.com/article/10.1007/s10586-021-03375-4). This solution addresses the following Quality Requirements:

Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid Cloud-Fog Computing.
Solution Architecture Hybrid Cloud-Fog Computing: described as: .
Reliability is reached/addressed using/by Fog layer in two levels: 1-Intra mobility (User network): Nodes (Data transfer, device management, temporary local storage, notifications, and data consistency). 2-Extra mobility (Different domains): Smart Gateways (Local processing and database) (No description.).
Performance/Efficiency is reached/addressed using/by Fog computing (No description.).
Reference: Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, https://link.springer.com/article/10.1007/s10586-021-03375-4.
The solution is composed by the architecture named as Industrial Internet of Things (IIoT), proposed in the paper A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Performance/Efficiency,Security,.
The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and Industrial Internet of Things (IIoT).
Solution Architecture Industrial Internet of Things (IIoT): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Flexibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Performance/Efficiency is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).
Security is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).
Reference: A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, https://doi.org/10.3390/s20195603.
The solution is composed by the architecture named as Layered, proposed in the paper An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, with D.O.I. doi.org/10.1016/j.future.2020.07.047 (link https://doi.org/10.1016/j.future.2020.07.047). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.
Solution Architecture Layered: described as: .
Flexibility is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).
Performance/Efficiency is reached/addressed using/by Big data analytics engine [Ali et al. 2021] (No description.).
Security is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).
Reference: An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, https://doi.org/10.1016/j.future.2020.07.047.
The solution is composed by the architecture named as Layered, proposed in the paper A novel architecture for the Internet of Things based E-Health systems, 2020, with D.O.I. doi.org/10.1166/jmihi.2020.3184 (link http://dx.doi.org/10.1166/jmihi.2020.3184). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.
Solution Architecture Layered: described as: .
Performance/Efficiency is reached/addressed using/by IoT objects connected at the edge of the network (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: A novel architecture for the Internet of Things based E-Health systems, 2020, http://dx.doi.org/10.1166/jmihi.2020.3184.
The solution is composed by the architecture named as Layered, proposed in the paper Software architecture quality attributes of a layered sensor-based IoT system, 2021, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Reliability,Security,.
The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and Layered.
Solution Architecture Layered: described as: .
Flexibility is reached/addressed using/by Messaging protocol brokers and at the client side, web-based applications (No description.).
Maintainability is reached/addressed using/by Loosely coupled components (No description.).
Reliability is reached/addressed using/by Open source software solutions and modular architecture design with clear relationships between components (No description.).
Security is reached/addressed using/by IoT devices authentication and data encryption (No description.).
Reference: Software architecture quality attributes of a layered sensor-based IoT system, 2021, https://doi.org/10.3390/s20195603.
The solution is composed by the architecture named as Layered, proposed in the paper Cloud-FOG assisted Energy Efficient Architectural Paradigm for disaster evacuation, 2022, with D.O.I. doi.org/10.1016/j.is.2021.101732 (link https://doi.org/10.1016/j.is.2021.101732). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Layered.
Solution Architecture Layered: described as: .
Performance/Efficiency is reached/addressed using/by Algorithm for Fog layer data selection (Data uniqueness and only important data) for transmission [Kaur et al. 2022] (No description.).
Reference: Cloud-FOG assisted Energy Efficient Architectural Paradigm for disaster evacuation, 2022, https://doi.org/10.1016/j.is.2021.101732.

The solution is composed by the architecture named as Layered, proposed in the paper An agile AI and IoT-Augmented Smart Farming: a Cost-Effective cognitive weather station, 2021, with D.O.I. doi.org/10.3390/agriculture12010035 (link https://doi.org/10.3390/agriculture12010035). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Interaction Capability,Maintainability,Performance/Efficiency,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Layered.
Solution Architecture Layered: described as: .
Compatibility is reached/addressed using/by Docker containers (No description.).
Flexibility is reached/addressed using/by Docker containers (No description.).
Interaction Capability is reached/addressed using/by Remote access to the deployed platform and its components (Web App and SSH) (No description.).
Maintainability is reached/addressed using/by Docker containers (No description.).
Performance/Efficiency is reached/addressed using/by Docker containers (No description.).
Reference: An agile AI and IoT-Augmented Smart Farming: a Cost-Effective cognitive weather station, 2021, https://doi.org/10.3390/agriculture12010035.
The solution is composed by the architecture named as Layered, proposed in the paper Smart Indoor Farms: Leveraging Technological Advancements to Power a Sustainable Agricultural Revolution, 2021, with D.O.I. doi.org/10.3390/agriengineering3040047 (link https://doi.org/10.3390/agriengineering3040047). This solution addresses the following Quality Requirements: .
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Layered.
Solution Architecture Layered: described as: .
Reference: Smart Indoor Farms: Leveraging Technological Advancements to Power a Sustainable Agricultural Revolution, 2021, https://doi.org/10.3390/agriengineering3040047.
The solution is composed by the architecture named as Layered Blockchain and AI-enabled, proposed in the paper A blockchain and artificial intelligence-enabled smart IoT framework for sustainable city, 2022, with D.O.I. doi.org/10.1002/int.22852 (link https://doi.org/10.1002/int.22852). This solution addresses the following Quality Requirements: Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Layered Blockchain and AI-enabled.
Solution Architecture Layered Blockchain and AI-enabled: described as: .
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: A blockchain and artificial intelligence-enabled smart IoT framework for sustainable city, 2022, https://doi.org/10.1002/int.22852.
The solution is composed by the architecture named as Layered Blockchain-Based SDN, proposed in the paper Intelligent Blockchain Based secure routing for multidomain SDN Enabled IoT networks, 2022, with D.O.I. doi.org/10.1155/2022/5693962 (link https://doi.org/10.1155/2022/5693962). This solution addresses the following Quality Requirements: Reliability,Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Layered Blockchain-Based SDN.
Solution Architecture Layered Blockchain-Based SDN: described as: .
Reliability is reached/addressed using/by Reputation concept (Blockchain-based SDN-enabled network architecture) [Zeng et al. 2022] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: Intelligent Blockchain Based secure routing for multidomain SDN Enabled IoT networks, 2022, https://doi.org/10.1155/2022/5693962.
The solution is composed by the architecture named as LoRaWAN, proposed in the paper A new Low-Cost LORAWAN power switch for smart farm applications, 2021, with D.O.I. doi.org/10.1109/SMC52423.2021.9659076 (link https://doi.org/10.1109/SMC52423.2021.9659076). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and LoRaWAN.
Solution Architecture LoRaWAN: described as: .
Performance/Efficiency is reached/addressed using/by Long-Range Wide Area Network (LoRaWAN) (No description.).
Reference: A new Low-Cost LORAWAN power switch for smart farm applications, 2021, https://doi.org/10.1109/SMC52423.2021.9659076.
The solution is composed by the architecture named as Loosely Coupled IoT Platform and Localization Module, proposed in the paper A Self-Adaptive Bluetooth Indoor Localization System using LSTM-based Distance Estimator, 2020, with D.O.I. 10.1109/ICCCN49398.2020.9209674 (link https://ieeexplore.ieee.org/document/9209674). This solution addresses the following Quality Requirements: Compatibility,Functional Suitability,Interaction Capability,Maintainability,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Loosely Coupled IoT Platform and Localization Module.
Solution Architecture Loosely Coupled IoT Platform and Localization Module: described as: The proposed system architecture consists of two main components: the IoT platform and the localization module, connected by middleware. The IoT platform follows a hierarchical structure with layers for hardware (Bluetooth devices), data (MQTT-based RSSI parsing), business logic, display, and user interface. The localization module enhances trilateration with self-adaptive mechanisms, including an LSTM-based distance estimator, elastic radius intersecting, weighted centroid localization, and self-adaptive Kalman filtering for trajectory optimization..
Maintainability is reached/addressed using/by Google Remote Procedure Calls (gRPC) (Modularity - The IoT platform (server side) and the localization module (client side) are connected via a middleware called Google Remote Procedure Calls (gRPC).").
Interaction Capability is reached/addressed using/by Web Interface (Operability - Web interface for visualizing localization and trajectory tracking.).
Compatibility is reached/addressed using/by Google Remote Procedure Calls (gRPC) (Interoperability - The gRPC middleware supports

communication between the IoT platform and the localization module, enabling system integration and compatibility with other platforms.).
Functional Suitability is reached/addressed using/by Kalman filtering (KF) (Kalman Filtering (KF) is an algorithm that estimates the state of a dynamic system by combining noisy sensor measurements with predictions. It recursively updates estimates based on new data, minimizing errors and improving accuracy, making it useful in tracking, navigation, and signal processing.).
Reference: A Self-Adaptive Bluetooth Indoor Localization System using LSTM-based Distance Estimator, 2020, https://ieeexplore.ieee.org/document/9209674.

The solution is composed by the architecture named as A4IoT (Anyplace 4.0 IoT), proposed in the paper The Anyplace 4.0 IoT Localization Architecture, 2020, with D.O.I. 10.1109/MDM48529.2020.00045 (link https://ieeexplore.ieee.org/document/9162205). This solution addresses the following Quality Requirements: Flexibility,Interaction Capability,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and A4IoT (Anyplace 4.0 IoT).
Solution Architecture A4IoT (Anyplace 4.0 IoT): described as: It is an open-source architecture designed for IoT-based indoor localization using technologies like Wi-Fi, BLE, Cellular, UWB, and Computer Vision (CV). The system supports edge computing, timeseries databases, and crowdsourcing to manage localization tasks across various smart spaces, such as factories, healthcare facilities, and emergency response environments. Its modular and scalable design ensures integration across platforms like Android, Linux, Windows, and Robot OS, making it adaptable to diverse IoT ecosystems..
Performance/Efficiency is reached/addressed using/by Time-Series Database (InfluxDB) (Time Behavior - With temporal queries we can now consume high-volume input streams to provide real-time IoT tracking.").
Reliability is reached/addressed using/by Multi-node Cluster Deployment (Fault Tolerance - The production environment uses a 3-node cluster with two replicas, for both the database engines and the DFS (Distributed Filesystem). This setup allows for a full operation with just a single node active at any time.").
Security is reached/addressed using/by SSL certificates - Encrypted Communications (Confidentiality - All communication is encrypted even in internal networks, as A4IoT image automatically creates and uses SSL certificates.).
Interaction Capability is reached/addressed using/by Web-based Dashboard (Operability - Web Apps built with HTML5, CSS3, and AngularJS.").
Flexibility is reached/addressed using/by Docker containers (Adaptability - It utilizes Docker to enable the deployment of single-node or multi-node configurations, regardless of any dependencies or the underlying OS.).
Reference: The Anyplace 4.0 IoT Localization Architecture, 2020, https://ieeexplore.ieee.org/document/9162205.
The solution is composed by the architecture named as Agricultural IoT Reference Architecture (AITRA), proposed in the paper Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3031634 (link https://doi.org/10.1109/ACCESS.2020.3031634). This solution addresses the following Quality Requirements: Compatibility,Reliability,Security,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Agricultural IoT Reference Architecture (AITRA).
Solution Architecture Agricultural IoT Reference Architecture (AITRA): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Reliability is reached/addressed using/by Edge gateway (No description.).
Security is reached/addressed using/by Authentication, authorization, and data access control (No description.).
Reference: Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, https://doi.org/10.1109/ACCESS.2020.3031634.
The solution is composed by the architecture named as Blockchain-based Zero Trust, proposed in the paper Blockchain-Based Zero Trust on the Edge, 2023, with D.O.I. 10.1109/CSCI62032.2023.00167 (link https://ieeexplore.ieee.org/document/10590526). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Performance/Efficiency,Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Blockchain-based Zero Trust.
Solution Architecture Blockchain-based Zero Trust: described as: The architecture proposed in "Blockchain-Based Zero Trust on the Edge" integrates Zero Trust principles with Blockchain to ensure secure access and trust management for Edge Computing in IoT environments. It focuses on continuous verification of users and devices, ensuring that all access is authenticated before being granted. Blockchain is used for decentralized trust management, maintaining an immutable record of activities, which enhances security and transparency. This setup is optimized for edge computing, reducing latency by processing data locally while maintaining the flexibility to scale across diverse IoT applications, such as smart cities and industrial IoT..
Security is reached/addressed using/by Zero Trust Architecture (ZTA) (Confidentiality - Zero Trust Architecture (ZTA)
ZTA ensures continuous verification and monitoring of all network traffic, restricting unauthorized access.).
Security is reached/addressed using/by Blockchain Technology (Integrity - Blockchain serves as an immutable ledger for storing requests, ensuring data integrity and trustworthiness.).
Performance/Efficiency is reached/addressed using/by Redis Message Broker (Time Behavior - Redis Message Broker for Asynchronous Processing [redis.io]
Asynchronous processing for write operations improves system responsiveness under high loads.).
Maintainability is reached/addressed using/by Component isolation through microservices (Modularity -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.
).
Flexibility is reached/addressed using/by Component isolation through microservices (Scalability -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.).

Reference: Blockchain-Based Zero Trust on the Edge, 2023, https://ieeexplore.ieee.org/document/10590526.
The solution is composed by the architecture named as Blockchain-enabled, proposed in the paper An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, with D.O.I. doi.org/10.1109/TSC.2020.2966970 (link https://doi.org/10.1109/TSC.2020.2966970). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Blockchain-enabled.
Solution Architecture Blockchain-enabled: described as: .
Performance/Efficiency is reached/addressed using/by Routing protocol designed for IoT devices in the SDN controller [Yazdinejad et al. 2020] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, https://doi.org/10.1109/TSC.2020.2966970.
The solution is composed by the architecture named as Cloud-based Secure Biometric, proposed in the paper BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, with D.O.I. doi.org/10.1016/j.jksuci.2017.07.001 (link https://doi.org/10.1016/j.jksuci.2017.07.001). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Cloud-based Secure Biometric.
Solution Architecture Cloud-based Secure Biometric: described as: .
Flexibility is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Reliability is reached/addressed using/by MapReduce framework (No description.).
Security is reached/addressed using/by Cloud-based biometric authentication [Shakil et al. 2020] (No description.).
Reference: BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, https://doi.org/10.1016/j.jksuci.2017.07.001.
The solution is composed by the architecture named as Deep Learning based, proposed in the paper HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, with D.O.I. doi.org/10.1016/j.future.2019.10.043 (link https://doi.org/10.1016/j.future.2019.10.043). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Deep Learning based.
Solution Architecture Deep Learning based: described as: .
Performance/Efficiency is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Security is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Reference: HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, https://doi.org/10.1016/j.future.2019.10.043.
The solution is composed by the architecture named as Distributed Blockchain-SDN based, proposed in the paper DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3039113 (link https://doi.org/10.1109/ACCESS.2020.3039113). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Distributed Blockchain-SDN based.
Solution Architecture Distributed Blockchain-SDN based: described as: .
Performance/Efficiency is reached/addressed using/by Decentralized blockchain nodes, dynamic SDN controllers, and edge computing (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, https://doi.org/10.1109/ACCESS.2020.3039113.
The solution is composed by the architecture named as Distributed Edge-Cloud, proposed in the paper Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, with D.O.I. doi.org/10.1016/j.future.2020.05.042 (link https://doi.org/10.1016/j.future.2020.05.042). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Distributed Edge-Cloud.
Solution Architecture Distributed Edge-Cloud: described as: .
Security is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Reference: Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, https://doi.org/10.1016/j.future.2020.05.042.
The solution is composed by the architecture named as Edge Computing-based Fault Tolerant Framework, proposed in the paper Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, with D.O.I. doi.org/10.1109/IWCMC48107.2020.9148269 (link https://doi.org/10.1109/IWCMC48107.2020.9148269). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Edge

Computing-based Fault Tolerant Framework.
Solution Architecture Edge Computing-based Fault Tolerant Framework: described as: .
Performance/Efficiency is reached/addressed using/by Kubernetes framework (No description.).
Flexibility is reached/addressed using/by Docker containers (No description.).
Reference: Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, https://doi.org/10.1109/IWCMC48107.2020.9148269.
The solution is composed by the architecture named as Event-driven IoT, proposed in the paper Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, with D.O.I. doi.org/10.1007/s10586-020-03189-w (link https://doi.org/10.1007/s10586-020-03189-w). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Event-driven IoT.
Solution Architecture Event-driven IoT: described as: .
Performance/Efficiency is reached/addressed using/by Complex Event Processing (CEP) [Rahmani et al. 2020] (No description.).
Reliability is reached/addressed using/by TCP-IP protocol on Client/Server architecture for data communication (No description.).
Reference: Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, https://doi.org/10.1007/s10586-020-03189-w.
The solution is composed by the architecture named as Fog Computing based, proposed in the paper FOG-based architecture and load balancing methodology for health monitoring systems, 2021, with D.O.I. doi.org/10.1109/ACCESS.2021.3094033 (link https://doi.org/10.1109/ACCESS.2021.3094033). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Fog Computing based.
Solution Architecture Fog Computing based: described as: .
Performance/Efficiency is reached/addressed using/by Load Balancing Scheme (LBS) - Distributed workload to all fog nodes [Asghar et al. 2021] (No description.).
Reference: FOG-based architecture and load balancing methodology for health monitoring systems, 2021, https://doi.org/10.1109/ACCESS.2021.3094033.
The solution is composed by the architecture named as Geographic-based, proposed in the paper A platform for integrating heterogeneous data and developing smart city applications, 2022, with D.O.I. doi.org/10.1016/j.future.2021.10.030 (link https://doi.org/10.1016/j.future.2021.10.030). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Geographic-based.
Solution Architecture Geographic-based: described as: .
Security is reached/addressed using/by FIWARE Platform [www.fiware.org] (Authentication, authorization, and data access control) (No description.).
Flexibility is reached/addressed using/by Virtualization (No description.).
Compatibility is reached/addressed using/by RESTFul APIs (No description.).
Reference: A platform for integrating heterogeneous data and developing smart city applications, 2022, https://doi.org/10.1016/j.future.2021.10.030.
The solution is composed by the architecture named as Hybrid (Centralized and Distributed), proposed in the paper Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, with D.O.I. doi.org/10.1016/j.procs.2021.03.030 (link https://doi.org/10.1016/j.procs.2021.03.030). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid (Centralized and Distributed).
Solution Architecture Hybrid (Centralized and Distributed): described as: .
Performance/Efficiency is reached/addressed using/by Dynamic allocation of the resources using Software Defined Networking (SDN) [Chudhary and Sharma 2021] (No description.).
Reference: Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, https://doi.org/10.1016/j.procs.2021.03.030.
The solution is composed by the architecture named as Hybrid Cloud-Fog Computing, proposed in the paper Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, with D.O.I. doi.org/10.1007/s10586-021-03375-4 (link https://link.springer.com/article/10.1007/s10586-021-03375-4). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid Cloud-Fog Computing.
Solution Architecture Hybrid Cloud-Fog Computing: described as: .
Reliability is reached/addressed using/by Fog layer in two levels: 1-Intra mobility (User network): Nodes (Data transfer, device management, temporary local storage, notifications, and data consistency). 2-Extra mobility (Different domains): Smart Gateways (Local processing and database) (No description.).
Performance/Efficiency is reached/addressed using/by Fog computing (No description.).
Reference: Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, https://link.springer.com/article/10.1007/s10586-021-03375-4.
The solution is composed by the architecture named as Industrial Internet of Things (IIoT), proposed in the paper A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Performance/Efficiency,Security,.
The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and

Industrial Internet of Things (IIoT).
Solution Architecture Industrial Internet of Things (IIoT): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Flexibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Performance/Efficiency is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).
Security is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).
Reference: A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, https://doi.org/10.3390/s20195603.
The solution is composed by the architecture named as Layered, proposed in the paper An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, with D.O.I. doi.org/10.1016/j.future.2020.07.047 (link https://doi.org/10.1016/j.future.2020.07.047). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.
Solution Architecture Layered: described as: .
Flexibility is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).
Performance/Efficiency is reached/addressed using/by Big data analytics engine [Ali et al. 2021] (No description.).
Security is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).
Reference: An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, https://doi.org/10.1016/j.future.2020.07.047.
The solution is composed by the architecture named as Layered, proposed in the paper A novel architecture for the Internet of Things based E-Health systems, 2020, with D.O.I. doi.org/10.1166/jmihi.2020.3184 (link http://dx.doi.org/10.1166/jmihi.2020.3184). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.
Solution Architecture Layered: described as: .
Performance/Efficiency is reached/addressed using/by IoT objects connected at the edge of the network (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: A novel architecture for the Internet of Things based E-Health systems, 2020, http://dx.doi.org/10.1166/jmihi.2020.3184.
The solution is composed by the architecture named as Layered, proposed in the paper Software architecture quality attributes of a layered sensor-based IoT system, 2021, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Reliability,Security,.
The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and Layered.
Solution Architecture Layered: described as: .
Flexibility is reached/addressed using/by Messaging protocol brokers and at the client side, web-based applications (No description.).
Maintainability is reached/addressed using/by Loosely coupled components (No description.).
Reliability is reached/addressed using/by Open source software solutions and modular architecture design with clear relationships between components (No description.).
Security is reached/addressed using/by IoT devices authentication and data encryption (No description.).
Reference: Software architecture quality attributes of a layered sensor-based IoT system, 2021, https://doi.org/10.3390/s20195603.
The solution is composed by the architecture named as Layered, proposed in the paper Cloud-FOG assisted Energy Efficient Architectural Paradigm for disaster evacuation, 2022, with D.O.I. doi.org/10.1016/j.is.2021.101732 (link https://doi.org/10.1016/j.is.2021.101732). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Layered.
Solution Architecture Layered: described as: .
Performance/Efficiency is reached/addressed using/by Algorithm for Fog layer data selection (Data uniqueness and only important data) for transmission [Kaur et al. 2022] (No description.).
Reference: Cloud-FOG assisted Energy Efficient Architectural Paradigm for disaster evacuation, 2022, https://doi.org/10.1016/j.is.2021.101732.
The solution is composed by the architecture named as Layered, proposed in the paper An agile AI and IoT-Augmented Smart Farming: a Cost-Effective cognitive weather station, 2021, with D.O.I. doi.org/10.3390/agriculture12010035 (link https://doi.org/10.3390/agriculture12010035). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Interaction Capability,Maintainability,Performance/Efficiency,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Layered.
Solution Architecture Layered: described as: .
Compatibility is reached/addressed using/by Docker containers (No description.).
Flexibility is reached/addressed using/by Docker containers (No description.).
Interaction Capability is reached/addressed using/by Remote access to the deployed platform and its components (Web App and SSH) (No description.).
Maintainability is reached/addressed using/by Docker containers (No description.).
Performance/Efficiency is reached/addressed using/by Docker containers (No description.).
Reference: An agile AI and IoT-Augmented Smart Farming: a Cost-Effective cognitive weather station, 2021, https://doi.org/10.3390/agriculture12010035.

The solution is composed by the architecture named as Layered, proposed in the paper Smart Indoor Farms: Leveraging Technological Advancements to Power a Sustainable Agricultural Revolution, 2021, with D.O.I. doi.org/10.3390/agriengineering3040047 (link https://doi.org/10.3390/agriengineering3040047). This solution addresses the following Quality Requirements: .
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Layered.
Solution Architecture Layered: described as: .
Reference: Smart Indoor Farms: Leveraging Technological Advancements to Power a Sustainable Agricultural Revolution, 2021, https://doi.org/10.3390/agriengineering3040047.
The solution is composed by the architecture named as Layered Blockchain and AI-enabled, proposed in the paper A blockchain and artificial intelligence-enabled smart IoT framework for sustainable city, 2022, with D.O.I. doi.org/10.1002/int.22852 (link https://doi.org/10.1002/int.22852). This solution addresses the following Quality Requirements: Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Layered Blockchain and AI-enabled.
Solution Architecture Layered Blockchain and AI-enabled: described as: .
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: A blockchain and artificial intelligence-enabled smart IoT framework for sustainable city, 2022, https://doi.org/10.1002/int.22852.
The solution is composed by the architecture named as Layered Blockchain-Based SDN, proposed in the paper Intelligent Blockchain Based secure routing for multidomain SDN Enabled IoT networks, 2022, with D.O.I. doi.org/10.1155/2022/5693962 (link https://doi.org/10.1155/2022/5693962). This solution addresses the following Quality Requirements: Reliability,Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Layered Blockchain-Based SDN.
Solution Architecture Layered Blockchain-Based SDN: described as: .
Reliability is reached/addressed using/by Reputation concept (Blockchain-based SDN-enabled network architecture) [Zeng et al. 2022] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: Intelligent Blockchain Based secure routing for multidomain SDN Enabled IoT networks, 2022, https://doi.org/10.1155/2022/5693962.
The solution is composed by the architecture named as LoRaWAN, proposed in the paper A new Low-Cost LORAWAN power switch for smart farm applications, 2021, with D.O.I. doi.org/10.1109/SMC52423.2021.9659076 (link https://doi.org/10.1109/SMC52423.2021.9659076). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and LoRaWAN.
Solution Architecture LoRaWAN: described as: .
Performance/Efficiency is reached/addressed using/by Long-Range Wide Area Network (LoRaWAN) (No description.).
Reference: A new Low-Cost LORAWAN power switch for smart farm applications, 2021, https://doi.org/10.1109/SMC52423.2021.9659076.
The solution is composed by the architecture named as Loosely Coupled IoT Platform and Localization Module, proposed in the paper A Self-Adaptive Bluetooth Indoor Localization System using LSTM-based Distance Estimator, 2020, with D.O.I. 10.1109/ICCCN49398.2020.9209674 (link https://ieeexplore.ieee.org/document/9209674). This solution addresses the following Quality Requirements: Compatibility,Functional Suitability,Interaction Capability,Maintainability,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Loosely Coupled IoT Platform and Localization Module.
Solution Architecture Loosely Coupled IoT Platform and Localization Module: described as: The proposed system architecture consists of two main components: the IoT platform and the localization module, connected by middleware. The IoT platform follows a hierarchical structure with layers for hardware (Bluetooth devices), data (MQTT-based RSSI parsing), business logic, display, and user interface. The localization module enhances trilateration with self-adaptive mechanisms, including an LSTM-based distance estimator, elastic radius intersecting, weighted centroid localization, and self-adaptive Kalman filtering for trajectory optimization..
Maintainability is reached/addressed using/by Google Remote Procedure Calls (gRPC) (Modularity - The IoT platform (server side) and the localization module (client side) are connected via a middleware called Google Remote Procedure Calls (gRPC).").
Interaction Capability is reached/addressed using/by Web Interface (Operability - Web interface for visualizing localization and trajectory tracking.).
Compatibility is reached/addressed using/by Google Remote Procedure Calls (gRPC) (Interoperability - The gRPC middleware supports communication between the IoT platform and the localization module, enabling system integration and compatibility with other platforms.).
Functional Suitability is reached/addressed using/by Kalman filtering (KF) (Kalman Filtering (KF) is an algorithm that estimates the state of a dynamic system by combining noisy sensor measurements with predictions. It recursively updates estimates based on new data, minimizing errors and improving accuracy, making it useful in tracking, navigation, and signal processing.).
Reference: A Self-Adaptive Bluetooth Indoor Localization System using LSTM-based Distance Estimator, 2020, https://ieeexplore.ieee.org/document/9209674.
The solution is composed by the architecture named as Microservices, proposed in the paper Design of scalable IoT architecture based on AWS for smart livestock, 2021, with D.O.I. doi.org/10.3390/ani11092697 (link https://doi.org/10.3390/ani11092697). This solution addresses the following Quality Requirements: Flexibility,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Microservices.
Solution Architecture Microservices: described as: .
Flexibility is reached/addressed using/by Amazon Web Services (AWS) (No description.).
Reference: Design of scalable IoT architecture based on AWS for smart livestock, 2021, https://doi.org/10.3390/ani11092697.

The solution is composed by the architecture named as A4IoT (Anyplace 4.0 IoT), proposed in the paper The Anyplace 4.0 IoT Localization Architecture, 2020, with D.O.I. 10.1109/MDM48529.2020.00045 (link https://ieeexplore.ieee.org/document/9162205). This solution addresses the following Quality Requirements: Flexibility,Interaction Capability,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and A4IoT (Anyplace 4.0 IoT).
Solution Architecture A4IoT (Anyplace 4.0 IoT): described as: It is an open-source architecture designed for IoT-based indoor localization using technologies like Wi-Fi, BLE, Cellular, UWB, and Computer Vision (CV). The system supports edge computing, timeseries databases, and crowdsourcing to manage localization tasks across various smart spaces, such as factories, healthcare facilities, and emergency response environments. Its modular and scalable design ensures integration across platforms like Android, Linux, Windows, and Robot OS, making it adaptable to diverse IoT ecosystems..
Performance/Efficiency is reached/addressed using/by Time-Series Database (InfluxDB) (Time Behavior - With temporal queries we can now consume high-volume input streams to provide real-time IoT tracking.").
Reliability is reached/addressed using/by Multi-node Cluster Deployment (Fault Tolerance - The production environment uses a 3-node cluster with two replicas, for both the database engines and the DFS (Distributed Filesystem). This setup allows for a full operation with just a single node active at any time.").
Security is reached/addressed using/by SSL certificates - Encrypted Communications (Confidentiality - All communication is encrypted even in internal networks, as A4IoT image automatically creates and uses SSL certificates.).
Interaction Capability is reached/addressed using/by Web-based Dashboard (Operability - Web Apps built with HTML5, CSS3, and AngularJS.").
Flexibility is reached/addressed using/by Docker containers (Adaptability - It utilizes Docker to enable the deployment of single-node or multi-node configurations, regardless of any dependencies or the underlying OS.).
Reference: The Anyplace 4.0 IoT Localization Architecture, 2020, https://ieeexplore.ieee.org/document/9162205.
The solution is composed by the architecture named as Agricultural IoT Reference Architecture (AITRA), proposed in the paper Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3031634 (link https://doi.org/10.1109/ACCESS.2020.3031634). This solution addresses the following Quality Requirements: Compatibility,Reliability,Security,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Agricultural IoT Reference Architecture (AITRA).
Solution Architecture Agricultural IoT Reference Architecture (AITRA): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Reliability is reached/addressed using/by Edge gateway (No description.).
Security is reached/addressed using/by Authentication, authorization, and data access control (No description.).
Reference: Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, https://doi.org/10.1109/ACCESS.2020.3031634.
The solution is composed by the architecture named as Blockchain-based Zero Trust, proposed in the paper Blockchain-Based Zero Trust on the Edge, 2023, with D.O.I. 10.1109/CSCI62032.2023.00167 (link https://ieeexplore.ieee.org/document/10590526). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Performance/Efficiency,Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Blockchain-based Zero Trust.
Solution Architecture Blockchain-based Zero Trust: described as: The architecture proposed in "Blockchain-Based Zero Trust on the Edge" integrates Zero Trust principles with Blockchain to ensure secure access and trust management for Edge Computing in IoT environments. It focuses on continuous verification of users and devices, ensuring that all access is authenticated before being granted. Blockchain is used for decentralized trust management, maintaining an immutable record of activities, which enhances security and transparency. This setup is optimized for edge computing, reducing latency by processing data locally while maintaining the flexibility to scale across diverse IoT applications, such as smart cities and industrial IoT..
Security is reached/addressed using/by Zero Trust Architecture (ZTA) (Confidentiality - Zero Trust Architecture (ZTA)
ZTA ensures continuous verification and monitoring of all network traffic, restricting unauthorized access.).
Security is reached/addressed using/by Blockchain Technology (Integrity - Blockchain serves as an immutable ledger for storing requests, ensuring data integrity and trustworthiness.).
Performance/Efficiency is reached/addressed using/by Redis Message Broker (Time Behavior - Redis Message Broker for Asynchronous Processing [redis.io]
Asynchronous processing for write operations improves system responsiveness under high loads.).
Maintainability is reached/addressed using/by Component isolation through microservices (Modularity -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.
).
Flexibility is reached/addressed using/by Component isolation through microservices (Scalability -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.).
Reference: Blockchain-Based Zero Trust on the Edge, 2023, https://ieeexplore.ieee.org/document/10590526.
The solution is composed by the architecture named as Blockchain-enabled, proposed in the paper An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, with D.O.I. doi.org/10.1109/TSC.2020.2966970 (link https://doi.org/10.1109/TSC.2020.2966970). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Blockchain-enabled.

Solution Architecture Blockchain-enabled: described as: .
Performance/Efficiency is reached/addressed using/by Routing protocol designed for IoT devices in the SDN controller [Yazdinejad et al. 2020] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, https://doi.org/10.1109/TSC.2020.2966970.
The solution is composed by the architecture named as Cloud-based Secure Biometric, proposed in the paper BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, with D.O.I. doi.org/10.1016/j.jksuci.2017.07.001 (link https://doi.org/10.1016/j.jksuci.2017.07.001). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Cloud-based Secure Biometric.
Solution Architecture Cloud-based Secure Biometric: described as: .
Flexibility is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Reliability is reached/addressed using/by MapReduce framework (No description.).
Security is reached/addressed using/by Cloud-based biometric authentication [Shakil et al. 2020] (No description.).
Reference: BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, https://doi.org/10.1016/j.jksuci.2017.07.001.
The solution is composed by the architecture named as Deep Learning based, proposed in the paper HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, with D.O.I. doi.org/10.1016/j.future.2019.10.043 (link https://doi.org/10.1016/j.future.2019.10.043). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Deep Learning based.
Solution Architecture Deep Learning based: described as: .
Performance/Efficiency is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Security is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Reference: HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, https://doi.org/10.1016/j.future.2019.10.043.
The solution is composed by the architecture named as Distributed Blockchain-SDN based, proposed in the paper DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3039113 (link https://doi.org/10.1109/ACCESS.2020.3039113). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Distributed Blockchain-SDN based.
Solution Architecture Distributed Blockchain-SDN based: described as: .
Performance/Efficiency is reached/addressed using/by Decentralized blockchain nodes, dynamic SDN controllers, and edge computing (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, https://doi.org/10.1109/ACCESS.2020.3039113.
The solution is composed by the architecture named as Distributed Edge-Cloud, proposed in the paper Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, with D.O.I. doi.org/10.1016/j.future.2020.05.042 (link https://doi.org/10.1016/j.future.2020.05.042). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Distributed Edge-Cloud.
Solution Architecture Distributed Edge-Cloud: described as: .
Security is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Reference: Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, https://doi.org/10.1016/j.future.2020.05.042.
The solution is composed by the architecture named as Edge Computing-based Fault Tolerant Framework, proposed in the paper Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, with D.O.I. doi.org/10.1109/IWCMC48107.2020.9148269 (link https://doi.org/10.1109/IWCMC48107.2020.9148269). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Edge Computing-based Fault Tolerant Framework.
Solution Architecture Edge Computing-based Fault Tolerant Framework: described as: .
Performance/Efficiency is reached/addressed using/by Kubernetes framework (No description.).
Flexibility is reached/addressed using/by Docker containers (No description.).
Reference: Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, https://doi.org/10.1109/IWCMC48107.2020.9148269.

The solution is composed by the architecture named as Event-driven IoT, proposed in the paper Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, with D.O.I. doi.org/10.1007/s10586-020-03189-w (link https://doi.org/10.1007/s10586-020-03189-w). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Event-driven IoT.
Solution Architecture Event-driven IoT: described as: .
Performance/Efficiency is reached/addressed using/by Complex Event Processing (CEP) [Rahmani et al. 2020] (No description.).
Reliability is reached/addressed using/by TCP-IP protocol on Client/Server architecture for data communication (No description.).
Reference: Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, https://doi.org/10.1007/s10586-020-03189-w.
The solution is composed by the architecture named as Fog Computing based, proposed in the paper FOG-based architecture and load balancing methodology for health monitoring systems, 2021, with D.O.I. doi.org/10.1109/ACCESS.2021.3094033 (link https://doi.org/10.1109/ACCESS.2021.3094033). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Fog Computing based.
Solution Architecture Fog Computing based: described as: .
Performance/Efficiency is reached/addressed using/by Load Balancing Scheme (LBS) - Distributed workload to all fog nodes [Asghar et al. 2021] (No description.).
Reference: FOG-based architecture and load balancing methodology for health monitoring systems, 2021, https://doi.org/10.1109/ACCESS.2021.3094033.
The solution is composed by the architecture named as Geographic-based, proposed in the paper A platform for integrating heterogeneous data and developing smart city applications, 2022, with D.O.I. doi.org/10.1016/j.future.2021.10.030 (link https://doi.org/10.1016/j.future.2021.10.030). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Geographic-based.
Solution Architecture Geographic-based: described as: .
Security is reached/addressed using/by FIWARE Platform [www.fiware.org] (Authentication, authorization, and data access control) (No description.).
Flexibility is reached/addressed using/by Virtualization (No description.).
Compatibility is reached/addressed using/by RESTFul APIs (No description.).
Reference: A platform for integrating heterogeneous data and developing smart city applications, 2022, https://doi.org/10.1016/j.future.2021.10.030.
The solution is composed by the architecture named as Hybrid (Centralized and Distributed), proposed in the paper Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, with D.O.I. doi.org/10.1016/j.procs.2021.03.030 (link https://doi.org/10.1016/j.procs.2021.03.030). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid (Centralized and Distributed).
Solution Architecture Hybrid (Centralized and Distributed): described as: .
Performance/Efficiency is reached/addressed using/by Dynamic allocation of the resources using Software Defined Networking (SDN) [Chudhary and Sharma 2021] (No description.).
Reference: Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, https://doi.org/10.1016/j.procs.2021.03.030.
The solution is composed by the architecture named as Hybrid Cloud-Fog Computing, proposed in the paper Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, with D.O.I. doi.org/10.1007/s10586-021-03375-4 (link https://link.springer.com/article/10.1007/s10586-021-03375-4). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid Cloud-Fog Computing.
Solution Architecture Hybrid Cloud-Fog Computing: described as: .
Reliability is reached/addressed using/by Fog layer in two levels: 1-Intra mobility (User network): Nodes (Data transfer, device management, temporary local storage, notifications, and data consistency). 2-Extra mobility (Different domains): Smart Gateways (Local processing and database) (No description.).
Performance/Efficiency is reached/addressed using/by Fog computing (No description.).
Reference: Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, https://link.springer.com/article/10.1007/s10586-021-03375-4.
The solution is composed by the architecture named as Industrial Internet of Things (IIoT), proposed in the paper A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Performance/Efficiency,Security,.
The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and Industrial Internet of Things (IIoT).
Solution Architecture Industrial Internet of Things (IIoT): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Flexibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Performance/Efficiency is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).
Security is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).

Reference: A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, https://doi.org/10.3390/s20195603.
The solution is composed by the architecture named as Layered, proposed in the paper An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, with D.O.I. doi.org/10.1016/j.future.2020.07.047 (link https://doi.org/10.1016/j.future.2020.07.047). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.
Solution Architecture Layered: described as: .
Flexibility is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).
Performance/Efficiency is reached/addressed using/by Big data analytics engine [Ali et al. 2021] (No description.).
Security is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).
Reference: An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, https://doi.org/10.1016/j.future.2020.07.047.
The solution is composed by the architecture named as Layered, proposed in the paper A novel architecture for the Internet of Things based E-Health systems, 2020, with D.O.I. doi.org/10.1166/jmihi.2020.3184 (link http://dx.doi.org/10.1166/jmihi.2020.3184). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.
Solution Architecture Layered: described as: .
Performance/Efficiency is reached/addressed using/by IoT objects connected at the edge of the network (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: A novel architecture for the Internet of Things based E-Health systems, 2020, http://dx.doi.org/10.1166/jmihi.2020.3184.
The solution is composed by the architecture named as Layered, proposed in the paper Software architecture quality attributes of a layered sensor-based IoT system, 2021, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Reliability,Security,.
The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and Layered.
Solution Architecture Layered: described as: .
Flexibility is reached/addressed using/by Messaging protocol brokers and at the client side, web-based applications (No description.).
Maintainability is reached/addressed using/by Loosely coupled components (No description.).
Reliability is reached/addressed using/by Open source software solutions and modular architecture design with clear relationships between components (No description.).
Security is reached/addressed using/by IoT devices authentication and data encryption (No description.).
Reference: Software architecture quality attributes of a layered sensor-based IoT system, 2021, https://doi.org/10.3390/s20195603.
The solution is composed by the architecture named as Layered, proposed in the paper Cloud-FOG assisted Energy Efficient Architectural Paradigm for disaster evacuation, 2022, with D.O.I. doi.org/10.1016/j.is.2021.101732 (link https://doi.org/10.1016/j.is.2021.101732). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Layered.
Solution Architecture Layered: described as: .
Performance/Efficiency is reached/addressed using/by Algorithm for Fog layer data selection (Data uniqueness and only important data) for transmission [Kaur et al. 2022] (No description.).
Reference: Cloud-FOG assisted Energy Efficient Architectural Paradigm for disaster evacuation, 2022, https://doi.org/10.1016/j.is.2021.101732.
The solution is composed by the architecture named as Layered, proposed in the paper An agile AI and IoT-Augmented Smart Farming: a Cost-Effective cognitive weather station, 2021, with D.O.I. doi.org/10.3390/agriculture12010035 (link https://doi.org/10.3390/agriculture12010035). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Interaction Capability,Maintainability,Performance/Efficiency,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Layered.
Solution Architecture Layered: described as: .
Compatibility is reached/addressed using/by Docker containers (No description.).
Flexibility is reached/addressed using/by Docker containers (No description.).
Interaction Capability is reached/addressed using/by Remote access to the deployed platform and its components (Web App and SSH) (No description.).
Maintainability is reached/addressed using/by Docker containers (No description.).
Performance/Efficiency is reached/addressed using/by Docker containers (No description.).
Reference: An agile AI and IoT-Augmented Smart Farming: a Cost-Effective cognitive weather station, 2021, https://doi.org/10.3390/agriculture12010035.
The solution is composed by the architecture named as Layered, proposed in the paper Smart Indoor Farms: Leveraging Technological Advancements to Power a Sustainable Agricultural Revolution, 2021, with D.O.I. doi.org/10.3390/agriengineering3040047 (link https://doi.org/10.3390/agriengineering3040047). This solution addresses the following Quality Requirements: .
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Layered.
Solution Architecture Layered: described as: .

Reference: Smart Indoor Farms: Leveraging Technological Advancements to Power a Sustainable Agricultural Revolution, 2021, https://doi.org/10.3390/agriengineering3040047.

The solution is composed by the architecture named as Layered Blockchain and AI-enabled, proposed in the paper A blockchain and artificial intelligence-enabled smart IoT framework for sustainable city, 2022, with D.O.I. doi.org/10.1002/int.22852 (link https://doi.org/10.1002/int.22852). This solution addresses the following Quality Requirements: Security,.

The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Layered Blockchain and AI-enabled.

Solution Architecture Layered Blockchain and AI-enabled: described as: .

Security is reached/addressed using/by Blockchain Technology (No description.).

Reference: A blockchain and artificial intelligence-enabled smart IoT framework for sustainable city, 2022, https://doi.org/10.1002/int.22852.

The solution is composed by the architecture named as Layered Blockchain-Based SDN, proposed in the paper Intelligent Blockchain Based secure routing for multidomain SDN Enabled IoT networks, 2022, with D.O.I. doi.org/10.1155/2022/5693962 (link https://doi.org/10.1155/2022/5693962). This solution addresses the following Quality Requirements: Reliability,Security,.

The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Layered Blockchain-Based SDN.

Solution Architecture Layered Blockchain-Based SDN: described as: .

Reliability is reached/addressed using/by Reputation concept (Blockchain-based SDN-enabled network architecture) [Zeng et al. 2022] (No description.).

Security is reached/addressed using/by Blockchain Technology (No description.).

Reference: Intelligent Blockchain Based secure routing for multidomain SDN Enabled IoT networks, 2022, https://doi.org/10.1155/2022/5693962.

The solution is composed by the architecture named as LoRaWAN, proposed in the paper A new Low-Cost LORAWAN power switch for smart farm applications, 2021, with D.O.I. doi.org/10.1109/SMC52423.2021.9659076 (link https://doi.org/10.1109/SMC52423.2021.9659076). This solution addresses the following Quality Requirements: Performance/Efficiency,.

The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and LoRaWAN.

Solution Architecture LoRaWAN: described as: .

Performance/Efficiency is reached/addressed using/by Long-Range Wide Area Network (LoRaWAN) (No description.).

Reference: A new Low-Cost LORAWAN power switch for smart farm applications, 2021, https://doi.org/10.1109/SMC52423.2021.9659076.

The solution is composed by the architecture named as Loosely Coupled IoT Platform and Localization Module, proposed in the paper A Self-Adaptive Bluetooth Indoor Localization System using LSTM-based Distance Estimator, 2020, with D.O.I. 10.1109/ICCCN49398.2020.9209674 (link https://ieeexplore.ieee.org/document/9209674). This solution addresses the following Quality Requirements: Compatibility,Functional Suitability,Interaction Capability,Maintainability,.

The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Loosely Coupled IoT Platform and Localization Module.

Solution Architecture Loosely Coupled IoT Platform and Localization Module: described as: The proposed system architecture consists of two main components: the IoT platform and the localization module, connected by middleware. The IoT platform follows a hierarchical structure with layers for hardware (Bluetooth devices), data (MQTT-based RSSI parsing), business logic, display, and user interface. The localization module enhances trilateration with self-adaptive mechanisms, including an LSTM-based distance estimator, elastic radius intersecting, weighted centroid localization, and self-adaptive Kalman filtering for trajectory optimization..

Maintainability is reached/addressed using/by Google Remote Procedure Calls (gRPC) (Modularity - The IoT platform (server side) and the localization module (client side) are connected via a middleware called Google Remote Procedure Calls (gRPC).").

Interaction Capability is reached/addressed using/by Web Interface (Operability - Web interface for visualizing localization and trajectory tracking.).

Compatibility is reached/addressed using/by Google Remote Procedure Calls (gRPC) (Interoperability - The gRPC middleware supports communication between the IoT platform and the localization module, enabling system integration and compatibility with other platforms.).

Functional Suitability is reached/addressed using/by Kalman filtering (KF) (Kalman Filtering (KF) is an algorithm that estimates the state of a dynamic system by combining noisy sensor measurements with predictions. It recursively updates estimates based on new data, minimizing errors and improving accuracy, making it useful in tracking, navigation, and signal processing.).

Reference: A Self-Adaptive Bluetooth Indoor Localization System using LSTM-based Distance Estimator, 2020, https://ieeexplore.ieee.org/document/9209674.

The solution is composed by the architecture named as Microservices, proposed in the paper Design of scalable IoT architecture based on AWS for smart livestock, 2021, with D.O.I. doi.org/10.3390/ani11092697 (link https://doi.org/10.3390/ani11092697). This solution addresses the following Quality Requirements: Flexibility,.

The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Microservices.

Solution Architecture Microservices: described as: .

Flexibility is reached/addressed using/by Amazon Web Services (AWS) (No description.).

Reference: Design of scalable IoT architecture based on AWS for smart livestock, 2021, https://doi.org/10.3390/ani11092697.

The solution is composed by the architecture named as Microservices, proposed in the paper A Lightweight User Interface for Smart Charging of Electric Vehicles: A Real-World Application, 2021, with D.O.I. 10.1109/ICSGCE52779.2021.9621604 (link https://ieeexplore.ieee.org/document/9621604). This solution addresses the following Quality Requirements: Compatibility,Reliability,.

The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Microservices.

Solution Architecture Microservices: described as: Microservice-Based IoT Architecture integrated with Edge-Cloud Computing. Key components

include: Charging Session Handler (CSH): Manages charging sessions through MQTT messages; PLC-Controlled Charging Station: Executes charging commands; Web-Based User Interface: Built with Streamlit for real-time user control; and Cloud Integration: Kubernetes ensures scalability and availability. This modular design supports distributed processing, real-time control, and seamless EV charging management..
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (Interoperability - The MQTT server stablishes communication between the web app, the Session Manager, and the Programmable Logic Controller (PLC) via JavaScript Object Notation (JSON) encoded messages.).
Reliability is reached/addressed using/by Docker containers (Availability - It is deployed in a Docker container on Kubernetes to provide a continuously available service.).
Reference: A Lightweight User Interface for Smart Charging of Electric Vehicles: A Real-World Application, 2021, https://ieeexplore.ieee.org/document/9621604.

The solution is composed by the architecture named as A4IoT (Anyplace 4.0 IoT), proposed in the paper The Anyplace 4.0 IoT Localization Architecture, 2020, with D.O.I. 10.1109/MDM48529.2020.00045 (link https://ieeexplore.ieee.org/document/9162205). This solution addresses the following Quality Requirements: Flexibility,Interaction Capability,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and A4IoT (Anyplace 4.0 IoT).
Solution Architecture A4IoT (Anyplace 4.0 IoT): described as: It is an open-source architecture designed for IoT-based indoor localization using technologies like Wi-Fi, BLE, Cellular, UWB, and Computer Vision (CV). The system supports edge computing, timeseries databases, and crowdsourcing to manage localization tasks across various smart spaces, such as factories, healthcare facilities, and emergency response environments. Its modular and scalable design ensures integration across platforms like Android, Linux, Windows, and Robot OS, making it adaptable to diverse IoT ecosystems..
Performance/Efficiency is reached/addressed using/by Time-Series Database (InfluxDB) (Time Behavior - With temporal queries we can now consume high-volume input streams to provide real-time IoT tracking.").
Reliability is reached/addressed using/by Multi-node Cluster Deployment (Fault Tolerance - The production environment uses a 3-node cluster with two replicas, for both the database engines and the DFS (Distributed Filesystem). This setup allows for a full operation with just a single node active at any time.").
Security is reached/addressed using/by SSL certificates - Encrypted Communications (Confidentiality - All communication is encrypted even in internal networks, as A4IoT image automatically creates and uses SSL certificates.).
Interaction Capability is reached/addressed using/by Web-based Dashboard (Operability - Web Apps built with HTML5, CSS3, and AngularJS.").
Flexibility is reached/addressed using/by Docker containers (Adaptability - It utilizes Docker to enable the deployment of single-node or multi-node configurations, regardless of any dependencies or the underlying OS.).
Reference: The Anyplace 4.0 IoT Localization Architecture, 2020, https://ieeexplore.ieee.org/document/9162205.
The solution is composed by the architecture named as Agricultural IoT Reference Architecture (AITRA), proposed in the paper Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3031634 (link https://doi.org/10.1109/ACCESS.2020.3031634). This solution addresses the following Quality Requirements: Compatibility,Reliability,Security,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Agricultural IoT Reference Architecture (AITRA).
Solution Architecture Agricultural IoT Reference Architecture (AITRA): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Reliability is reached/addressed using/by Edge gateway (No description.).
Security is reached/addressed using/by Authentication, authorization, and data access control (No description.).
Reference: Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, https://doi.org/10.1109/ACCESS.2020.3031634.
The solution is composed by the architecture named as Blockchain-based Zero Trust, proposed in the paper Blockchain-Based Zero Trust on the Edge, 2023, with D.O.I. 10.1109/CSCI62032.2023.00167 (link https://ieeexplore.ieee.org/document/10590526). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Performance/Efficiency,Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Blockchain-based Zero Trust.
Solution Architecture Blockchain-based Zero Trust: described as: The architecture proposed in "Blockchain-Based Zero Trust on the Edge" integrates Zero Trust principles with Blockchain to ensure secure access and trust management for Edge Computing in IoT environments. It focuses on continuous verification of users and devices, ensuring that all access is authenticated before being granted. Blockchain is used for decentralized trust management, maintaining an immutable record of activities, which enhances security and transparency. This setup is optimized for edge computing, reducing latency by processing data locally while maintaining the flexibility to scale across diverse IoT applications, such as smart cities and industrial IoT..
Security is reached/addressed using/by Zero Trust Architecture (ZTA) (Confidentiality - Zero Trust Architecture (ZTA)
ZTA ensures continuous verification and monitoring of all network traffic, restricting unauthorized access.).
Security is reached/addressed using/by Blockchain Technology (Integrity - Blockchain serves as an immutable ledger for storing requests, ensuring data integrity and trustworthiness.).
Performance/Efficiency is reached/addressed using/by Redis Message Broker (Time Behavior - Redis Message Broker for Asynchronous Processing [redis.io]
Asynchronous processing for write operations improves system responsiveness under high loads.).
Maintainability is reached/addressed using/by Component isolation through microservices (Modularity -

The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.
).
Flexibility is reached/addressed using/by Component isolation through microservices (Scalability -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.).
Reference: Blockchain-Based Zero Trust on the Edge, 2023, https://ieeexplore.ieee.org/document/10590526.
The solution is composed by the architecture named as Blockchain-enabled, proposed in the paper An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, with D.O.I. doi.org/10.1109/TSC.2020.2966970 (link https://doi.org/10.1109/TSC.2020.2966970). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Blockchain-enabled.
Solution Architecture Blockchain-enabled: described as: .
Performance/Efficiency is reached/addressed using/by Routing protocol designed for IoT devices in the SDN controller [Yazdinejad et al. 2020] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, https://doi.org/10.1109/TSC.2020.2966970.
The solution is composed by the architecture named as Cloud-based Secure Biometric, proposed in the paper BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, with D.O.I. doi.org/10.1016/j.jksuci.2017.07.001 (link https://doi.org/10.1016/j.jksuci.2017.07.001). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Cloud-based Secure Biometric.
Solution Architecture Cloud-based Secure Biometric: described as: .
Flexibility is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Reliability is reached/addressed using/by MapReduce framework (No description.).
Security is reached/addressed using/by Cloud-based biometric authentication [Shakil et al. 2020] (No description.).
Reference: BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, https://doi.org/10.1016/j.jksuci.2017.07.001.
The solution is composed by the architecture named as Deep Learning based, proposed in the paper HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, with D.O.I. doi.org/10.1016/j.future.2019.10.043 (link https://doi.org/10.1016/j.future.2019.10.043). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Deep Learning based.
Solution Architecture Deep Learning based: described as: .
Performance/Efficiency is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Security is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Reference: HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, https://doi.org/10.1016/j.future.2019.10.043.
The solution is composed by the architecture named as Distributed Blockchain-SDN based, proposed in the paper DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3039113 (link https://doi.org/10.1109/ACCESS.2020.3039113). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Distributed Blockchain-SDN based.
Solution Architecture Distributed Blockchain-SDN based: described as: .
Performance/Efficiency is reached/addressed using/by Decentralized blockchain nodes, dynamic SDN controllers, and edge computing (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, https://doi.org/10.1109/ACCESS.2020.3039113.
The solution is composed by the architecture named as Distributed Edge-Cloud, proposed in the paper Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, with D.O.I. doi.org/10.1016/j.future.2020.05.042 (link https://doi.org/10.1016/j.future.2020.05.042). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Distributed Edge-Cloud.
Solution Architecture Distributed Edge-Cloud: described as: .
Security is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Reference: Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, https://doi.org/10.1016/j.future.2020.05.042.
The solution is composed by the architecture named as Edge Computing-based Fault Tolerant Framework, proposed in the paper Edge Computing

based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, with D.O.I. doi.org/10.1109/IWCMC48107.2020.9148269 (link https://doi.org/10.1109/IWCMC48107.2020.9148269). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Edge Computing-based Fault Tolerant Framework.
Solution Architecture Edge Computing-based Fault Tolerant Framework: described as: .
Performance/Efficiency is reached/addressed using/by Kubernetes framework (No description.).
Flexibility is reached/addressed using/by Docker containers (No description.).
Reference: Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, https://doi.org/10.1109/IWCMC48107.2020.9148269.
The solution is composed by the architecture named as Event-driven IoT, proposed in the paper Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, with D.O.I. doi.org/10.1007/s10586-020-03189-w (link https://doi.org/10.1007/s10586-020-03189-w). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Event-driven IoT.
Solution Architecture Event-driven IoT: described as: .
Performance/Efficiency is reached/addressed using/by Complex Event Processing (CEP) [Rahmani et al. 2020] (No description.).
Reliability is reached/addressed using/by TCP-IP protocol on Client/Server architecture for data communication (No description.).
Reference: Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, https://doi.org/10.1007/s10586-020-03189-w.
The solution is composed by the architecture named as Fog Computing based, proposed in the paper FOG-based architecture and load balancing methodology for health monitoring systems, 2021, with D.O.I. doi.org/10.1109/ACCESS.2021.3094033 (link https://doi.org/10.1109/ACCESS.2021.3094033). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Fog Computing based.
Solution Architecture Fog Computing based: described as: .
Performance/Efficiency is reached/addressed using/by Load Balancing Scheme (LBS) - Distributed workload to all fog nodes [Asghar et al. 2021] (No description.).
Reference: FOG-based architecture and load balancing methodology for health monitoring systems, 2021, https://doi.org/10.1109/ACCESS.2021.3094033.
The solution is composed by the architecture named as Geographic-based, proposed in the paper A platform for integrating heterogeneous data and developing smart city applications, 2022, with D.O.I. doi.org/10.1016/j.future.2021.10.030 (link https://doi.org/10.1016/j.future.2021.10.030). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Geographic-based.
Solution Architecture Geographic-based: described as: .
Security is reached/addressed using/by FIWARE Platform [www.fiware.org] (Authentication, authorization, and data access control) (No description.).
Flexibility is reached/addressed using/by Virtualization (No description.).
Compatibility is reached/addressed using/by RESTFul APIs (No description.).
Reference: A platform for integrating heterogeneous data and developing smart city applications, 2022, https://doi.org/10.1016/j.future.2021.10.030.
The solution is composed by the architecture named as Hybrid (Centralized and Distributed), proposed in the paper Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, with D.O.I. doi.org/10.1016/j.procs.2021.03.030 (link https://doi.org/10.1016/j.procs.2021.03.030). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid (Centralized and Distributed).
Solution Architecture Hybrid (Centralized and Distributed): described as: .
Performance/Efficiency is reached/addressed using/by Dynamic allocation of the resources using Software Defined Networking (SDN) [Chudhary and Sharma 2021] (No description.).
Reference: Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, https://doi.org/10.1016/j.procs.2021.03.030.
The solution is composed by the architecture named as Hybrid Cloud-Fog Computing, proposed in the paper Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, with D.O.I. doi.org/10.1007/s10586-021-03375-4 (link https://link.springer.com/article/10.1007/s10586-021-03375-4). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid Cloud-Fog Computing.
Solution Architecture Hybrid Cloud-Fog Computing: described as: .
Reliability is reached/addressed using/by Fog layer in two levels: 1-Intra mobility (User network): Nodes (Data transfer, device management, temporary local storage, notifications, and data consistency). 2-Extra mobility (Different domains): Smart Gateways (Local processing and database) (No description.).
Performance/Efficiency is reached/addressed using/by Fog computing (No description.).
Reference: Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, https://link.springer.com/article/10.1007/s10586-021-03375-4.

The solution is composed by the architecture named as Industrial Internet of Things (IIoT), proposed in the paper A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Performance/Efficiency,Security,.

The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and Industrial Internet of Things (IIoT).

Solution Architecture Industrial Internet of Things (IIoT): described as: .

Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).

Flexibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).

Performance/Efficiency is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).

Security is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).

Reference: A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, https://doi.org/10.3390/s20195603.

The solution is composed by the architecture named as Layered, proposed in the paper An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, with D.O.I. doi.org/10.1016/j.future.2020.07.047 (link https://doi.org/10.1016/j.future.2020.07.047). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Security,.

The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.

Solution Architecture Layered: described as: .

Flexibility is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).

Performance/Efficiency is reached/addressed using/by Big data analytics engine [Ali et al. 2021] (No description.).

Security is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).

Reference: An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, https://doi.org/10.1016/j.future.2020.07.047.

The solution is composed by the architecture named as Layered, proposed in the paper A novel architecture for the Internet of Things based E-Health systems, 2020, with D.O.I. doi.org/10.1166/jmihi.2020.3184 (link http://dx.doi.org/10.1166/jmihi.2020.3184). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.

The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.

Solution Architecture Layered: described as: .

Performance/Efficiency is reached/addressed using/by IoT objects connected at the edge of the network (No description.).

Security is reached/addressed using/by Blockchain Technology (No description.).

Reference: A novel architecture for the Internet of Things based E-Health systems, 2020, http://dx.doi.org/10.1166/jmihi.2020.3184.

The solution is composed by the architecture named as Layered, proposed in the paper Software architecture quality attributes of a layered sensor-based IoT system, 2021, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Reliability,Security,.

The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and Layered.

Solution Architecture Layered: described as: .

Flexibility is reached/addressed using/by Messaging protocol brokers and at the client side, web-based applications (No description.).

Maintainability is reached/addressed using/by Loosely coupled components (No description.).

Reliability is reached/addressed using/by Open source software solutions and modular architecture design with clear relationships between components (No description.).

Security is reached/addressed using/by IoT devices authentication and data encryption (No description.).

Reference: Software architecture quality attributes of a layered sensor-based IoT system, 2021, https://doi.org/10.3390/s20195603.

The solution is composed by the architecture named as Layered, proposed in the paper Cloud-FOG assisted Energy Efficient Architectural Paradigm for disaster evacuation, 2022, with D.O.I. doi.org/10.1016/j.is.2021.101732 (link https://doi.org/10.1016/j.is.2021.101732). This solution addresses the following Quality Requirements: Performance/Efficiency,.

The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Layered.

Solution Architecture Layered: described as: .

Performance/Efficiency is reached/addressed using/by Algorithm for Fog layer data selection (Data uniqueness and only important data) for transmission [Kaur et al. 2022] (No description.).

Reference: Cloud-FOG assisted Energy Efficient Architectural Paradigm for disaster evacuation, 2022, https://doi.org/10.1016/j.is.2021.101732.

The solution is composed by the architecture named as Layered, proposed in the paper An agile AI and IoT-Augmented Smart Farming: a Cost-Effective cognitive weather station, 2021, with D.O.I. doi.org/10.3390/agriculture12010035 (link https://doi.org/10.3390/agriculture12010035). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Interaction Capability,Maintainability,Performance/Efficiency,.

The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Layered.

Solution Architecture Layered: described as: .

Compatibility is reached/addressed using/by Docker containers (No description.).

Flexibility is reached/addressed using/by Docker containers (No description.).

Interaction Capability is reached/addressed using/by Remote access to the deployed platform and its components (Web App and SSH) (No description.).

Maintainability is reached/addressed using/by Docker containers (No description.).
Performance/Efficiency is reached/addressed using/by Docker containers (No description.).
Reference: An agile AI and IoT-Augmented Smart Farming: a Cost-Effective cognitive weather station, 2021, https://doi.org/10.3390/agriculture12010035.
The solution is composed by the architecture named as Layered, proposed in the paper Smart Indoor Farms: Leveraging Technological Advancements to Power a Sustainable Agricultural Revolution, 2021, with D.O.I. doi.org/10.3390/agriengineering3040047 (link https://doi.org/10.3390/agriengineering3040047). This solution addresses the following Quality Requirements: .
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Layered.
Solution Architecture Layered: described as: .
Reference: Smart Indoor Farms: Leveraging Technological Advancements to Power a Sustainable Agricultural Revolution, 2021, https://doi.org/10.3390/agriengineering3040047.
The solution is composed by the architecture named as Layered Blockchain and AI-enabled, proposed in the paper A blockchain and artificial intelligence-enabled smart IoT framework for sustainable city, 2022, with D.O.I. doi.org/10.1002/int.22852 (link https://doi.org/10.1002/int.22852). This solution addresses the following Quality Requirements: Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Layered Blockchain and AI-enabled.
Solution Architecture Layered Blockchain and AI-enabled: described as: .
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: A blockchain and artificial intelligence-enabled smart IoT framework for sustainable city, 2022, https://doi.org/10.1002/int.22852.
The solution is composed by the architecture named as Layered Blockchain-Based SDN, proposed in the paper Intelligent Blockchain Based secure routing for multidomain SDN Enabled IoT networks, 2022, with D.O.I. doi.org/10.1155/2022/5693962 (link https://doi.org/10.1155/2022/5693962). This solution addresses the following Quality Requirements: Reliability,Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Layered Blockchain-Based SDN.
Solution Architecture Layered Blockchain-Based SDN: described as: .
Reliability is reached/addressed using/by Reputation concept (Blockchain-based SDN-enabled network architecture) [Zeng et al. 2022] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: Intelligent Blockchain Based secure routing for multidomain SDN Enabled IoT networks, 2022, https://doi.org/10.1155/2022/5693962.
The solution is composed by the architecture named as LoRaWAN, proposed in the paper A new Low-Cost LORAWAN power switch for smart farm applications, 2021, with D.O.I. doi.org/10.1109/SMC52423.2021.9659076 (link https://doi.org/10.1109/SMC52423.2021.9659076). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and LoRaWAN.
Solution Architecture LoRaWAN: described as: .
Performance/Efficiency is reached/addressed using/by Long-Range Wide Area Network (LoRaWAN) (No description.).
Reference: A new Low-Cost LORAWAN power switch for smart farm applications, 2021, https://doi.org/10.1109/SMC52423.2021.9659076.
The solution is composed by the architecture named as Loosely Coupled IoT Platform and Localization Module, proposed in the paper A Self-Adaptive Bluetooth Indoor Localization System using LSTM-based Distance Estimator, 2020, with D.O.I. 10.1109/ICCCN49398.2020.9209674 (link https://ieeexplore.ieee.org/document/9209674). This solution addresses the following Quality Requirements: Compatibility,Functional Suitability,Interaction Capability,Maintainability,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Loosely Coupled IoT Platform and Localization Module.
Solution Architecture Loosely Coupled IoT Platform and Localization Module: described as: The proposed system architecture consists of two main components: the IoT platform and the localization module, connected by middleware. The IoT platform follows a hierarchical structure with layers for hardware (Bluetooth devices), data (MQTT-based RSSI parsing), business logic, display, and user interface. The localization module enhances trilateration with self-adaptive mechanisms, including an LSTM-based distance estimator, elastic radius intersecting, weighted centroid localization, and self-adaptive Kalman filtering for trajectory optimization..
Maintainability is reached/addressed using/by Google Remote Procedure Calls (gRPC) (Modularity - The IoT platform (server side) and the localization module (client side) are connected via a middleware called Google Remote Procedure Calls (gRPC).").
Interaction Capability is reached/addressed using/by Web Interface (Operability - Web interface for visualizing localization and trajectory tracking.).
Compatibility is reached/addressed using/by Google Remote Procedure Calls (gRPC) (Interoperability - The gRPC middleware supports communication between the IoT platform and the localization module, enabling system integration and compatibility with other platforms.).
Functional Suitability is reached/addressed using/by Kalman filtering (KF) (Kalman Filtering (KF) is an algorithm that estimates the state of a dynamic system by combining noisy sensor measurements with predictions. It recursively updates estimates based on new data, minimizing errors and improving accuracy, making it useful in tracking, navigation, and signal processing.).
Reference: A Self-Adaptive Bluetooth Indoor Localization System using LSTM-based Distance Estimator, 2020, https://ieeexplore.ieee.org/document/9209674.
The solution is composed by the architecture named as Microservices, proposed in the paper Design of scalable IoT architecture based on AWS for smart livestock, 2021, with D.O.I. doi.org/10.3390/ani11092697 (link https://doi.org/10.3390/ani11092697). This solution addresses the following Quality Requirements: Flexibility,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and

Microservices.
Solution Architecture Microservices: described as: .
Flexibility is reached/addressed using/by Amazon Web Services (AWS) (No description.).
Reference: Design of scalable IoT architecture based on AWS for smart livestock, 2021, https://doi.org/10.3390/ani11092697.
The solution is composed by the architecture named as Microservices, proposed in the paper A Lightweight User Interface for Smart Charging of Electric Vehicles: A Real-World Application, 2021, with D.O.I. 10.1109/ICSGCE52779.2021.9621604 (link https://ieeexplore.ieee.org/document/9621604). This solution addresses the following Quality Requirements: Compatibility,Reliability,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Microservices.
Solution Architecture Microservices: described as: Microservice-Based IoT Architecture integrated with Edge-Cloud Computing. Key components include: Charging Session Handler (CSH): Manages charging sessions through MQTT messages; PLC-Controlled Charging Station: Executes charging commands; Web-Based User Interface: Built with Streamlit for real-time user control; and Cloud Integration: Kubernetes ensures scalability and availability. This modular design supports distributed processing, real-time control, and seamless EV charging management..
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (Interoperability - The MQTT server stablishes communication between the web app, the Session Manager, and the Programmable Logic Controller (PLC) via JavaScript Object Notation (JSON) encoded messages.).
Reliability is reached/addressed using/by Docker containers (Availability - It is deployed in a Docker container on Kubernetes to provide a continuously available service.).
Reference: A Lightweight User Interface for Smart Charging of Electric Vehicles: A Real-World Application, 2021, https://ieeexplore.ieee.org/document/9621604.
The solution is composed by the architecture named as Microservices, proposed in the paper An open source software architecture and ready-to-use components for health IoT, 2020, with D.O.I. 10.1109/CBMS49503.2020.00077 (link https://ieeexplore.ieee.org/document/9183119). This solution addresses the following Quality Requirements: Interaction Capability,Maintainability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Microservices.
Solution Architecture Microservices: described as: Microservices-based software architecture for Health IoT, inspired by best practices from sources like Microsoft's microservices guidelines. The architecture utilizes open-source components for various functionalities. Key components include an API Gateway (Express Gateway) acting as a single entry point, a messaging channel (RabbitMQ) for asynchronous communication between microservices, and a security solution leveraging Vault and Consul for key management and high availability. The architecture is instantiated in the OCARIoT project, where it manages data from wearables, sensors, and mobile apps, processing data and triggering personalized interventions..
Interaction Capability is reached/addressed using/by User-friendly dashboard (Operability - User-friendly dashboard for health professionals: Health professionals are able to do a follow-up through the dashboard and also to receive tips and recommendations. Accessibility Open access through mobile and web platforms. The OCARIoT Dashboard visualizes real-time updates from mobile devices and web interfaces.).
Interaction Capability is reached/addressed using/by Intuitive mobile apps (Learnability - Intuitive mobile apps for health monitoring The Data Acquisition App manages the collection of data from different devices and shows key information for health professionals.).
Maintainability is reached/addressed using/by Component isolation through microservices (Modularity - Open Source Software Architecture Component isolation through microservices. Microservices can be implemented in different languages, using different databases, ensuring modular architecture.).
Reference: An open source software architecture and ready-to-use components for health IoT, 2020, https://ieeexplore.ieee.org/document/9183119.

The solution is composed by the architecture named as A4IoT (Anyplace 4.0 IoT), proposed in the paper The Anyplace 4.0 IoT Localization Architecture, 2020, with D.O.I. 10.1109/MDM48529.2020.00045 (link https://ieeexplore.ieee.org/document/9162205). This solution addresses the following Quality Requirements: Flexibility,Interaction Capability,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and A4IoT (Anyplace 4.0 IoT).
Solution Architecture A4IoT (Anyplace 4.0 IoT): described as: It is an open-source architecture designed for IoT-based indoor localization using technologies like Wi-Fi, BLE, Cellular, UWB, and Computer Vision (CV). The system supports edge computing, timeseries databases, and crowdsourcing to manage localization tasks across various smart spaces, such as factories, healthcare facilities, and emergency response environments. Its modular and scalable design ensures integration across platforms like Android, Linux, Windows, and Robot OS, making it adaptable to diverse IoT ecosystems..
Performance/Efficiency is reached/addressed using/by Time-Series Database (InfluxDB) (Time Behavior - With temporal queries we can now consume high-volume input streams to provide real-time IoT tracking.").
Reliability is reached/addressed using/by Multi-node Cluster Deployment (Fault Tolerance - The production environment uses a 3-node cluster with two replicas, for both the database engines and the DFS (Distributed Filesystem). This setup allows for a full operation with just a single node active at any time.").
Security is reached/addressed using/by SSL certificates - Encrypted Communications (Confidentiality - All communication is encrypted even in internal networks, as A4IoT image automatically creates and uses SSL certificates.).
Interaction Capability is reached/addressed using/by Web-based Dashboard (Operability - Web Apps built with HTML5, CSS3, and AngularJS.").
Flexibility is reached/addressed using/by Docker containers (Adaptability - It utilizes Docker to enable the deployment of single-node or multi-node configurations, regardless of any dependencies or the underlying OS.).
Reference: The Anyplace 4.0 IoT Localization Architecture, 2020, https://ieeexplore.ieee.org/document/9162205.

The solution is composed by the architecture named as Agricultural IoT Reference Architecture (AITRA), proposed in the paper Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3031634 (link https://doi.org/10.1109/ACCESS.2020.3031634). This solution addresses the following Quality Requirements: Compatibility,Reliability,Security,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Agricultural IoT Reference Architecture (AITRA).
Solution Architecture Agricultural IoT Reference Architecture (AITRA): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Reliability is reached/addressed using/by Edge gateway (No description.).
Security is reached/addressed using/by Authentication, authorization, and data access control (No description.).
Reference: Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, https://doi.org/10.1109/ACCESS.2020.3031634.
The solution is composed by the architecture named as Blockchain-based Zero Trust, proposed in the paper Blockchain-Based Zero Trust on the Edge, 2023, with D.O.I. 10.1109/CSCI62032.2023.00167 (link https://ieeexplore.ieee.org/document/10590526). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Performance/Efficiency,Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Blockchain-based Zero Trust.
Solution Architecture Blockchain-based Zero Trust: described as: The architecture proposed in "Blockchain-Based Zero Trust on the Edge" integrates Zero Trust principles with Blockchain to ensure secure access and trust management for Edge Computing in IoT environments. It focuses on continuous verification of users and devices, ensuring that all access is authenticated before being granted. Blockchain is used for decentralized trust management, maintaining an immutable record of activities, which enhances security and transparency. This setup is optimized for edge computing, reducing latency by processing data locally while maintaining the flexibility to scale across diverse IoT applications, such as smart cities and industrial IoT..
Security is reached/addressed using/by Zero Trust Architecture (ZTA) (Confidentiality - Zero Trust Architecture (ZTA)
ZTA ensures continuous verification and monitoring of all network traffic, restricting unauthorized access.).
Security is reached/addressed using/by Blockchain Technology (Integrity - Blockchain serves as an immutable ledger for storing requests, ensuring data integrity and trustworthiness.).
Performance/Efficiency is reached/addressed using/by Redis Message Broker (Time Behavior - Redis Message Broker for Asynchronous Processing [redis.io]
Asynchronous processing for write operations improves system responsiveness under high loads.).
Maintainability is reached/addressed using/by Component isolation through microservices (Modularity -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.
).
Flexibility is reached/addressed using/by Component isolation through microservices (Scalability -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.).
Reference: Blockchain-Based Zero Trust on the Edge, 2023, https://ieeexplore.ieee.org/document/10590526.
The solution is composed by the architecture named as Blockchain-enabled, proposed in the paper An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, with D.O.I. doi.org/10.1109/TSC.2020.2966970 (link https://doi.org/10.1109/TSC.2020.2966970). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Blockchain-enabled.
Solution Architecture Blockchain-enabled: described as: .
Performance/Efficiency is reached/addressed using/by Routing protocol designed for IoT devices in the SDN controller [Yazdinejad et al. 2020] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, https://doi.org/10.1109/TSC.2020.2966970.
The solution is composed by the architecture named as Cloud-based Secure Biometric, proposed in the paper BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, with D.O.I. doi.org/10.1016/j.jksuci.2017.07.001 (link https://doi.org/10.1016/j.jksuci.2017.07.001). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Cloud-based Secure Biometric.
Solution Architecture Cloud-based Secure Biometric: described as: .
Flexibility is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Reliability is reached/addressed using/by MapReduce framework (No description.).
Security is reached/addressed using/by Cloud-based biometric authentication [Shakil et al. 2020] (No description.).
Reference: BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, https://doi.org/10.1016/j.jksuci.2017.07.001.
The solution is composed by the architecture named as Deep Learning based, proposed in the paper HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, with D.O.I. doi.org/10.1016/j.future.2019.10.043 (link https://doi.org/10.1016/j.future.2019.10.043). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.

The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Deep Learning based.
Solution Architecture Deep Learning based: described as: .
Performance/Efficiency is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Security is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Reference: HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, https://doi.org/10.1016/j.future.2019.10.043.
The solution is composed by the architecture named as Distributed Blockchain-SDN based, proposed in the paper DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3039113 (link https://doi.org/10.1109/ACCESS.2020.3039113). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Distributed Blockchain-SDN based.
Solution Architecture Distributed Blockchain-SDN based: described as: .
Performance/Efficiency is reached/addressed using/by Decentralized blockchain nodes, dynamic SDN controllers, and edge computing (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, https://doi.org/10.1109/ACCESS.2020.3039113.
The solution is composed by the architecture named as Distributed Edge-Cloud, proposed in the paper Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, with D.O.I. doi.org/10.1016/j.future.2020.05.042 (link https://doi.org/10.1016/j.future.2020.05.042). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Distributed Edge-Cloud.
Solution Architecture Distributed Edge-Cloud: described as: .
Security is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Reference: Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, https://doi.org/10.1016/j.future.2020.05.042.
The solution is composed by the architecture named as Edge Computing-based Fault Tolerant Framework, proposed in the paper Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, with D.O.I. doi.org/10.1109/IWCMC48107.2020.9148269 (link https://doi.org/10.1109/IWCMC48107.2020.9148269). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Edge Computing-based Fault Tolerant Framework.
Solution Architecture Edge Computing-based Fault Tolerant Framework: described as: .
Performance/Efficiency is reached/addressed using/by Kubernetes framework (No description.).
Flexibility is reached/addressed using/by Docker containers (No description.).
Reference: Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, https://doi.org/10.1109/IWCMC48107.2020.9148269.
The solution is composed by the architecture named as Event-driven IoT, proposed in the paper Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, with D.O.I. doi.org/10.1007/s10586-020-03189-w (link https://doi.org/10.1007/s10586-020-03189-w). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Event-driven IoT.
Solution Architecture Event-driven IoT: described as: .
Performance/Efficiency is reached/addressed using/by Complex Event Processing (CEP) [Rahmani et al. 2020] (No description.).
Reliability is reached/addressed using/by TCP-IP protocol on Client/Server architecture for data communication (No description.).
Reference: Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, https://doi.org/10.1007/s10586-020-03189-w.
The solution is composed by the architecture named as Fog Computing based, proposed in the paper FOG-based architecture and load balancing methodology for health monitoring systems, 2021, with D.O.I. doi.org/10.1109/ACCESS.2021.3094033 (link https://doi.org/10.1109/ACCESS.2021.3094033). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Fog Computing based.
Solution Architecture Fog Computing based: described as: .
Performance/Efficiency is reached/addressed using/by Load Balancing Scheme (LBS) - Distributed workload to all fog nodes [Asghar et al. 2021] (No description.).
Reference: FOG-based architecture and load balancing methodology for health monitoring systems, 2021, https://doi.org/10.1109/ACCESS.2021.3094033.
The solution is composed by the architecture named as Geographic-based, proposed in the paper A platform for integrating heterogeneous data and developing smart city applications, 2022, with D.O.I. doi.org/10.1016/j.future.2021.10.030 (link https://doi.org/10.1016/j.future.2021.10.030). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Security,.

The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Geographic-based.
Solution Architecture Geographic-based: described as: .
Security is reached/addressed using/by FIWARE Platform [www.fiware.org] (Authentication, authorization, and data access control) (No description.).
Flexibility is reached/addressed using/by Virtualization (No description.).
Compatibility is reached/addressed using/by RESTFul APIs (No description.).
Reference: A platform for integrating heterogeneous data and developing smart city applications, 2022, https://doi.org/10.1016/j.future.2021.10.030.
The solution is composed by the architecture named as Hybrid (Centralized and Distributed), proposed in the paper Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, with D.O.I. doi.org/10.1016/j.procs.2021.03.030 (link https://doi.org/10.1016/j.procs.2021.03.030). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid (Centralized and Distributed).
Solution Architecture Hybrid (Centralized and Distributed): described as: .
Performance/Efficiency is reached/addressed using/by Dynamic allocation of the resources using Software Defined Networking (SDN) [Chudhary and Sharma 2021] (No description.).
Reference: Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, https://doi.org/10.1016/j.procs.2021.03.030.
The solution is composed by the architecture named as Hybrid Cloud-Fog Computing, proposed in the paper Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, with D.O.I. doi.org/10.1007/s10586-021-03375-4 (link https://link.springer.com/article/10.1007/s10586-021-03375-4). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid Cloud-Fog Computing.
Solution Architecture Hybrid Cloud-Fog Computing: described as: .
Reliability is reached/addressed using/by Fog layer in two levels: 1-Intra mobility (User network): Nodes (Data transfer, device management, temporary local storage, notifications, and data consistency). 2-Extra mobility (Different domains): Smart Gateways (Local processing and database) (No description.).
Performance/Efficiency is reached/addressed using/by Fog computing (No description.).
Reference: Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, https://link.springer.com/article/10.1007/s10586-021-03375-4.
The solution is composed by the architecture named as Industrial Internet of Things (IIoT), proposed in the paper A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Performance/Efficiency,Security,.
The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and Industrial Internet of Things (IIoT).
Solution Architecture Industrial Internet of Things (IIoT): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Flexibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Performance/Efficiency is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).
Security is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).
Reference: A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, https://doi.org/10.3390/s20195603.
The solution is composed by the architecture named as Layered, proposed in the paper An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, with D.O.I. doi.org/10.1016/j.future.2020.07.047 (link https://doi.org/10.1016/j.future.2020.07.047). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.
Solution Architecture Layered: described as: .
Flexibility is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).
Performance/Efficiency is reached/addressed using/by Big data analytics engine [Ali et al. 2021] (No description.).
Security is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).
Reference: An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, https://doi.org/10.1016/j.future.2020.07.047.
The solution is composed by the architecture named as Layered, proposed in the paper A novel architecture for the Internet of Things based E-Health systems, 2020, with D.O.I. doi.org/10.1166/jmihi.2020.3184 (link http://dx.doi.org/10.1166/jmihi.2020.3184). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.
Solution Architecture Layered: described as: .
Performance/Efficiency is reached/addressed using/by IoT objects connected at the edge of the network (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: A novel architecture for the Internet of Things based E-Health systems, 2020, http://dx.doi.org/10.1166/jmihi.2020.3184.
The solution is composed by the architecture named as Layered, proposed in the paper Software architecture quality attributes of a layered sensor-

based IoT system, 2021, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Reliability,Security,.

The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and Layered.

Solution Architecture Layered: described as: .

Flexibility is reached/addressed using/by Messaging protocol brokers and at the client side, web-based applications (No description.).

Maintainability is reached/addressed using/by Loosely coupled components (No description.).

Reliability is reached/addressed using/by Open source software solutions and modular architecture design with clear relationships between components (No description.).

Security is reached/addressed using/by IoT devices authentication and data encryption (No description.).

Reference: Software architecture quality attributes of a layered sensor-based IoT system, 2021, https://doi.org/10.3390/s20195603.

The solution is composed by the architecture named as Layered, proposed in the paper Cloud-FOG assisted Energy Efficient Architectural Paradigm for disaster evacuation, 2022, with D.O.I. doi.org/10.1016/j.is.2021.101732 (link https://doi.org/10.1016/j.is.2021.101732). This solution addresses the following Quality Requirements: Performance/Efficiency,.

The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Layered.

Solution Architecture Layered: described as: .

Performance/Efficiency is reached/addressed using/by Algorithm for Fog layer data selection (Data uniqueness and only important data) for transmission [Kaur et al. 2022] (No description.).

Reference: Cloud-FOG assisted Energy Efficient Architectural Paradigm for disaster evacuation, 2022, https://doi.org/10.1016/j.is.2021.101732.

The solution is composed by the architecture named as Layered, proposed in the paper An agile AI and IoT-Augmented Smart Farming: a Cost-Effective cognitive weather station, 2021, with D.O.I. doi.org/10.3390/agriculture12010035 (link https://doi.org/10.3390/agriculture12010035). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Interaction Capability,Maintainability,Performance/Efficiency,.

The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Layered.

Solution Architecture Layered: described as: .

Compatibility is reached/addressed using/by Docker containers (No description.).

Flexibility is reached/addressed using/by Docker containers (No description.).

Interaction Capability is reached/addressed using/by Remote access to the deployed platform and its components (Web App and SSH) (No description.).

Maintainability is reached/addressed using/by Docker containers (No description.).

Performance/Efficiency is reached/addressed using/by Docker containers (No description.).

Reference: An agile AI and IoT-Augmented Smart Farming: a Cost-Effective cognitive weather station, 2021, https://doi.org/10.3390/agriculture12010035.

The solution is composed by the architecture named as Layered, proposed in the paper Smart Indoor Farms: Leveraging Technological Advancements to Power a Sustainable Agricultural Revolution, 2021, with D.O.I. doi.org/10.3390/agriengineering3040047 (link https://doi.org/10.3390/agriengineering3040047). This solution addresses the following Quality Requirements: .

The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Layered.

Solution Architecture Layered: described as: .

Reference: Smart Indoor Farms: Leveraging Technological Advancements to Power a Sustainable Agricultural Revolution, 2021, https://doi.org/10.3390/agriengineering3040047.

The solution is composed by the architecture named as Layered Blockchain and AI-enabled, proposed in the paper A blockchain and artificial intelligence-enabled smart IoT framework for sustainable city, 2022, with D.O.I. doi.org/10.1002/int.22852 (link https://doi.org/10.1002/int.22852). This solution addresses the following Quality Requirements: Security,.

The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Layered Blockchain and AI-enabled.

Solution Architecture Layered Blockchain and AI-enabled: described as: .

Security is reached/addressed using/by Blockchain Technology (No description.).

Reference: A blockchain and artificial intelligence-enabled smart IoT framework for sustainable city, 2022, https://doi.org/10.1002/int.22852.

The solution is composed by the architecture named as Layered Blockchain-Based SDN, proposed in the paper Intelligent Blockchain Based secure routing for multidomain SDN Enabled IoT networks, 2022, with D.O.I. doi.org/10.1155/2022/5693962 (link https://doi.org/10.1155/2022/5693962). This solution addresses the following Quality Requirements: Reliability,Security,.

The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Layered Blockchain-Based SDN.

Solution Architecture Layered Blockchain-Based SDN: described as: .

Reliability is reached/addressed using/by Reputation concept (Blockchain-based SDN-enabled network architecture) [Zeng et al. 2022] (No description.).

Security is reached/addressed using/by Blockchain Technology (No description.).

Reference: Intelligent Blockchain Based secure routing for multidomain SDN Enabled IoT networks, 2022, https://doi.org/10.1155/2022/5693962.

The solution is composed by the architecture named as LoRaWAN, proposed in the paper A new Low-Cost LORAWAN power switch for smart farm applications, 2021, with D.O.I. doi.org/10.1109/SMC52423.2021.9659076 (link https://doi.org/10.1109/SMC52423.2021.9659076). This solution addresses the following Quality Requirements: Performance/Efficiency,.

The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and LoRaWAN.
Solution Architecture LoRaWAN: described as: .
Performance/Efficiency is reached/addressed using/by Long-Range Wide Area Network (LoRaWAN) (No description.).
Reference: A new Low-Cost LORAWAN power switch for smart farm applications, 2021, https://doi.org/10.1109/SMC52423.2021.9659076.
The solution is composed by the architecture named as Loosely Coupled IoT Platform and Localization Module, proposed in the paper A Self-Adaptive Bluetooth Indoor Localization System using LSTM-based Distance Estimator, 2020, with D.O.I. 10.1109/ICCCN49398.2020.9209674 (link https://ieeexplore.ieee.org/document/9209674). This solution addresses the following Quality Requirements: Compatibility,Functional Suitability,Interaction Capability,Maintainability,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Loosely Coupled IoT Platform and Localization Module.
Solution Architecture Loosely Coupled IoT Platform and Localization Module: described as: The proposed system architecture consists of two main components: the IoT platform and the localization module, connected by middleware. The IoT platform follows a hierarchical structure with layers for hardware (Bluetooth devices), data (MQTT-based RSSI parsing), business logic, display, and user interface. The localization module enhances trilateration with self-adaptive mechanisms, including an LSTM-based distance estimator, elastic radius intersecting, weighted centroid localization, and self-adaptive Kalman filtering for trajectory optimization..
Maintainability is reached/addressed using/by Google Remote Procedure Calls (gRPC) (Modularity - The IoT platform (server side) and the localization module (client side) are connected via a middleware called Google Remote Procedure Calls (gRPC).").
Interaction Capability is reached/addressed using/by Web Interface (Operability - Web interface for visualizing localization and trajectory tracking.).
Compatibility is reached/addressed using/by Google Remote Procedure Calls (gRPC) (Interoperability - The gRPC middleware supports communication between the IoT platform and the localization module, enabling system integration and compatibility with other platforms.).
Functional Suitability is reached/addressed using/by Kalman filtering (KF) (Kalman Filtering (KF) is an algorithm that estimates the state of a dynamic system by combining noisy sensor measurements with predictions. It recursively updates estimates based on new data, minimizing errors and improving accuracy, making it useful in tracking, navigation, and signal processing.).
Reference: A Self-Adaptive Bluetooth Indoor Localization System using LSTM-based Distance Estimator, 2020, https://ieeexplore.ieee.org/document/9209674.
The solution is composed by the architecture named as Microservices, proposed in the paper Design of scalable IoT architecture based on AWS for smart livestock, 2021, with D.O.I. doi.org/10.3390/ani11092697 (link https://doi.org/10.3390/ani11092697). This solution addresses the following Quality Requirements: Flexibility,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Microservices.
Solution Architecture Microservices: described as: .
Flexibility is reached/addressed using/by Amazon Web Services (AWS) (No description.).
Reference: Design of scalable IoT architecture based on AWS for smart livestock, 2021, https://doi.org/10.3390/ani11092697.
The solution is composed by the architecture named as Microservices, proposed in the paper A Lightweight User Interface for Smart Charging of Electric Vehicles: A Real-World Application, 2021, with D.O.I. 10.1109/ICSGCE52779.2021.9621604 (link https://ieeexplore.ieee.org/document/9621604). This solution addresses the following Quality Requirements: Compatibility,Reliability,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Microservices.
Solution Architecture Microservices: described as: Microservice-Based IoT Architecture integrated with Edge-Cloud Computing. Key components include: Charging Session Handler (CSH): Manages charging sessions through MQTT messages; PLC-Controlled Charging Station: Executes charging commands; Web-Based User Interface: Built with Streamlit for real-time user control; and Cloud Integration: Kubernetes ensures scalability and availability. This modular design supports distributed processing, real-time control, and seamless EV charging management..
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (Interoperability - The MQTT server stablishes communication between the web app, the Session Manager, and the Programmable Logic Controller (PLC) via JavaScript Object Notation (JSON) encoded messages.).
Reliability is reached/addressed using/by Docker containers (Availability - It is deployed in a Docker container on Kubernetes to provide a continuously available service.).
Reference: A Lightweight User Interface for Smart Charging of Electric Vehicles: A Real-World Application, 2021, https://ieeexplore.ieee.org/document/9621604.
The solution is composed by the architecture named as Microservices, proposed in the paper An open source software architecture and ready-to-use components for health IoT, 2020, with D.O.I. 10.1109/CBMS49503.2020.00077 (link https://ieeexplore.ieee.org/document/9183119). This solution addresses the following Quality Requirements: Interaction Capability,Maintainability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Microservices.
Solution Architecture Microservices: described as: Microservices-based software architecture for Health IoT, inspired by best practices from sources like Microsoft's microservices guidelines. The architecture utilizes open-source components for various functionalities. Key components include an API Gateway (Express Gateway) acting as a single entry point, a messaging channel (RabbitMQ) for asynchronous communication between microservices, and a security solution leveraging Vault and Consul for key management and high availability. The architecture is instantiated in the OCARIoT project, where it manages data from wearables, sensors, and mobile apps, processing data and triggering personalized interventions..
Interaction Capability is reached/addressed using/by User-friendly dashboard (Operability - User-friendly dashboard for health professionals: Health professionals are able to do a follow-up through the dashboard and also to receive tips and recommendations. Accessibility Open access through mobile and web platforms. The OCARIoT Dashboard visualizes real-time updates from mobile devices and web interfaces.).

Interaction Capability is reached/addressed using/by Intuitive mobile apps (Learnability - Intuitive mobile apps for health monitoring The Data Acquisition App manages the collection of data from different devices and shows key information for health professionals.).
Maintainability is reached/addressed using/by Component isolation through microservices (Modularity - Open Source Software Architecture Component isolation through microservices. Microservices can be implemented in different languages, using different databases, ensuring modular architecture.).
Reference: An open source software architecture and ready-to-use components for health IoT, 2020, https://ieeexplore.ieee.org/document/9183119.
The solution is composed by the architecture named as Multi-layered Cloud-Edge, proposed in the paper Securing smart healthcare system with edge computing, 2021, with D.O.I. doi.org/10.1016/j.cose.2021.102353 (link https://doi.org/10.1016/j.cose.2021.102353). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Multi-layered Cloud-Edge.
Solution Architecture Multi-layered Cloud-Edge: described as: .
Performance/Efficiency is reached/addressed using/by Edge computing (No description.).
Security is reached/addressed using/by Access Control System (ACS) and Searchable Encryption Module [Singh and Chatterjee 2021] (No description.).
Reference: Securing smart healthcare system with edge computing, 2021, https://doi.org/10.1016/j.cose.2021.102353.

The solution is composed by the architecture named as A4IoT (Anyplace 4.0 IoT), proposed in the paper The Anyplace 4.0 IoT Localization Architecture, 2020, with D.O.I. 10.1109/MDM48529.2020.00045 (link https://ieeexplore.ieee.org/document/9162205). This solution addresses the following Quality Requirements: Flexibility,Interaction Capability,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and A4IoT (Anyplace 4.0 IoT).
Solution Architecture A4IoT (Anyplace 4.0 IoT): described as: It is an open-source architecture designed for IoT-based indoor localization using technologies like Wi-Fi, BLE, Cellular, UWB, and Computer Vision (CV). The system supports edge computing, timeseries databases, and crowdsourcing to manage localization tasks across various smart spaces, such as factories, healthcare facilities, and emergency response environments. Its modular and scalable design ensures integration across platforms like Android, Linux, Windows, and Robot OS, making it adaptable to diverse IoT ecosystems..
Performance/Efficiency is reached/addressed using/by Time-Series Database (InfluxDB) (Time Behavior - With temporal queries we can now consume high-volume input streams to provide real-time IoT tracking.").
Reliability is reached/addressed using/by Multi-node Cluster Deployment (Fault Tolerance - The production environment uses a 3-node cluster with two replicas, for both the database engines and the DFS (Distributed Filesystem). This setup allows for a full operation with just a single node active at any time.").
Security is reached/addressed using/by SSL certificates - Encrypted Communications (Confidentiality - All communication is encrypted even in internal networks, as A4IoT image automatically creates and uses SSL certificates.).
Interaction Capability is reached/addressed using/by Web-based Dashboard (Operability - Web Apps built with HTML5, CSS3, and AngularJS.").
Flexibility is reached/addressed using/by Docker containers (Adaptability - It utilizes Docker to enable the deployment of single-node or multi-node configurations, regardless of any dependencies or the underlying OS.).
Reference: The Anyplace 4.0 IoT Localization Architecture, 2020, https://ieeexplore.ieee.org/document/9162205.
The solution is composed by the architecture named as Agricultural IoT Reference Architecture (AITRA), proposed in the paper Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3031634 (link https://doi.org/10.1109/ACCESS.2020.3031634). This solution addresses the following Quality Requirements: Compatibility,Reliability,Security,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Agricultural IoT Reference Architecture (AITRA).
Solution Architecture Agricultural IoT Reference Architecture (AITRA): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Reliability is reached/addressed using/by Edge gateway (No description.).
Security is reached/addressed using/by Authentication, authorization, and data access control (No description.).
Reference: Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, https://doi.org/10.1109/ACCESS.2020.3031634.
The solution is composed by the architecture named as Blockchain-based Zero Trust, proposed in the paper Blockchain-Based Zero Trust on the Edge, 2023, with D.O.I. 10.1109/CSCI62032.2023.00167 (link https://ieeexplore.ieee.org/document/10590526). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Performance/Efficiency,Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Blockchain-based Zero Trust.
Solution Architecture Blockchain-based Zero Trust: described as: The architecture proposed in "Blockchain-Based Zero Trust on the Edge" integrates Zero Trust principles with Blockchain to ensure secure access and trust management for Edge Computing in IoT environments. It focuses on continuous verification of users and devices, ensuring that all access is authenticated before being granted. Blockchain is used for decentralized trust management, maintaining an immutable record of activities, which enhances security and transparency. This setup is optimized for edge computing, reducing latency by processing data locally while maintaining the flexibility to scale across diverse IoT applications, such as smart cities and industrial IoT..
Security is reached/addressed using/by Zero Trust Architecture (ZTA) (Confidentiality - Zero Trust Architecture (ZTA)

ZTA ensures continuous verification and monitoring of all network traffic, restricting unauthorized access.).
Security is reached/addressed using/by Blockchain Technology (Integrity - Blockchain serves as an immutable ledger for storing requests, ensuring data integrity and trustworthiness.).
Performance/Efficiency is reached/addressed using/by Redis Message Broker (Time Behavior - Redis Message Broker for Asynchronous Processing [redis.io]
Asynchronous processing for write operations improves system responsiveness under high loads.).
Maintainability is reached/addressed using/by Component isolation through microservices (Modularity -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.
).
Flexibility is reached/addressed using/by Component isolation through microservices (Scalability -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.).
Reference: Blockchain-Based Zero Trust on the Edge, 2023, https://ieeexplore.ieee.org/document/10590526.
The solution is composed by the architecture named as Blockchain-enabled, proposed in the paper An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, with D.O.I. doi.org/10.1109/TSC.2020.2966970 (link https://doi.org/10.1109/TSC.2020.2966970). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Blockchain-enabled.
Solution Architecture Blockchain-enabled: described as: .
Performance/Efficiency is reached/addressed using/by Routing protocol designed for IoT devices in the SDN controller [Yazdinejad et al. 2020] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, https://doi.org/10.1109/TSC.2020.2966970.
The solution is composed by the architecture named as Cloud-based Secure Biometric, proposed in the paper BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, with D.O.I. doi.org/10.1016/j.jksuci.2017.07.001 (link https://doi.org/10.1016/j.jksuci.2017.07.001). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Cloud-based Secure Biometric.
Solution Architecture Cloud-based Secure Biometric: described as: .
Flexibility is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Reliability is reached/addressed using/by MapReduce framework (No description.).
Security is reached/addressed using/by Cloud-based biometric authentication [Shakil et al. 2020] (No description.).
Reference: BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, https://doi.org/10.1016/j.jksuci.2017.07.001.
The solution is composed by the architecture named as Deep Learning based, proposed in the paper HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, with D.O.I. doi.org/10.1016/j.future.2019.10.043 (link https://doi.org/10.1016/j.future.2019.10.043). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Deep Learning based.
Solution Architecture Deep Learning based: described as: .
Performance/Efficiency is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Security is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Reference: HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, https://doi.org/10.1016/j.future.2019.10.043.
The solution is composed by the architecture named as Distributed Blockchain-SDN based, proposed in the paper DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3039113 (link https://doi.org/10.1109/ACCESS.2020.3039113). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Distributed Blockchain-SDN based.
Solution Architecture Distributed Blockchain-SDN based: described as: .
Performance/Efficiency is reached/addressed using/by Decentralized blockchain nodes, dynamic SDN controllers, and edge computing (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, https://doi.org/10.1109/ACCESS.2020.3039113.
The solution is composed by the architecture named as Distributed Edge-Cloud, proposed in the paper Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, with D.O.I. doi.org/10.1016/j.future.2020.05.042 (link https://doi.org/10.1016/j.future.2020.05.042). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Distributed Edge-Cloud.

Solution Architecture Distributed Edge-Cloud: described as: .
Security is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Reference: Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, https://doi.org/10.1016/j.future.2020.05.042.
The solution is composed by the architecture named as Edge Computing-based Fault Tolerant Framework, proposed in the paper Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, with D.O.I. doi.org/10.1109/IWCMC48107.2020.9148269 (link https://doi.org/10.1109/IWCMC48107.2020.9148269). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Edge Computing-based Fault Tolerant Framework.
Solution Architecture Edge Computing-based Fault Tolerant Framework: described as: .
Performance/Efficiency is reached/addressed using/by Kubernetes framework (No description.).
Flexibility is reached/addressed using/by Docker containers (No description.).
Reference: Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, https://doi.org/10.1109/IWCMC48107.2020.9148269.
The solution is composed by the architecture named as Event-driven IoT, proposed in the paper Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, with D.O.I. doi.org/10.1007/s10586-020-03189-w (link https://doi.org/10.1007/s10586-020-03189-w). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Event-driven IoT.
Solution Architecture Event-driven IoT: described as: .
Performance/Efficiency is reached/addressed using/by Complex Event Processing (CEP) [Rahmani et al. 2020] (No description.).
Reliability is reached/addressed using/by TCP-IP protocol on Client/Server architecture for data communication (No description.).
Reference: Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, https://doi.org/10.1007/s10586-020-03189-w.
The solution is composed by the architecture named as Fog Computing based, proposed in the paper FOG-based architecture and load balancing methodology for health monitoring systems, 2021, with D.O.I. doi.org/10.1109/ACCESS.2021.3094033 (link https://doi.org/10.1109/ACCESS.2021.3094033). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Fog Computing based.
Solution Architecture Fog Computing based: described as: .
Performance/Efficiency is reached/addressed using/by Load Balancing Scheme (LBS) - Distributed workload to all fog nodes [Asghar et al. 2021] (No description.).
Reference: FOG-based architecture and load balancing methodology for health monitoring systems, 2021, https://doi.org/10.1109/ACCESS.2021.3094033.
The solution is composed by the architecture named as Geographic-based, proposed in the paper A platform for integrating heterogeneous data and developing smart city applications, 2022, with D.O.I. doi.org/10.1016/j.future.2021.10.030 (link https://doi.org/10.1016/j.future.2021.10.030). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Geographic-based.
Solution Architecture Geographic-based: described as: .
Security is reached/addressed using/by FIWARE Platform [www.fiware.org] (Authentication, authorization, and data access control) (No description.).
Flexibility is reached/addressed using/by Virtualization (No description.).
Compatibility is reached/addressed using/by RESTFul APIs (No description.).
Reference: A platform for integrating heterogeneous data and developing smart city applications, 2022, https://doi.org/10.1016/j.future.2021.10.030.
The solution is composed by the architecture named as Hybrid (Centralized and Distributed), proposed in the paper Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, with D.O.I. doi.org/10.1016/j.procs.2021.03.030 (link https://doi.org/10.1016/j.procs.2021.03.030). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid (Centralized and Distributed).
Solution Architecture Hybrid (Centralized and Distributed): described as: .
Performance/Efficiency is reached/addressed using/by Dynamic allocation of the resources using Software Defined Networking (SDN) [Chudhary and Sharma 2021] (No description.).
Reference: Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, https://doi.org/10.1016/j.procs.2021.03.030.
The solution is composed by the architecture named as Hybrid Cloud-Fog Computing, proposed in the paper Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, with D.O.I. doi.org/10.1007/s10586-021-03375-4 (link https://link.springer.com/article/10.1007/s10586-021-03375-4). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid Cloud-Fog Computing.

Solution Architecture Hybrid Cloud-Fog Computing: described as: .
Reliability is reached/addressed using/by Fog layer in two levels: 1-Intra mobility (User network): Nodes (Data transfer, device management, temporary local storage, notifications, and data consistency). 2-Extra mobility (Different domains): Smart Gateways (Local processing and database) (No description.).
Performance/Efficiency is reached/addressed using/by Fog computing (No description.).
Reference: Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, https://link.springer.com/article/10.1007/s10586-021-03375-4.
The solution is composed by the architecture named as Industrial Internet of Things (IIoT), proposed in the paper A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Performance/Efficiency,Security,.
The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and Industrial Internet of Things (IIoT).
Solution Architecture Industrial Internet of Things (IIoT): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Flexibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Performance/Efficiency is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).
Security is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).
Reference: A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, https://doi.org/10.3390/s20195603.
The solution is composed by the architecture named as Layered, proposed in the paper An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, with D.O.I. doi.org/10.1016/j.future.2020.07.047 (link https://doi.org/10.1016/j.future.2020.07.047). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.
Solution Architecture Layered: described as: .
Flexibility is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).
Performance/Efficiency is reached/addressed using/by Big data analytics engine [Ali et al. 2021] (No description.).
Security is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).
Reference: An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, https://doi.org/10.1016/j.future.2020.07.047.
The solution is composed by the architecture named as Layered, proposed in the paper A novel architecture for the Internet of Things based E-Health systems, 2020, with D.O.I. doi.org/10.1166/jmihi.2020.3184 (link http://dx.doi.org/10.1166/jmihi.2020.3184). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.
Solution Architecture Layered: described as: .
Performance/Efficiency is reached/addressed using/by IoT objects connected at the edge of the network (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: A novel architecture for the Internet of Things based E-Health systems, 2020, http://dx.doi.org/10.1166/jmihi.2020.3184.
The solution is composed by the architecture named as Layered, proposed in the paper Software architecture quality attributes of a layered sensor-based IoT system, 2021, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Reliability,Security,.
The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and Layered.
Solution Architecture Layered: described as: .
Flexibility is reached/addressed using/by Messaging protocol brokers and at the client side, web-based applications (No description.).
Maintainability is reached/addressed using/by Loosely coupled components (No description.).
Reliability is reached/addressed using/by Open source software solutions and modular architecture design with clear relationships between components (No description.).
Security is reached/addressed using/by IoT devices authentication and data encryption (No description.).
Reference: Software architecture quality attributes of a layered sensor-based IoT system, 2021, https://doi.org/10.3390/s20195603.
The solution is composed by the architecture named as Layered, proposed in the paper Cloud-FOG assisted Energy Efficient Architectural Paradigm for disaster evacuation, 2022, with D.O.I. doi.org/10.1016/j.is.2021.101732 (link https://doi.org/10.1016/j.is.2021.101732). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Layered.
Solution Architecture Layered: described as: .
Performance/Efficiency is reached/addressed using/by Algorithm for Fog layer data selection (Data uniqueness and only important data) for transmission [Kaur et al. 2022] (No description.).
Reference: Cloud-FOG assisted Energy Efficient Architectural Paradigm for disaster evacuation, 2022, https://doi.org/10.1016/j.is.2021.101732.
The solution is composed by the architecture named as Layered, proposed in the paper An agile AI and IoT-Augmented Smart Farming: a Cost-Effective cognitive weather station, 2021, with D.O.I. doi.org/10.3390/agriculture12010035 (link https://doi.org/10.3390/agriculture12010035). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Interaction Capability,Maintainability,Performance/Efficiency,.

The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Layered.
Solution Architecture Layered: described as: .
Compatibility is reached/addressed using/by Docker containers (No description.).
Flexibility is reached/addressed using/by Docker containers (No description.).
Interaction Capability is reached/addressed using/by Remote access to the deployed platform and its components (Web App and SSH) (No description.).
Maintainability is reached/addressed using/by Docker containers (No description.).
Performance/Efficiency is reached/addressed using/by Docker containers (No description.).
Reference: An agile AI and IoT-Augmented Smart Farming: a Cost-Effective cognitive weather station, 2021, https://doi.org/10.3390/agriculture12010035.
The solution is composed by the architecture named as Layered, proposed in the paper Smart Indoor Farms: Leveraging Technological Advancements to Power a Sustainable Agricultural Revolution, 2021, with D.O.I. doi.org/10.3390/agriengineering3040047 (link https://doi.org/10.3390/agriengineering3040047). This solution addresses the following Quality Requirements: .
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Layered.
Solution Architecture Layered: described as: .
Reference: Smart Indoor Farms: Leveraging Technological Advancements to Power a Sustainable Agricultural Revolution, 2021, https://doi.org/10.3390/agriengineering3040047.
The solution is composed by the architecture named as Layered Blockchain and AI-enabled, proposed in the paper A blockchain and artificial intelligence-enabled smart IoT framework for sustainable city, 2022, with D.O.I. doi.org/10.1002/int.22852 (link https://doi.org/10.1002/int.22852). This solution addresses the following Quality Requirements: Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Layered Blockchain and AI-enabled.
Solution Architecture Layered Blockchain and AI-enabled: described as: .
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: A blockchain and artificial intelligence-enabled smart IoT framework for sustainable city, 2022, https://doi.org/10.1002/int.22852.
The solution is composed by the architecture named as Layered Blockchain-Based SDN, proposed in the paper Intelligent Blockchain Based secure routing for multidomain SDN Enabled IoT networks, 2022, with D.O.I. doi.org/10.1155/2022/5693962 (link https://doi.org/10.1155/2022/5693962). This solution addresses the following Quality Requirements: Reliability,Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Layered Blockchain-Based SDN.
Solution Architecture Layered Blockchain-Based SDN: described as: .
Reliability is reached/addressed using/by Reputation concept (Blockchain-based SDN-enabled network architecture) [Zeng et al. 2022] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: Intelligent Blockchain Based secure routing for multidomain SDN Enabled IoT networks, 2022, https://doi.org/10.1155/2022/5693962.
The solution is composed by the architecture named as LoRaWAN, proposed in the paper A new Low-Cost LORAWAN power switch for smart farm applications, 2021, with D.O.I. doi.org/10.1109/SMC52423.2021.9659076 (link https://doi.org/10.1109/SMC52423.2021.9659076). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and LoRaWAN.
Solution Architecture LoRaWAN: described as: .
Performance/Efficiency is reached/addressed using/by Long-Range Wide Area Network (LoRaWAN) (No description.).
Reference: A new Low-Cost LORAWAN power switch for smart farm applications, 2021, https://doi.org/10.1109/SMC52423.2021.9659076.
The solution is composed by the architecture named as Loosely Coupled IoT Platform and Localization Module, proposed in the paper A Self-Adaptive Bluetooth Indoor Localization System using LSTM-based Distance Estimator, 2020, with D.O.I. 10.1109/ICCCN49398.2020.9209674 (link https://ieeexplore.ieee.org/document/9209674). This solution addresses the following Quality Requirements: Compatibility,Functional Suitability,Interaction Capability,Maintainability,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Loosely Coupled IoT Platform and Localization Module.
Solution Architecture Loosely Coupled IoT Platform and Localization Module: described as: The proposed system architecture consists of two main components: the IoT platform and the localization module, connected by middleware. The IoT platform follows a hierarchical structure with layers for hardware (Bluetooth devices), data (MQTT-based RSSI parsing), business logic, display, and user interface. The localization module enhances trilateration with self-adaptive mechanisms, including an LSTM-based distance estimator, elastic radius intersecting, weighted centroid localization, and self-adaptive Kalman filtering for trajectory optimization..
Maintainability is reached/addressed using/by Google Remote Procedure Calls (gRPC) (Modularity - The IoT platform (server side) and the localization module (client side) are connected via a middleware called Google Remote Procedure Calls (gRPC).").
Interaction Capability is reached/addressed using/by Web Interface (Operability - Web interface for visualizing localization and trajectory tracking.).
Compatibility is reached/addressed using/by Google Remote Procedure Calls (gRPC) (Interoperability - The gRPC middleware supports communication between the IoT platform and the localization module, enabling system integration and compatibility with other platforms.).
Functional Suitability is reached/addressed using/by Kalman filtering (KF) (Kalman Filtering (KF) is an algorithm that estimates the state of a dynamic system by combining noisy sensor measurements with predictions. It recursively updates estimates based on new data, minimizing errors

and improving accuracy, making it useful in tracking, navigation, and signal processing.).
Reference: A Self-Adaptive Bluetooth Indoor Localization System using LSTM-based Distance Estimator, 2020, https://ieeexplore.ieee.org/document/9209674.
The solution is composed by the architecture named as Microservices, proposed in the paper Design of scalable IoT architecture based on AWS for smart livestock, 2021, with D.O.I. doi.org/10.3390/ani11092697 (link https://doi.org/10.3390/ani11092697). This solution addresses the following Quality Requirements: Flexibility,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Microservices.
Solution Architecture Microservices: described as: .
Flexibility is reached/addressed using/by Amazon Web Services (AWS) (No description.).
Reference: Design of scalable IoT architecture based on AWS for smart livestock, 2021, https://doi.org/10.3390/ani11092697.
The solution is composed by the architecture named as Microservices, proposed in the paper A Lightweight User Interface for Smart Charging of Electric Vehicles: A Real-World Application, 2021, with D.O.I. 10.1109/ICSGCE52779.2021.9621604 (link https://ieeexplore.ieee.org/document/9621604). This solution addresses the following Quality Requirements: Compatibility,Reliability,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Microservices.
Solution Architecture Microservices: described as: Microservice-Based IoT Architecture integrated with Edge-Cloud Computing. Key components include: Charging Session Handler (CSH): Manages charging sessions through MQTT messages; PLC-Controlled Charging Station: Executes charging commands; Web-Based User Interface: Built with Streamlit for real-time user control; and Cloud Integration: Kubernetes ensures scalability and availability. This modular design supports distributed processing, real-time control, and seamless EV charging management..
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (Interoperability - The MQTT server stablishes communication between the web app, the Session Manager, and the Programmable Logic Controller (PLC) via JavaScript Object Notation (JSON) encoded messages.).
Reliability is reached/addressed using/by Docker containers (Availability - It is deployed in a Docker container on Kubernetes to provide a continuously available service.).
Reference: A Lightweight User Interface for Smart Charging of Electric Vehicles: A Real-World Application, 2021, https://ieeexplore.ieee.org/document/9621604.
The solution is composed by the architecture named as Microservices, proposed in the paper An open source software architecture and ready-to-use components for health IoT, 2020, with D.O.I. 10.1109/CBMS49503.2020.00077 (link https://ieeexplore.ieee.org/document/9183119). This solution addresses the following Quality Requirements: Interaction Capability,Maintainability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Microservices.
Solution Architecture Microservices: described as: Microservices-based software architecture for Health IoT, inspired by best practices from sources like Microsoft's microservices guidelines. The architecture utilizes open-source components for various functionalities. Key components include an API Gateway (Express Gateway) acting as a single entry point, a messaging channel (RabbitMQ) for asynchronous communication between microservices, and a security solution leveraging Vault and Consul for key management and high availability. The architecture is instantiated in the OCARIoT project, where it manages data from wearables, sensors, and mobile apps, processing data and triggering personalized interventions..
Interaction Capability is reached/addressed using/by User-friendly dashboard (Operability - User-friendly dashboard for health professionals: Health professionals are able to do a follow-up through the dashboard and also to receive tips and recommendations. Accessibility Open access through mobile and web platforms. The OCARIoT Dashboard visualizes real-time updates from mobile devices and web interfaces.).
Interaction Capability is reached/addressed using/by Intuitive mobile apps (Learnability - Intuitive mobile apps for health monitoring The Data Acquisition App manages the collection of data from different devices and shows key information for health professionals.).
Maintainability is reached/addressed using/by Component isolation through microservices (Modularity - Open Source Software Architecture Component isolation through microservices. Microservices can be implemented in different languages, using different databases, ensuring modular architecture.).
Reference: An open source software architecture and ready-to-use components for health IoT, 2020, https://ieeexplore.ieee.org/document/9183119.
The solution is composed by the architecture named as Multi-layered Cloud-Edge, proposed in the paper Securing smart healthcare system with edge computing, 2021, with D.O.I. doi.org/10.1016/j.cose.2021.102353 (link https://doi.org/10.1016/j.cose.2021.102353). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Multi-layered Cloud-Edge.
Solution Architecture Multi-layered Cloud-Edge: described as: .
Performance/Efficiency is reached/addressed using/by Edge computing (No description.).
Security is reached/addressed using/by Access Control System (ACS) and Searchable Encryption Module [Singh and Chatterjee 2021] (No description.).
Reference: Securing smart healthcare system with edge computing, 2021, https://doi.org/10.1016/j.cose.2021.102353.
The solution is composed by the architecture named as SAPPARCHI (Smart City), proposed in the paper SAPPARCHI: an Osmotic Platform to Execute Scalable Applications on Smart City Environments, 2021, with D.O.I. doi.org/10.1109/CLOUD55607.2022.00051 (link https://doi.org/10.1109/CLOUD55607.2022.00051). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and SAPPARCHI (Smart City).
Solution Architecture SAPPARCHI (Smart City): described as: .

Flexibility is reached/addressed using/by Serverless on Edge and Microservice in Fog (No description.).
Maintainability is reached/addressed using/by Serverless on Edge and Microservice in Fog (No description.).
Security is reached/addressed using/by Serverless on Edge and Microservice in Fog (No description.).
Reference: SAPPARCHI: an Osmotic Platform to Execute Scalable Applications on Smart City Environments, 2021, https://doi.org/10.1109/CLOUD55607.2022.00051.

The solution is composed by the architecture named as A4IoT (Anyplace 4.0 IoT), proposed in the paper The Anyplace 4.0 IoT Localization Architecture, 2020, with D.O.I. 10.1109/MDM48529.2020.00045 (link https://ieeexplore.ieee.org/document/9162205). This solution addresses the following Quality Requirements: Flexibility,Interaction Capability,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and A4IoT (Anyplace 4.0 IoT).
Solution Architecture A4IoT (Anyplace 4.0 IoT): described as: It is an open-source architecture designed for IoT-based indoor localization using technologies like Wi-Fi, BLE, Cellular, UWB, and Computer Vision (CV). The system supports edge computing, timeseries databases, and crowdsourcing to manage localization tasks across various smart spaces, such as factories, healthcare facilities, and emergency response environments. Its modular and scalable design ensures integration across platforms like Android, Linux, Windows, and Robot OS, making it adaptable to diverse IoT ecosystems..
Performance/Efficiency is reached/addressed using/by Time-Series Database (InfluxDB) (Time Behavior - With temporal queries we can now consume high-volume input streams to provide real-time IoT tracking.").
Reliability is reached/addressed using/by Multi-node Cluster Deployment (Fault Tolerance - The production environment uses a 3-node cluster with two replicas, for both the database engines and the DFS (Distributed Filesystem). This setup allows for a full operation with just a single node active at any time.").
Security is reached/addressed using/by SSL certificates - Encrypted Communications (Confidentiality - All communication is encrypted even in internal networks, as A4IoT image automatically creates and uses SSL certificates.).
Interaction Capability is reached/addressed using/by Web-based Dashboard (Operability - Web Apps built with HTML5, CSS3, and AngularJS.").
Flexibility is reached/addressed using/by Docker containers (Adaptability - It utilizes Docker to enable the deployment of single-node or multi-node configurations, regardless of any dependencies or the underlying OS.).
Reference: The Anyplace 4.0 IoT Localization Architecture, 2020, https://ieeexplore.ieee.org/document/9162205.
The solution is composed by the architecture named as Agricultural IoT Reference Architecture (AITRA), proposed in the paper Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3031634 (link https://doi.org/10.1109/ACCESS.2020.3031634). This solution addresses the following Quality Requirements: Compatibility,Reliability,Security,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Agricultural IoT Reference Architecture (AITRA).
Solution Architecture Agricultural IoT Reference Architecture (AITRA): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Reliability is reached/addressed using/by Edge gateway (No description.).
Security is reached/addressed using/by Authentication, authorization, and data access control (No description.).
Reference: Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, https://doi.org/10.1109/ACCESS.2020.3031634.
The solution is composed by the architecture named as Blockchain-based Zero Trust, proposed in the paper Blockchain-Based Zero Trust on the Edge, 2023, with D.O.I. 10.1109/CSCI62032.2023.00167 (link https://ieeexplore.ieee.org/document/10590526). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Performance/Efficiency,Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Blockchain-based Zero Trust.
Solution Architecture Blockchain-based Zero Trust: described as: The architecture proposed in "Blockchain-Based Zero Trust on the Edge" integrates Zero Trust principles with Blockchain to ensure secure access and trust management for Edge Computing in IoT environments. It focuses on continuous verification of users and devices, ensuring that all access is authenticated before being granted. Blockchain is used for decentralized trust management, maintaining an immutable record of activities, which enhances security and transparency. This setup is optimized for edge computing, reducing latency by processing data locally while maintaining the flexibility to scale across diverse IoT applications, such as smart cities and industrial IoT..
Security is reached/addressed using/by Zero Trust Architecture (ZTA) (Confidentiality - Zero Trust Architecture (ZTA)
ZTA ensures continuous verification and monitoring of all network traffic, restricting unauthorized access.).
Security is reached/addressed using/by Blockchain Technology (Integrity - Blockchain serves as an immutable ledger for storing requests, ensuring data integrity and trustworthiness.).
Performance/Efficiency is reached/addressed using/by Redis Message Broker (Time Behavior - Redis Message Broker for Asynchronous Processing [redis.io]
Asynchronous processing for write operations improves system responsiveness under high loads.).
Maintainability is reached/addressed using/by Component isolation through microservices (Modularity -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.
).
Flexibility is reached/addressed using/by Component isolation through microservices (Scalability -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.).
Reference: Blockchain-Based Zero Trust on the Edge, 2023, https://ieeexplore.ieee.org/document/10590526.

The solution is composed by the architecture named as Blockchain-enabled, proposed in the paper An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, with D.O.I. doi.org/10.1109/TSC.2020.2966970 (link https://doi.org/10.1109/TSC.2020.2966970). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Blockchain-enabled.
Solution Architecture Blockchain-enabled: described as: .
Performance/Efficiency is reached/addressed using/by Routing protocol designed for IoT devices in the SDN controller [Yazdinejad et al. 2020] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, https://doi.org/10.1109/TSC.2020.2966970.
The solution is composed by the architecture named as Cloud-based Secure Biometric, proposed in the paper BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, with D.O.I. doi.org/10.1016/j.jksuci.2017.07.001 (link https://doi.org/10.1016/j.jksuci.2017.07.001). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Cloud-based Secure Biometric.
Solution Architecture Cloud-based Secure Biometric: described as: .
Flexibility is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Reliability is reached/addressed using/by MapReduce framework (No description.).
Security is reached/addressed using/by Cloud-based biometric authentication [Shakil et al. 2020] (No description.).
Reference: BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, https://doi.org/10.1016/j.jksuci.2017.07.001.
The solution is composed by the architecture named as Deep Learning based, proposed in the paper HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, with D.O.I. doi.org/10.1016/j.future.2019.10.043 (link https://doi.org/10.1016/j.future.2019.10.043). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Deep Learning based.
Solution Architecture Deep Learning based: described as: .
Performance/Efficiency is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Security is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Reference: HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, https://doi.org/10.1016/j.future.2019.10.043.
The solution is composed by the architecture named as Distributed Blockchain-SDN based, proposed in the paper DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3039113 (link https://doi.org/10.1109/ACCESS.2020.3039113). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Distributed Blockchain-SDN based.
Solution Architecture Distributed Blockchain-SDN based: described as: .
Performance/Efficiency is reached/addressed using/by Decentralized blockchain nodes, dynamic SDN controllers, and edge computing (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, https://doi.org/10.1109/ACCESS.2020.3039113.
The solution is composed by the architecture named as Distributed Edge-Cloud, proposed in the paper Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, with D.O.I. doi.org/10.1016/j.future.2020.05.042 (link https://doi.org/10.1016/j.future.2020.05.042). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Distributed Edge-Cloud.
Solution Architecture Distributed Edge-Cloud: described as: .
Security is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Reference: Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, https://doi.org/10.1016/j.future.2020.05.042.
The solution is composed by the architecture named as Edge Computing-based Fault Tolerant Framework, proposed in the paper Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, with D.O.I. doi.org/10.1109/IWCMC48107.2020.9148269 (link https://doi.org/10.1109/IWCMC48107.2020.9148269). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Edge Computing-based Fault Tolerant Framework.

Solution Architecture Edge Computing-based Fault Tolerant Framework: described as: .
Performance/Efficiency is reached/addressed using/by Kubernetes framework (No description.).
Flexibility is reached/addressed using/by Docker containers (No description.).
Reference: Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, https://doi.org/10.1109/IWCMC48107.2020.9148269.
The solution is composed by the architecture named as Event-driven IoT, proposed in the paper Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, with D.O.I. doi.org/10.1007/s10586-020-03189-w (link https://doi.org/10.1007/s10586-020-03189-w). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Event-driven IoT.
Solution Architecture Event-driven IoT: described as: .
Performance/Efficiency is reached/addressed using/by Complex Event Processing (CEP) [Rahmani et al. 2020] (No description.).
Reliability is reached/addressed using/by TCP-IP protocol on Client/Server architecture for data communication (No description.).
Reference: Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, https://doi.org/10.1007/s10586-020-03189-w.
The solution is composed by the architecture named as Fog Computing based, proposed in the paper FOG-based architecture and load balancing methodology for health monitoring systems, 2021, with D.O.I. doi.org/10.1109/ACCESS.2021.3094033 (link https://doi.org/10.1109/ACCESS.2021.3094033). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Fog Computing based.
Solution Architecture Fog Computing based: described as: .
Performance/Efficiency is reached/addressed using/by Load Balancing Scheme (LBS) - Distributed workload to all fog nodes [Asghar et al. 2021] (No description.).
Reference: FOG-based architecture and load balancing methodology for health monitoring systems, 2021, https://doi.org/10.1109/ACCESS.2021.3094033.
The solution is composed by the architecture named as Geographic-based, proposed in the paper A platform for integrating heterogeneous data and developing smart city applications, 2022, with D.O.I. doi.org/10.1016/j.future.2021.10.030 (link https://doi.org/10.1016/j.future.2021.10.030). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Geographic-based.
Solution Architecture Geographic-based: described as: .
Security is reached/addressed using/by FIWARE Platform [www.fiware.org] (Authentication, authorization, and data access control) (No description.).
Flexibility is reached/addressed using/by Virtualization (No description.).
Compatibility is reached/addressed using/by RESTFul APIs (No description.).
Reference: A platform for integrating heterogeneous data and developing smart city applications, 2022, https://doi.org/10.1016/j.future.2021.10.030.
The solution is composed by the architecture named as Hybrid (Centralized and Distributed), proposed in the paper Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, with D.O.I. doi.org/10.1016/j.procs.2021.03.030 (link https://doi.org/10.1016/j.procs.2021.03.030). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid (Centralized and Distributed).
Solution Architecture Hybrid (Centralized and Distributed): described as: .
Performance/Efficiency is reached/addressed using/by Dynamic allocation of the resources using Software Defined Networking (SDN) [Chudhary and Sharma 2021] (No description.).
Reference: Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, https://doi.org/10.1016/j.procs.2021.03.030.
The solution is composed by the architecture named as Hybrid Cloud-Fog Computing, proposed in the paper Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, with D.O.I. doi.org/10.1007/s10586-021-03375-4 (link https://link.springer.com/article/10.1007/s10586-021-03375-4). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid Cloud-Fog Computing.
Solution Architecture Hybrid Cloud-Fog Computing: described as: .
Reliability is reached/addressed using/by Fog layer in two levels: 1-Intra mobility (User network): Nodes (Data transfer, device management, temporary local storage, notifications, and data consistency). 2-Extra mobility (Different domains): Smart Gateways (Local processing and database) (No description.).
Performance/Efficiency is reached/addressed using/by Fog computing (No description.).
Reference: Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, https://link.springer.com/article/10.1007/s10586-021-03375-4.
The solution is composed by the architecture named as Industrial Internet of Things (IIoT), proposed in the paper A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Performance/Efficiency,Security,.
The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and Industrial Internet of Things (IIoT).

Solution Architecture Industrial Internet of Things (IIoT): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Flexibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Performance/Efficiency is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).
Security is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).
Reference: A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, https://doi.org/10.3390/s20195603.
The solution is composed by the architecture named as Layered, proposed in the paper An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, with D.O.I. doi.org/10.1016/j.future.2020.07.047 (link https://doi.org/10.1016/j.future.2020.07.047). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.
Solution Architecture Layered: described as: .
Flexibility is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).
Performance/Efficiency is reached/addressed using/by Big data analytics engine [Ali et al. 2021] (No description.).
Security is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).
Reference: An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, https://doi.org/10.1016/j.future.2020.07.047.
The solution is composed by the architecture named as Layered, proposed in the paper A novel architecture for the Internet of Things based E-Health systems, 2020, with D.O.I. doi.org/10.1166/jmihi.2020.3184 (link http://dx.doi.org/10.1166/jmihi.2020.3184). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.
Solution Architecture Layered: described as: .
Performance/Efficiency is reached/addressed using/by IoT objects connected at the edge of the network (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: A novel architecture for the Internet of Things based E-Health systems, 2020, http://dx.doi.org/10.1166/jmihi.2020.3184.
The solution is composed by the architecture named as Layered, proposed in the paper Software architecture quality attributes of a layered sensor-based IoT system, 2021, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Reliability,Security,.
The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and Layered.
Solution Architecture Layered: described as: .
Flexibility is reached/addressed using/by Messaging protocol brokers and at the client side, web-based applications (No description.).
Maintainability is reached/addressed using/by Loosely coupled components (No description.).
Reliability is reached/addressed using/by Open source software solutions and modular architecture design with clear relationships between components (No description.).
Security is reached/addressed using/by IoT devices authentication and data encryption (No description.).
Reference: Software architecture quality attributes of a layered sensor-based IoT system, 2021, https://doi.org/10.3390/s20195603.
The solution is composed by the architecture named as Layered, proposed in the paper Cloud-FOG assisted Energy Efficient Architectural Paradigm for disaster evacuation, 2022, with D.O.I. doi.org/10.1016/j.is.2021.101732 (link https://doi.org/10.1016/j.is.2021.101732). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Layered.
Solution Architecture Layered: described as: .
Performance/Efficiency is reached/addressed using/by Algorithm for Fog layer data selection (Data uniqueness and only important data) for transmission [Kaur et al. 2022] (No description.).
Reference: Cloud-FOG assisted Energy Efficient Architectural Paradigm for disaster evacuation, 2022, https://doi.org/10.1016/j.is.2021.101732.
The solution is composed by the architecture named as Layered, proposed in the paper An agile AI and IoT-Augmented Smart Farming: a Cost-Effective cognitive weather station, 2021, with D.O.I. doi.org/10.3390/agriculture12010035 (link https://doi.org/10.3390/agriculture12010035). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Interaction Capability,Maintainability,Performance/Efficiency,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Layered.
Solution Architecture Layered: described as: .
Compatibility is reached/addressed using/by Docker containers (No description.).
Flexibility is reached/addressed using/by Docker containers (No description.).
Interaction Capability is reached/addressed using/by Remote access to the deployed platform and its components (Web App and SSH) (No description.).
Maintainability is reached/addressed using/by Docker containers (No description.).
Performance/Efficiency is reached/addressed using/by Docker containers (No description.).
Reference: An agile AI and IoT-Augmented Smart Farming: a Cost-Effective cognitive weather station, 2021, https://doi.org/10.3390/agriculture12010035.
The solution is composed by the architecture named as Layered, proposed in the paper Smart Indoor Farms: Leveraging Technological

Advancements to Power a Sustainable Agricultural Revolution, 2021, with D.O.I. doi.org/10.3390/agriengineering3040047 (link https://doi.org/10.3390/agriengineering3040047). This solution addresses the following Quality Requirements: .
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Layered.
Solution Architecture Layered: described as: .
Reference: Smart Indoor Farms: Leveraging Technological Advancements to Power a Sustainable Agricultural Revolution, 2021, https://doi.org/10.3390/agriengineering3040047.
The solution is composed by the architecture named as Layered Blockchain and AI-enabled, proposed in the paper A blockchain and artificial intelligence-enabled smart IoT framework for sustainable city, 2022, with D.O.I. doi.org/10.1002/int.22852 (link https://doi.org/10.1002/int.22852). This solution addresses the following Quality Requirements: Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Layered Blockchain and AI-enabled.
Solution Architecture Layered Blockchain and AI-enabled: described as: .
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: A blockchain and artificial intelligence-enabled smart IoT framework for sustainable city, 2022, https://doi.org/10.1002/int.22852.
The solution is composed by the architecture named as Layered Blockchain-Based SDN, proposed in the paper Intelligent Blockchain Based secure routing for multidomain SDN Enabled IoT networks, 2022, with D.O.I. doi.org/10.1155/2022/5693962 (link https://doi.org/10.1155/2022/5693962). This solution addresses the following Quality Requirements: Reliability,Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Layered Blockchain-Based SDN.
Solution Architecture Layered Blockchain-Based SDN: described as: .
Reliability is reached/addressed using/by Reputation concept (Blockchain-based SDN-enabled network architecture) [Zeng et al. 2022] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: Intelligent Blockchain Based secure routing for multidomain SDN Enabled IoT networks, 2022, https://doi.org/10.1155/2022/5693962.
The solution is composed by the architecture named as LoRaWAN, proposed in the paper A new Low-Cost LORAWAN power switch for smart farm applications, 2021, with D.O.I. doi.org/10.1109/SMC52423.2021.9659076 (link https://doi.org/10.1109/SMC52423.2021.9659076). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and LoRaWAN.
Solution Architecture LoRaWAN: described as: .
Performance/Efficiency is reached/addressed using/by Long-Range Wide Area Network (LoRaWAN) (No description.).
Reference: A new Low-Cost LORAWAN power switch for smart farm applications, 2021, https://doi.org/10.1109/SMC52423.2021.9659076.
The solution is composed by the architecture named as Loosely Coupled IoT Platform and Localization Module, proposed in the paper A Self-Adaptive Bluetooth Indoor Localization System using LSTM-based Distance Estimator, 2020, with D.O.I. 10.1109/ICCCN49398.2020.9209674 (link https://ieeexplore.ieee.org/document/9209674). This solution addresses the following Quality Requirements: Compatibility,Functional Suitability,Interaction Capability,Maintainability,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Loosely Coupled IoT Platform and Localization Module.
Solution Architecture Loosely Coupled IoT Platform and Localization Module: described as: The proposed system architecture consists of two main components: the IoT platform and the localization module, connected by middleware. The IoT platform follows a hierarchical structure with layers for hardware (Bluetooth devices), data (MQTT-based RSSI parsing), business logic, display, and user interface. The localization module enhances trilateration with self-adaptive mechanisms, including an LSTM-based distance estimator, elastic radius intersecting, weighted centroid localization, and self-adaptive Kalman filtering for trajectory optimization..
Maintainability is reached/addressed using/by Google Remote Procedure Calls (gRPC) (Modularity - The IoT platform (server side) and the localization module (client side) are connected via a middleware called Google Remote Procedure Calls (gRPC).").
Interaction Capability is reached/addressed using/by Web Interface (Operability - Web interface for visualizing localization and trajectory tracking.).
Compatibility is reached/addressed using/by Google Remote Procedure Calls (gRPC) (Interoperability - The gRPC middleware supports communication between the IoT platform and the localization module, enabling system integration and compatibility with other platforms.).
Functional Suitability is reached/addressed using/by Kalman filtering (KF) (Kalman Filtering (KF) is an algorithm that estimates the state of a dynamic system by combining noisy sensor measurements with predictions. It recursively updates estimates based on new data, minimizing errors and improving accuracy, making it useful in tracking, navigation, and signal processing.).
Reference: A Self-Adaptive Bluetooth Indoor Localization System using LSTM-based Distance Estimator, 2020, https://ieeexplore.ieee.org/document/9209674.
The solution is composed by the architecture named as Microservices, proposed in the paper Design of scalable IoT architecture based on AWS for smart livestock, 2021, with D.O.I. doi.org/10.3390/ani11092697 (link https://doi.org/10.3390/ani11092697). This solution addresses the following Quality Requirements: Flexibility,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Microservices.
Solution Architecture Microservices: described as: .
Flexibility is reached/addressed using/by Amazon Web Services (AWS) (No description.).
Reference: Design of scalable IoT architecture based on AWS for smart livestock, 2021, https://doi.org/10.3390/ani11092697.
The solution is composed by the architecture named as Microservices, proposed in the paper A Lightweight User Interface for Smart Charging of

Electric Vehicles: A Real-World Application, 2021, with D.O.I. 10.1109/ICSGCE52779.2021.9621604 (link https://ieeexplore.ieee.org/document/9621604). This solution addresses the following Quality Requirements: Compatibility,Reliability,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Microservices.
Solution Architecture Microservices: described as: Microservice-Based IoT Architecture integrated with Edge-Cloud Computing. Key components include: Charging Session Handler (CSH): Manages charging sessions through MQTT messages; PLC-Controlled Charging Station: Executes charging commands; Web-Based User Interface: Built with Streamlit for real-time user control; and Cloud Integration: Kubernetes ensures scalability and availability. This modular design supports distributed processing, real-time control, and seamless EV charging management..
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (Interoperability - The MQTT server stablishes communication between the web app, the Session Manager, and the Programmable Logic Controller (PLC) via JavaScript Object Notation (JSON) encoded messages.).
Reliability is reached/addressed using/by Docker containers (Availability - It is deployed in a Docker container on Kubernetes to provide a continuously available service.).
Reference: A Lightweight User Interface for Smart Charging of Electric Vehicles: A Real-World Application, 2021, https://ieeexplore.ieee.org/document/9621604.
The solution is composed by the architecture named as Microservices, proposed in the paper An open source software architecture and ready-to-use components for health IoT, 2020, with D.O.I. 10.1109/CBMS49503.2020.00077 (link https://ieeexplore.ieee.org/document/9183119). This solution addresses the following Quality Requirements: Interaction Capability,Maintainability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Microservices.
Solution Architecture Microservices: described as: Microservices-based software architecture for Health IoT, inspired by best practices from sources like Microsoft's microservices guidelines. The architecture utilizes open-source components for various functionalities. Key components include an API Gateway (Express Gateway) acting as a single entry point, a messaging channel (RabbitMQ) for asynchronous communication between microservices, and a security solution leveraging Vault and Consul for key management and high availability. The architecture is instantiated in the OCARIoT project, where it manages data from wearables, sensors, and mobile apps, processing data and triggering personalized interventions..
Interaction Capability is reached/addressed using/by User-friendly dashboard (Operability - User-friendly dashboard for health professionals: Health professionals are able to do a follow-up through the dashboard and also to receive tips and recommendations. Accessibility Open access through mobile and web platforms. The OCARIoT Dashboard visualizes real-time updates from mobile devices and web interfaces.).
Interaction Capability is reached/addressed using/by Intuitive mobile apps (Learnability - Intuitive mobile apps for health monitoring The Data Acquisition App manages the collection of data from different devices and shows key information for health professionals.).
Maintainability is reached/addressed using/by Component isolation through microservices (Modularity - Open Source Software Architecture Component isolation through microservices. Microservices can be implemented in different languages, using different databases, ensuring modular architecture.).
Reference: An open source software architecture and ready-to-use components for health IoT, 2020, https://ieeexplore.ieee.org/document/9183119.
The solution is composed by the architecture named as Multi-layered Cloud-Edge, proposed in the paper Securing smart healthcare system with edge computing, 2021, with D.O.I. doi.org/10.1016/j.cose.2021.102353 (link https://doi.org/10.1016/j.cose.2021.102353). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Multi-layered Cloud-Edge.
Solution Architecture Multi-layered Cloud-Edge: described as: .
Performance/Efficiency is reached/addressed using/by Edge computing (No description.).
Security is reached/addressed using/by Access Control System (ACS) and Searchable Encryption Module [Singh and Chatterjee 2021] (No description.).
Reference: Securing smart healthcare system with edge computing, 2021, https://doi.org/10.1016/j.cose.2021.102353.
The solution is composed by the architecture named as SAPPARCHI (Smart City), proposed in the paper SAPPARCHI: an Osmotic Platform to Execute Scalable Applications on Smart City Environments, 2021, with D.O.I. doi.org/10.1109/CLOUD55607.2022.00051 (link https://doi.org/10.1109/CLOUD55607.2022.00051). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and SAPPARCHI (Smart City).
Solution Architecture SAPPARCHI (Smart City): described as: .
Flexibility is reached/addressed using/by Serverless on Edge and Microservice in Fog (No description.).
Maintainability is reached/addressed using/by Serverless on Edge and Microservice in Fog (No description.).
Security is reached/addressed using/by Serverless on Edge and Microservice in Fog (No description.).
Reference: SAPPARCHI: an Osmotic Platform to Execute Scalable Applications on Smart City Environments, 2021, https://doi.org/10.1109/CLOUD55607.2022.00051.
The solution is composed by the architecture named as SDN based, proposed in the paper An efficient Counter Based DDOS attack detection framework leveraging software defined IoT (SD-IoT), 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3043082 (link https://doi.org/10.1109/ACCESS.2020.3043082). This solution addresses the following Quality Requirements: Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and SDN based.
Solution Architecture SDN based: described as: .
Security is reached/addressed using/by Counter-based DDoS Attack Detection (C-DAD) application for Software Defined Networks (SDN) [Bhayo et al. 2020] (No description.).

Reference: An efficient Counter Based DDOS attack detection framework leveraging software defined IoT (SD-IoT), 2020, https://doi.org/10.1109/ACCESS.2020.3043082.

The solution is composed by the architecture named as A4IoT (Anyplace 4.0 IoT), proposed in the paper The Anyplace 4.0 IoT Localization Architecture, 2020, with D.O.I. 10.1109/MDM48529.2020.00045 (link https://ieeexplore.ieee.org/document/9162205). This solution addresses the following Quality Requirements: Flexibility,Interaction Capability,Performance/Efficiency,Reliability,Security,.

The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and A4IoT (Anyplace 4.0 IoT).

Solution Architecture A4IoT (Anyplace 4.0 IoT): described as: It is an open-source architecture designed for IoT-based indoor localization using technologies like Wi-Fi, BLE, Cellular, UWB, and Computer Vision (CV). The system supports edge computing, timeseries databases, and crowdsourcing to manage localization tasks across various smart spaces, such as factories, healthcare facilities, and emergency response environments. Its modular and scalable design ensures integration across platforms like Android, Linux, Windows, and Robot OS, making it adaptable to diverse IoT ecosystems..

Performance/Efficiency is reached/addressed using/by Time-Series Database (InfluxDB) (Time Behavior - With temporal queries we can now consume high-volume input streams to provide real-time IoT tracking.").

Reliability is reached/addressed using/by Multi-node Cluster Deployment (Fault Tolerance - The production environment uses a 3-node cluster with two replicas, for both the database engines and the DFS (Distributed Filesystem). This setup allows for a full operation with just a single node active at any time.").

Security is reached/addressed using/by SSL certificates - Encrypted Communications (Confidentiality - All communication is encrypted even in internal networks, as A4IoT image automatically creates and uses SSL certificates.).

Interaction Capability is reached/addressed using/by Web-based Dashboard (Operability - Web Apps built with HTML5, CSS3, and AngularJS.").

Flexibility is reached/addressed using/by Docker containers (Adaptability - It utilizes Docker to enable the deployment of single-node or multi-node configurations, regardless of any dependencies or the underlying OS.).

Reference: The Anyplace 4.0 IoT Localization Architecture, 2020, https://ieeexplore.ieee.org/document/9162205.

The solution is composed by the architecture named as Agricultural IoT Reference Architecture (AITRA), proposed in the paper Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3031634 (link https://doi.org/10.1109/ACCESS.2020.3031634). This solution addresses the following Quality Requirements: Compatibility,Reliability,Security,.

The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Agricultural IoT Reference Architecture (AITRA).

Solution Architecture Agricultural IoT Reference Architecture (AITRA): described as: .

Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).

Reliability is reached/addressed using/by Edge gateway (No description.).

Security is reached/addressed using/by Authentication, authorization, and data access control (No description.).

Reference: Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, https://doi.org/10.1109/ACCESS.2020.3031634.

The solution is composed by the architecture named as Blockchain-based Zero Trust, proposed in the paper Blockchain-Based Zero Trust on the Edge, 2023, with D.O.I. 10.1109/CSCI62032.2023.00167 (link https://ieeexplore.ieee.org/document/10590526). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Performance/Efficiency,Security,.

The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Blockchain-based Zero Trust.

Solution Architecture Blockchain-based Zero Trust: described as: The architecture proposed in "Blockchain-Based Zero Trust on the Edge" integrates Zero Trust principles with Blockchain to ensure secure access and trust management for Edge Computing in IoT environments. It focuses on continuous verification of users and devices, ensuring that all access is authenticated before being granted. Blockchain is used for decentralized trust management, maintaining an immutable record of activities, which enhances security and transparency. This setup is optimized for edge computing, reducing latency by processing data locally while maintaining the flexibility to scale across diverse IoT applications, such as smart cities and industrial IoT..

Security is reached/addressed using/by Zero Trust Architecture (ZTA) (Confidentiality - Zero Trust Architecture (ZTA)
ZTA ensures continuous verification and monitoring of all network traffic, restricting unauthorized access.).

Security is reached/addressed using/by Blockchain Technology (Integrity - Blockchain serves as an immutable ledger for storing requests, ensuring data integrity and trustworthiness.).

Performance/Efficiency is reached/addressed using/by Redis Message Broker (Time Behavior - Redis Message Broker for Asynchronous Processing [redis.io]
Asynchronous processing for write operations improves system responsiveness under high loads.).

Maintainability is reached/addressed using/by Component isolation through microservices (Modularity -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.
).

Flexibility is reached/addressed using/by Component isolation through microservices (Scalability -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.).

Reference: Blockchain-Based Zero Trust on the Edge, 2023, https://ieeexplore.ieee.org/document/10590526.

The solution is composed by the architecture named as Blockchain-enabled, proposed in the paper An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, with D.O.I. doi.org/10.1109/TSC.2020.2966970 (link https://doi.org/10.1109/TSC.2020.2966970). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.

The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Blockchain-enabled.
Solution Architecture Blockchain-enabled: described as: .
Performance/Efficiency is reached/addressed using/by Routing protocol designed for IoT devices in the SDN controller [Yazdinejad et al. 2020] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, https://doi.org/10.1109/TSC.2020.2966970.
The solution is composed by the architecture named as Cloud-based Secure Biometric, proposed in the paper BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, with D.O.I. doi.org/10.1016/j.jksuci.2017.07.001 (link https://doi.org/10.1016/j.jksuci.2017.07.001). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Cloud-based Secure Biometric.
Solution Architecture Cloud-based Secure Biometric: described as: .
Flexibility is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Reliability is reached/addressed using/by MapReduce framework (No description.).
Security is reached/addressed using/by Cloud-based biometric authentication [Shakil et al. 2020] (No description.).
Reference: BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, https://doi.org/10.1016/j.jksuci.2017.07.001.
The solution is composed by the architecture named as Deep Learning based, proposed in the paper HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, with D.O.I. doi.org/10.1016/j.future.2019.10.043 (link https://doi.org/10.1016/j.future.2019.10.043). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Deep Learning based.
Solution Architecture Deep Learning based: described as: .
Performance/Efficiency is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Security is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Reference: HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, https://doi.org/10.1016/j.future.2019.10.043.
The solution is composed by the architecture named as Distributed Blockchain-SDN based, proposed in the paper DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3039113 (link https://doi.org/10.1109/ACCESS.2020.3039113). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Distributed Blockchain-SDN based.
Solution Architecture Distributed Blockchain-SDN based: described as: .
Performance/Efficiency is reached/addressed using/by Decentralized blockchain nodes, dynamic SDN controllers, and edge computing (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, https://doi.org/10.1109/ACCESS.2020.3039113.
The solution is composed by the architecture named as Distributed Edge-Cloud, proposed in the paper Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, with D.O.I. doi.org/10.1016/j.future.2020.05.042 (link https://doi.org/10.1016/j.future.2020.05.042). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Distributed Edge-Cloud.
Solution Architecture Distributed Edge-Cloud: described as: .
Security is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Reference: Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, https://doi.org/10.1016/j.future.2020.05.042.
The solution is composed by the architecture named as Edge Computing-based Fault Tolerant Framework, proposed in the paper Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, with D.O.I. doi.org/10.1109/IWCMC48107.2020.9148269 (link https://doi.org/10.1109/IWCMC48107.2020.9148269). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Edge Computing-based Fault Tolerant Framework.
Solution Architecture Edge Computing-based Fault Tolerant Framework: described as: .
Performance/Efficiency is reached/addressed using/by Kubernetes framework (No description.).
Flexibility is reached/addressed using/by Docker containers (No description.).

Reference: Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, https://doi.org/10.1109/IWCMC48107.2020.9148269.
The solution is composed by the architecture named as Event-driven IoT, proposed in the paper Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, with D.O.I. doi.org/10.1007/s10586-020-03189-w (link https://doi.org/10.1007/s10586-020-03189-w). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Event-driven IoT.
Solution Architecture Event-driven IoT: described as: .
Performance/Efficiency is reached/addressed using/by Complex Event Processing (CEP) [Rahmani et al. 2020] (No description.).
Reliability is reached/addressed using/by TCP-IP protocol on Client/Server architecture for data communication (No description.).
Reference: Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, https://doi.org/10.1007/s10586-020-03189-w.
The solution is composed by the architecture named as Fog Computing based, proposed in the paper FOG-based architecture and load balancing methodology for health monitoring systems, 2021, with D.O.I. doi.org/10.1109/ACCESS.2021.3094033 (link https://doi.org/10.1109/ACCESS.2021.3094033). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Fog Computing based.
Solution Architecture Fog Computing based: described as: .
Performance/Efficiency is reached/addressed using/by Load Balancing Scheme (LBS) - Distributed workload to all fog nodes [Asghar et al. 2021] (No description.).
Reference: FOG-based architecture and load balancing methodology for health monitoring systems, 2021, https://doi.org/10.1109/ACCESS.2021.3094033.
The solution is composed by the architecture named as Geographic-based, proposed in the paper A platform for integrating heterogeneous data and developing smart city applications, 2022, with D.O.I. doi.org/10.1016/j.future.2021.10.030 (link https://doi.org/10.1016/j.future.2021.10.030). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Geographic-based.
Solution Architecture Geographic-based: described as: .
Security is reached/addressed using/by FIWARE Platform [www.fiware.org] (Authentication, authorization, and data access control) (No description.).
Flexibility is reached/addressed using/by Virtualization (No description.).
Compatibility is reached/addressed using/by RESTFul APIs (No description.).
Reference: A platform for integrating heterogeneous data and developing smart city applications, 2022, https://doi.org/10.1016/j.future.2021.10.030.
The solution is composed by the architecture named as Hybrid (Centralized and Distributed), proposed in the paper Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, with D.O.I. doi.org/10.1016/j.procs.2021.03.030 (link https://doi.org/10.1016/j.procs.2021.03.030). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid (Centralized and Distributed).
Solution Architecture Hybrid (Centralized and Distributed): described as: .
Performance/Efficiency is reached/addressed using/by Dynamic allocation of the resources using Software Defined Networking (SDN) [Chudhary and Sharma 2021] (No description.).
Reference: Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, https://doi.org/10.1016/j.procs.2021.03.030.
The solution is composed by the architecture named as Hybrid Cloud-Fog Computing, proposed in the paper Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, with D.O.I. doi.org/10.1007/s10586-021-03375-4 (link https://link.springer.com/article/10.1007/s10586-021-03375-4). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid Cloud-Fog Computing.
Solution Architecture Hybrid Cloud-Fog Computing: described as: .
Reliability is reached/addressed using/by Fog layer in two levels: 1-Intra mobility (User network): Nodes (Data transfer, device management, temporary local storage, notifications, and data consistency). 2-Extra mobility (Different domains): Smart Gateways (Local processing and database) (No description.).
Performance/Efficiency is reached/addressed using/by Fog computing (No description.).
Reference: Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, https://link.springer.com/article/10.1007/s10586-021-03375-4.
The solution is composed by the architecture named as Industrial Internet of Things (IIoT), proposed in the paper A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Performance/Efficiency,Security,.
The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and Industrial Internet of Things (IIoT).
Solution Architecture Industrial Internet of Things (IIoT): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Flexibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).

Performance/Efficiency is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).
Security is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).
Reference: A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, https://doi.org/10.3390/s20195603.
The solution is composed by the architecture named as Layered, proposed in the paper An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, with D.O.I. doi.org/10.1016/j.future.2020.07.047 (link https://doi.org/10.1016/j.future.2020.07.047). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.
Solution Architecture Layered: described as: .
Flexibility is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).
Performance/Efficiency is reached/addressed using/by Big data analytics engine [Ali et al. 2021] (No description.).
Security is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).
Reference: An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, https://doi.org/10.1016/j.future.2020.07.047.
The solution is composed by the architecture named as Layered, proposed in the paper A novel architecture for the Internet of Things based E-Health systems, 2020, with D.O.I. doi.org/10.1166/jmihi.2020.3184 (link http://dx.doi.org/10.1166/jmihi.2020.3184). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.
Solution Architecture Layered: described as: .
Performance/Efficiency is reached/addressed using/by IoT objects connected at the edge of the network (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: A novel architecture for the Internet of Things based E-Health systems, 2020, http://dx.doi.org/10.1166/jmihi.2020.3184.
The solution is composed by the architecture named as Layered, proposed in the paper Software architecture quality attributes of a layered sensor-based IoT system, 2021, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Reliability,Security,.
The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and Layered.
Solution Architecture Layered: described as: .
Flexibility is reached/addressed using/by Messaging protocol brokers and at the client side, web-based applications (No description.).
Maintainability is reached/addressed using/by Loosely coupled components (No description.).
Reliability is reached/addressed using/by Open source software solutions and modular architecture design with clear relationships between components (No description.).
Security is reached/addressed using/by IoT devices authentication and data encryption (No description.).
Reference: Software architecture quality attributes of a layered sensor-based IoT system, 2021, https://doi.org/10.3390/s20195603.
The solution is composed by the architecture named as Layered, proposed in the paper Cloud-FOG assisted Energy Efficient Architectural Paradigm for disaster evacuation, 2022, with D.O.I. doi.org/10.1016/j.is.2021.101732 (link https://doi.org/10.1016/j.is.2021.101732). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Layered.
Solution Architecture Layered: described as: .
Performance/Efficiency is reached/addressed using/by Algorithm for Fog layer data selection (Data uniqueness and only important data) for transmission [Kaur et al. 2022] (No description.).
Reference: Cloud-FOG assisted Energy Efficient Architectural Paradigm for disaster evacuation, 2022, https://doi.org/10.1016/j.is.2021.101732.
The solution is composed by the architecture named as Layered, proposed in the paper An agile AI and IoT-Augmented Smart Farming: a Cost-Effective cognitive weather station, 2021, with D.O.I. doi.org/10.3390/agriculture12010035 (link https://doi.org/10.3390/agriculture12010035). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Interaction Capability,Maintainability,Performance/Efficiency,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Layered.
Solution Architecture Layered: described as: .
Compatibility is reached/addressed using/by Docker containers (No description.).
Flexibility is reached/addressed using/by Docker containers (No description.).
Interaction Capability is reached/addressed using/by Remote access to the deployed platform and its components (Web App and SSH) (No description.).
Maintainability is reached/addressed using/by Docker containers (No description.).
Performance/Efficiency is reached/addressed using/by Docker containers (No description.).
Reference: An agile AI and IoT-Augmented Smart Farming: a Cost-Effective cognitive weather station, 2021, https://doi.org/10.3390/agriculture12010035.
The solution is composed by the architecture named as Layered, proposed in the paper Smart Indoor Farms: Leveraging Technological Advancements to Power a Sustainable Agricultural Revolution, 2021, with D.O.I. doi.org/10.3390/agriengineering3040047 (link https://doi.org/10.3390/agriengineering3040047). This solution addresses the following Quality Requirements: .
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and

Layered.
Solution Architecture Layered: described as: .
Reference: Smart Indoor Farms: Leveraging Technological Advancements to Power a Sustainable Agricultural Revolution, 2021, https://doi.org/10.3390/agriengineering3040047.
The solution is composed by the architecture named as Layered Blockchain and AI-enabled, proposed in the paper A blockchain and artificial intelligence-enabled smart IoT framework for sustainable city, 2022, with D.O.I. doi.org/10.1002/int.22852 (link https://doi.org/10.1002/int.22852). This solution addresses the following Quality Requirements: Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Layered Blockchain and AI-enabled.
Solution Architecture Layered Blockchain and AI-enabled: described as: .
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: A blockchain and artificial intelligence-enabled smart IoT framework for sustainable city, 2022, https://doi.org/10.1002/int.22852.
The solution is composed by the architecture named as Layered Blockchain-Based SDN, proposed in the paper Intelligent Blockchain Based secure routing for multidomain SDN Enabled IoT networks, 2022, with D.O.I. doi.org/10.1155/2022/5693962 (link https://doi.org/10.1155/2022/5693962). This solution addresses the following Quality Requirements: Reliability,Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Layered Blockchain-Based SDN.
Solution Architecture Layered Blockchain-Based SDN: described as: .
Reliability is reached/addressed using/by Reputation concept (Blockchain-based SDN-enabled network architecture) [Zeng et al. 2022] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: Intelligent Blockchain Based secure routing for multidomain SDN Enabled IoT networks, 2022, https://doi.org/10.1155/2022/5693962.
The solution is composed by the architecture named as LoRaWAN, proposed in the paper A new Low-Cost LORAWAN power switch for smart farm applications, 2021, with D.O.I. doi.org/10.1109/SMC52423.2021.9659076 (link https://doi.org/10.1109/SMC52423.2021.9659076). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and LoRaWAN.
Solution Architecture LoRaWAN: described as: .
Performance/Efficiency is reached/addressed using/by Long-Range Wide Area Network (LoRaWAN) (No description.).
Reference: A new Low-Cost LORAWAN power switch for smart farm applications, 2021, https://doi.org/10.1109/SMC52423.2021.9659076.
The solution is composed by the architecture named as Loosely Coupled IoT Platform and Localization Module, proposed in the paper A Self-Adaptive Bluetooth Indoor Localization System using LSTM-based Distance Estimator, 2020, with D.O.I. 10.1109/ICCCN49398.2020.9209674 (link https://ieeexplore.ieee.org/document/9209674). This solution addresses the following Quality Requirements: Compatibility,Functional Suitability,Interaction Capability,Maintainability,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Loosely Coupled IoT Platform and Localization Module.
Solution Architecture Loosely Coupled IoT Platform and Localization Module: described as: The proposed system architecture consists of two main components: the IoT platform and the localization module, connected by middleware. The IoT platform follows a hierarchical structure with layers for hardware (Bluetooth devices), data (MQTT-based RSSI parsing), business logic, display, and user interface. The localization module enhances trilateration with self-adaptive mechanisms, including an LSTM-based distance estimator, elastic radius intersecting, weighted centroid localization, and self-adaptive Kalman filtering for trajectory optimization..
Maintainability is reached/addressed using/by Google Remote Procedure Calls (gRPC) (Modularity - The IoT platform (server side) and the localization module (client side) are connected via a middleware called Google Remote Procedure Calls (gRPC).").
Interaction Capability is reached/addressed using/by Web Interface (Operability - Web interface for visualizing localization and trajectory tracking.).
Compatibility is reached/addressed using/by Google Remote Procedure Calls (gRPC) (Interoperability - The gRPC middleware supports communication between the IoT platform and the localization module, enabling system integration and compatibility with other platforms.).
Functional Suitability is reached/addressed using/by Kalman filtering (KF) (Kalman Filtering (KF) is an algorithm that estimates the state of a dynamic system by combining noisy sensor measurements with predictions. It recursively updates estimates based on new data, minimizing errors and improving accuracy, making it useful in tracking, navigation, and signal processing.).
Reference: A Self-Adaptive Bluetooth Indoor Localization System using LSTM-based Distance Estimator, 2020, https://ieeexplore.ieee.org/document/9209674.
The solution is composed by the architecture named as Microservices, proposed in the paper Design of scalable IoT architecture based on AWS for smart livestock, 2021, with D.O.I. doi.org/10.3390/ani11092697 (link https://doi.org/10.3390/ani11092697). This solution addresses the following Quality Requirements: Flexibility,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Microservices.
Solution Architecture Microservices: described as: .
Flexibility is reached/addressed using/by Amazon Web Services (AWS) (No description.).
Reference: Design of scalable IoT architecture based on AWS for smart livestock, 2021, https://doi.org/10.3390/ani11092697.
The solution is composed by the architecture named as Microservices, proposed in the paper A Lightweight User Interface for Smart Charging of Electric Vehicles: A Real-World Application, 2021, with D.O.I. 10.1109/ICSGCE52779.2021.9621604 (link https://ieeexplore.ieee.org/document/9621604). This solution addresses the following Quality Requirements: Compatibility,Reliability,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and

Microservices.
Solution Architecture Microservices: described as: Microservice-Based IoT Architecture integrated with Edge-Cloud Computing. Key components include: Charging Session Handler (CSH): Manages charging sessions through MQTT messages; PLC-Controlled Charging Station: Executes charging commands; Web-Based User Interface: Built with Streamlit for real-time user control; and Cloud Integration: Kubernetes ensures scalability and availability. This modular design supports distributed processing, real-time control, and seamless EV charging management..
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (Interoperability - The MQTT server stablishes communication between the web app, the Session Manager, and the Programmable Logic Controller (PLC) via JavaScript Object Notation (JSON) encoded messages.).
Reliability is reached/addressed using/by Docker containers (Availability - It is deployed in a Docker container on Kubernetes to provide a continuously available service.).
Reference: A Lightweight User Interface for Smart Charging of Electric Vehicles: A Real-World Application, 2021, https://ieeexplore.ieee.org/document/9621604.
The solution is composed by the architecture named as Microservices, proposed in the paper An open source software architecture and ready-to-use components for health IoT, 2020, with D.O.I. 10.1109/CBMS49503.2020.00077 (link https://ieeexplore.ieee.org/document/9183119). This solution addresses the following Quality Requirements: Interaction Capability,Maintainability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Microservices.
Solution Architecture Microservices: described as: Microservices-based software architecture for Health IoT, inspired by best practices from sources like Microsoft's microservices guidelines. The architecture utilizes open-source components for various functionalities. Key components include an API Gateway (Express Gateway) acting as a single entry point, a messaging channel (RabbitMQ) for asynchronous communication between microservices, and a security solution leveraging Vault and Consul for key management and high availability. The architecture is instantiated in the OCARIoT project, where it manages data from wearables, sensors, and mobile apps, processing data and triggering personalized interventions..
Interaction Capability is reached/addressed using/by User-friendly dashboard (Operability - User-friendly dashboard for health professionals: Health professionals are able to do a follow-up through the dashboard and also to receive tips and recommendations. Accessibility Open access through mobile and web platforms. The OCARIoT Dashboard visualizes real-time updates from mobile devices and web interfaces.).
Interaction Capability is reached/addressed using/by Intuitive mobile apps (Learnability - Intuitive mobile apps for health monitoring The Data Acquisition App manages the collection of data from different devices and shows key information for health professionals.).
Maintainability is reached/addressed using/by Component isolation through microservices (Modularity - Open Source Software Architecture Component isolation through microservices. Microservices can be implemented in different languages, using different databases, ensuring modular architecture.).
Reference: An open source software architecture and ready-to-use components for health IoT, 2020, https://ieeexplore.ieee.org/document/9183119.
The solution is composed by the architecture named as Multi-layered Cloud-Edge, proposed in the paper Securing smart healthcare system with edge computing, 2021, with D.O.I. doi.org/10.1016/j.cose.2021.102353 (link https://doi.org/10.1016/j.cose.2021.102353). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Multi-layered Cloud-Edge.
Solution Architecture Multi-layered Cloud-Edge: described as: .
Performance/Efficiency is reached/addressed using/by Edge computing (No description.).
Security is reached/addressed using/by Access Control System (ACS) and Searchable Encryption Module [Singh and Chatterjee 2021] (No description.).
Reference: Securing smart healthcare system with edge computing, 2021, https://doi.org/10.1016/j.cose.2021.102353.
The solution is composed by the architecture named as SAPPARCHI (Smart City), proposed in the paper SAPPARCHI: an Osmotic Platform to Execute Scalable Applications on Smart City Environments, 2021, with D.O.I. doi.org/10.1109/CLOUD55607.2022.00051 (link https://doi.org/10.1109/CLOUD55607.2022.00051). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and SAPPARCHI (Smart City).
Solution Architecture SAPPARCHI (Smart City): described as: .
Flexibility is reached/addressed using/by Serverless on Edge and Microservice in Fog (No description.).
Maintainability is reached/addressed using/by Serverless on Edge and Microservice in Fog (No description.).
Security is reached/addressed using/by Serverless on Edge and Microservice in Fog (No description.).
Reference: SAPPARCHI: an Osmotic Platform to Execute Scalable Applications on Smart City Environments, 2021, https://doi.org/10.1109/CLOUD55607.2022.00051.
The solution is composed by the architecture named as SDN based, proposed in the paper An efficient Counter Based DDOS attack detection framework leveraging software defined IoT (SD-IoT), 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3043082 (link https://doi.org/10.1109/ACCESS.2020.3043082). This solution addresses the following Quality Requirements: Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and SDN based.
Solution Architecture SDN based: described as: .
Security is reached/addressed using/by Counter-based DDoS Attack Detection (C-DAD) application for Software Defined Networks (SDN) [Bhayo et al. 2020] (No description.).
Reference: An efficient Counter Based DDOS attack detection framework leveraging software defined IoT (SD-IoT), 2020, https://doi.org/10.1109/ACCESS.2020.3043082.
The solution is composed by the architecture named as Secure Publish-Subscribe, proposed in the paper Publish Subscribe System Security

Requirement: A Case Study for V2V Communication, 2024, with D.O.I. 10.1109/OJCS.2024.3442921 (link https://www.scopus.com/record/display.uri?eid=2-s2.0-85201290507&doi=10.1109%2fOJCS.2024.3442921&origin=inward&txGid=43ee5fbd5bae9bbdf927871aab51aa4d). This solution addresses the following Quality Requirements: Reliability,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Secure Publish-Subscribe.
Solution Architecture Secure Publish-Subscribe: described as: This solution focuses on ensuring secure, reliable, and efficient communication between vehicles using a publish-subscribe messaging system. It integrates brokers, clients, and publishers/subscribers with advanced security features such as access control, encryption, authentication, and message integrity verification..
Reliability is reached/addressed using/by Duplicate ClientID Handling (Availability - The broker incorporates a security measure called 'Handling Duplicate ClientID' to guarantee the continuous availability of services.).
Security is reached/addressed using/by Cryptography (Encryption) (Confidentiality - The publisher will employ cryptographic techniques and key sizes to ensure the confidentiality of the payload data throughout the whole communication process.).
Security is reached/addressed using/by Data Origin Authentication (Authenticity - The subscriber should have the capability to offer the end-user a method, such as a digital signature or Message Authentication Code (MAC), to authenticate the origin of information.).
Security is reached/addressed using/by Digital Signatures and Hashing (Integrity - The publisher transmits the information to the subscriber... using digital signature, MAC, and hashing to identify any abnormalities.).
Reference: Publish Subscribe System Security Requirement: A Case Study for V2V Communication, 2024, https://www.scopus.com/record/display.uri?eid=2-s2.0-85201290507&doi=10.1109%2fOJCS.2024.3442921&origin=inward&txGid=43ee5fbd5bae9bbdf927871aab51aa4d.

The solution is composed by the architecture named as A4IoT (Anyplace 4.0 IoT), proposed in the paper The Anyplace 4.0 IoT Localization Architecture, 2020, with D.O.I. 10.1109/MDM48529.2020.00045 (link https://ieeexplore.ieee.org/document/9162205). This solution addresses the following Quality Requirements: Flexibility,Interaction Capability,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and A4IoT (Anyplace 4.0 IoT).
Solution Architecture A4IoT (Anyplace 4.0 IoT): described as: It is an open-source architecture designed for IoT-based indoor localization using technologies like Wi-Fi, BLE, Cellular, UWB, and Computer Vision (CV). The system supports edge computing, timeseries databases, and crowdsourcing to manage localization tasks across various smart spaces, such as factories, healthcare facilities, and emergency response environments. Its modular and scalable design ensures integration across platforms like Android, Linux, Windows, and Robot OS, making it adaptable to diverse IoT ecosystems..
Performance/Efficiency is reached/addressed using/by Time-Series Database (InfluxDB) (Time Behavior - With temporal queries we can now consume high-volume input streams to provide real-time IoT tracking.").
Reliability is reached/addressed using/by Multi-node Cluster Deployment (Fault Tolerance - The production environment uses a 3-node cluster with two replicas, for both the database engines and the DFS (Distributed Filesystem). This setup allows for a full operation with just a single node active at any time.").
Security is reached/addressed using/by SSL certificates - Encrypted Communications (Confidentiality - All communication is encrypted even in internal networks, as A4IoT image automatically creates and uses SSL certificates.).
Interaction Capability is reached/addressed using/by Web-based Dashboard (Operability - Web Apps built with HTML5, CSS3, and AngularJS.").
Flexibility is reached/addressed using/by Docker containers (Adaptability - It utilizes Docker to enable the deployment of single-node or multi-node configurations, regardless of any dependencies or the underlying OS.).
Reference: The Anyplace 4.0 IoT Localization Architecture, 2020, https://ieeexplore.ieee.org/document/9162205.
The solution is composed by the architecture named as Agricultural IoT Reference Architecture (AITRA), proposed in the paper Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3031634 (link https://doi.org/10.1109/ACCESS.2020.3031634). This solution addresses the following Quality Requirements: Compatibility,Reliability,Security,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Agricultural IoT Reference Architecture (AITRA).
Solution Architecture Agricultural IoT Reference Architecture (AITRA): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Reliability is reached/addressed using/by Edge gateway (No description.).
Security is reached/addressed using/by Authentication, authorization, and data access control (No description.).
Reference: Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, https://doi.org/10.1109/ACCESS.2020.3031634.
The solution is composed by the architecture named as Blockchain-based Zero Trust, proposed in the paper Blockchain-Based Zero Trust on the Edge, 2023, with D.O.I. 10.1109/CSCI62032.2023.00167 (link https://ieeexplore.ieee.org/document/10590526). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Performance/Efficiency,Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Blockchain-based Zero Trust.
Solution Architecture Blockchain-based Zero Trust: described as: The architecture proposed in "Blockchain-Based Zero Trust on the Edge" integrates Zero Trust principles with Blockchain to ensure secure access and trust management for Edge Computing in IoT environments. It focuses on continuous verification of users and devices, ensuring that all access is authenticated before being granted. Blockchain is used for decentralized trust management, maintaining an immutable record of activities, which enhances security and transparency. This setup is optimized

for edge computing, reducing latency by processing data locally while maintaining the flexibility to scale across diverse IoT applications, such as smart cities and industrial IoT..
Security is reached/addressed using/by Zero Trust Architecture (ZTA) (Confidentiality - Zero Trust Architecture (ZTA)
ZTA ensures continuous verification and monitoring of all network traffic, restricting unauthorized access.).
Security is reached/addressed using/by Blockchain Technology (Integrity - Blockchain serves as an immutable ledger for storing requests, ensuring data integrity and trustworthiness.).
Performance/Efficiency is reached/addressed using/by Redis Message Broker (Time Behavior - Redis Message Broker for Asynchronous Processing [redis.io]
Asynchronous processing for write operations improves system responsiveness under high loads.).
Maintainability is reached/addressed using/by Component isolation through microservices (Modularity -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.
).
Flexibility is reached/addressed using/by Component isolation through microservices (Scalability -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.).
Reference: Blockchain-Based Zero Trust on the Edge, 2023, https://ieeexplore.ieee.org/document/10590526.
The solution is composed by the architecture named as Blockchain-enabled, proposed in the paper An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, with D.O.I. doi.org/10.1109/TSC.2020.2966970 (link https://doi.org/10.1109/TSC.2020.2966970). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Blockchain-enabled.
Solution Architecture Blockchain-enabled: described as: .
Performance/Efficiency is reached/addressed using/by Routing protocol designed for IoT devices in the SDN controller [Yazdinejad et al. 2020] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, https://doi.org/10.1109/TSC.2020.2966970.
The solution is composed by the architecture named as Cloud-based Secure Biometric, proposed in the paper BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, with D.O.I. doi.org/10.1016/j.jksuci.2017.07.001 (link https://doi.org/10.1016/j.jksuci.2017.07.001). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Cloud-based Secure Biometric.
Solution Architecture Cloud-based Secure Biometric: described as: .
Flexibility is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Reliability is reached/addressed using/by MapReduce framework (No description.).
Security is reached/addressed using/by Cloud-based biometric authentication [Shakil et al. 2020] (No description.).
Reference: BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, https://doi.org/10.1016/j.jksuci.2017.07.001.
The solution is composed by the architecture named as Deep Learning based, proposed in the paper HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, with D.O.I. doi.org/10.1016/j.future.2019.10.043 (link https://doi.org/10.1016/j.future.2019.10.043). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Deep Learning based.
Solution Architecture Deep Learning based: described as: .
Performance/Efficiency is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Security is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Reference: HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, https://doi.org/10.1016/j.future.2019.10.043.
The solution is composed by the architecture named as Distributed Blockchain-SDN based, proposed in the paper DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3039113 (link https://doi.org/10.1109/ACCESS.2020.3039113). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Distributed Blockchain-SDN based.
Solution Architecture Distributed Blockchain-SDN based: described as: .
Performance/Efficiency is reached/addressed using/by Decentralized blockchain nodes, dynamic SDN controllers, and edge computing (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, https://doi.org/10.1109/ACCESS.2020.3039113.
The solution is composed by the architecture named as Distributed Edge-Cloud, proposed in the paper Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, with D.O.I. doi.org/10.1016/j.future.2020.05.042 (link

https://doi.org/10.1016/j.future.2020.05.042). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Distributed Edge-Cloud.
Solution Architecture Distributed Edge-Cloud: described as: .
Security is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Reference: Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, https://doi.org/10.1016/j.future.2020.05.042.
The solution is composed by the architecture named as Edge Computing-based Fault Tolerant Framework, proposed in the paper Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, with D.O.I. doi.org/10.1109/IWCMC48107.2020.9148269 (link https://doi.org/10.1109/IWCMC48107.2020.9148269). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Edge Computing-based Fault Tolerant Framework.
Solution Architecture Edge Computing-based Fault Tolerant Framework: described as: .
Performance/Efficiency is reached/addressed using/by Kubernetes framework (No description.).
Flexibility is reached/addressed using/by Docker containers (No description.).
Reference: Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, https://doi.org/10.1109/IWCMC48107.2020.9148269.
The solution is composed by the architecture named as Event-driven IoT, proposed in the paper Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, with D.O.I. doi.org/10.1007/s10586-020-03189-w (link https://doi.org/10.1007/s10586-020-03189-w). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Event-driven IoT.
Solution Architecture Event-driven IoT: described as: .
Performance/Efficiency is reached/addressed using/by Complex Event Processing (CEP) [Rahmani et al. 2020] (No description.).
Reliability is reached/addressed using/by TCP-IP protocol on Client/Server architecture for data communication (No description.).
Reference: Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, https://doi.org/10.1007/s10586-020-03189-w.
The solution is composed by the architecture named as Fog Computing based, proposed in the paper FOG-based architecture and load balancing methodology for health monitoring systems, 2021, with D.O.I. doi.org/10.1109/ACCESS.2021.3094033 (link https://doi.org/10.1109/ACCESS.2021.3094033). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Fog Computing based.
Solution Architecture Fog Computing based: described as: .
Performance/Efficiency is reached/addressed using/by Load Balancing Scheme (LBS) - Distributed workload to all fog nodes [Asghar et al. 2021] (No description.).
Reference: FOG-based architecture and load balancing methodology for health monitoring systems, 2021, https://doi.org/10.1109/ACCESS.2021.3094033.
The solution is composed by the architecture named as Geographic-based, proposed in the paper A platform for integrating heterogeneous data and developing smart city applications, 2022, with D.O.I. doi.org/10.1016/j.future.2021.10.030 (link https://doi.org/10.1016/j.future.2021.10.030). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Geographic-based.
Solution Architecture Geographic-based: described as: .
Security is reached/addressed using/by FIWARE Platform [www.fiware.org] (Authentication, authorization, and data access control) (No description.).
Flexibility is reached/addressed using/by Virtualization (No description.).
Compatibility is reached/addressed using/by RESTFul APIs (No description.).
Reference: A platform for integrating heterogeneous data and developing smart city applications, 2022, https://doi.org/10.1016/j.future.2021.10.030.
The solution is composed by the architecture named as Hybrid (Centralized and Distributed), proposed in the paper Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, with D.O.I. doi.org/10.1016/j.procs.2021.03.030 (link https://doi.org/10.1016/j.procs.2021.03.030). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid (Centralized and Distributed).
Solution Architecture Hybrid (Centralized and Distributed): described as: .
Performance/Efficiency is reached/addressed using/by Dynamic allocation of the resources using Software Defined Networking (SDN) [Chudhary and Sharma 2021] (No description.).
Reference: Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, https://doi.org/10.1016/j.procs.2021.03.030.
The solution is composed by the architecture named as Hybrid Cloud-Fog Computing, proposed in the paper Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, with D.O.I. doi.org/10.1007/s10586-021-03375-4 (link https://link.springer.com/article/10.1007/s10586-021-03375-4). This solution addresses the following Quality Requirements:

Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid Cloud-Fog Computing.
Solution Architecture Hybrid Cloud-Fog Computing: described as: .
Reliability is reached/addressed using/by Fog layer in two levels: 1-Intra mobility (User network): Nodes (Data transfer, device management, temporary local storage, notifications, and data consistency). 2-Extra mobility (Different domains): Smart Gateways (Local processing and database) (No description.).
Performance/Efficiency is reached/addressed using/by Fog computing (No description.).
Reference: Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, https://link.springer.com/article/10.1007/s10586-021-03375-4.
The solution is composed by the architecture named as Industrial Internet of Things (IIoT), proposed in the paper A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Performance/Efficiency,Security,.
The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and Industrial Internet of Things (IIoT).
Solution Architecture Industrial Internet of Things (IIoT): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Flexibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Performance/Efficiency is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).
Security is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).
Reference: A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, https://doi.org/10.3390/s20195603.
The solution is composed by the architecture named as Layered, proposed in the paper An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, with D.O.I. doi.org/10.1016/j.future.2020.07.047 (link https://doi.org/10.1016/j.future.2020.07.047). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.
Solution Architecture Layered: described as: .
Flexibility is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).
Performance/Efficiency is reached/addressed using/by Big data analytics engine [Ali et al. 2021] (No description.).
Security is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).
Reference: An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, https://doi.org/10.1016/j.future.2020.07.047.
The solution is composed by the architecture named as Layered, proposed in the paper A novel architecture for the Internet of Things based E-Health systems, 2020, with D.O.I. doi.org/10.1166/jmihi.2020.3184 (link http://dx.doi.org/10.1166/jmihi.2020.3184). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.
Solution Architecture Layered: described as: .
Performance/Efficiency is reached/addressed using/by IoT objects connected at the edge of the network (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: A novel architecture for the Internet of Things based E-Health systems, 2020, http://dx.doi.org/10.1166/jmihi.2020.3184.
The solution is composed by the architecture named as Layered, proposed in the paper Software architecture quality attributes of a layered sensor-based IoT system, 2021, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Reliability,Security,.
The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and Layered.
Solution Architecture Layered: described as: .
Flexibility is reached/addressed using/by Messaging protocol brokers and at the client side, web-based applications (No description.).
Maintainability is reached/addressed using/by Loosely coupled components (No description.).
Reliability is reached/addressed using/by Open source software solutions and modular architecture design with clear relationships between components (No description.).
Security is reached/addressed using/by IoT devices authentication and data encryption (No description.).
Reference: Software architecture quality attributes of a layered sensor-based IoT system, 2021, https://doi.org/10.3390/s20195603.
The solution is composed by the architecture named as Layered, proposed in the paper Cloud-FOG assisted Energy Efficient Architectural Paradigm for disaster evacuation, 2022, with D.O.I. doi.org/10.1016/j.is.2021.101732 (link https://doi.org/10.1016/j.is.2021.101732). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Layered.
Solution Architecture Layered: described as: .
Performance/Efficiency is reached/addressed using/by Algorithm for Fog layer data selection (Data uniqueness and only important data) for transmission [Kaur et al. 2022] (No description.).
Reference: Cloud-FOG assisted Energy Efficient Architectural Paradigm for disaster evacuation, 2022, https://doi.org/10.1016/j.is.2021.101732.

The solution is composed by the architecture named as Layered, proposed in the paper An agile AI and IoT-Augmented Smart Farming: a Cost-Effective cognitive weather station, 2021, with D.O.I. doi.org/10.3390/agriculture12010035 (link https://doi.org/10.3390/agriculture12010035). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Interaction Capability,Maintainability,Performance/Efficiency,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Layered.
Solution Architecture Layered: described as: .
Compatibility is reached/addressed using/by Docker containers (No description.).
Flexibility is reached/addressed using/by Docker containers (No description.).
Interaction Capability is reached/addressed using/by Remote access to the deployed platform and its components (Web App and SSH) (No description.).
Maintainability is reached/addressed using/by Docker containers (No description.).
Performance/Efficiency is reached/addressed using/by Docker containers (No description.).
Reference: An agile AI and IoT-Augmented Smart Farming: a Cost-Effective cognitive weather station, 2021, https://doi.org/10.3390/agriculture12010035.
The solution is composed by the architecture named as Layered, proposed in the paper Smart Indoor Farms: Leveraging Technological Advancements to Power a Sustainable Agricultural Revolution, 2021, with D.O.I. doi.org/10.3390/agriengineering3040047 (link https://doi.org/10.3390/agriengineering3040047). This solution addresses the following Quality Requirements: .
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Layered.
Solution Architecture Layered: described as: .
Reference: Smart Indoor Farms: Leveraging Technological Advancements to Power a Sustainable Agricultural Revolution, 2021, https://doi.org/10.3390/agriengineering3040047.
The solution is composed by the architecture named as Layered Blockchain and AI-enabled, proposed in the paper A blockchain and artificial intelligence-enabled smart IoT framework for sustainable city, 2022, with D.O.I. doi.org/10.1002/int.22852 (link https://doi.org/10.1002/int.22852). This solution addresses the following Quality Requirements: Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Layered Blockchain and AI-enabled.
Solution Architecture Layered Blockchain and AI-enabled: described as: .
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: A blockchain and artificial intelligence-enabled smart IoT framework for sustainable city, 2022, https://doi.org/10.1002/int.22852.
The solution is composed by the architecture named as Layered Blockchain-Based SDN, proposed in the paper Intelligent Blockchain Based secure routing for multidomain SDN Enabled IoT networks, 2022, with D.O.I. doi.org/10.1155/2022/5693962 (link https://doi.org/10.1155/2022/5693962). This solution addresses the following Quality Requirements: Reliability,Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Layered Blockchain-Based SDN.
Solution Architecture Layered Blockchain-Based SDN: described as: .
Reliability is reached/addressed using/by Reputation concept (Blockchain-based SDN-enabled network architecture) [Zeng et al. 2022] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: Intelligent Blockchain Based secure routing for multidomain SDN Enabled IoT networks, 2022, https://doi.org/10.1155/2022/5693962.
The solution is composed by the architecture named as LoRaWAN, proposed in the paper A new Low-Cost LORAWAN power switch for smart farm applications, 2021, with D.O.I. doi.org/10.1109/SMC52423.2021.9659076 (link https://doi.org/10.1109/SMC52423.2021.9659076). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and LoRaWAN.
Solution Architecture LoRaWAN: described as: .
Performance/Efficiency is reached/addressed using/by Long-Range Wide Area Network (LoRaWAN) (No description.).
Reference: A new Low-Cost LORAWAN power switch for smart farm applications, 2021, https://doi.org/10.1109/SMC52423.2021.9659076.
The solution is composed by the architecture named as Loosely Coupled IoT Platform and Localization Module, proposed in the paper A Self-Adaptive Bluetooth Indoor Localization System using LSTM-based Distance Estimator, 2020, with D.O.I. 10.1109/ICCCN49398.2020.9209674 (link https://ieeexplore.ieee.org/document/9209674). This solution addresses the following Quality Requirements: Compatibility,Functional Suitability,Interaction Capability,Maintainability,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Loosely Coupled IoT Platform and Localization Module.
Solution Architecture Loosely Coupled IoT Platform and Localization Module: described as: The proposed system architecture consists of two main components: the IoT platform and the localization module, connected by middleware. The IoT platform follows a hierarchical structure with layers for hardware (Bluetooth devices), data (MQTT-based RSSI parsing), business logic, display, and user interface. The localization module enhances trilateration with self-adaptive mechanisms, including an LSTM-based distance estimator, elastic radius intersecting, weighted centroid localization, and self-adaptive Kalman filtering for trajectory optimization..
Maintainability is reached/addressed using/by Google Remote Procedure Calls (gRPC) (Modularity - The IoT platform (server side) and the localization module (client side) are connected via a middleware called Google Remote Procedure Calls (gRPC).").
Interaction Capability is reached/addressed using/by Web Interface (Operability - Web interface for visualizing localization and trajectory tracking.).
Compatibility is reached/addressed using/by Google Remote Procedure Calls (gRPC) (Interoperability - The gRPC middleware supports

communication between the IoT platform and the localization module, enabling system integration and compatibility with other platforms.).
Functional Suitability is reached/addressed using/by Kalman filtering (KF) (Kalman Filtering (KF) is an algorithm that estimates the state of a dynamic system by combining noisy sensor measurements with predictions. It recursively updates estimates based on new data, minimizing errors and improving accuracy, making it useful in tracking, navigation, and signal processing.).
Reference: A Self-Adaptive Bluetooth Indoor Localization System using LSTM-based Distance Estimator, 2020, https://ieeexplore.ieee.org/document/9209674.
The solution is composed by the architecture named as Microservices, proposed in the paper Design of scalable IoT architecture based on AWS for smart livestock, 2021, with D.O.I. doi.org/10.3390/ani11092697 (link https://doi.org/10.3390/ani11092697). This solution addresses the following Quality Requirements: Flexibility,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Microservices.
Solution Architecture Microservices: described as: .
Flexibility is reached/addressed using/by Amazon Web Services (AWS) (No description.).
Reference: Design of scalable IoT architecture based on AWS for smart livestock, 2021, https://doi.org/10.3390/ani11092697.
The solution is composed by the architecture named as Microservices, proposed in the paper A Lightweight User Interface for Smart Charging of Electric Vehicles: A Real-World Application, 2021, with D.O.I. 10.1109/ICSGCE52779.2021.9621604 (link https://ieeexplore.ieee.org/document/9621604). This solution addresses the following Quality Requirements: Compatibility,Reliability,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Microservices.
Solution Architecture Microservices: described as: Microservice-Based IoT Architecture integrated with Edge-Cloud Computing. Key components include: Charging Session Handler (CSH): Manages charging sessions through MQTT messages; PLC-Controlled Charging Station: Executes charging commands; Web-Based User Interface: Built with Streamlit for real-time user control; and Cloud Integration: Kubernetes ensures scalability and availability. This modular design supports distributed processing, real-time control, and seamless EV charging management..
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (Interoperability - The MQTT server stablishes communication between the web app, the Session Manager, and the Programmable Logic Controller (PLC) via JavaScript Object Notation (JSON) encoded messages.).
Reliability is reached/addressed using/by Docker containers (Availability - It is deployed in a Docker container on Kubernetes to provide a continuously available service.).
Reference: A Lightweight User Interface for Smart Charging of Electric Vehicles: A Real-World Application, 2021, https://ieeexplore.ieee.org/document/9621604.
The solution is composed by the architecture named as Microservices, proposed in the paper An open source software architecture and ready-to-use components for health IoT, 2020, with D.O.I. 10.1109/CBMS49503.2020.00077 (link https://ieeexplore.ieee.org/document/9183119). This solution addresses the following Quality Requirements: Interaction Capability,Maintainability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Microservices.
Solution Architecture Microservices: described as: Microservices-based software architecture for Health IoT, inspired by best practices from sources like Microsoft's microservices guidelines. The architecture utilizes open-source components for various functionalities. Key components include an API Gateway (Express Gateway) acting as a single entry point, a messaging channel (RabbitMQ) for asynchronous communication between microservices, and a security solution leveraging Vault and Consul for key management and high availability. The architecture is instantiated in the OCARIoT project, where it manages data from wearables, sensors, and mobile apps, processing data and triggering personalized interventions..
Interaction Capability is reached/addressed using/by User-friendly dashboard (Operability - User-friendly dashboard for health professionals: Health professionals are able to do a follow-up through the dashboard and also to receive tips and recommendations. Accessibility Open access through mobile and web platforms. The OCARIoT Dashboard visualizes real-time updates from mobile devices and web interfaces.).
Interaction Capability is reached/addressed using/by Intuitive mobile apps (Learnability - Intuitive mobile apps for health monitoring The Data Acquisition App manages the collection of data from different devices and shows key information for health professionals.).
Maintainability is reached/addressed using/by Component isolation through microservices (Modularity - Open Source Software Architecture Component isolation through microservices. Microservices can be implemented in different languages, using different databases, ensuring modular architecture.).
Reference: An open source software architecture and ready-to-use components for health IoT, 2020, https://ieeexplore.ieee.org/document/9183119.
The solution is composed by the architecture named as Multi-layered Cloud-Edge, proposed in the paper Securing smart healthcare system with edge computing, 2021, with D.O.I. doi.org/10.1016/j.cose.2021.102353 (link https://doi.org/10.1016/j.cose.2021.102353). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Multi-layered Cloud-Edge.
Solution Architecture Multi-layered Cloud-Edge: described as: .
Performance/Efficiency is reached/addressed using/by Edge computing (No description.).
Security is reached/addressed using/by Access Control System (ACS) and Searchable Encryption Module [Singh and Chatterjee 2021] (No description.).
Reference: Securing smart healthcare system with edge computing, 2021, https://doi.org/10.1016/j.cose.2021.102353.
The solution is composed by the architecture named as SAPPARCHI (Smart City), proposed in the paper SAPPARCHI: an Osmotic Platform to Execute Scalable Applications on Smart City Environments, 2021, with D.O.I. doi.org/10.1109/CLOUD55607.2022.00051 (link https://doi.org/10.1109/CLOUD55607.2022.00051). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Security,.

The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and SAPPARCHI (Smart City).
Solution Architecture SAPPARCHI (Smart City): described as: .
Flexibility is reached/addressed using/by Serverless on Edge and Microservice in Fog (No description.).
Maintainability is reached/addressed using/by Serverless on Edge and Microservice in Fog (No description.).
Security is reached/addressed using/by Serverless on Edge and Microservice in Fog (No description.).
Reference: SAPPARCHI: an Osmotic Platform to Execute Scalable Applications on Smart City Environments, 2021, https://doi.org/10.1109/CLOUD55607.2022.00051.
The solution is composed by the architecture named as SDN based, proposed in the paper An efficient Counter Based DDOS attack detection framework leveraging software defined IoT (SD-IoT), 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3043082 (link https://doi.org/10.1109/ACCESS.2020.3043082). This solution addresses the following Quality Requirements: Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and SDN based.
Solution Architecture SDN based: described as: .
Security is reached/addressed using/by Counter-based DDoS Attack Detection (C-DAD) application for Software Defined Networks (SDN) [Bhayo et al. 2020] (No description.).
Reference: An efficient Counter Based DDOS attack detection framework leveraging software defined IoT (SD-IoT), 2020, https://doi.org/10.1109/ACCESS.2020.3043082.
The solution is composed by the architecture named as Secure Publish-Subscribe, proposed in the paper Publish Subscribe System Security Requirement: A Case Study for V2V Communication, 2024, with D.O.I. 10.1109/OJCS.2024.3442921 (link https://www.scopus.com/record/display.uri?eid=2-s2.0-85201290507&doi=10.1109%2fOJCS.2024.3442921&origin=inward&txGid=43ee5fbd5bae9bbdf927871aab51aa4d). This solution addresses the following Quality Requirements: Reliability,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Secure Publish-Subscribe.
Solution Architecture Secure Publish-Subscribe: described as: This solution focuses on ensuring secure, reliable, and efficient communication between vehicles using a publish-subscribe messaging system. It integrates brokers, clients, and publishers/subscribers with advanced security features such as access control, encryption, authentication, and message integrity verification..
Reliability is reached/addressed using/by Duplicate ClientID Handling (Availability - The broker incorporates a security measure called 'Handling Duplicate ClientID' to guarantee the continuous availability of services.).
Security is reached/addressed using/by Cryptography (Encryption) (Confidentiality - The publisher will employ cryptographic techniques and key sizes to ensure the confidentiality of the payload data throughout the whole communication process.).
Security is reached/addressed using/by Data Origin Authentication (Authenticity - The subscriber should have the capability to offer the end-user a method, such as a digital signature or Message Authentication Code (MAC), to authenticate the origin of information.).
Security is reached/addressed using/by Digital Signatures and Hashing (Integrity - The publisher transmits the information to the subscriber... using digital signature, MAC, and hashing to identify any abnormalities.).
Reference: Publish Subscribe System Security Requirement: A Case Study for V2V Communication, 2024, https://www.scopus.com/record/display.uri?eid=2-s2.0-85201290507&doi=10.1109%2fOJCS.2024.3442921&origin=inward&txGid=43ee5fbd5bae9bbdf927871aab51aa4d.
The solution is composed by the architecture named as Smart City, proposed in the paper SmartGC: a software architecture for garbage collection in smart cities, 2020, with D.O.I. dx.doi.org/10.1504/IJBIC.2020.109675 (link https://dx.doi.org/10.1504/IJBIC.2020.109675). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Maintainability,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Smart City.
Solution Architecture Smart City: described as: .
Compatibility is reached/addressed using/by REST services (No description.).
Flexibility is reached/addressed using/by REST services (No description.).
Maintainability is reached/addressed using/by FIWARE Platform [www.fiware.org] (No description.).
Security is reached/addressed using/by HTTP reverse proxy to protect REST services (No description.).
Reference: SmartGC: a software architecture for garbage collection in smart cities, 2020, https://dx.doi.org/10.1504/IJBIC.2020.109675.

The solution is composed by the architecture named as A4IoT (Anyplace 4.0 IoT), proposed in the paper The Anyplace 4.0 IoT Localization Architecture, 2020, with D.O.I. 10.1109/MDM48529.2020.00045 (link https://ieeexplore.ieee.org/document/9162205). This solution addresses the following Quality Requirements: Flexibility,Interaction Capability,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and A4IoT (Anyplace 4.0 IoT).
Solution Architecture A4IoT (Anyplace 4.0 IoT): described as: It is an open-source architecture designed for IoT-based indoor localization using technologies like Wi-Fi, BLE, Cellular, UWB, and Computer Vision (CV). The system supports edge computing, timeseries databases, and crowdsourcing to manage localization tasks across various smart spaces, such as factories, healthcare facilities, and emergency response environments. Its modular and scalable design ensures integration across platforms like Android, Linux, Windows, and Robot OS, making it adaptable to diverse IoT ecosystems..
Performance/Efficiency is reached/addressed using/by Time-Series Database (InfluxDB) (Time Behavior - With temporal queries we can now consume high-volume input streams to provide real-time IoT tracking.").

Reliability is reached/addressed using/by Multi-node Cluster Deployment (Fault Tolerance - The production environment uses a 3-node cluster with two replicas, for both the database engines and the DFS (Distributed Filesystem). This setup allows for a full operation with just a single node active at any time.").
Security is reached/addressed using/by SSL certificates - Encrypted Communications (Confidentiality - All communication is encrypted even in internal networks, as A4IoT image automatically creates and uses SSL certificates.).
Interaction Capability is reached/addressed using/by Web-based Dashboard (Operability - Web Apps built with HTML5, CSS3, and AngularJS.").
Flexibility is reached/addressed using/by Docker containers (Adaptability - It utilizes Docker to enable the deployment of single-node or multi-node configurations, regardless of any dependencies or the underlying OS.).
Reference: The Anyplace 4.0 IoT Localization Architecture, 2020, https://ieeexplore.ieee.org/document/9162205.
The solution is composed by the architecture named as Agricultural IoT Reference Architecture (AITRA), proposed in the paper Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3031634 (link https://doi.org/10.1109/ACCESS.2020.3031634). This solution addresses the following Quality Requirements: Compatibility,Reliability,Security,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Agricultural IoT Reference Architecture (AITRA).
Solution Architecture Agricultural IoT Reference Architecture (AITRA): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Reliability is reached/addressed using/by Edge gateway (No description.).
Security is reached/addressed using/by Authentication, authorization, and data access control (No description.).
Reference: Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, https://doi.org/10.1109/ACCESS.2020.3031634.
The solution is composed by the architecture named as Blockchain-based Zero Trust, proposed in the paper Blockchain-Based Zero Trust on the Edge, 2023, with D.O.I. 10.1109/CSCI62032.2023.00167 (link https://ieeexplore.ieee.org/document/10590526). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Performance/Efficiency,Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Blockchain-based Zero Trust.
Solution Architecture Blockchain-based Zero Trust: described as: The architecture proposed in "Blockchain-Based Zero Trust on the Edge" integrates Zero Trust principles with Blockchain to ensure secure access and trust management for Edge Computing in IoT environments. It focuses on continuous verification of users and devices, ensuring that all access is authenticated before being granted. Blockchain is used for decentralized trust management, maintaining an immutable record of activities, which enhances security and transparency. This setup is optimized for edge computing, reducing latency by processing data locally while maintaining the flexibility to scale across diverse IoT applications, such as smart cities and industrial IoT..
Security is reached/addressed using/by Zero Trust Architecture (ZTA) (Confidentiality - Zero Trust Architecture (ZTA)
ZTA ensures continuous verification and monitoring of all network traffic, restricting unauthorized access.).
Security is reached/addressed using/by Blockchain Technology (Integrity - Blockchain serves as an immutable ledger for storing requests, ensuring data integrity and trustworthiness.).
Performance/Efficiency is reached/addressed using/by Redis Message Broker (Time Behavior - Redis Message Broker for Asynchronous Processing [redis.io]
Asynchronous processing for write operations improves system responsiveness under high loads.).
Maintainability is reached/addressed using/by Component isolation through microservices (Modularity -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.
).
Flexibility is reached/addressed using/by Component isolation through microservices (Scalability -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.).
Reference: Blockchain-Based Zero Trust on the Edge, 2023, https://ieeexplore.ieee.org/document/10590526.
The solution is composed by the architecture named as Blockchain-enabled, proposed in the paper An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, with D.O.I. doi.org/10.1109/TSC.2020.2966970 (link https://doi.org/10.1109/TSC.2020.2966970). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Blockchain-enabled.
Solution Architecture Blockchain-enabled: described as: .
Performance/Efficiency is reached/addressed using/by Routing protocol designed for IoT devices in the SDN controller [Yazdinejad et al. 2020] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, https://doi.org/10.1109/TSC.2020.2966970.
The solution is composed by the architecture named as Cloud-based Secure Biometric, proposed in the paper BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, with D.O.I. doi.org/10.1016/j.jksuci.2017.07.001 (link https://doi.org/10.1016/j.jksuci.2017.07.001). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Cloud-based Secure Biometric.
Solution Architecture Cloud-based Secure Biometric: described as: .
Flexibility is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).

Performance/Efficiency is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Reliability is reached/addressed using/by MapReduce framework (No description.).
Security is reached/addressed using/by Cloud-based biometric authentication [Shakil et al. 2020] (No description.).
Reference: BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, https://doi.org/10.1016/j.jksuci.2017.07.001.
The solution is composed by the architecture named as Deep Learning based, proposed in the paper HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, with D.O.I. doi.org/10.1016/j.future.2019.10.043 (link https://doi.org/10.1016/j.future.2019.10.043). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Deep Learning based.
Solution Architecture Deep Learning based: described as: .
Performance/Efficiency is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Security is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Reference: HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, https://doi.org/10.1016/j.future.2019.10.043.
The solution is composed by the architecture named as Distributed Blockchain-SDN based, proposed in the paper DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3039113 (link https://doi.org/10.1109/ACCESS.2020.3039113). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Distributed Blockchain-SDN based.
Solution Architecture Distributed Blockchain-SDN based: described as: .
Performance/Efficiency is reached/addressed using/by Decentralized blockchain nodes, dynamic SDN controllers, and edge computing (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, https://doi.org/10.1109/ACCESS.2020.3039113.
The solution is composed by the architecture named as Distributed Edge-Cloud, proposed in the paper Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, with D.O.I. doi.org/10.1016/j.future.2020.05.042 (link https://doi.org/10.1016/j.future.2020.05.042). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Distributed Edge-Cloud.
Solution Architecture Distributed Edge-Cloud: described as: .
Security is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Reference: Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, https://doi.org/10.1016/j.future.2020.05.042.
The solution is composed by the architecture named as Edge Computing-based Fault Tolerant Framework, proposed in the paper Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, with D.O.I. doi.org/10.1109/IWCMC48107.2020.9148269 (link https://doi.org/10.1109/IWCMC48107.2020.9148269). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Edge Computing-based Fault Tolerant Framework.
Solution Architecture Edge Computing-based Fault Tolerant Framework: described as: .
Performance/Efficiency is reached/addressed using/by Kubernetes framework (No description.).
Flexibility is reached/addressed using/by Docker containers (No description.).
Reference: Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, https://doi.org/10.1109/IWCMC48107.2020.9148269.
The solution is composed by the architecture named as Event-driven IoT, proposed in the paper Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, with D.O.I. doi.org/10.1007/s10586-020-03189-w (link https://doi.org/10.1007/s10586-020-03189-w). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Event-driven IoT.
Solution Architecture Event-driven IoT: described as: .
Performance/Efficiency is reached/addressed using/by Complex Event Processing (CEP) [Rahmani et al. 2020] (No description.).
Reliability is reached/addressed using/by TCP-IP protocol on Client/Server architecture for data communication (No description.).
Reference: Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, https://doi.org/10.1007/s10586-020-03189-w.
The solution is composed by the architecture named as Fog Computing based, proposed in the paper FOG-based architecture and load balancing methodology for health monitoring systems, 2021, with D.O.I. doi.org/10.1109/ACCESS.2021.3094033 (link https://doi.org/10.1109/ACCESS.2021.3094033). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Fog

Computing based.
Solution Architecture Fog Computing based: described as: .
Performance/Efficiency is reached/addressed using/by Load Balancing Scheme (LBS) - Distributed workload to all fog nodes [Asghar et al. 2021] (No description.).
Reference: FOG-based architecture and load balancing methodology for health monitoring systems, 2021, https://doi.org/10.1109/ACCESS.2021.3094033.
The solution is composed by the architecture named as Geographic-based, proposed in the paper A platform for integrating heterogeneous data and developing smart city applications, 2022, with D.O.I. doi.org/10.1016/j.future.2021.10.030 (link https://doi.org/10.1016/j.future.2021.10.030). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Geographic-based.
Solution Architecture Geographic-based: described as: .
Security is reached/addressed using/by FIWARE Platform [www.fiware.org] (Authentication, authorization, and data access control) (No description.).
Flexibility is reached/addressed using/by Virtualization (No description.).
Compatibility is reached/addressed using/by RESTFul APIs (No description.).
Reference: A platform for integrating heterogeneous data and developing smart city applications, 2022, https://doi.org/10.1016/j.future.2021.10.030.
The solution is composed by the architecture named as Hybrid (Centralized and Distributed), proposed in the paper Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, with D.O.I. doi.org/10.1016/j.procs.2021.03.030 (link https://doi.org/10.1016/j.procs.2021.03.030). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid (Centralized and Distributed).
Solution Architecture Hybrid (Centralized and Distributed): described as: .
Performance/Efficiency is reached/addressed using/by Dynamic allocation of the resources using Software Defined Networking (SDN) [Chudhary and Sharma 2021] (No description.).
Reference: Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, https://doi.org/10.1016/j.procs.2021.03.030.
The solution is composed by the architecture named as Hybrid Cloud-Fog Computing, proposed in the paper Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, with D.O.I. doi.org/10.1007/s10586-021-03375-4 (link https://link.springer.com/article/10.1007/s10586-021-03375-4). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid Cloud-Fog Computing.
Solution Architecture Hybrid Cloud-Fog Computing: described as: .
Reliability is reached/addressed using/by Fog layer in two levels: 1-Intra mobility (User network): Nodes (Data transfer, device management, temporary local storage, notifications, and data consistency). 2-Extra mobility (Different domains): Smart Gateways (Local processing and database) (No description.).
Performance/Efficiency is reached/addressed using/by Fog computing (No description.).
Reference: Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, https://link.springer.com/article/10.1007/s10586-021-03375-4.
The solution is composed by the architecture named as Industrial Internet of Things (IIoT), proposed in the paper A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Performance/Efficiency,Security,.
The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and Industrial Internet of Things (IIoT).
Solution Architecture Industrial Internet of Things (IIoT): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Flexibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Performance/Efficiency is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).
Security is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).
Reference: A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, https://doi.org/10.3390/s20195603.
The solution is composed by the architecture named as Layered, proposed in the paper An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, with D.O.I. doi.org/10.1016/j.future.2020.07.047 (link https://doi.org/10.1016/j.future.2020.07.047). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.
Solution Architecture Layered: described as: .
Flexibility is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).
Performance/Efficiency is reached/addressed using/by Big data analytics engine [Ali et al. 2021] (No description.).
Security is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).
Reference: An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, https://doi.org/10.1016/j.future.2020.07.047.
The solution is composed by the architecture named as Layered, proposed in the paper A novel architecture for the Internet of Things based E-

Health systems, 2020, with D.O.I. doi.org/10.1166/jmihi.2020.3184 (link http://dx.doi.org/10.1166/jmihi.2020.3184). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.

The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.

Solution Architecture Layered: described as: .

Performance/Efficiency is reached/addressed using/by IoT objects connected at the edge of the network (No description.).

Security is reached/addressed using/by Blockchain Technology (No description.).

Reference: A novel architecture for the Internet of Things based E-Health systems, 2020, http://dx.doi.org/10.1166/jmihi.2020.3184.

The solution is composed by the architecture named as Layered, proposed in the paper Software architecture quality attributes of a layered sensor-based IoT system, 2021, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Reliability,Security,.

The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and Layered.

Solution Architecture Layered: described as: .

Flexibility is reached/addressed using/by Messaging protocol brokers and at the client side, web-based applications (No description.).

Maintainability is reached/addressed using/by Loosely coupled components (No description.).

Reliability is reached/addressed using/by Open source software solutions and modular architecture design with clear relationships between components (No description.).

Security is reached/addressed using/by IoT devices authentication and data encryption (No description.).

Reference: Software architecture quality attributes of a layered sensor-based IoT system, 2021, https://doi.org/10.3390/s20195603.

The solution is composed by the architecture named as Layered, proposed in the paper Cloud-FOG assisted Energy Efficient Architectural Paradigm for disaster evacuation, 2022, with D.O.I. doi.org/10.1016/j.is.2021.101732 (link https://doi.org/10.1016/j.is.2021.101732). This solution addresses the following Quality Requirements: Performance/Efficiency,.

The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Layered.

Solution Architecture Layered: described as: .

Performance/Efficiency is reached/addressed using/by Algorithm for Fog layer data selection (Data uniqueness and only important data) for transmission [Kaur et al. 2022] (No description.).

Reference: Cloud-FOG assisted Energy Efficient Architectural Paradigm for disaster evacuation, 2022, https://doi.org/10.1016/j.is.2021.101732.

The solution is composed by the architecture named as Layered, proposed in the paper An agile AI and IoT-Augmented Smart Farming: a Cost-Effective cognitive weather station, 2021, with D.O.I. doi.org/10.3390/agriculture12010035 (link https://doi.org/10.3390/agriculture12010035). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Interaction Capability,Maintainability,Performance/Efficiency,.

The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Layered.

Solution Architecture Layered: described as: .

Compatibility is reached/addressed using/by Docker containers (No description.).

Flexibility is reached/addressed using/by Docker containers (No description.).

Interaction Capability is reached/addressed using/by Remote access to the deployed platform and its components (Web App and SSH) (No description.).

Maintainability is reached/addressed using/by Docker containers (No description.).

Performance/Efficiency is reached/addressed using/by Docker containers (No description.).

Reference: An agile AI and IoT-Augmented Smart Farming: a Cost-Effective cognitive weather station, 2021, https://doi.org/10.3390/agriculture12010035.

The solution is composed by the architecture named as Layered, proposed in the paper Smart Indoor Farms: Leveraging Technological Advancements to Power a Sustainable Agricultural Revolution, 2021, with D.O.I. doi.org/10.3390/agriengineering3040047 (link https://doi.org/10.3390/agriengineering3040047). This solution addresses the following Quality Requirements: .

The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Layered.

Solution Architecture Layered: described as: .

Reference: Smart Indoor Farms: Leveraging Technological Advancements to Power a Sustainable Agricultural Revolution, 2021, https://doi.org/10.3390/agriengineering3040047.

The solution is composed by the architecture named as Layered Blockchain and AI-enabled, proposed in the paper A blockchain and artificial intelligence-enabled smart IoT framework for sustainable city, 2022, with D.O.I. doi.org/10.1002/int.22852 (link https://doi.org/10.1002/int.22852). This solution addresses the following Quality Requirements: Security,.

The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Layered Blockchain and AI-enabled.

Solution Architecture Layered Blockchain and AI-enabled: described as: .

Security is reached/addressed using/by Blockchain Technology (No description.).

Reference: A blockchain and artificial intelligence-enabled smart IoT framework for sustainable city, 2022, https://doi.org/10.1002/int.22852.

The solution is composed by the architecture named as Layered Blockchain-Based SDN, proposed in the paper Intelligent Blockchain Based secure routing for multidomain SDN Enabled IoT networks, 2022, with D.O.I. doi.org/10.1155/2022/5693962 (link https://doi.org/10.1155/2022/5693962). This solution addresses the following Quality Requirements: Reliability,Security,.

The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Layered

Blockchain-Based SDN.
Solution Architecture Layered Blockchain-Based SDN: described as: .
Reliability is reached/addressed using/by Reputation concept (Blockchain-based SDN-enabled network architecture) [Zeng et al. 2022] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: Intelligent Blockchain Based secure routing for multidomain SDN Enabled IoT networks, 2022, https://doi.org/10.1155/2022/5693962.
The solution is composed by the architecture named as LoRaWAN, proposed in the paper A new Low-Cost LORAWAN power switch for smart farm applications, 2021, with D.O.I. doi.org/10.1109/SMC52423.2021.9659076 (link https://doi.org/10.1109/SMC52423.2021.9659076). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and LoRaWAN.
Solution Architecture LoRaWAN: described as: .
Performance/Efficiency is reached/addressed using/by Long-Range Wide Area Network (LoRaWAN) (No description.).
Reference: A new Low-Cost LORAWAN power switch for smart farm applications, 2021, https://doi.org/10.1109/SMC52423.2021.9659076.
The solution is composed by the architecture named as Loosely Coupled IoT Platform and Localization Module, proposed in the paper A Self-Adaptive Bluetooth Indoor Localization System using LSTM-based Distance Estimator, 2020, with D.O.I. 10.1109/ICCCN49398.2020.9209674 (link https://ieeexplore.ieee.org/document/9209674). This solution addresses the following Quality Requirements: Compatibility,Functional Suitability,Interaction Capability,Maintainability,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Loosely Coupled IoT Platform and Localization Module.
Solution Architecture Loosely Coupled IoT Platform and Localization Module: described as: The proposed system architecture consists of two main components: the IoT platform and the localization module, connected by middleware. The IoT platform follows a hierarchical structure with layers for hardware (Bluetooth devices), data (MQTT-based RSSI parsing), business logic, display, and user interface. The localization module enhances trilateration with self-adaptive mechanisms, including an LSTM-based distance estimator, elastic radius intersecting, weighted centroid localization, and self-adaptive Kalman filtering for trajectory optimization..
Maintainability is reached/addressed using/by Google Remote Procedure Calls (gRPC) (Modularity - The IoT platform (server side) and the localization module (client side) are connected via a middleware called Google Remote Procedure Calls (gRPC).").
Interaction Capability is reached/addressed using/by Web Interface (Operability - Web interface for visualizing localization and trajectory tracking.).
Compatibility is reached/addressed using/by Google Remote Procedure Calls (gRPC) (Interoperability - The gRPC middleware supports communication between the IoT platform and the localization module, enabling system integration and compatibility with other platforms.).
Functional Suitability is reached/addressed using/by Kalman filtering (KF) (Kalman Filtering (KF) is an algorithm that estimates the state of a dynamic system by combining noisy sensor measurements with predictions. It recursively updates estimates based on new data, minimizing errors and improving accuracy, making it useful in tracking, navigation, and signal processing.).
Reference: A Self-Adaptive Bluetooth Indoor Localization System using LSTM-based Distance Estimator, 2020, https://ieeexplore.ieee.org/document/9209674.
The solution is composed by the architecture named as Microservices, proposed in the paper Design of scalable IoT architecture based on AWS for smart livestock, 2021, with D.O.I. doi.org/10.3390/ani11092697 (link https://doi.org/10.3390/ani11092697). This solution addresses the following Quality Requirements: Flexibility,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Microservices.
Solution Architecture Microservices: described as: .
Flexibility is reached/addressed using/by Amazon Web Services (AWS) (No description.).
Reference: Design of scalable IoT architecture based on AWS for smart livestock, 2021, https://doi.org/10.3390/ani11092697.
The solution is composed by the architecture named as Microservices, proposed in the paper A Lightweight User Interface for Smart Charging of Electric Vehicles: A Real-World Application, 2021, with D.O.I. 10.1109/ICSGCE52779.2021.9621604 (link https://ieeexplore.ieee.org/document/9621604). This solution addresses the following Quality Requirements: Compatibility,Reliability,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Microservices.
Solution Architecture Microservices: described as: Microservice-Based IoT Architecture integrated with Edge-Cloud Computing. Key components include: Charging Session Handler (CSH): Manages charging sessions through MQTT messages; PLC-Controlled Charging Station: Executes charging commands; Web-Based User Interface: Built with Streamlit for real-time user control; and Cloud Integration: Kubernetes ensures scalability and availability. This modular design supports distributed processing, real-time control, and seamless EV charging management..
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (Interoperability - The MQTT server stablishes communication between the web app, the Session Manager, and the Programmable Logic Controller (PLC) via JavaScript Object Notation (JSON) encoded messages.).
Reliability is reached/addressed using/by Docker containers (Availability - It is deployed in a Docker container on Kubernetes to provide a continuously available service.).
Reference: A Lightweight User Interface for Smart Charging of Electric Vehicles: A Real-World Application, 2021, https://ieeexplore.ieee.org/document/9621604.
The solution is composed by the architecture named as Microservices, proposed in the paper An open source software architecture and ready-to-use components for health IoT, 2020, with D.O.I. 10.1109/CBMS49503.2020.00077 (link https://ieeexplore.ieee.org/document/9183119). This solution addresses the following Quality Requirements: Interaction Capability,Maintainability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and

Microservices.
Solution Architecture Microservices: described as: Microservices-based software architecture for Health IoT, inspired by best practices from sources like Microsoft's microservices guidelines. The architecture utilizes open-source components for various functionalities. Key components include an API Gateway (Express Gateway) acting as a single entry point, a messaging channel (RabbitMQ) for asynchronous communication between microservices, and a security solution leveraging Vault and Consul for key management and high availability. The architecture is instantiated in the OCARIoT project, where it manages data from wearables, sensors, and mobile apps, processing data and triggering personalized interventions..
Interaction Capability is reached/addressed using/by User-friendly dashboard (Operability - User-friendly dashboard for health professionals: Health professionals are able to do a follow-up through the dashboard and also to receive tips and recommendations. Accessibility Open access through mobile and web platforms. The OCARIoT Dashboard visualizes real-time updates from mobile devices and web interfaces.).
Interaction Capability is reached/addressed using/by Intuitive mobile apps (Learnability - Intuitive mobile apps for health monitoring The Data Acquisition App manages the collection of data from different devices and shows key information for health professionals.).
Maintainability is reached/addressed using/by Component isolation through microservices (Modularity - Open Source Software Architecture Component isolation through microservices. Microservices can be implemented in different languages, using different databases, ensuring modular architecture.).
Reference: An open source software architecture and ready-to-use components for health IoT, 2020, https://ieeexplore.ieee.org/document/9183119.
The solution is composed by the architecture named as Multi-layered Cloud-Edge, proposed in the paper Securing smart healthcare system with edge computing, 2021, with D.O.I. doi.org/10.1016/j.cose.2021.102353 (link https://doi.org/10.1016/j.cose.2021.102353). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Multi-layered Cloud-Edge.
Solution Architecture Multi-layered Cloud-Edge: described as: .
Performance/Efficiency is reached/addressed using/by Edge computing (No description.).
Security is reached/addressed using/by Access Control System (ACS) and Searchable Encryption Module [Singh and Chatterjee 2021] (No description.).
Reference: Securing smart healthcare system with edge computing, 2021, https://doi.org/10.1016/j.cose.2021.102353.
The solution is composed by the architecture named as SAPPARCHI (Smart City), proposed in the paper SAPPARCHI: an Osmotic Platform to Execute Scalable Applications on Smart City Environments, 2021, with D.O.I. doi.org/10.1109/CLOUD55607.2022.00051 (link https://doi.org/10.1109/CLOUD55607.2022.00051). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and SAPPARCHI (Smart City).
Solution Architecture SAPPARCHI (Smart City): described as: .
Flexibility is reached/addressed using/by Serverless on Edge and Microservice in Fog (No description.).
Maintainability is reached/addressed using/by Serverless on Edge and Microservice in Fog (No description.).
Security is reached/addressed using/by Serverless on Edge and Microservice in Fog (No description.).
Reference: SAPPARCHI: an Osmotic Platform to Execute Scalable Applications on Smart City Environments, 2021, https://doi.org/10.1109/CLOUD55607.2022.00051.
The solution is composed by the architecture named as SDN based, proposed in the paper An efficient Counter Based DDOS attack detection framework leveraging software defined IoT (SD-IoT), 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3043082 (link https://doi.org/10.1109/ACCESS.2020.3043082). This solution addresses the following Quality Requirements: Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and SDN based.
Solution Architecture SDN based: described as: .
Security is reached/addressed using/by Counter-based DDoS Attack Detection (C-DAD) application for Software Defined Networks (SDN) [Bhayo et al. 2020] (No description.).
Reference: An efficient Counter Based DDOS attack detection framework leveraging software defined IoT (SD-IoT), 2020, https://doi.org/10.1109/ACCESS.2020.3043082.
The solution is composed by the architecture named as Secure Publish-Subscribe, proposed in the paper Publish Subscribe System Security Requirement: A Case Study for V2V Communication, 2024, with D.O.I. 10.1109/OJCS.2024.3442921 (link https://www.scopus.com/record/display.uri?eid=2-s2.0-85201290507&doi=10.1109%2fOJCS.2024.3442921&origin=inward&txGid=43ee5fbd5bae9bbdf927871aab51aa4d). This solution addresses the following Quality Requirements: Reliability,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Secure Publish-Subscribe.
Solution Architecture Secure Publish-Subscribe: described as: This solution focuses on ensuring secure, reliable, and efficient communication between vehicles using a publish-subscribe messaging system. It integrates brokers, clients, and publishers/subscribers with advanced security features such as access control, encryption, authentication, and message integrity verification..
Reliability is reached/addressed using/by Duplicate ClientID Handling (Availability - The broker incorporates a security measure called 'Handling Duplicate ClientID' to guarantee the continuous availability of services.).
Security is reached/addressed using/by Cryptography (Encryption) (Confidentiality - The publisher will employ cryptographic techniques and key sizes to ensure the confidentiality of the payload data throughout the whole communication process.).
Security is reached/addressed using/by Data Origin Authentication (Authenticity - The subscriber should have the capability to offer the end-user a method, such as a digital signature or Message Authentication Code (MAC), to authenticate the origin of information.).
Security is reached/addressed using/by Digital Signatures and Hashing (Integrity - The publisher transmits the information to the subscriber... using

digital signature, MAC, and hashing to identify any abnormalities.).
Reference: Publish Subscribe System Security Requirement: A Case Study for V2V Communication, 2024, https://www.scopus.com/record/display.uri?eid=2-s2.0-85201290507&doi=10.1109%2fOJCS.2024.3442921&origin=inward&txGid=43ee5fbd5bae9bbdf927871aab51aa4d.
The solution is composed by the architecture named as Smart City, proposed in the paper SmartGC: a software architecture for garbage collection in smart cities, 2020, with D.O.I. dx.doi.org/10.1504/IJBIC.2020.109675 (link https://dx.doi.org/10.1504/IJBIC.2020.109675). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Maintainability,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Smart City.
Solution Architecture Smart City: described as: .
Compatibility is reached/addressed using/by REST services (No description.).
Flexibility is reached/addressed using/by REST services (No description.).
Maintainability is reached/addressed using/by FIWARE Platform [www.fiware.org] (No description.).
Security is reached/addressed using/by HTTP reverse proxy to protect REST services (No description.).
Reference: SmartGC: a software architecture for garbage collection in smart cities, 2020, https://dx.doi.org/10.1504/IJBIC.2020.109675.
The solution is composed by the architecture named as Smart City, proposed in the paper Design and application of fog computing and Internet of Things service platform for smart city, 2020, with D.O.I. doi.org/10.1016/j.future.2020.06.016 (link https://doi.org/10.1016/j.future.2020.06.016). This solution addresses the following Quality Requirements: Flexibility,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Smart City.
Solution Architecture Smart City: described as: .
Security is reached/addressed using/by Proxy Virtual Processor[Zhang 2020] (No description.).
Flexibility is reached/addressed using/by The implementation of a software-defined network (SDN) to enable the dynamic allocation of network resources according to demand [Zhang 2020] (No description.).
Reference: Design and application of fog computing and Internet of Things service platform for smart city, 2020, https://doi.org/10.1016/j.future.2020.06.016.

The solution is composed by the architecture named as A4IoT (Anyplace 4.0 IoT), proposed in the paper The Anyplace 4.0 IoT Localization Architecture, 2020, with D.O.I. 10.1109/MDM48529.2020.00045 (link https://ieeexplore.ieee.org/document/9162205). This solution addresses the following Quality Requirements: Flexibility,Interaction Capability,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and A4IoT (Anyplace 4.0 IoT).
Solution Architecture A4IoT (Anyplace 4.0 IoT): described as: It is an open-source architecture designed for IoT-based indoor localization using technologies like Wi-Fi, BLE, Cellular, UWB, and Computer Vision (CV). The system supports edge computing, timeseries databases, and crowdsourcing to manage localization tasks across various smart spaces, such as factories, healthcare facilities, and emergency response environments. Its modular and scalable design ensures integration across platforms like Android, Linux, Windows, and Robot OS, making it adaptable to diverse IoT ecosystems..
Performance/Efficiency is reached/addressed using/by Time-Series Database (InfluxDB) (Time Behavior - With temporal queries we can now consume high-volume input streams to provide real-time IoT tracking.").
Reliability is reached/addressed using/by Multi-node Cluster Deployment (Fault Tolerance - The production environment uses a 3-node cluster with two replicas, for both the database engines and the DFS (Distributed Filesystem). This setup allows for a full operation with just a single node active at any time.").
Security is reached/addressed using/by SSL certificates - Encrypted Communications (Confidentiality - All communication is encrypted even in internal networks, as A4IoT image automatically creates and uses SSL certificates.).
Interaction Capability is reached/addressed using/by Web-based Dashboard (Operability - Web Apps built with HTML5, CSS3, and AngularJS.").
Flexibility is reached/addressed using/by Docker containers (Adaptability - It utilizes Docker to enable the deployment of single-node or multi-node configurations, regardless of any dependencies or the underlying OS.).
Reference: The Anyplace 4.0 IoT Localization Architecture, 2020, https://ieeexplore.ieee.org/document/9162205.
The solution is composed by the architecture named as Agricultural IoT Reference Architecture (AITRA), proposed in the paper Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3031634 (link https://doi.org/10.1109/ACCESS.2020.3031634). This solution addresses the following Quality Requirements: Compatibility,Reliability,Security,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Agricultural IoT Reference Architecture (AITRA).
Solution Architecture Agricultural IoT Reference Architecture (AITRA): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Reliability is reached/addressed using/by Edge gateway (No description.).
Security is reached/addressed using/by Authentication, authorization, and data access control (No description.).
Reference: Laying the foundations for an IoT reference architecture for the agricultural application domain, 2020, https://doi.org/10.1109/ACCESS.2020.3031634.
The solution is composed by the architecture named as Blockchain-based Zero Trust, proposed in the paper Blockchain-Based Zero Trust on the Edge, 2023, with D.O.I. 10.1109/CSCI62032.2023.00167 (link https://ieeexplore.ieee.org/document/10590526). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Performance/Efficiency,Security,.

The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Blockchain-based Zero Trust.
Solution Architecture Blockchain-based Zero Trust: described as: The architecture proposed in "Blockchain-Based Zero Trust on the Edge" integrates Zero Trust principles with Blockchain to ensure secure access and trust management for Edge Computing in IoT environments. It focuses on continuous verification of users and devices, ensuring that all access is authenticated before being granted. Blockchain is used for decentralized trust management, maintaining an immutable record of activities, which enhances security and transparency. This setup is optimized for edge computing, reducing latency by processing data locally while maintaining the flexibility to scale across diverse IoT applications, such as smart cities and industrial IoT..
Security is reached/addressed using/by Zero Trust Architecture (ZTA) (Confidentiality - Zero Trust Architecture (ZTA)
ZTA ensures continuous verification and monitoring of all network traffic, restricting unauthorized access.).
Security is reached/addressed using/by Blockchain Technology (Integrity - Blockchain serves as an immutable ledger for storing requests, ensuring data integrity and trustworthiness.).
Performance/Efficiency is reached/addressed using/by Redis Message Broker (Time Behavior - Redis Message Broker for Asynchronous Processing [redis.io]
Asynchronous processing for write operations improves system responsiveness under high loads.).
Maintainability is reached/addressed using/by Component isolation through microservices (Modularity -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.
).
Flexibility is reached/addressed using/by Component isolation through microservices (Scalability -
The system is built as modular microservices, adding complexity but enhancing scalability and flexibility.).
Reference: Blockchain-Based Zero Trust on the Edge, 2023, https://ieeexplore.ieee.org/document/10590526.
The solution is composed by the architecture named as Blockchain-enabled, proposed in the paper An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, with D.O.I. doi.org/10.1109/TSC.2020.2966970 (link https://doi.org/10.1109/TSC.2020.2966970). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Blockchain-enabled.
Solution Architecture Blockchain-enabled: described as: .
Performance/Efficiency is reached/addressed using/by Routing protocol designed for IoT devices in the SDN controller [Yazdinejad et al. 2020] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: An Energy Efficient SDN controller architecture for IoT networks with Blockchain Based security, 2020, https://doi.org/10.1109/TSC.2020.2966970.
The solution is composed by the architecture named as Cloud-based Secure Biometric, proposed in the paper BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, with D.O.I. doi.org/10.1016/j.jksuci.2017.07.001 (link https://doi.org/10.1016/j.jksuci.2017.07.001). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Cloud-based Secure Biometric.
Solution Architecture Cloud-based Secure Biometric: described as: .
Flexibility is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Parallelized MapReduce programming model [Shakil et al. 2020] (No description.).
Reliability is reached/addressed using/by MapReduce framework (No description.).
Security is reached/addressed using/by Cloud-based biometric authentication [Shakil et al. 2020] (No description.).
Reference: BAMHealthCloud: A biometric authentication and data management system for healthcare data in the cloud, 2020, https://doi.org/10.1016/j.jksuci.2017.07.001.
The solution is composed by the architecture named as Deep Learning based, proposed in the paper HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, with D.O.I. doi.org/10.1016/j.future.2019.10.043 (link https://doi.org/10.1016/j.future.2019.10.043). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Deep Learning based.
Solution Architecture Deep Learning based: described as: .
Performance/Efficiency is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Security is reached/addressed using/by FogBus framework [Tuli et al. 2019] (No description.).
Reference: HealthFog: An ensemble deep learning-based Smart Healthcare System for Automatic Diagnosis of Heart Diseases in integrated IoT and fog computing environments., 2020, https://doi.org/10.1016/j.future.2019.10.043.
The solution is composed by the architecture named as Distributed Blockchain-SDN based, proposed in the paper DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3039113 (link https://doi.org/10.1109/ACCESS.2020.3039113). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Distributed Blockchain-SDN based.
Solution Architecture Distributed Blockchain-SDN based: described as: .
Performance/Efficiency is reached/addressed using/by Decentralized blockchain nodes, dynamic SDN controllers, and edge computing (No

description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: DISTB-ConDo: Distributed Blockchain Based IoT SDN model for smart condominium, 2020, https://doi.org/10.1109/ACCESS.2020.3039113.
The solution is composed by the architecture named as Distributed Edge-Cloud, proposed in the paper Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, with D.O.I. doi.org/10.1016/j.future.2020.05.042 (link https://doi.org/10.1016/j.future.2020.05.042). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Distributed Edge-Cloud.
Solution Architecture Distributed Edge-Cloud: described as: .
Security is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Performance/Efficiency is reached/addressed using/by Data communication framework integrating NDN-based IoT with Edge cloud (SHNIE) [Wang and Cai 2020] (No description.).
Reference: Secure healthcare monitoring framework integrating NDN-based IoT with edge cloud, 2020, https://doi.org/10.1016/j.future.2020.05.042.
The solution is composed by the architecture named as Edge Computing-based Fault Tolerant Framework, proposed in the paper Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, with D.O.I. doi.org/10.1109/IWCMC48107.2020.9148269 (link https://doi.org/10.1109/IWCMC48107.2020.9148269). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Edge Computing-based Fault Tolerant Framework.
Solution Architecture Edge Computing-based Fault Tolerant Framework: described as: .
Performance/Efficiency is reached/addressed using/by Kubernetes framework (No description.).
Flexibility is reached/addressed using/by Docker containers (No description.).
Reference: Edge Computing based Fault Tolerant Framework: A Case Study on Vehicular Networks. International Wireless Communications, 2020, https://doi.org/10.1109/IWCMC48107.2020.9148269.
The solution is composed by the architecture named as Event-driven IoT, proposed in the paper Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, with D.O.I. doi.org/10.1007/s10586-020-03189-w (link https://doi.org/10.1007/s10586-020-03189-w). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Event-driven IoT.
Solution Architecture Event-driven IoT: described as: .
Performance/Efficiency is reached/addressed using/by Complex Event Processing (CEP) [Rahmani et al. 2020] (No description.).
Reliability is reached/addressed using/by TCP-IP protocol on Client/Server architecture for data communication (No description.).
Reference: Event-driven IoT architecture for data analysis of reliable healthcare applications using complex event processing, 2020, https://doi.org/10.1007/s10586-020-03189-w.
The solution is composed by the architecture named as Fog Computing based, proposed in the paper FOG-based architecture and load balancing methodology for health monitoring systems, 2021, with D.O.I. doi.org/10.1109/ACCESS.2021.3094033 (link https://doi.org/10.1109/ACCESS.2021.3094033). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Fog Computing based.
Solution Architecture Fog Computing based: described as: .
Performance/Efficiency is reached/addressed using/by Load Balancing Scheme (LBS) - Distributed workload to all fog nodes [Asghar et al. 2021] (No description.).
Reference: FOG-based architecture and load balancing methodology for health monitoring systems, 2021, https://doi.org/10.1109/ACCESS.2021.3094033.
The solution is composed by the architecture named as Geographic-based, proposed in the paper A platform for integrating heterogeneous data and developing smart city applications, 2022, with D.O.I. doi.org/10.1016/j.future.2021.10.030 (link https://doi.org/10.1016/j.future.2021.10.030). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Geographic-based.
Solution Architecture Geographic-based: described as: .
Security is reached/addressed using/by FIWARE Platform [www.fiware.org] (Authentication, authorization, and data access control) (No description.).
Flexibility is reached/addressed using/by Virtualization (No description.).
Compatibility is reached/addressed using/by RESTFul APIs (No description.).
Reference: A platform for integrating heterogeneous data and developing smart city applications, 2022, https://doi.org/10.1016/j.future.2021.10.030.
The solution is composed by the architecture named as Hybrid (Centralized and Distributed), proposed in the paper Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, with D.O.I. doi.org/10.1016/j.procs.2021.03.030 (link https://doi.org/10.1016/j.procs.2021.03.030). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid (Centralized and Distributed).
Solution Architecture Hybrid (Centralized and Distributed): described as: .

Performance/Efficiency is reached/addressed using/by Dynamic allocation of the resources using Software Defined Networking (SDN) [Chudhary and Sharma 2021] (No description.).
Reference: Fog-cloud assisted framework for Heterogeneous Internet of Healthcare Things, 2021, https://doi.org/10.1016/j.procs.2021.03.030.
The solution is composed by the architecture named as Hybrid Cloud-Fog Computing, proposed in the paper Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, with D.O.I. doi.org/10.1007/s10586-021-03375-4 (link https://link.springer.com/article/10.1007/s10586-021-03375-4). This solution addresses the following Quality Requirements: Performance/Efficiency,Reliability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Hybrid Cloud-Fog Computing.
Solution Architecture Hybrid Cloud-Fog Computing: described as: .
Reliability is reached/addressed using/by Fog layer in two levels: 1-Intra mobility (User network): Nodes (Data transfer, device management, temporary local storage, notifications, and data consistency). 2-Extra mobility (Different domains): Smart Gateways (Local processing and database) (No description.).
Performance/Efficiency is reached/addressed using/by Fog computing (No description.).
Reference: Software architecture for IoT-based health-care systems with cloud/fog service model, 2021, https://link.springer.com/article/10.1007/s10586-021-03375-4.
The solution is composed by the architecture named as Industrial Internet of Things (IIoT), proposed in the paper A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Performance/Efficiency,Security,.
The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and Industrial Internet of Things (IIoT).
Solution Architecture Industrial Internet of Things (IIoT): described as: .
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Flexibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (No description.).
Performance/Efficiency is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).
Security is reached/addressed using/by Data Distribution Protocol (DDS) (No description.).
Reference: A software architecture for the Industrial Internet of Things - A Conceptual model, 2020, https://doi.org/10.3390/s20195603.
The solution is composed by the architecture named as Layered, proposed in the paper An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, with D.O.I. doi.org/10.1016/j.future.2020.07.047 (link https://doi.org/10.1016/j.future.2020.07.047). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.
Solution Architecture Layered: described as: .
Flexibility is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).
Performance/Efficiency is reached/addressed using/by Big data analytics engine [Ali et al. 2021] (No description.).
Security is reached/addressed using/by Amazon Simple Storage Service (Amazon S3) (No description.).
Reference: An intelligent healthcare monitoring framework using wearable sensors and social networking data, 2021, https://doi.org/10.1016/j.future.2020.07.047.
The solution is composed by the architecture named as Layered, proposed in the paper A novel architecture for the Internet of Things based E-Health systems, 2020, with D.O.I. doi.org/10.1166/jmihi.2020.3184 (link http://dx.doi.org/10.1166/jmihi.2020.3184). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Layered.
Solution Architecture Layered: described as: .
Performance/Efficiency is reached/addressed using/by IoT objects connected at the edge of the network (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: A novel architecture for the Internet of Things based E-Health systems, 2020, http://dx.doi.org/10.1166/jmihi.2020.3184.
The solution is composed by the architecture named as Layered, proposed in the paper Software architecture quality attributes of a layered sensor-based IoT system, 2021, with D.O.I. doi.org/10.3390/s20195603 (link https://doi.org/10.3390/s20195603). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Reliability,Security,.
The IoT domain of this solution is Industry 4.0. You can find all information about this solution in Knowledge base, filtering by Industry 4.0 and Layered.
Solution Architecture Layered: described as: .
Flexibility is reached/addressed using/by Messaging protocol brokers and at the client side, web-based applications (No description.).
Maintainability is reached/addressed using/by Loosely coupled components (No description.).
Reliability is reached/addressed using/by Open source software solutions and modular architecture design with clear relationships between components (No description.).
Security is reached/addressed using/by IoT devices authentication and data encryption (No description.).
Reference: Software architecture quality attributes of a layered sensor-based IoT system, 2021, https://doi.org/10.3390/s20195603.
The solution is composed by the architecture named as Layered, proposed in the paper Cloud-FOG assisted Energy Efficient Architectural Paradigm for disaster evacuation, 2022, with D.O.I. doi.org/10.1016/j.is.2021.101732 (link https://doi.org/10.1016/j.is.2021.101732). This solution addresses the following Quality Requirements: Performance/Efficiency,.

The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Layered.
Solution Architecture Layered: described as: .
Performance/Efficiency is reached/addressed using/by Algorithm for Fog layer data selection (Data uniqueness and only important data) for transmission [Kaur et al. 2022] (No description.).
Reference: Cloud-FOG assisted Energy Efficient Architectural Paradigm for disaster evacuation, 2022, https://doi.org/10.1016/j.is.2021.101732.
The solution is composed by the architecture named as Layered, proposed in the paper An agile AI and IoT-Augmented Smart Farming: a Cost-Effective cognitive weather station, 2021, with D.O.I. doi.org/10.3390/agriculture12010035 (link https://doi.org/10.3390/agriculture12010035). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Interaction Capability,Maintainability,Performance/Efficiency,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Layered.
Solution Architecture Layered: described as: .
Compatibility is reached/addressed using/by Docker containers (No description.).
Flexibility is reached/addressed using/by Docker containers (No description.).
Interaction Capability is reached/addressed using/by Remote access to the deployed platform and its components (Web App and SSH) (No description.).
Maintainability is reached/addressed using/by Docker containers (No description.).
Performance/Efficiency is reached/addressed using/by Docker containers (No description.).
Reference: An agile AI and IoT-Augmented Smart Farming: a Cost-Effective cognitive weather station, 2021, https://doi.org/10.3390/agriculture12010035.
The solution is composed by the architecture named as Layered, proposed in the paper Smart Indoor Farms: Leveraging Technological Advancements to Power a Sustainable Agricultural Revolution, 2021, with D.O.I. doi.org/10.3390/agriengineering3040047 (link https://doi.org/10.3390/agriengineering3040047). This solution addresses the following Quality Requirements: .
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Layered.
Solution Architecture Layered: described as: .
Reference: Smart Indoor Farms: Leveraging Technological Advancements to Power a Sustainable Agricultural Revolution, 2021, https://doi.org/10.3390/agriengineering3040047.
The solution is composed by the architecture named as Layered Blockchain and AI-enabled, proposed in the paper A blockchain and artificial intelligence-enabled smart IoT framework for sustainable city, 2022, with D.O.I. doi.org/10.1002/int.22852 (link https://doi.org/10.1002/int.22852). This solution addresses the following Quality Requirements: Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Layered Blockchain and AI-enabled.
Solution Architecture Layered Blockchain and AI-enabled: described as: .
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: A blockchain and artificial intelligence-enabled smart IoT framework for sustainable city, 2022, https://doi.org/10.1002/int.22852.
The solution is composed by the architecture named as Layered Blockchain-Based SDN, proposed in the paper Intelligent Blockchain Based secure routing for multidomain SDN Enabled IoT networks, 2022, with D.O.I. doi.org/10.1155/2022/5693962 (link https://doi.org/10.1155/2022/5693962). This solution addresses the following Quality Requirements: Reliability,Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Layered Blockchain-Based SDN.
Solution Architecture Layered Blockchain-Based SDN: described as: .
Reliability is reached/addressed using/by Reputation concept (Blockchain-based SDN-enabled network architecture) [Zeng et al. 2022] (No description.).
Security is reached/addressed using/by Blockchain Technology (No description.).
Reference: Intelligent Blockchain Based secure routing for multidomain SDN Enabled IoT networks, 2022, https://doi.org/10.1155/2022/5693962.
The solution is composed by the architecture named as LoRaWAN, proposed in the paper A new Low-Cost LORAWAN power switch for smart farm applications, 2021, with D.O.I. doi.org/10.1109/SMC52423.2021.9659076 (link https://doi.org/10.1109/SMC52423.2021.9659076). This solution addresses the following Quality Requirements: Performance/Efficiency,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and LoRaWAN.
Solution Architecture LoRaWAN: described as: .
Performance/Efficiency is reached/addressed using/by Long-Range Wide Area Network (LoRaWAN) (No description.).
Reference: A new Low-Cost LORAWAN power switch for smart farm applications, 2021, https://doi.org/10.1109/SMC52423.2021.9659076.
The solution is composed by the architecture named as Loosely Coupled IoT Platform and Localization Module, proposed in the paper A Self-Adaptive Bluetooth Indoor Localization System using LSTM-based Distance Estimator, 2020, with D.O.I. 10.1109/ICCCN49398.2020.9209674 (link https://ieeexplore.ieee.org/document/9209674). This solution addresses the following Quality Requirements: Compatibility,Functional Suitability,Interaction Capability,Maintainability,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and Loosely Coupled IoT Platform and Localization Module.
Solution Architecture Loosely Coupled IoT Platform and Localization Module: described as: The proposed system architecture consists of two main components: the IoT platform and the localization module, connected by middleware. The IoT platform follows a hierarchical structure with layers for hardware (Bluetooth devices), data (MQTT-based RSSI parsing), business logic, display, and user interface. The localization module enhances

trilateration with self-adaptive mechanisms, including an LSTM-based distance estimator, elastic radius intersecting, weighted centroid localization, and self-adaptive Kalman filtering for trajectory optimization..
Maintainability is reached/addressed using/by Google Remote Procedure Calls (gRPC) (Modularity - The IoT platform (server side) and the localization module (client side) are connected via a middleware called Google Remote Procedure Calls (gRPC).").
Interaction Capability is reached/addressed using/by Web Interface (Operability - Web interface for visualizing localization and trajectory tracking.).
Compatibility is reached/addressed using/by Google Remote Procedure Calls (gRPC) (Interoperability - The gRPC middleware supports communication between the IoT platform and the localization module, enabling system integration and compatibility with other platforms.).
Functional Suitability is reached/addressed using/by Kalman filtering (KF) (Kalman Filtering (KF) is an algorithm that estimates the state of a dynamic system by combining noisy sensor measurements with predictions. It recursively updates estimates based on new data, minimizing errors and improving accuracy, making it useful in tracking, navigation, and signal processing.).
Reference: A Self-Adaptive Bluetooth Indoor Localization System using LSTM-based Distance Estimator, 2020, https://ieeexplore.ieee.org/document/9209674.
The solution is composed by the architecture named as Microservices, proposed in the paper Design of scalable IoT architecture based on AWS for smart livestock, 2021, with D.O.I. doi.org/10.3390/ani11092697 (link https://doi.org/10.3390/ani11092697). This solution addresses the following Quality Requirements: Flexibility,.
The IoT domain of this solution is Smart Farm. You can find all information about this solution in Knowledge base, filtering by Smart Farm and Microservices.
Solution Architecture Microservices: described as: .
Flexibility is reached/addressed using/by Amazon Web Services (AWS) (No description.).
Reference: Design of scalable IoT architecture based on AWS for smart livestock, 2021, https://doi.org/10.3390/ani11092697.
The solution is composed by the architecture named as Microservices, proposed in the paper A Lightweight User Interface for Smart Charging of Electric Vehicles: A Real-World Application, 2021, with D.O.I. 10.1109/ICSGCE52779.2021.9621604 (link https://ieeexplore.ieee.org/document/9621604). This solution addresses the following Quality Requirements: Compatibility,Reliability,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Microservices.
Solution Architecture Microservices: described as: Microservice-Based IoT Architecture integrated with Edge-Cloud Computing. Key components include: Charging Session Handler (CSH): Manages charging sessions through MQTT messages; PLC-Controlled Charging Station: Executes charging commands; Web-Based User Interface: Built with Streamlit for real-time user control; and Cloud Integration: Kubernetes ensures scalability and availability. This modular design supports distributed processing, real-time control, and seamless EV charging management..
Compatibility is reached/addressed using/by Publish/Subscribe to standard messaging pattern (Interoperability - The MQTT server stablishes communication between the web app, the Session Manager, and the Programmable Logic Controller (PLC) via JavaScript Object Notation (JSON) encoded messages.).
Reliability is reached/addressed using/by Docker containers (Availability - It is deployed in a Docker container on Kubernetes to provide a continuously available service.).
Reference: A Lightweight User Interface for Smart Charging of Electric Vehicles: A Real-World Application, 2021, https://ieeexplore.ieee.org/document/9621604.
The solution is composed by the architecture named as Microservices, proposed in the paper An open source software architecture and ready-to-use components for health IoT, 2020, with D.O.I. 10.1109/CBMS49503.2020.00077 (link https://ieeexplore.ieee.org/document/9183119). This solution addresses the following Quality Requirements: Interaction Capability,Maintainability,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Microservices.
Solution Architecture Microservices: described as: Microservices-based software architecture for Health IoT, inspired by best practices from sources like Microsoft's microservices guidelines. The architecture utilizes open-source components for various functionalities. Key components include an API Gateway (Express Gateway) acting as a single entry point, a messaging channel (RabbitMQ) for asynchronous communication between microservices, and a security solution leveraging Vault and Consul for key management and high availability. The architecture is instantiated in the OCARIoT project, where it manages data from wearables, sensors, and mobile apps, processing data and triggering personalized interventions..
Interaction Capability is reached/addressed using/by User-friendly dashboard (Operability - User-friendly dashboard for health professionals: Health professionals are able to do a follow-up through the dashboard and also to receive tips and recommendations. Accessibility Open access through mobile and web platforms. The OCARIoT Dashboard visualizes real-time updates from mobile devices and web interfaces.).
Interaction Capability is reached/addressed using/by Intuitive mobile apps (Learnability - Intuitive mobile apps for health monitoring The Data Acquisition App manages the collection of data from different devices and shows key information for health professionals.).
Maintainability is reached/addressed using/by Component isolation through microservices (Modularity - Open Source Software Architecture Component isolation through microservices. Microservices can be implemented in different languages, using different databases, ensuring modular architecture.).
Reference: An open source software architecture and ready-to-use components for health IoT, 2020, https://ieeexplore.ieee.org/document/9183119.
The solution is composed by the architecture named as Multi-layered Cloud-Edge, proposed in the paper Securing smart healthcare system with edge computing, 2021, with D.O.I. doi.org/10.1016/j.cose.2021.102353 (link https://doi.org/10.1016/j.cose.2021.102353). This solution addresses the following Quality Requirements: Performance/Efficiency,Security,.
The IoT domain of this solution is Healthcare. You can find all information about this solution in Knowledge base, filtering by Healthcare and Multi-layered Cloud-Edge.
Solution Architecture Multi-layered Cloud-Edge: described as: .
Performance/Efficiency is reached/addressed using/by Edge computing (No description.).

Security is reached/addressed using/by Access Control System (ACS) and Searchable Encryption Module [Singh and Chatterjee 2021] (No description.).
Reference: Securing smart healthcare system with edge computing, 2021, https://doi.org/10.1016/j.cose.2021.102353.
The solution is composed by the architecture named as SAPPARCHI (Smart City), proposed in the paper SAPPARCHI: an Osmotic Platform to Execute Scalable Applications on Smart City Environments, 2021, with D.O.I. doi.org/10.1109/CLOUD55607.2022.00051 (link https://doi.org/10.1109/CLOUD55607.2022.00051). This solution addresses the following Quality Requirements: Flexibility,Maintainability,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and SAPPARCHI (Smart City).
Solution Architecture SAPPARCHI (Smart City): described as: .
Flexibility is reached/addressed using/by Serverless on Edge and Microservice in Fog (No description.).
Maintainability is reached/addressed using/by Serverless on Edge and Microservice in Fog (No description.).
Security is reached/addressed using/by Serverless on Edge and Microservice in Fog (No description.).
Reference: SAPPARCHI: an Osmotic Platform to Execute Scalable Applications on Smart City Environments, 2021, https://doi.org/10.1109/CLOUD55607.2022.00051.
The solution is composed by the architecture named as SDN based, proposed in the paper An efficient Counter Based DDOS attack detection framework leveraging software defined IoT (SD-IoT), 2020, with D.O.I. doi.org/10.1109/ACCESS.2020.3043082 (link https://doi.org/10.1109/ACCESS.2020.3043082). This solution addresses the following Quality Requirements: Security,.
The IoT domain of this solution is Generic. You can find all information about this solution in Knowledge base, filtering by Generic and SDN based.
Solution Architecture SDN based: described as: .
Security is reached/addressed using/by Counter-based DDoS Attack Detection (C-DAD) application for Software Defined Networks (SDN) [Bhayo et al. 2020] (No description.).
Reference: An efficient Counter Based DDOS attack detection framework leveraging software defined IoT (SD-IoT), 2020, https://doi.org/10.1109/ACCESS.2020.3043082.
The solution is composed by the architecture named as Secure Publish-Subscribe, proposed in the paper Publish Subscribe System Security Requirement: A Case Study for V2V Communication, 2024, with D.O.I. 10.1109/OJCS.2024.3442921 (link https://www.scopus.com/record/display.uri?eid=2-s2.0-85201290507&doi=10.1109%2fOJCS.2024.3442921&origin=inward&txGid=43ee5fbd5bae9bbdf927871aab51aa4d). This solution addresses the following Quality Requirements: Reliability,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Secure Publish-Subscribe.
Solution Architecture Secure Publish-Subscribe: described as: This solution focuses on ensuring secure, reliable, and efficient communication between vehicles using a publish-subscribe messaging system. It integrates brokers, clients, and publishers/subscribers with advanced security features such as access control, encryption, authentication, and message integrity verification..
Reliability is reached/addressed using/by Duplicate ClientID Handling (Availability - The broker incorporates a security measure called 'Handling Duplicate ClientID' to guarantee the continuous availability of services.).
Security is reached/addressed using/by Cryptography (Encryption) (Confidentiality - The publisher will employ cryptographic techniques and key sizes to ensure the confidentiality of the payload data throughout the whole communication process.).
Security is reached/addressed using/by Data Origin Authentication (Authenticity - The subscriber should have the capability to offer the end-user a method, such as a digital signature or Message Authentication Code (MAC), to authenticate the origin of information.).
Security is reached/addressed using/by Digital Signatures and Hashing (Integrity - The publisher transmits the information to the subscriber... using digital signature, MAC, and hashing to identify any abnormalities.).
Reference: Publish Subscribe System Security Requirement: A Case Study for V2V Communication, 2024, https://www.scopus.com/record/display.uri?eid=2-s2.0-85201290507&doi=10.1109%2fOJCS.2024.3442921&origin=inward&txGid=43ee5fbd5bae9bbdf927871aab51aa4d.
The solution is composed by the architecture named as Smart City, proposed in the paper SmartGC: a software architecture for garbage collection in smart cities, 2020, with D.O.I. dx.doi.org/10.1504/IJBIC.2020.109675 (link https://dx.doi.org/10.1504/IJBIC.2020.109675). This solution addresses the following Quality Requirements: Compatibility,Flexibility,Maintainability,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Smart City.
Solution Architecture Smart City: described as: .
Compatibility is reached/addressed using/by REST services (No description.).
Flexibility is reached/addressed using/by REST services (No description.).
Maintainability is reached/addressed using/by FIWARE Platform [www.fiware.org] (No description.).
Security is reached/addressed using/by HTTP reverse proxy to protect REST services (No description.).
Reference: SmartGC: a software architecture for garbage collection in smart cities, 2020, https://dx.doi.org/10.1504/IJBIC.2020.109675.
The solution is composed by the architecture named as Smart City, proposed in the paper Design and application of fog computing and Internet of Things service platform for smart city, 2020, with D.O.I. doi.org/10.1016/j.future.2020.06.016 (link https://doi.org/10.1016/j.future.2020.06.016). This solution addresses the following Quality Requirements: Flexibility,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Smart City.
Solution Architecture Smart City: described as: .
Security is reached/addressed using/by Proxy Virtual Processor[Zhang 2020] (No description.).
Flexibility is reached/addressed using/by The implementation of a software-defined network (SDN) to enable the dynamic allocation of network

resources according to demand [Zhang 2020] (No description.).
Reference: Design and application of fog computing and Internet of Things service platform for smart city, 2020, https://doi.org/10.1016/j.future.2020.06.016.
The solution is composed by the architecture named as Unicorn, proposed in the paper An Architecture for a Large-Scale IoT e-Mobility Solution, 2023, with D.O.I. 10.5220/0011827100003467 (link https://www.scitepress.org/Link.aspx?doi=10.5220/0011827100003467). This solution addresses the following Quality Requirements: Flexibility,Performance/Efficiency,Reliability,Security,.
The IoT domain of this solution is Smart City. You can find all information about this solution in Knowledge base, filtering by Smart City and Unicorn.
Solution Architecture Unicorn: described as: An Edge-Based Microservices Architecture. This design leverages microservices deployed across edge devices to ensure distributed processing, scalability, and efficient resource utilization. The architecture supports real-time data analysis and mobility management while reducing latency through local edge-based processing. It integrates services using REST APIs and enables dynamic service scaling to handle large-scale deployments in e-mobility environments..
Reliability is reached/addressed using/by Unicorn Cloud Framework (uuCloud) (Availability - Application availability requirement is 99.97% with a 2-hour RTO (Recovery Time Objective). This requirement is achieved using the Unicorn Cloud
Framework (uuCloud) that isolates failures to a specific microservice and supports automatic failover.).
Flexibility is reached/addressed using/by Unicorn Cloud Framework (uuCloud) (Adaptability - Cloud-Agnostic Deployment: Individual microservices are provisioned in the form of Docker containers and deployed to the Microsoft Azure public cloud infrastructure, but could be deployed to another compatible cloud platform.).
Performance/Efficiency is reached/addressed using/by Unicorn Cloud Framework (uuCloud) (Time Behavior - Load Balancer (uuGateway).).
Security is reached/addressed using/by Unicorn Cloud Framework (uuCloud) (Identity and Access Management - Access to individual microservices is controlled using identity (uuIdentity) associated with user roles.).
Reference: An Architecture for a Large-Scale IoT e-Mobility Solution, 2023, https://www.scitepress.org/Link.aspx?doi=10.5220/0011827100003467.

ALL SOLUTIONS

Total of 35 solutions found.

A4IoT (Anyplace 4.0 IoT): It is an open-source architecture designed for IoT-based indoor localization using technologies like Wi-Fi, BLE, Cellular, UWB, and Computer Vision (CV). The system supports edge computing, timeseries databases, and crowdsourcing to manage localization tasks across various smart spaces, such as factories, healthcare facilities, and emergency response environments. Its modular and scalable design ensures integration across platforms like Android, Linux, Windows, and Robot OS, making it adaptable to diverse IoT ecosystems.

Total of 35 solutions found.

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Agricultural IoT Reference Architecture (AITRA):

Total of 35 solutions found.

A4IoT (Anyplace 4.0 IoT): It is an open-source architecture designed for IoT-based indoor localization using technologies like Wi-Fi, BLE, Cellular, UWB, and Computer Vision (CV). The system supports edge computing, timeseries databases, and crowdsourcing to manage localization tasks across various smart spaces, such as factories, healthcare facilities, and emergency response environments. Its modular and scalable design ensures integration across platforms like Android, Linux, Windows, and Robot OS, making it adaptable to diverse IoT ecosystems.

Agricultural IoT Reference Architecture (AITRA):

Blockchain-based Zero Trust: The architecture proposed in "Blockchain-Based Zero Trust on the Edge" integrates Zero Trust principles with Blockchain to ensure secure access and trust management for Edge Computing in IoT environments. It focuses on continuous verification of users and devices, ensuring that all access is authenticated before being granted. Blockchain is used for decentralized trust management, maintaining an immutable record of activities, which enhances security and transparency. This setup is optimized for edge computing, reducing latency by processing data locally while maintaining the flexibility to scale across diverse IoT applications, such as smart cities and industrial IoT.

Total of 35 solutions found.

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Blockchain-enabled:

Total of 35 solutions found.

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Blockchain-enabled:

Cloud-based Secure Biometric:

Total of 35 solutions found.

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Blockchain-based Zero Trust: The architecture proposed in "Blockchain-Based Zero Trust on the Edge" integrates Zero Trust principles with Blockchain to ensure secure access and trust management for Edge Computing in IoT environments. It focuses on continuous verification of users and devices, ensuring that all access is authenticated before being granted. Blockchain is used for decentralized trust management, maintaining an immutable record of activities, which enhances security and transparency. This setup is optimized for edge computing, reducing latency by processing data locally while maintaining the flexibility to scale across diverse IoT applications, such as smart cities and industrial IoT.

Blockchain-enabled:

Cloud-based Secure Biometric:

Deep Learning based:

Total of 35 solutions found.

A4IoT (Anyplace 4.0 IoT): It is an open-source architecture designed for IoT-based indoor localization using technologies like Wi-Fi, BLE, Cellular, UWB, and Computer Vision (CV). The system supports edge computing, timeseries databases, and crowdsourcing to manage localization tasks across various smart spaces, such as factories, healthcare facilities, and emergency response environments. Its modular and scalable design ensures integration across platforms like Android, Linux, Windows, and Robot OS, making it adaptable to diverse IoT ecosystems.

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Blockchain-enabled:

Cloud-based Secure Biometric:

Deep Learning based:

Distributed Blockchain-SDN based:

Total of 35 solutions found.

A4IoT (Anyplace 4.0 IoT): It is an open-source architecture designed for IoT-based indoor localization using technologies like Wi-Fi, BLE, Cellular, UWB, and Computer Vision (CV). The system supports edge computing, timeseries databases, and crowdsourcing to manage localization tasks across various smart spaces, such as factories, healthcare facilities, and emergency response environments. Its modular and scalable design ensures integration across platforms like Android, Linux, Windows, and Robot OS, making it adaptable to diverse IoT ecosystems.

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Blockchain-enabled:

Cloud-based Secure Biometric:

Deep Learning based:

Distributed Blockchain-SDN based:

Distributed Edge-Cloud:

Total of 35 solutions found.

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Blockchain-enabled:

Cloud-based Secure Biometric:

Deep Learning based:

Distributed Blockchain-SDN based:

Distributed Edge-Cloud:

Edge Computing-based Fault Tolerant Framework:

Total of 35 solutions found.

A4IoT (Anyplace 4.0 IoT): It is an open-source architecture designed for IoT-based indoor localization using technologies like Wi-Fi, BLE, Cellular, UWB, and Computer Vision (CV). The system supports edge computing, timeseries databases, and crowdsourcing to manage localization tasks across various smart spaces, such as factories, healthcare facilities, and emergency response environments. Its modular and scalable design ensures integration across platforms like Android, Linux, Windows, and Robot OS, making it adaptable to diverse IoT ecosystems.

Agricultural IoT Reference Architecture (AITRA):

Blockchain-based Zero Trust: The architecture proposed in "Blockchain-Based Zero Trust on the Edge" integrates Zero Trust principles with

Blockchain to ensure secure access and trust management for Edge Computing in IoT environments. It focuses on continuous verification of users and devices, ensuring that all access is authenticated before being granted. Blockchain is used for decentralized trust management, maintaining an immutable record of activities, which enhances security and transparency. This setup is optimized for edge computing, reducing latency by processing data locally while maintaining the flexibility to scale across diverse IoT applications, such as smart cities and industrial IoT.

Blockchain-enabled:

Cloud-based Secure Biometric:

Deep Learning based:

Distributed Blockchain-SDN based:

Distributed Edge-Cloud:

Edge Computing-based Fault Tolerant Framework:

Event-driven IoT:

Total of 35 solutions found.

A4IoT (Anyplace 4.0 IoT): It is an open-source architecture designed for IoT-based indoor localization using technologies like Wi-Fi, BLE, Cellular, UWB, and Computer Vision (CV). The system supports edge computing, timeseries databases, and crowdsourcing to manage localization tasks across various smart spaces, such as factories, healthcare facilities, and emergency response environments. Its modular and scalable design ensures integration across platforms like Android, Linux, Windows, and Robot OS, making it adaptable to diverse IoT ecosystems.

Agricultural IoT Reference Architecture (AITRA):

Blockchain-based Zero Trust: The architecture proposed in "Blockchain-Based Zero Trust on the Edge" integrates Zero Trust principles with Blockchain to ensure secure access and trust management for Edge Computing in IoT environments. It focuses on continuous verification of users and devices, ensuring that all access is authenticated before being granted. Blockchain is used for decentralized trust management, maintaining an immutable record of activities, which enhances security and transparency. This setup is optimized for edge computing, reducing latency by processing data locally while maintaining the flexibility to scale across diverse IoT applications, such as smart cities and industrial IoT.

Blockchain-enabled:

Cloud-based Secure Biometric:

Deep Learning based:

Distributed Blockchain-SDN based:

Distributed Edge-Cloud:

Edge Computing-based Fault Tolerant Framework:

Event-driven IoT:

Fog Computing based:

Total of 35 solutions found.

A4IoT (Anyplace 4.0 IoT): It is an open-source architecture designed for IoT-based indoor localization using technologies like Wi-Fi, BLE, Cellular, UWB, and Computer Vision (CV). The system supports edge computing, timeseries databases, and crowdsourcing to manage localization tasks across various smart spaces, such as factories, healthcare facilities, and emergency response environments. Its modular and scalable design ensures integration across platforms like Android, Linux, Windows, and Robot OS, making it adaptable to diverse IoT ecosystems.

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Blockchain-enabled:

Cloud-based Secure Biometric:

Deep Learning based:

Distributed Blockchain-SDN based:

Distributed Edge-Cloud:

Edge Computing-based Fault Tolerant Framework:

Event-driven IoT:

Fog Computing based:

Geographic-based:

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Geographic-based:

Hybrid (Centralized and Distributed):

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Fog Computing based:

Geographic-based:

Hybrid (Centralized and Distributed):

Hybrid Cloud-Fog Computing:

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Fog Computing based:

Geographic-based:

Hybrid (Centralized and Distributed):

Hybrid Cloud-Fog Computing:

Industrial Internet of Things (IIoT):

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Hybrid Cloud-Fog Computing:

Industrial Internet of Things (IIoT):

Layered:

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Industrial Internet of Things (IIoT):

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Layered:

Layered:

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Layered:

Layered:

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Layered:

Layered:

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Industrial Internet of Things (IIoT):

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Layered:

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Hybrid (Centralized and Distributed):

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Industrial Internet of Things (IIoT):

Layered:

Layered:

Layered:

Layered:

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Layered Blockchain and AI-enabled:

Layered Blockchain-Based SDN:

LoRaWAN:

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Layered:

Layered:

Layered:

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Layered Blockchain-Based SDN:

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Microservices:

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Microservices: Microservices-based software architecture for Health IoT, inspired by best practices from sources like Microsoft's microservices guidelines. The architecture utilizes open-source components for various functionalities. Key components include an API Gateway (Express Gateway) acting as a single entry point, a messaging channel (RabbitMQ) for asynchronous communication between microservices, and a security solution leveraging Vault and Consul for key management and high availability. The architecture is instantiated in the OCARIoT project, where it manages data from wearables, sensors, and mobile apps, processing data and triggering personalized interventions.

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Multi-layered Cloud-Edge:

Total of 35 solutions found.

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Blockchain-enabled:

Cloud-based Secure Biometric:

Deep Learning based:

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Edge Computing-based Fault Tolerant Framework:

Event-driven IoT:

Fog Computing based:

Geographic-based:

Hybrid (Centralized and Distributed):

Hybrid Cloud-Fog Computing:

Industrial Internet of Things (IIoT):

Layered:

Layered:

Layered:

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Loosely Coupled IoT Platform and Localization Module: The proposed system architecture consists of two main components: the IoT platform and the localization module, connected by middleware. The IoT platform follows a hierarchical structure with layers for hardware (Bluetooth devices), data (MQTT-based RSSI parsing), business logic, display, and user interface. The localization module enhances trilateration with self-adaptive mechanisms, including an LSTM-based distance estimator, elastic radius intersecting, weighted centroid localization, and self-adaptive Kalman filtering for trajectory optimization.

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Smart City:

Smart City:

Unicorn: An Edge-Based Microservices Architecture. This design leverages microservices deployed across edge devices to ensure distributed processing, scalability, and efficient resource utilization. The architecture supports real-time data analysis and mobility management while reducing latency through local edge-based processing. It integrates services using REST APIs and enables dynamic service scaling to handle large-scale deployments in e-mobility environments.

ALL IoT DOMAINS

Generic: null

Generic: null

Healthcare: null

Generic: null

Healthcare: null

Industry 4.0: null

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Smart City: null

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ALL ARCHITECTURES

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Blockchain-enabled:

Cloud-based Secure Biometric:

Deep Learning based:

Distributed Blockchain-SDN based:

Distributed Edge-Cloud:

Edge Computing-based Fault Tolerant Framework:

Event-driven IoT:

Fog Computing based:

Geographic-based:

Hybrid (Centralized and Distributed):

Hybrid Cloud-Fog Computing:

Industrial Internet of Things (IIoT):

Layered:

Layered:

Layered:

Layered:

Layered:

Layered:

Layered Blockchain and AI-enabled:

Layered Blockchain-Based SDN:

LoRaWAN:

Loosely Coupled IoT Platform and Localization Module: The proposed system architecture consists of two main components: the IoT platform and the localization module, connected by middleware. The IoT platform follows a hierarchical structure with layers for hardware (Bluetooth devices), data (MQTT-based RSSI parsing), business logic, display, and user interface. The localization module enhances trilateration with self-adaptive mechanisms, including an LSTM-based distance estimator, elastic radius intersecting, weighted centroid localization, and self-adaptive Kalman filtering for trajectory optimization.

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Smart City:

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Unicorn: An Edge-Based Microservices Architecture. This design leverages microservices deployed across edge devices to ensure distributed processing, scalability, and efficient resource utilization. The architecture supports real-time data analysis and mobility management while reducing latency through local edge-based processing. It integrates services using REST APIs and enables dynamic service scaling to handle large-scale deployments in e-mobility environments.

ALL QUALITY REQUIREMENTS

Total of 9 quality requirements.

Compatibility: Based on ISO/IEC 25010:2023

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ALL TECHNOLOGIES/FEATURES

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FIWARE Platform [www.fiware.org]: null

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Loosely coupled components: null

MapReduce framework: null

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Long-Range Wide Area Network (LoRaWAN): null

Loosely coupled components: null

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Loosely coupled components: null

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ALL TECHNOLOGIES/FEATURES

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